Architecture Diagrams Compendium

System diagrams extracted from the five architecture atlases

1. Recommender Systems — Architecture Diagrams

01. The Canonical Multi-Stage Recommender Architecture
02. Candidate Generation and Retrieval: Two-Tower Models, Negative Sampling, ANN Indexes and Multi-Source Retrieval
03. The Evolution of Ranking and CTR Models: From Logistic Regression to Scaling-Law Architectures
04. Multi-Task and Multi-Objective Ranking: From Shared-Bottom to Value Models
05. Sequential and User-History Modeling: From GRU4Rec to Lifelong Sequences and Foundation User Models
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17. Google & YouTube Case Studies: From Co-Visitation Graphs to Semantic IDs and LLM-Native Recommenders
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21. Twitter/X Case Study: From WTF and GraphJet to the 2023 the-algorithm and the 2026 Phoenix Transformer
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26. Marketplace and Travel Recommenders: Airbnb, Booking.com, Uber Eats, DoorDash and Others
27. The Chinese Ecosystem: Alibaba, Tencent, Meituan, Baidu, Huawei, JD, Xiaohongshu and the Architecture of Hyperscale Commerce and Content
28. News, Social and Content Platforms: From Click Co-visitation to Open Feed Protocols
29. LLMs in Recommender Systems (2023-2026): Rankers, Features, Agents, Distillation and Production Deployments
30. Scaling Laws and Hardware for Recommenders: Compute, Embeddings, Accelerators and Cost
31. Search Meets Recommendation: Learning-to-Rank, Personalized and Semantic Search, Hybrid Retrieval and Unified Models
32. Notification, Push and Email Recommendation: Volume, Fatigue and Timing
33. Content Understanding and Integrity in Recommender Systems: Multimodal Item Embeddings, Integrity Filters, Borderline Demotion, Spam and the Policy Layer
34. Open-Source Frameworks, Tooling, Benchmarks and Public Datasets
35. The Recommender System Design Interview Playbook: Feeds, PYMK, Ads, E-commerce and Notifications

36. Classic Algorithms Foundations: Neighborhoods, Factorization, Linear Autoencoders, Embeddings, and Why Baselines Still Win
37. Real-Time and Session-Based Recommendation: Streaming Updates, Online Learning, Nearline Pipelines and In-Session Intent

2. RAG — Architecture Diagrams

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03. Embedding Models for Retrieval: Commercial APIs, Open Families, Compression and Adaptation
04. Vector Databases and ANN Indexing: From In-Memory Graphs to Object-Storage-Native Engines
05. Hybrid Search and Reranking: Fusion, Learned Sparse, Late Interaction and Multi-Stage Cascades
06. Query Understanding and Transformation: HyDE, Multi-Query, RAG-Fusion, Step-Back, Decomposition, Conversational Rewriting, Routing and Self-Query
07. GraphRAG and Knowledge Graphs: Community Summaries, Graph Memory, and When Graphs Beat Vectors
08. Agentic and Iterative RAG: Self-RAG, CRAG, FLARE, IRCoT, Adaptive-RAG, ReAct, RL-Trained Search Agents and Agentic Search
09. Long Context vs RAG: Cache-Augmented Generation, Prompt and KV Caching, Context Rot, and Context Engineering
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13. Code Retrieval and RAG: Codebase Indexes, Repo Maps, and Agentic Search in Coding Assistants
14. Answer Engines and Web-Grounded LLMs: Perplexity, Google AI Overviews and AI Mode, Bing/Copilot, ChatGPT Search, Search APIs for Agents, Deep Research and Citation Systems
15. RAG Frameworks and Managed Services: From Orchestration Libraries to Retrieval-as-a-Service
16. RAG Security and Governance: Permission-Aware Retrieval, Injection, Poisoning, Privacy and Audit
17. Fine-tuning vs RAG and Training the Retrieval Stack
18. Long-Term Memory Systems for LLMs and Agents: MemGPT/Letta, Mem0, Zep/Graphiti, ChatGPT and Claude Memory, LangMem, and the Write/Read Policies Behind Them
19. Structured-Data RAG: Text-to-SQL Architectures, Semantic Layers, Table QA and Schema Retrieval
20. Production RAG Operations: Ingestion, Freshness, Migrations, Latency, Cost, Observability, Caching and Debugging
21. RAG System Design Interview Playbook: Enterprise Q&A, Support Bots, Code Assistants, and Web Answer Engines
22. Master Index of RAG Engineering Blogs, Talks and Guides
23. Canonical RAG and Retrieval Paper Reading List: 100+ Papers (2017-2026) by Theme, with Venue, Link and System-Design Takeaway
24. Domain-Specific RAG for Regulated and Expert Fields: Medicine, Law, Finance and Science
25. Generative Retrieval and Model-Based Indexing: DSI, NCI, SEAL, TIGER, CorpusBrain and Their Hybrids
26. Tool Retrieval: RAG over APIs, Function Catalogs and MCP Servers
27. Multilingual and Cross-Lingual RAG: Benchmarks, Embeddings, Language Bias, Tokenization and Sovereign Deployments
28. On-Device, Edge and Air-Gapped RAG: Small Models, Embedded Indexes and Private Compute
29. The Generation Side of RAG: Knowledge Conflicts, Robustness, Compression, Attribution, and Abstention
30. Algorithm-System Co-Design for RAG Inference: Overlap, Reuse, Compression, and Placement
31. Real-Time and Time-Aware RAG: Streaming Ingestion, Temporal Retrieval and Staleness Evaluation
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3. LLM Evaluation — Architecture Diagrams

01. Evaluation Foundations and Taxonomy: Purposes, Modes, and the Eval Stack
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04. LLM-as-a-Judge: Protocols, Biases, Calibration, Ensembles, Fine-Tuned Judges and Judge Alignment
05. Human Evaluation and Arenas: Pairwise Preference Platforms, Their Statistics, Their Critics, and Expert Annotation Pipelines
06. Agent Evaluation Architectures: Trajectories, Outcomes, Sandboxes and Reliability
07. Product Eval Practice: Eval-Driven Development, Error Analysis, Golden Datasets, Evals in CI and Validating the Validators
08. Online Evaluation and Production Monitoring: Tracing, Sampled Grading, Experiments, Feedback Signals, Guardrails and Drift
09. Safety and Dangerous-Capability Evaluation: Frontier Safety Frameworks, Third-Party Testing, Red-Teaming Pipelines, Jailbreak Benchmarks, Sandbagging and Evaluation Awareness
10. Benchmark Validity and Statistical Rigor: Contamination, Saturation, Construct Validity, Error Bars, and Reproducible Leaderboards
11. Factuality, Hallucination and RAG Evaluation: From Atomic Facts to Grounding Judges
12. Code Evaluation in Depth: From Unit-Tested Functions to Repository Agents and In-IDE Assistants
13. Multimodal and Speech Evaluation: MMMU, MathVista, Video Benchmarks, Image Generation Evals, and Speech, ASR, TTS and Voice-Agent Evals
14. Evals as Training Signal: Reward Models, Verifiers, Rubric Rewards, Process Supervision and Reward-Hacking Detection
15. Company Eval Systems in Practice: What Labs and Product Companies Have Published (2023-2026)
16. Synthetic Data for Evaluation: LLM-Generated Test Cases, Adversarial Generation, Persona and User Simulation, and Dataset Curation
17. Eval System Design Interview Playbook: Platforms, Product Evals, Metrics, Sampling, and Cost

4. Closed-Loop AI Systems — Architecture Diagrams

01. Foundations: Feedback Loops, Technical Debt, and the Anatomy of Closed-Loop ML Systems
02. Data Engines and Flywheels: Fleet Learning Architectures from Tesla to NVIDIA
03. Preference-Learning Loops: RLHF, Constitutional AI/RLAIF, DPO, and Online Reward-Model Refresh
04. Self-Improvement via Verifiable Rewards: From STaR and Self-Play to GRPO, Reasoning RL Infrastructure, and Self-Verifying Provers
05. Online and Continual Learning in Production: Streaming Recommenders, Bandit Loops, Forgetting, Continual Pretraining and Test-Time Training
06. Human-in-the-Loop and Active Learning: Labeling Platforms, Weak Supervision, Annotation Workflows, and Expert Feedback Pipelines
07. Production Traces to Fine-Tuning Loops: Logging, Evaluation, Curation, Distillation and Routing
08. Self-Improving Agents and Automated Optimization: Reflection, Skill Libraries, Evolutionary Program Search, Self-Modifying Agents and Prompt Optimizers
09. Closed-Loop Robotics and Control: Sim-to-Real Loops, Robot Foundation Model Data Flywheels, Teleoperation, DAgger, and World Models
10. Closed-Loop Scientific Discovery: Self-Driving Labs, Active Learning, AI Scientists and Drug Design Engines
11. Risks of Closed Loops: Model Collapse, Reward Hacking, Filter Bubbles, Goodhart's Law and Sycophancy
12. Product Case Studies: Closed Loops in Shipped Products, from Copilot and Cursor to Netflix, Duolingo and Gmail
13. Infrastructure for Closed Loops: Continuous Training, Logging, Monitoring, Registries and Safe Rollout
14. Master Index of Blogs and Talks on Data Flywheels, Feedback Loops, RLHF and Self-Improving Systems: 170 Annotated Entries
15. The Canonical Closed-Loop Reading List: 110+ Papers on RLHF, Self-Improvement, Verifiable Rewards and Online Learning

5. AI Agents — Architecture Diagrams

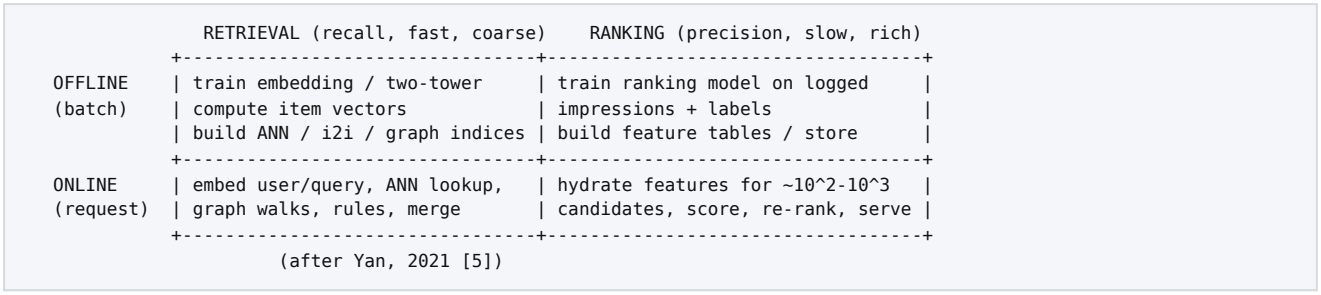
01. Agent Foundations and Taxonomy: Workflows, Agents, ReAct, and Levels of Autonomy
02. The Agent Loop and Tool Use: Function Calling, Tool Design, Selection at Scale, and Code-as-Action
03. Planning and Reasoning Architectures: Plan-and-Execute, ReWOO, LLMCompiler, Tree and Graph Search, LATS, Reflection, and Reasoning Models with Interleaved Thinking
04. Multi-Agent Systems: Orchestrators, Teams, Handoffs, and Why They Fail

05. Context Engineering and Memory: Caches, Compaction, Scratchpads, and Long-Horizon State
06. Coding Agents: Architectures, Sandboxes, and Verification Loops
07. Computer and Browser Use Agents: Pixels, Accessibility Trees, and the Race from 15% to Human-Level on OSWorld
08. Deep Research Agents: Architectures, Products, and Training
09. Agent Frameworks and SDKs: LangGraph, OpenAI Agents SDK, Claude Agent SDK, Google ADK, Pydantic AI, smolagents, Mastra, Strands, Microsoft Agent Framework, LlamaIndex, DSPy
10. Agent Infrastructure and Runtime: Sandboxes, Durable Execution, State, Observability, Deployment, and Cost
11. Agent Protocols: MCP, A2A, ACP, AGNTCY, Agent Payments, Tool Registries, and Interop Patterns
12. Agent Security and Safety: Indirect Prompt Injection, the Lethal Trifecta, CaMeL and Dual-LLM Patterns, Permissions, Sandboxing, Approval Gates, OWASP, and MCP Threats
13. Agent Evaluation: Benchmarks, Trajectory Grading, pass^k Reliability, and Eval Harness Design
14. Training Agents: Agentic RL, Environment Design, Tool-Use Fine-Tuning, RL for Software Engineering, and RL Gyms
15. Customer Support and Enterprise Agents: Sierra, Decagon, Fin, Klarna, Agentforce, ServiceNow, Zendesk, and the Guardrail Stack
16. Agent Case Studies from Engineering Blogs: Replit, Vercel v0, Ramp, Shopify, Uber, LinkedIn, Airbnb, Stripe, Notion, Canva, Booking.com, DoorDash and Anthropic
17. Voice and Realtime Agents: Speech-to-Speech vs Cascaded Architectures, Turn-Taking and Latency Budgets
18. Agent System Design Interview Playbook: Coding, Customer Support, Research and Browser Agents

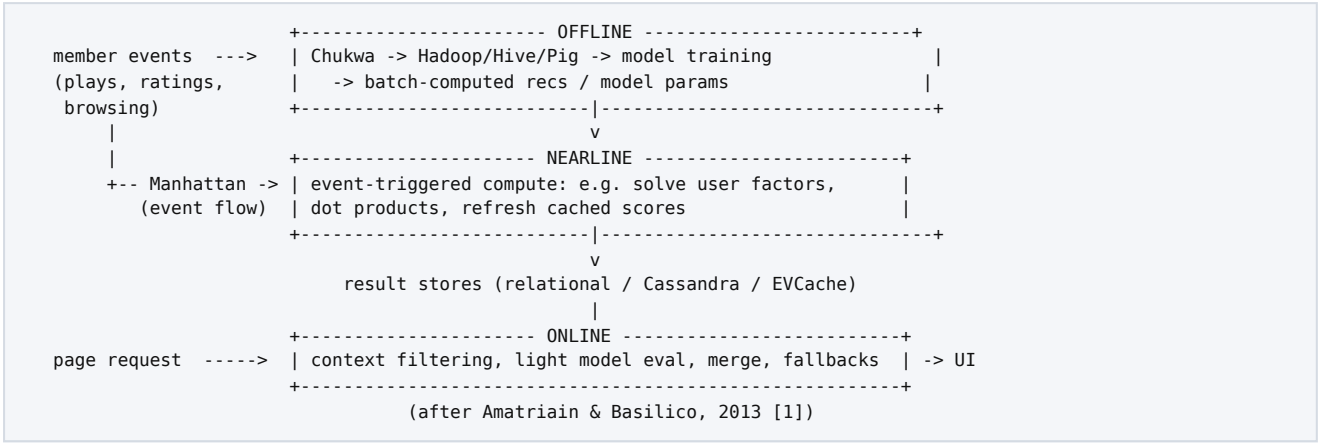
1. Recommender Systems — Architecture Diagrams

01. The Canonical Multi-Stage Recommender Architecture

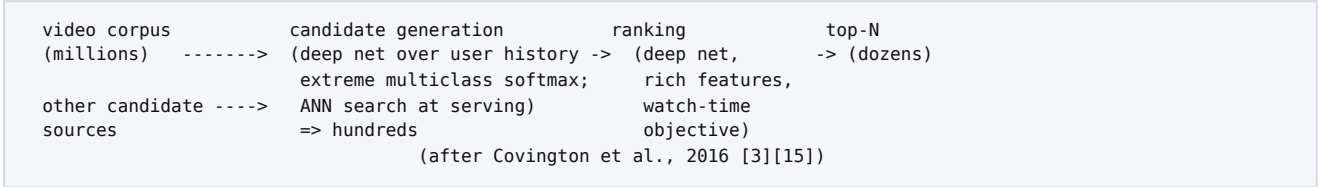
1.1 The cost argument



2.2 Architecture



3.2 Architecture



4.1 Instagram Explore, 2019: the three-pass distillation funnel

```
ig2vec account KNN + other sources
          |
          v
 ~500 candidates
          |   pass 1: distillation model (minimal features) mimics later stages
          v
 ~150 candidates
          |   pass 2: lightweight neural network
          v
 ~50 candidates
          |   pass 3: full deep neural network (all dense features)
          v
 ~25 shown first (after integrity/diversity rules)
                              (after Instagram, 2019 [7]; counts approximate)
```

4.2 Instagram Explore, 2023: four stages, caching everywhere

```
 +-------------+   up to   +-----------------+   top    +-----------------+   +--------+
 | Retrieval   |  ~1,500   | Early-stage     |  ~100    | Late-stage      |-->| Final  |
 | heuristics, |---------->| ranking (two-   |--------->| ranking (MTML,  |   | rerank |
 | two-tower   |           | tower, distilled|          | value model)    |   | rules, |
 | ANN, ...    |           | from LSR)       |          |                 |   | divers.|
 +-------------+           +-----------------+          +-----------------+   +--------+
   item tower: daily offline embeddings; user tower: online, fresh features
                                  (after Meta 2023 [8] and transparency page [9])
```

5.1 2023 architecture

```
request -> Home Mixer (Product Mixer framework)
             |
             +-- in-network:  Earlybird index + light ranker (RealGraph) --+
             +-- out-of-net:  UTEG/GraphJet, SimClusters ANN, TwHIN ANN ---+--> ~1,500
             |                                                              |
             +--> feature hydration --> Heavy Ranker (~48M param MTL NN) <-+
             |
             +--> visibility filtering, author diversity, dedup, fatigue
             +--> mix ads / WTF modules --> For You timeline
                              (after Twitter 2023 [11][12])
```

6. Pinterest: candidate generators, a light-weight scorer and a blender

```
Pixie walks | HNSW embeds | token idx | explore  --> merge+dedup --> LWS (light)
                                                                      |
                                    Utility Prediction Scorer (MTL) <-+
                                                  |
                               Multi-Objective Blender --> home feed
                                   (after Pinterest WWW 2022 [14], [16])
```

8.3 Kuaishou: a six-stage cascade, quantified

```
 corpus (10^8+)
    | retrieval: many channels (i2i, u2i, transformer retrieval e.g. KuaiFormer)
    v
 ~8,000 merged candidates
    | pre-ranking / coarse ranking
    v
 ~500
    | cascade ranking -> full (fine) ranking (multi-task)
    v
 ~tens
    | re-ranking (listwise, diversity, business)
    v
 ~6-10 exposed
                            (after RecFlow [22], UniRec [23]; approximate)
```

10. Request flow, end to end

```
client --> API/gateway --> mixer/orchestrator (per-surface pipeline)
   1. fetch user context: profile, recent actions (real-time feature store)
   2. fan out to N candidate sources in parallel, each with its own timeout
        two-tower ANN | i2i / graph walk | follow graph | trending | rules
   3. merge, dedup, eligibility & integrity filter          (~10^3-10^4)
   4. pre-rank: cached / distilled / two-tower or small DNN (-> ~10^2-10^3)
   5. hydrate rich features (batch RPC to feature store, embeddings)
   6. rank: multi-task NN -> P(event_i); value model sum w_i * P_i (-> ~10^1-10^2)
   7. re-rank: diversity, freshness, fatigue, policy, ads/module insertion
   8. log impressions + features served ("log-and-wait") for training
   9. respond; async side effects (cache next page, update seen-sets)
```

02. Candidate Generation and Retrieval: Two-Tower Models, Negative Sampling, ANN Indexes and Multi-Source Retrieval

Overview

```
                  THE CANONICAL RETRIEVAL FUNNEL
corpus: 1e6 .. 1e10 items
  |
  |  [Retrieval / candidate generation]   many sources in parallel
  |    - two-tower EBR (ANN over item embeddings)
  |    - item-to-item CF / co-engagement tables
  |    - graph walks (Pixie, GraphJet/UTEG, PPR)
  |    - social graph (friends-of-friends, in-network)
  |    - heuristic (trending, fresh, location)
  v
~1e3 .. 1e4 candidates (dedup + merge + filter)
  |
  |  [Pre-rank / lightweight scoring]  often a two-tower too
  v
~1e2 .. 1e3
  |
  |  [Heavy ranking]  multi-task DNN with cross features
  v
~1e1 .. 1e2 shown, after re-ranking for diversity, policy, ads
```

1.1 YouTube DNN candidate generation (Google, 2016)

```
YouTube DNN candidate generation (2016)
user history + context features --> MLP (ReLU stack) --> user vector u
                                                              |
training:  softmax over all videos (approximated by sampled softmax)
serving :  top-N by dot(u, v_j) via nearest-neighbor index over v_j
```

1.2 The two-tower model with in-batch negatives and logQ correction (YouTube, 2019)

```
Two-tower training with in-batch softmax + logQ   (batch of B pairs)

query tower  q_i = f(user_i, context_i)          i = 1..B
item tower   v_j = g(item_j features)            j = 1..B
logits       s_ij = <q_i, v_j> / temperature  -  log p_j     (logQ)
loss         -sum_i log softmax_j(s_ij)[j = i]
p_j          streaming estimate from hashed "last-seen step" arrays
```

1.6 Facebook Search EBR: triplet loss, hard negatives and ANN tuning (KDD 2020)

```
Facebook EBR inside Unicorn (2020)

query + searcher context --> query tower --> q (e.g. 128-d)
                                                |
          Unicorn query:  (nn  q :radius ...)  AND  (term filters)
                                                |
inverted index:  term = coarse cluster id (IVF centroid)
                 payload = PQ code of residual
                                                |
           candidates --> later-stage ranking (embedding as a feature)
```

2.1 Instagram Explore retrieval (2023)

```
Instagram Explore (2023, per Meta)
 sources (weighted mix): heuristic | item-CF | PPR | two-tower | ...
      |  hundreds each, from billions
      v
 retrieval output: <= ~1500 media
      |  first-stage ranker = two-tower (cached user + item embeddings)
      v
 top 100 --> second-stage ranker (heavy MTML) --> final re-rank
```

Scale.

```
Pixie (graph retriever)
 query pins (weighted) --> N short random walks w/ restart on Pin<->Board graph
                           (single-machine RAM, ~120 GB pruned graph)
                       --> visit counts (early stop when top-k stabilizes)
                       --> top-k pins as candidates   (p99 < 60 ms)
```

4.1 LiNR: model-based neural retrieval on GPUs (2024)

```
LiNR (GPU "retrieval as a model")
 model binary = { user/query tower weights , item matrix [N x d] , attrs }
 request:  q = tower(member)          (GPU)
           mask = prefilter(attrs)    (GPU, exact)
           scores = q . items^T  over masked rows  (dense matmul / quantized)
           top-k --> ranking
```

5. Twitter/X: a portfolio of graph and embedding retrievers

```
Twitter "For You" candidate sourcing (2023)
 in-network  : follow graph --> RealGraph-weighted (LR) ---------\
 out-network : UTEG/GraphJet traversal ----------------------------+--> ~1500
               SimClusters sparse community vectors -------------|     merged
               TwHIN dense KG embeddings (ANN) ------------------/     to ranker
```

10. Multi-source retrieval and merging

```
Generic multi-source retrieval service
request --> fan-out (parallel, per-source timeout)
   |-- EBR source(s): user tower -> ANN (k per embedding) -----\
   |-- item-CF: recent items -> co-engagement table lookups ---|
   |-- graph walk: seeds -> random walks / traversals ---------+--> union
   |-- social: follow / FoF expansion --------------------------|   dedup
   |-- heuristic: trending / fresh / local ---------------------/   filter
                                                                     |
                     quotas / weights / round-robin / pre-ranker ------+
                                                                     v
                                                          1e2..1e3 to ranking
```

03. The Evolution of Ranking and CTR Models: From Logistic Regression to Scaling-Law Architectures

Overview

```
Evolution of CTR / ranking architectures (simplified)

2010  FM (Rendle) ---------------------------------------+
2013  LR + FTRL-Proximal (Google ads)                    |
2014  GBDT -> LR (Facebook ads)      FFM (Kaggle/Criteo) |
2016  Wide & Deep (Google Play)                          |
2017  DeepFM (Huawei)  DCN (Google)  DIN (Alibaba) <-----+ factorized / attention
2018  xDeepFM (CIN)  AutoInt (self-attn)  DIEN (AUGRU)
2019  DLRM (Facebook, open source)
2020  DCN-V2 (Google)  SIM (Alibaba, 54k-length sequences)
2021  MaskNet (Weibo) -> Twitter heavy ranker (2023)
2022  DHEN (Meta)
2023  TWIN (Kuaishou)
2024  Wukong (Meta)  HSTU (Meta)  LiRank (LinkedIn)  InterFormer (Meta)
2025  RankMixer (ByteDance)  GEM (Meta ads foundation model)
2026  TokenMixer-Large (ByteDance)  ML-DCN (Pinterest)  GEM training at LLM scale
```

Architecture and algorithm

```
impression stream --> feature extraction --> hashed / sparse feature ids
                                                  |
                                                  v
                       +------------------------------------------+
                       |  FTRL-Proximal logistic regression       |
                       |  per-coordinate state: z_i, n_i          |
                       |  L1 => exact zeros => sparse model       |
                       +------------------------------------------+
                                 |                    |
                            weights (q2.13, 16-bit)   calibration layer
                                 |                    |
                                 v                    v
                            serving replicas  -->  pCTR to auction
```

Architecture

```
raw features (dense + categorical)
         |
         v
+----------------------------+     trained in batch (e.g. daily)
|  Boosted trees T1 ... TK   |
+----------------------------+
  | leaf id   | leaf id    | leaf id
  v           v            v
[one-hot(T1) | one-hot(T2) | ... | one-hot(TK)]  + other sparse features
         |
         v
+----------------------------+     trained online, near real time
| Logistic regression (SGD)  |
+----------------------------+
         |
         v
    pCTR  --> recalibrate for negative downsampling --> auction
```

4.1 Wide & Deep (Google Play, 2016)

```
     wide (memorization)                    deep (generalization)
sparse ids + cross products         sparse ids -> embeddings (~32-d each)
           |                                  + dense features
           |                                      |
           |                                MLP (ReLU layers)
           |                                      |
           +----------- sum of logits -----------+
                              |
                        sigmoid -> P(app acquisition)
```

4.6 DLRM (Facebook/Meta, 2019)

```
dense features --> bottom MLP --> d-dim vector ----+
                                                   |
sparse feat 1 --> embedding table 1 --> d-dim -----+
sparse feat 2 --> embedding table 2 --> d-dim -----+--> pairwise dot products
...                                                |    (all pairs of vectors)
sparse feat N --> embedding table N --> d-dim -----+          |
                                                              v
                              concat(dot products, bottom-MLP output)
                                                              |
                                                         top MLP
                                                              |
                                                           sigmoid
```

5.1 DIN: Deep Interest Network (KDD 2018)

```
candidate ad emb ------------------------------+
                                               |
behavior_1 emb --[act unit(b1, ad)]-- w1 --+   |
behavior_2 emb --[act unit(b2, ad)]-- w2 --+-> sum_i w_i * b_i --> user interest
...                                        |                          |
behavior_T emb --[act unit(bT, ad)]-- wT --+                          |
                                                                       v
          concat(user interest, user profile, ad, context) --> MLP --> pCTR
```

5.3 SIM: Search-based Interest Model (CIKM 2020)

```
lifelong behaviors (up to ~54,000)
         |
         v
+----------------------------------------+
| General Search Unit (GSU)              |
|   hard search: behaviors with same     |
|     category as candidate (index lookup) |
|   soft search: embedding MIPS          |
+----------------------------------------+
         | top-K relevant behaviors (K ~ hundreds)
         v
+----------------------------------------+
| Exact Search Unit (ESU)                |
|   multi-head target attention          |
+----------------------------------------+
         |
         v
  concat with short-term DIN/DIEN features --> MLP --> pCTR
```

6.1 DHEN: Deep and Hierarchical Ensemble Network (Meta, 2022)

```
feature embeddings
      |
+---------------------------------------------+
| Layer 1: [DotProd] [SelfAttn] [DCN] [Linear]  |  -> ensemble combine + residual
+---------------------------------------------+
      |
+---------------------------------------------+
| Layer 2: [DotProd] [SelfAttn] [DCN] [Linear]  |
+---------------------------------------------+
      |
     ...  -> top MLP -> task heads
```

6.5 Meta GEM (2025-2026)

```
            +---------------------------------------+
            |  GEM foundation model (offline, huge)   |
            |  sequence + non-sequence interaction    |
            +---------------------------------------+
                |  knowledge transfer / distillation
      +---------+----------+----------------+
      v                    v                v
vertical model A      vertical model B   vertical model C   (served online,
(e.g. surface/CVR)    (e.g. CTR)         (...)              latency-bounded)
```

04. Multi-Task and Multi-Objective Ranking: From Shared-Bottom to Value Models

Overview

```
                +-------------------------------------------+
features  ---> |  multi-task ranker (shared-bottom / MMoE / |
(user, item,   |  PLE / MaskNet / transformer)             |
 context)      +-------------------------------------------+
                 | pClick | pLike | pShare | pHide | E[dwell] ...
                 v        v       v        v       v
                +-------------------------------------------+
                | calibration (per task: Platt / temperature |
                | / isotonic / learned isotonic layer)       |
                +-------------------------------------------+
                                |
                                v
                +-------------------------------------------+
                | value model / fusion:                     |
                |   score = sum_i w_i * p_i   (linear)      |
                |   score = prod_i p_i ^ w_i  (log-linear)  |
                |   w = f(user, request)      (personalized) |
                +-------------------------------------------+
                                |
                                v
                diversity / out-of-network / freshness
                multipliers, DPP rerank, integrity demotions
```

1.1 Shared-bottom (hard parameter sharing)

```
input embeddings -> [shared MLP] -> h
                                   |-> tower_click -> pClick
                                   |-> tower_like  -> pLike
                                   |-> tower_share -> pShare
```

1.2 MMoE: Multi-gate Mixture-of-Experts (Google, KDD 2018)

```
                 +----------+  +----------+       +----------+
x -------------> | expert 1 |  | expert 2 |  ...  | expert n |
|                +----------+  +----------+       +----------+
|                       \            |            /
|-> gate_A (softmax) ---> weighted sum_A --> tower_A --> y_A
|-> gate_B (softmax) ---> weighted sum_B --> tower_B --> y_B
```

1.3 YouTube "What Video to Watch Next" (RecSys 2019)

```
user/video/context features --> [MMoE: experts + per-objective gates]
                                       |              |
                                  engagement towers  satisfaction towers
                                       |              |
position, device -> [shallow tower] --+ (bias logit added at train only)
                                       v
            per-objective predictions --> weighted combination --> rank
```

Architecture.

```
Level 1:  [A-experts] [shared-experts] [B-experts]
            gate_A      gate_shared      gate_B
              |             |              |
Level 2:  [A-experts] [shared-experts] [B-experts]
            gate_A                       gate_B
              |                             |
           tower_A -> VTR (AUC)          tower_B -> VCR (MSE)
```

1.7 Sequential-dependency models: ESMM and successors

```
impression features
      |
 [shared embeddings]
   /           \
[CTR tower]   [CVR tower]
   |               |
 pCTR ----(x)---- pCVR
   |         |
loss(click)  pCTCVR -> loss(click & conversion)
```

2.2 X / xAI (2026): Phoenix transformer and the `xai-value-model`

```
Phoenix (transformer, candidate-isolated) -> ~25 heads
      -> weighted sum (param.rs weights) -> offset negatives
      -> + exploration bonus -> x author-diversity decay
      -> x out-of-network factor -> cold-start lift
      -> vm-ranker DPP rerank -> filters -> timeline
```

2.8 Kuaishou: from formula fusion to learned ensemble sort

```
ranking model --> pLTR, pWTR, pFTR, pCMTR, E[watch] ...
                       |
   +--------------------+--------------------+
   | 2018-era: hand formula, global weights  |
   | 2022+:    RL agent picks weights/user   |
   | 2025+:    Pareto policy trained jointly |
   +--------------------+--------------------+
                       v
                top-k by fused score
```

5.1 The workflow most companies converge on

```
1. Train calibrated MTL heads (one model, many heads).
2. Define the value function: score = sum w_i * p_i (+ multipliers).
3. Initialize w_i ~ 1 / mean(p_i) * importance_i  (equal-contribution
   heuristic, as in Twitter 2023 [11]).
4. Offline: replay logged sessions; sweep w; trace a Pareto front of
   proxy metrics (e.g. predicted likes vs predicted hides) [21][32].
5. Online: constrained Bayesian optimization or GP Thompson sampling
   over w, primary metric s.t. guardrails [22][26].
6. Personalize: learn w = f(user, request) with RL / policy
   optimization, keep a Pareto frontier for governance [32][35][36].
7. Re-check calibration on every model change; re-tune w on every
   product-goal change.
```

05. Sequential and User-History Modeling: From GRU4Rec to Lifelong Sequences and Foundation User Models

Overview

```
                  The user-history design space
+-------------------------------------------------------------------------+
|                 short (10^1-10^2)    long (10^3-10^4)    lifelong (10^5+) |
| retrieval       GRU4Rec, SASRec,     PinnerFormer,       PinFM, HSTU,     |
| (user tower)    BERT4Rec, KuaiFormer Netflix FM          Netflix FM       |
| ranking         DIN, BST, TransAct   MIMN, LONGER,       SIM, ETA, SDIM,  |
| (target-aware)                       TransAct V2         TWIN, TWIN V2    |
+-------------------------------------------------------------------------+
   cost grows with length x candidates x model depth -> most tricks attack this
```

Architecture.

```
other features (user profile, context, cross)        target item
        |                                                    |
   embeddings                      behaviour seq: item_1 .. item_n
        |                                      |  (item_id, category_id,
        |                                      |   position = t_now - t_click)
        |                          +-----------v-------------------+
        |                          | Transformer layer             |
        |                          | (self-attn over seq+target)|
        |                          +-------+-------------------+
        +-------------+------------------+
                      |  concat
                MLP (3 layers) -> sigmoid -> pCTR
```

3.2 SIM: Search-based Interest Model (Alibaba, CIKM 2020)

```
lifelong behaviours (10^4 - 10^5)
         |
+--------v------------------------------+
| GSU: General Search Unit               |
|  hard search: behaviours with the same  |
|   category as the candidate             |   offline: User Behavior Tree
|  soft search: MIPS on embeddings        |<--  key1 = user_id
+--------+------------------------------+     key2 = category_id
         | top-K sub-sequence (SBS)            value = behaviour items
+--------v------------------------------+     (distributed, ~22 TB)
| ESU: Exact Search Unit                 |
|  multi-head target attention on SBS     |
|  + time-interval embeddings             |
+--------+------------------------------+
         v
  concat with short-term DIN features -> MLP -> pCTR
```

3.4 TWIN: TWo-stage Interest Network (Kuaishou, KDD 2023)

```
per behaviour j:  K_j = W * inherent_j     (cached offline per item)
                  b_j = w . cross_j         (cheap, per user)
per candidate c:  score_j = (Q(c) . K_j) / sqrt(d) + b_j
CP-GSU: top-100 by score_j  ->  ESU: multi-head attention over top-100
```

Architecture.

```
last N actions (pin, action type, surface, time features)
     |  each pin: 256-d PinSage embedding (frozen) + learned feature embeddings
+----v-----------------------------+
| causal Transformer encoder         |
+----+-----------------------------+
     v
user embedding (L2-normalised), computed daily for every user
     |
     v  ANN retrieval / ranking feature, matched against pin embeddings
```

4.3 TransAct (KDD 2023): real-time actions inside the ranker

```
  real-time action stream (Kafka-like)                  candidate pin
        |                                                     |
  latest ~100 actions: PinSage emb + action type + time       |
        |                                                     |
  early fusion: concat each action with candidate embedding <-+
        |
  Transformer encoder (small: ~2 layers, reported)
        |
  output compression: first K tokens + max pooling -> flatten
        |
  concat with PinnerFormer user emb + other features -> DCN-style ranker
        |
  multi-head outputs (repin, click, hide, ...) -> utility score
```

Architecture.

```
lifelong seq (~16k, int8 PinSage, offline store)  real-time seq   impression seq
        \                                            |             /
         +------- nearest-neighbour selection by candidate -------+
                 (dot product in PinSage space; always keep
                  the most recent real-time actions)
                           |
                     short selected sequence
                           |
              Transformer (fused single-kernel)
                           |
            features -> ranker heads  +  Next Action Loss (aux)
```

8.1 Split the user part from the candidate part

```
request -> [ user history encoder ]  ---- once per request (or cached / batch)
                  |   KV cache / embedding
        +---------+----------+----------+
        v         v          v          v
     cand_1    cand_2  ...  cand_1000      cheap cross-attention or dot product
```

06. Generative Recommenders (2023-2026): HSTU, Semantic IDs, OneRec, MTGR and the Foundation-Model Turn

1. The conceptual shift: from DLRM to generative recommenders

```
Classical cascade (DLRM era)            Generative recommender (GR era)
----------------------------            -------------------------------
user feats ─┐                           [item0, act0, item1, act1, ... , itemN]
item feats ─┼─> Retrieval (two-tower)             | causal transformer (HSTU,
cross feats ┘      | ~10^3-10^4                    |  decoder-only, enc-dec)
                   v                               v
            Pre-rank (light DNN)        retrieval head: p(next item | history)
                   | ~10^2-10^3         ranking head:  p(action | history, item)
                   v                    or: decode semantic-ID tokens directly
            Rank (DLRM, MTL)                       |
                   | ~10^1-10^2                    v
                   v                    (optional) RL post-training vs reward model
            Re-rank / blend
```

2.3 The HSTU block

```
  X ──> Linear ──> SiLU ──> split ──> U, V, Q, K
                                        |
          SiLU(Q·K^T + rab(pos,time)) ⊙ V      <- no softmax; rel. pos+time bias
                                        |
                     LayerNorm(·) ⊙ U          <- gating replaces the FFN
                                        |
                          Linear ──> + X (residual)
```

3.2 Pipeline

```
item text/metadata ──> frozen text encoder (e.g., Sentence-T5) ──> dense emb e
                                                                      |
                      RQ-VAE: encoder -> z
                      level 1: c1 = argmin ||z - C1[k]||, r1 = z - C1[c1]
                      level 2: c2 = argmin ||r1 - C2[k]||, r2 = r1 - C2[c2]
                      level 3: c3 = ...
                                                                      |
                    Semantic ID = (c1, c2, c3 [, dedup token])  e.g. (10, 5, 42)
                                                                      |
user history = [SID(item1), SID(item2), ...] ──> Transformer seq2seq (T5-style)
                                                 ──> beam-decode SID of next item
                                                 ──> map SID -> item (lookup)
```

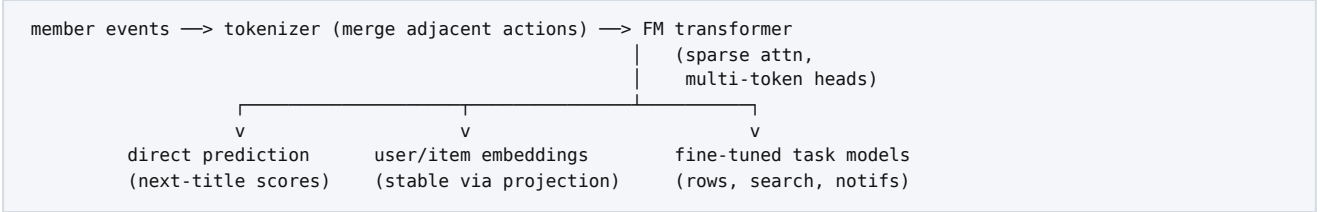
Tokenizer.

```
user static ─┐
short seq 20 ├─> Encoder (Transformer, RMSNorm) ──> 405 ctx tokens
pos-fb 256  ─┤                                          | cross-attn
lifelong 1e5 ─> hier-kmeans 2000 ─> QFormer 128 ─┘      v
                        [BOS] s1 s2 s3 ──> Decoder (causal + MoE FFN)
                                                        | beam search over SIDs
                                                        v
                      candidate sessions ──> Reward model (P-Score: CTR/LTR/VTR..)
                                                        |
                        preference alignment (DPO-style IPA in V1 paper;
                        ECPO = early-clipped GRPO in technical report)
```

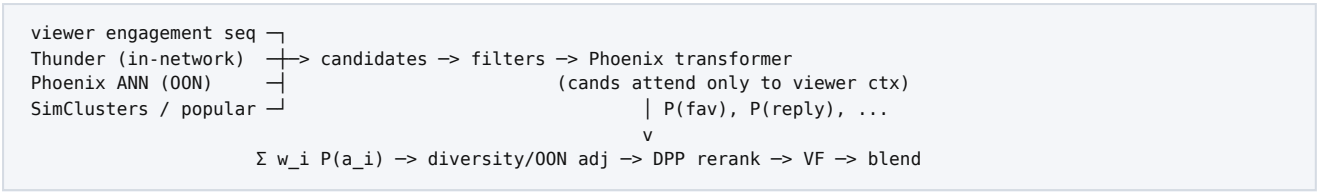
5.2 Architecture

```
[U1..Uk | S1 ... Sn | R1 ... Rr | C1  C2  ...  Cm]
  static   long seq   real-time   candidates (item ⊕ cross feats)
mask: U,S -> visible to all; R causal & time-gated per candidate; Ci ⊥ Cj
heads: CTR / CTCVR on C positions only
```

7.2 Design elements (as described in the blog; summarized via a public reimplementation's mapping)

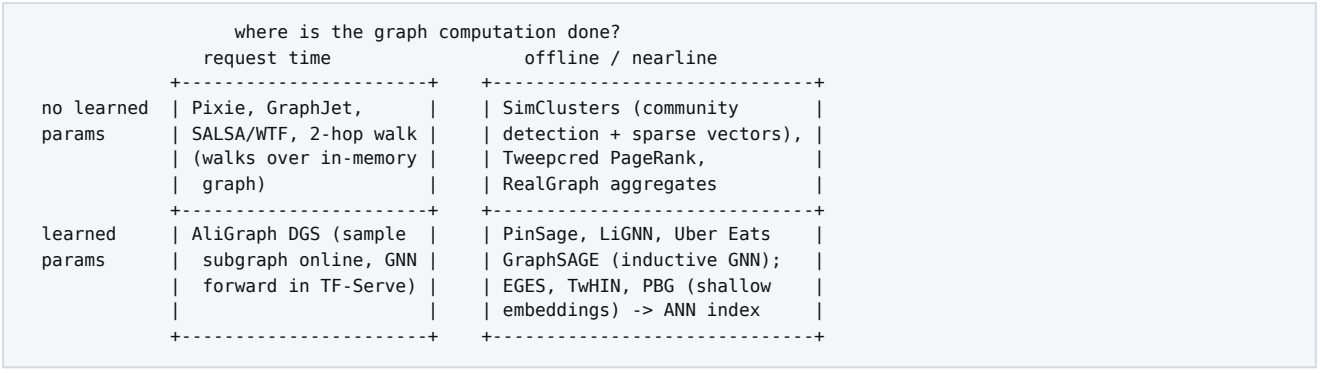


10. X (Twitter) For You: Phoenix transformer ranker

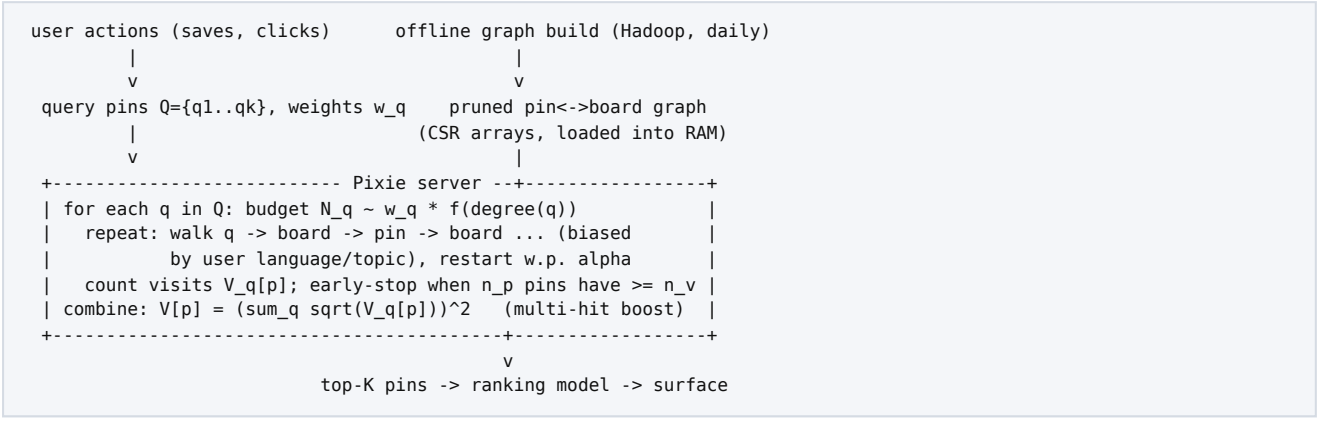


07. Graph-Based Recommendation in Production: Random Walks, Graph Embeddings and GNNs at Web Scale

1. A taxonomy of graph recommenders in production



Architecture



Architecture

```
          TRAINING (GPU, producer-consumer)
 CPU producers: sample minibatch of (query, positive) pin pairs
                -> random-walk neighbourhoods (top-T by visit count)
                -> gather features (visual, text, ...)  ---+
                                                           v
 GPU consumer:  K=2 conv layers; importance pooling (weights =
                normalised visit counts); max-margin loss with
                random + curriculum "hard" negatives
                                                           |
          INFERENCE (MapReduce, no repeated computation)  v
 layer-1 embeddings for all nodes -> join -> layer-2 embeddings
                 -> pin embeddings -> ANN index (related pins, homefeed)
```

4. Twitter/X: a full graph recommendation stack

```
unified-user-actions / event bus (fav, RT, follow, reply, click ...)
       |                |                          |
       v                v                          v
 Recos-Injector     SimClusters Heron jobs     batch (Scalding/GCP)
 (filter, publish   (tweet embeddings updated   KnownFor, InterestedIn,
  per-service Kafka  on every fav)              producer/topic embeds,
  topics)               |                       TwHIN, RealGraph,
       |                v                       Tweepcred PageRank
       v           SimClusters-ANN                    |
 GraphJet services  (cluster->top tweets index)       v
 UTEG UTG UUG UVG        |                  Representation Manager
       |                 |                  / Representation Scorer
       +--------+--------+-------------------------+
                v
         CR-Mixer / tweet-mixer (fetch, mix, filter, light rank)
                v
         Home Mixer -> heavy ranker -> For You timeline
```

4.3 SimClusters: sparse community embeddings

```
source embedding SV (top N=50 clusters)
     |  fetch top M=400 tweets per cluster   (cluster->tweet index)
     v
sparse approx matrix R (M*N tweets x N clusters)
     |  score = R . SV^T  (dot product on truncated vectors)
     v
top X=200 tweets -> fetch full tweet embeddings (only ~6% of M*N)
     |  exact cosine vs SV, keep top Y above threshold Z
     v
candidates (cosine >= ~0.7 treated as highly related in prod)
```

Architecture and techniques (as reported in [19])

```
heterogeneous member/job/post/ad/company graph (multi-edge-type, timestamps)
       |
       v
graph densification for cold start (add edges from low-degree nodes
       to similar/content-matched nodes)
       |
       v
GNN training: temporal neighbourhood sampling (multi-hop),
  learnable ID embeddings + content features, long-term losses,
  adaptive neighbour sampling, grouped/sliced batches,
  shared-memory sampling queue, local gradient optimisation
       |
       v
embeddings refreshed (batch + nearline) -> feature store / ANN
       |
       v
downstream rankers & retrievers: Feed, Jobs, PYMK, Ads
```

Design (as reported in [20])

```
orders (per city) -> user-dish / user-restaurant weighted bipartite graphs
       -> GraphSAGE (2-hop sampled, max-margin + low-rank positives)
       -> user, dish, restaurant embeddings (daily)
       -> similarity features -> ranking model -> Eats homefeed carousels
```

7.2 AliGraph / Graph-Learn and the Dynamic Graph Service

```
graph updates (stream/bulk)
   -> Data Loader (build Vertex/EdgeUpdateRecord, partition)
   -> Sampling Workers (decompose n-hop query into 1-hop samplers,
      Sample Store + Subscription Table, Subgraph Router)
   -> Serving Workers (all n-hop samples for a vertex co-located;
      SubGraph Store)                      <- client request (Java GSL)
   -> sampled subgraph -> TF model service -> prediction
(all hops connected via fault-tolerant queues)
```

08. Feature Engineering and Feature Stores: Point-in-Time Correctness, Streaming Features, and Serving-Time Logging

Architecture.

```
               client events (Kafka "julep" client-event topics)
                        |                                |
        daily DataRecord dumps (HDFS)        Storm/Heron (summingbird)
                        |                                |
  summingbird-scalding batch job          real-time aggregate job
  (DataRecordAggregationMonoid)           (same AggregateGroup config)
                        |                                |
      offline aggregate store (HDFS)         memcached (RTA cache)
                        |                                |
           upload -> Manhattan KV                        |
                        \_______________  _______________/
                                        \/
                  Home Mixer / pushservice feature hydration
                        (~6000 features per ranked candidate)
```

Architecture (as publicly described).

```
 Offline (batch)                          Online / near-real-time
 ---------------                          -----------------------
 HDFS / Hive data lake                    Kafka event streams
       |                                        |
 Spark / Hive SQL jobs                    streaming jobs (Samza, later Flink)
       |                                        |
 Feature tables (Hive) --- batch upload --->  Cassandra online feature store
       |                                        |
 training data generation                prediction service fetches features
       \_____________________   _____________________/
                             \ /
              Feature DSL (select + transform, applied
               identically at training and serving)
```

5.1 Data model

```
v1 = GroupBy(
    sources=[Source(events=EventSource(table="data.purchases",
                                       topic="events/purchases",
                                       query=Query(selects=..., time_column="ts")))],
    keys=["user_id"], online=True,
    aggregations=[Aggregation(input_column="price", operation=Operation.SUM,
                              windows=[Window(d, TimeUnit.DAYS) for d in [3, 14, 30]]),
                  Aggregation(input_column="price", operation=Operation.LAST_K(10))])
```

5.3 The backfill algorithm: from quadratic joins to sawtooth windows

```
window W, hop h:
|<------------- tail: precomputed per hop bucket ------------->|<- head ->|
[round(ts - W, h) ............................. round(ts, h)) [.. ts)
  shared by all queries whose ts falls in the same h-bucket      exact, per query
```

5.4 Online architecture: untiled vs tiled (Flink)

```
Kafka -> Flink: [Source] -> [Spark-SQL eval (CatalystUtil)] -> [Window/tiling op]
                                                                 |  IRs per tile,
                                                                 v  emitted per event
                                       [Avro encode] -> [async KV sink]
Fetcher: get(key, tiles in [now-W, now)) -> merge IRs -> finalize -> feature values
e.g. COUNT/LAST over 3h, 1h tiles: [2,"B"] + [9,"A"] + [3,"C"] => [14, "C"]
```

6. LinkedIn: Feathr (and Venice as the serving substrate)

```
 HDFS / Spark (offline features) ----bulk push----\
                                                   >---> Venice (derived data store)
 Kafka -> Samza (nearline features) ---stream-----/        |  remote reads or
                                                            |  Da Vinci in-process cache
 Feathr definitions (anchors, derived, registry)            v
       |-- offline: PIT join onto observation data --> training sets
       `-- online: feature fetch for inference <------ ranking services
```

12.1 Serving-time feature logging ("log what you serve")

```
request --> retrieval --> feature fetch (KV + request ctx + on-demand fns)
                          |                          |
                          |                    ranker scores
                          v                          v
                log {request_id, ts, keys,      impressions shown
                     feature_vector, model_ver}        |
                          |                          v
                          |              engagement events (Kafka)
                          v                          |
                  warehouse: served_features     labels (delayed join
                          \                     with attribution window)
                           \_______  __________/
                                   \/
                    training rows = logged features + labels
                    (+ bootstrap/backfill only for NEW features via PITC)
```

09. Training Infrastructure at Scale: Sparse Embeddings, Sharding, Parameter Servers and Online Learning

1.1 Model anatomy

```
           dense features            sparse features (multi-hot IDs)
                |                        |        |        |
           +---------+             +------+ +------+ +------+
           | bottom  |             | EMB  | | EMB  | | EMB  |   <- 99%+ of params
           |  MLP    |             | T1   | | T2   | | Tk   |      (sharded)
           +---------+             +------+ +------+ +------+
                |                      | pool   | pool   | pool
                +----------+-----------+--------+--------+
                           |  feature interaction (dot / cat)
                     +---------+
                     | top MLP |   <- small, data-parallel
                     +---------+
                          |
                       p(click)
```

1.2 Why it is hard to train efficiently

```
forward:  sample-sharded batch -> [input dist: all-to-all of IDs]
          -> table-sharded lookup + pooling
          -> [output dist: all-to-all of pooled embeddings]
          -> data-parallel interaction + MLP on each GPU
backward: MLP grads -> all-reduce (dense)
          pooled-emb grads -> all-to-all back to owners
          -> fused sparse optimizer update in place on owner GPU
```

2.3 TorchRec and FBGEMM_GPU

```
TorchRec TrainPipelineSparseDist (conceptual)
 time ->
 batch i   : [H2D copy][ID a2a][lookup][emb a2a][fwd/bwd MLP][emb grad a2a+update]
 batch i+1 :           [H2D copy][ID a2a]......(overlapped with batch i compute)
 batch i+2 :                     [H2D copy]....
```

3.3 Real-time (online) training pipeline

```
 user actions (clicks, plays, ...) ---> Kafka (action log)
 serving features ------------------>  Kafka (feature log)
                                          |
                                  Flink online joiner
                    (join by request key, in-memory cache + on-disk KV
                     for late actions; negative sampling)
                                          |
                    +---------------------+------------------+
                    |                                        |
           Kafka (training examples)                  HDFS (batch dumps)
                    |                                        |
            online training workers  <--- same model --->  batch training
                    |
            training parameter servers (Cuckoo hash tables)
                    |  periodic sync: sparse incremental (minutes), dense (slower)
                    v
            serving parameter servers -> online serving
```

4.2 Kuaishou PERSIA: hybrid synchronous/asynchronous training

```
  data loaders --> embedding workers <--> embedding parameter servers (CPU, huge)
                        |   (async: get/put embeddings, stale-tolerant)
                        v
                NN workers (GPU) -- sync all-reduce of dense grads (Bagua)
```

6.1 Patterns

```
 stream (Kafka) -> joiner (Flink/Storm-like) -> online trainer
      |                                            |
      |                          incremental export (delta of touched IDs)
      v                                            v
 offline log (HDFS) -> periodic full retrain   serving PS / embedding cache
                       (bootstrap / recovery)
```

10. Online Serving and Inference: Latency Budgets, Model Servers, Caches and Cascades

10.1 The shape of the problem: funnels and latency budgets

```
corpus (10^6 - 10^10 items)
  |
  |  RETRIEVAL / candidate generation   ANN, inverted index, graph walks, in-mem
  v                                     follow-graph stores; cost ~O(log N) per source
10^3 - 10^4 candidates
  |
  |  FEATURE HYDRATION                  fan-out to feature stores, embedding servers,
  v                                     counters; often the latency long pole
  |
  |  LIGHT / EARLY-STAGE RANKING        LR, GBDT, two-tower dot product, distilled nets
  v
10^2 - 10^3 candidates
  |
  |  HEAVY RANKING                      large MTML DNN / transformer, GPU or accelerator
  v
10^1 - 10^2 candidates
  |
  |  RE-RANKING / POLICY                diversity (DPP, author decay), filters, ads mix
  v
response (page of K items)
```

10.2.1 Architecture (2023 Twitter stack)

```
                    +-------------------------------+
 client request ----> | Home Mixer (Product Mixer,     |
                     | Scala)                         |
                     +---+-----------+-----------+---+
                         |           |           |
            in-network   |   OON     |  graph    |  follow recs
          +--------------+  +--------+------+  +--+-----------------+
          | Earlybird     |  | CR-Mixer /    |  | UTEG (GraphJet,    |
          | search index  |  | tweet-mixer   |  | in-memory graph)   |
          | + LR light    |  | + light rank  |  | FRS                |
          |   ranker      |  | + result cache|  |                    |
          +-------+-------+  +-------+-------+  +----------+---------+
                  \                  |                   /
                   v                 v                  v
             +-------------------------------------------------+
             | Feature hydration (~6000 features): user-signal |
             | service, graph-feature-service, real-graph,     |
             | representation-manager (SimClusters/TwHIN emb.) |
             +-------------------------+-----------------------+
                                       v
             +-------------------------------------------------+
             | Heavy ranker (parallel MaskNet, multi-task)     |
             | served by Navi (Rust; TF / ONNX runtimes)       |
             +-------------------------+-----------------------+
                                       v
             filters / heuristics / visibility / ads + WTF mixing
```

10.3.1 Instagram Explore (2023 post)

```
sources: two-tower retrieval, user-interaction histories, heuristics
          |   (item embeddings precomputed offline, ANN at request time)
          v
first-stage ranking: lightweight two-tower-style model, trained partly to
          |          reproduce the late-stage ranker's output (distillation)
          v
second-stage ranking: multi-task multi-label (MTML) DNN predicting many
          |           engagement events
          v
value model: weighted combination of MTML outputs (weights tuned online)
          |
          v
final re-ranking: integrity filters, diversity / business rules
```

10.5.1 Vespa phased ranking

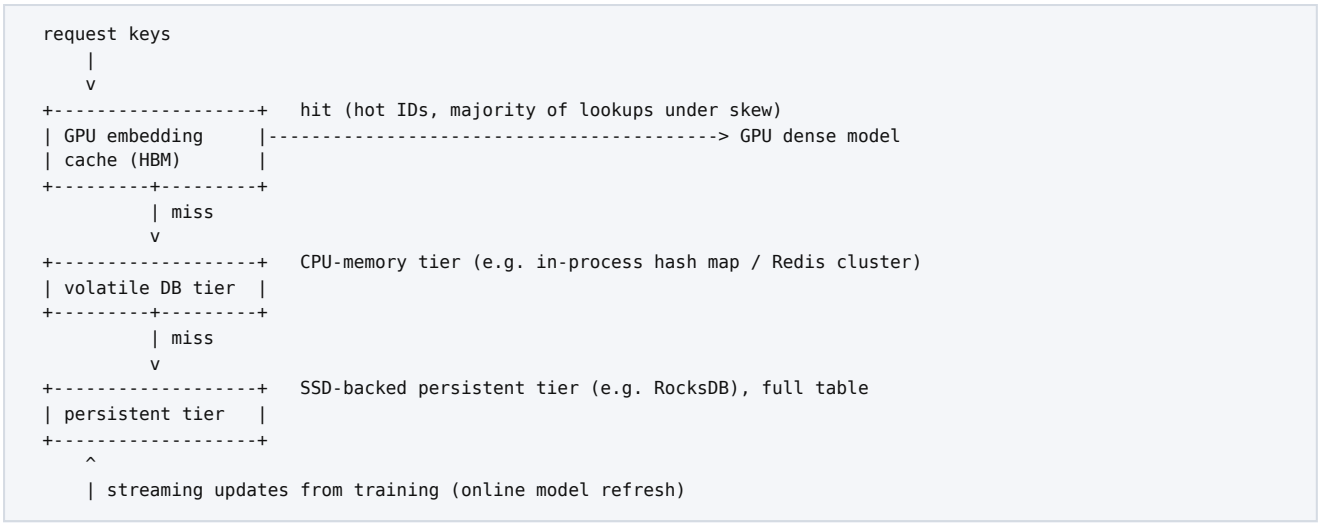
```
             query (user vector, filters, terms)
                          |
                +---------v---------+
                | stateless         |  scatter
                | container         |-----------+-------------+
                +---------^---------+           |             |
                          | gather          +---v---+     +---v---+
                          |                 | content |   | content |
                          |                 | node 1  |   | node N  |
                          |                 |         |   |         |
                          |                 | match:  |   | match:  |
                          |                 | WAND /  |   | WAND /  |
                          |                 | ANN     |   | ANN     |
                          |                 | 1st ph. |   | 1st ph. |
                          |                 | (all    |   | (all    |
                          |                 |  hits)  |   |  hits)  |
                          |                 | 2nd ph. |   | 2nd ph. |
                          |                 | (top    |   | (top    |
                          |                 |  ~100)  |   |  ~100)  |
                          |                 +---+---+   +---+---+
                          +---------------------+-------------+
                global-phase (top ~100 merged, ONNX, optional GPU)
```

10.6 Generative recommenders at serving time: HSTU and M-FALCON

```
naive: for each candidate c_j:  encode([history ; c_j])     cost ~ m * O(n^2)

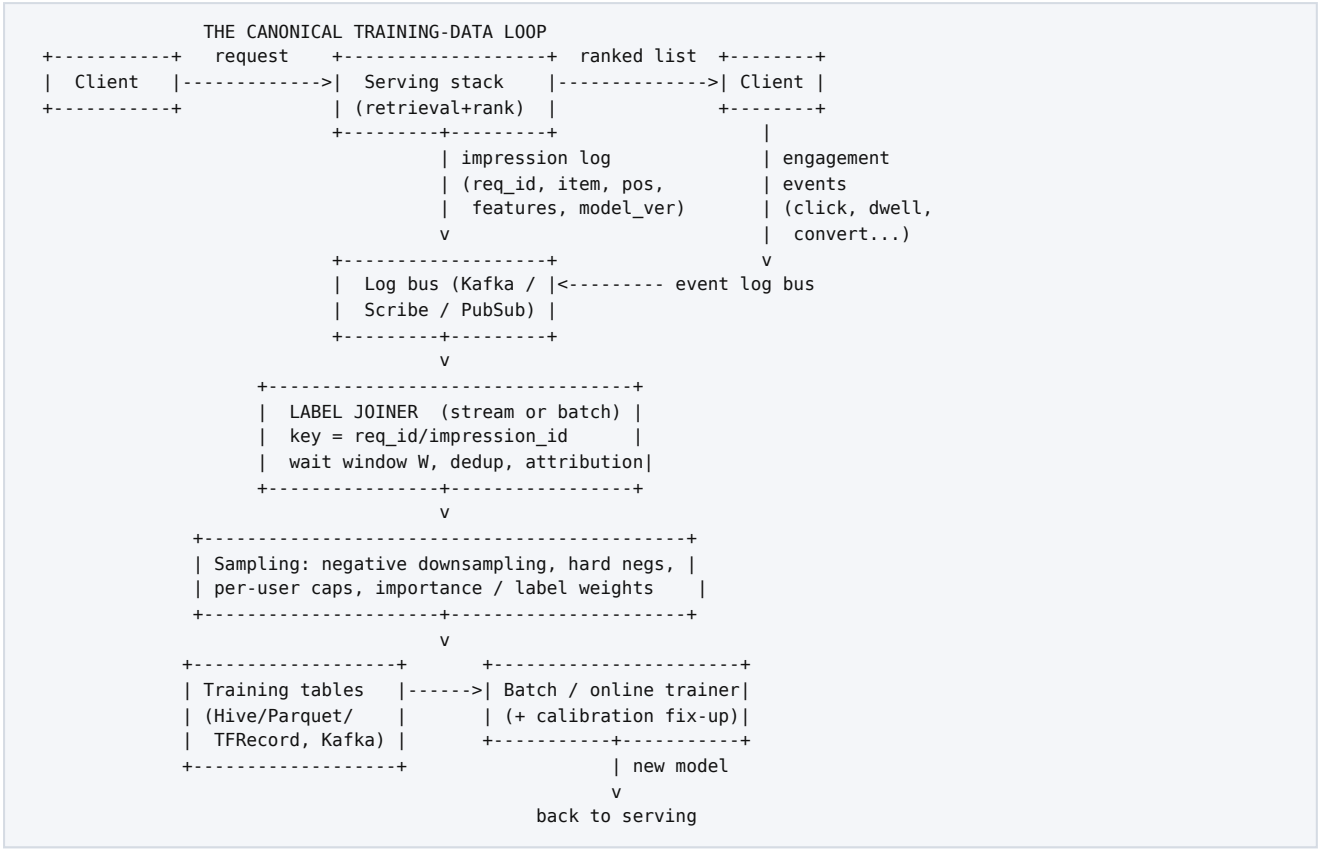
M-FALCON:
  encode history once ---------------> cached keys/values (n tokens)
  split m candidates into microbatches of size b
  for each microbatch: attend [b candidates] -> cached history,
                       candidates masked from each other
                                                          cost ~ O(n^2) + m*O(n)
```

10.7.2 NVIDIA Merlin HugeCTR Hierarchical Parameter Server



11. Training Data and Labeling: Impression Logs, Label Joins, Delayed Feedback, Negatives and Pipelines

Overview



1.1 Facebook Ads (2014): the Online Joiner

```
impressions --(req_id)--+
                        |   +----------------------+   labeled   +---------+
                        +-->| Online joiner        |------------>| Online  |
clicks -----(req_id)----+   | hold until click or  |   examples  | trainer |
                            | window W expires     |             +----+----+
                            +----------------------+                  |
                                   ^                                  v
                                   |      models <-------------- ranker
```

1.3 ByteDance Monolith (2022): streaming joiner for online training

```
serving --> [Kafka: features] --+
                                +--> Flink joiner --> [Kafka: examples] --> online
client  --> [Kafka: actions] ---+    (mem cache +                           trainer
                                      disk KV for                             |
                                     pending feats)                          | param sync
                                       |                                     v  (minutes)
                                        +--> HDFS (batch training)       serving PS
```

6.2 Training/serving skew and leakage

```
LEAKAGE TIMELINE (what a point-in-time join must enforce)

  feature_ts <= impression_ts  <  label_ts <= impression_ts + attribution_window
  |------- allowed features ---|-- label collection --|
                               ^ impression            ^ label final
  Using any feature value with ts > impression_ts = leakage.
```

7.1 Meta: Data Storage and Ingestion (DSI), ISCA 2022

```
serving --> Scribe --> streaming/batch ETL (label join) --> Hive tables (DWRF)
                                                                on Tectonic
                                                                    |
                                      +-----------------------------+
                                      v
                        DPP workers (CPU fleet: read cols, decode,
                        transform, batch)  ==tensors==>  GPU trainers
```

7.5 Reference pipeline (synthesis)

```
ONLINE                                NEARLINE                      OFFLINE
------                                --------                      -------
ranker logs per request:              Flink/Beam streaming joiner:  Spark/Beam daily:
 req_id, user_id, ts, model_ver,       - key by req_id/imp_id       - re-join with full
 [item_id, position, features,        - short window (min-hours)     attribution window
  scores] per impression              - emit early (fake neg) +     - dedup, bot filter
      |                                  re-emit on label change     - neg downsampling
      v                                - importance weight column    - weights, splits
Kafka/PubSub/Scribe topics  -------->  -> online trainer topic       - write Parquet /
(impressions, engagements,             -> warehouse sink               TFRecord / Hive
 conversions, surveys)                                               - data validation
                                                                      -> batch trainer
```

12. Evaluation and Experimentation: Offline Metrics, Off-Policy Evaluation, A/B Platforms, Interleaving and Long-Term Measurement

Overview

```
COST / LATENCY OF A DECISION                          FIDELITY TO BUSINESS GOAL
low                                                                    high
+-----------+   +-------------+   +--------------+   +---------+   +-----------+
| Offline   |-->| Off-policy  |-->| Interleaving |-->|  A/B    |-->| Long-term |
| metrics   |   | evaluation  |   | (ranking)    |   |  test   |   | holdout / |
| NDCG,AUC  |   | IPS, DR     |   | team-draft   |   | 1-4 wks |   | surrogate |
+-----------+   +-------------+   +--------------+   +---------+   +-----------+
minutes-hours     hours             days               weeks          months
1000s of ideas    100s              10s                handful        portfolio
```

12.1.3 Pitfalls of offline evaluation

```
OFFLINE EVAL PIPELINE (typical)
impression+engagement logs ---> time-based split ----> train window | eval window
                                                                     |
   candidate model --(score full catalogue or logged slate)----------+
                                                                     v
         +----------------+--------------+---------------+------------------+
         | retrieval:     | ranking:     | calibration:  | beyond-accuracy: |
         | recall@K       | NDCG, AUC,   | sum(p)/sum(y) | coverage, KL     |
         | (full catalog) | NE/logloss   | by slice      | genre calib.,    |
         |                | per slice    | reliability   | diversity, nov.  |
         +----------------+--------------+---------------+------------------+
                   \_____________________ gate: "worth an online test?" ______/
```

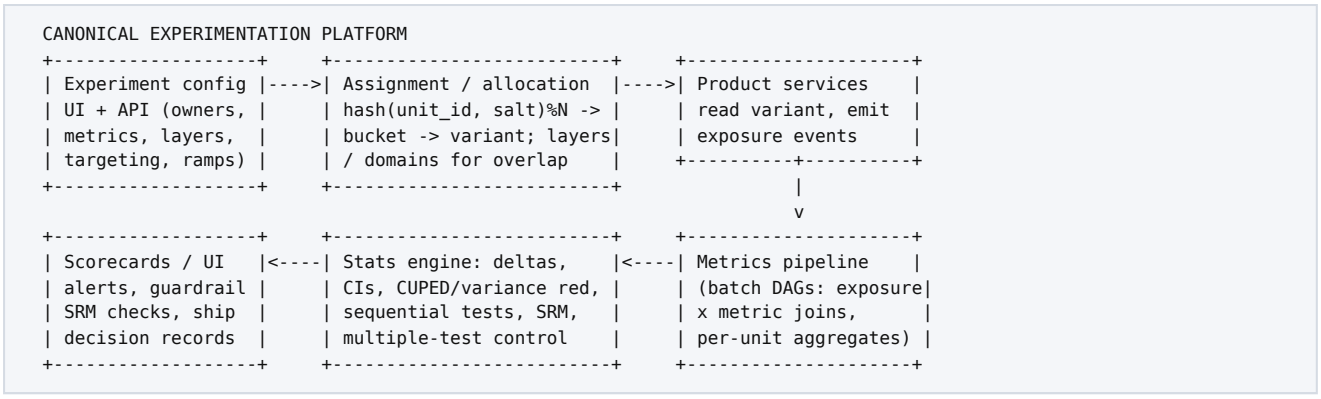
12.2.5 Tooling and benchmarks

```
OPE DATA FLOW
 request --> logging policy pi_0 (ranker + sampler) --> slate shown
                       |  logs: x, slate, per-slot p(a|x), model_version
                       v
               [ propensity-aware log store ]
                       |
   candidate pi ------>+-----> IPS / SNIPS / DR / PI estimators
   reward model r_hat->+            |
                                    +--> V_hat(pi), CI, ESS
                                    v
                gate: V_hat significantly > V_hat(pi_0) and ESS ok
                       --> interleaving / A/B
```

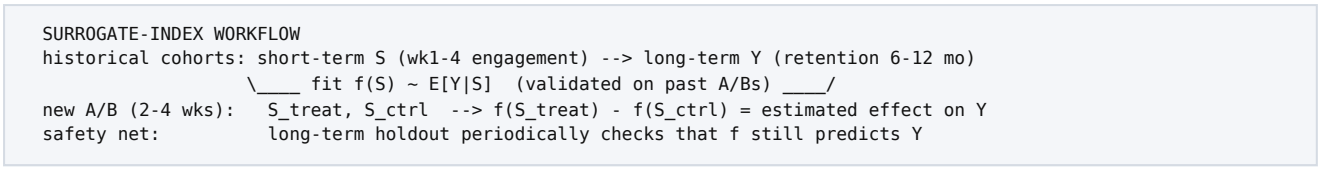
12.3.1 Method

```
TEAM-DRAFT INTERLEAVING
Ranker A: a1 a2 a3 a4 ...           Ranker B: b1 a1 b2 b3 ...
round 1: coin=A -> A picks a1 ; B picks b1
round 2: coin=B -> B picks b2 (a1 taken) ; A picks a2
...
merged list: [a1(A), b1(B), b2(B), a2(A), ...]
user engages with b1, b2  --> preference(B) for this user
aggregate over users: Delta = (#users prefer A - #users prefer B) / N  -> sign test
```

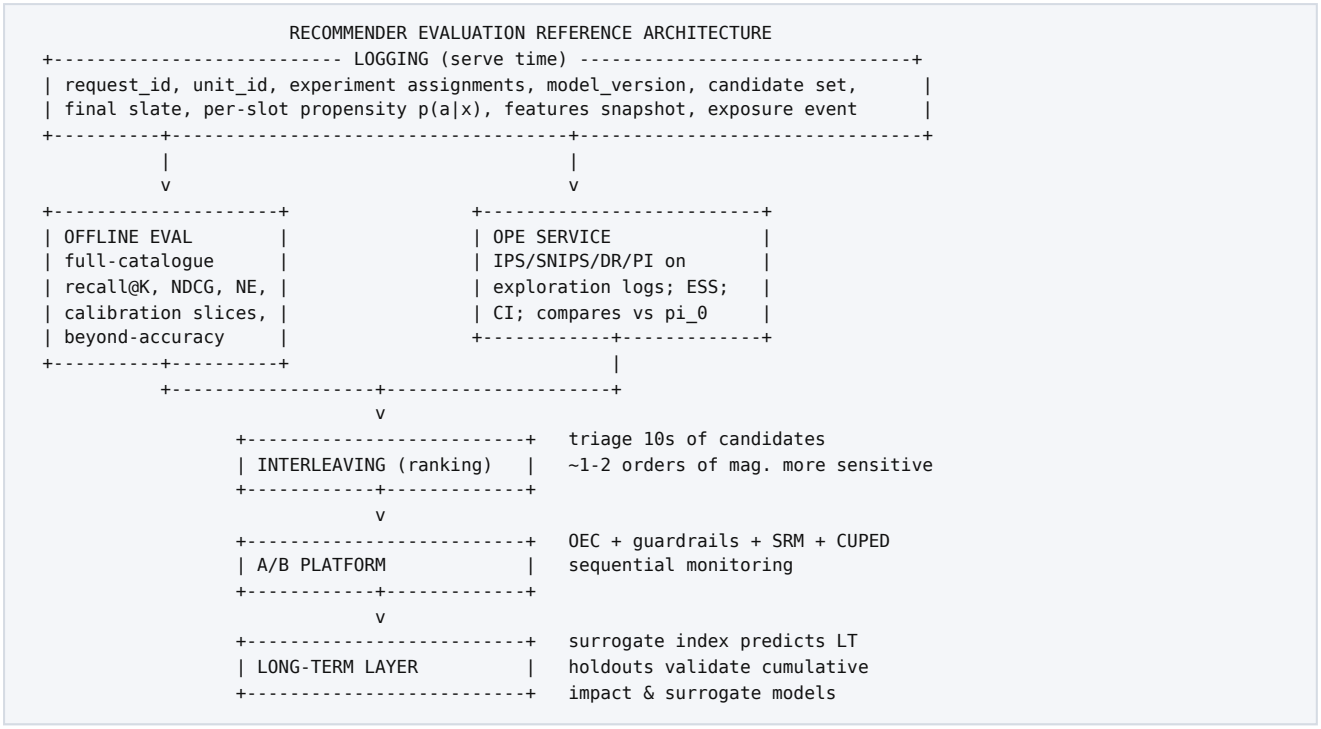
12.4 A/B testing platforms



12.5.2 Surrogate index

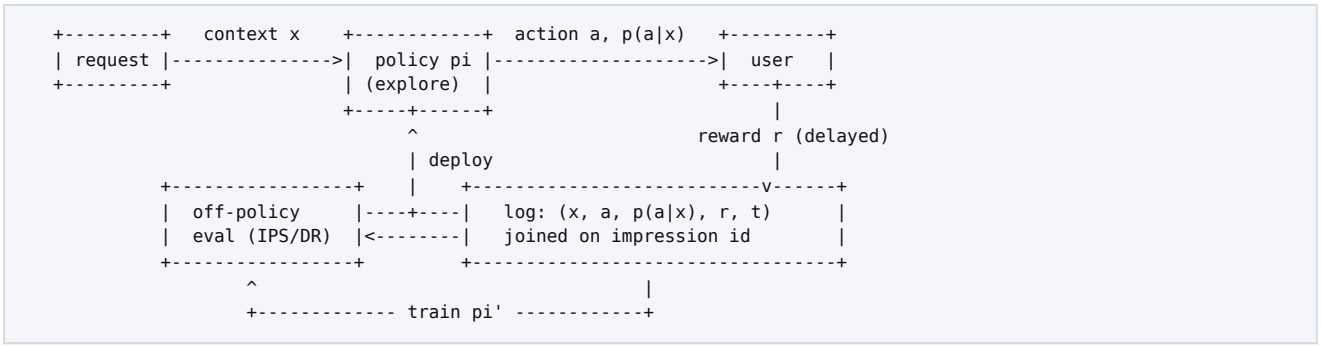


12.6 Putting it together: a reference evaluation architecture

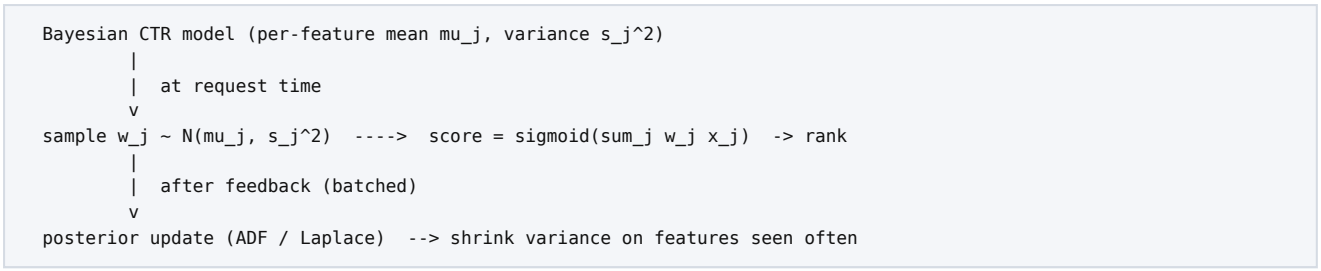


13. Exploration, Bandits and Reinforcement Learning in Recommendation

1. The problem: closed loops and missing counterfactuals



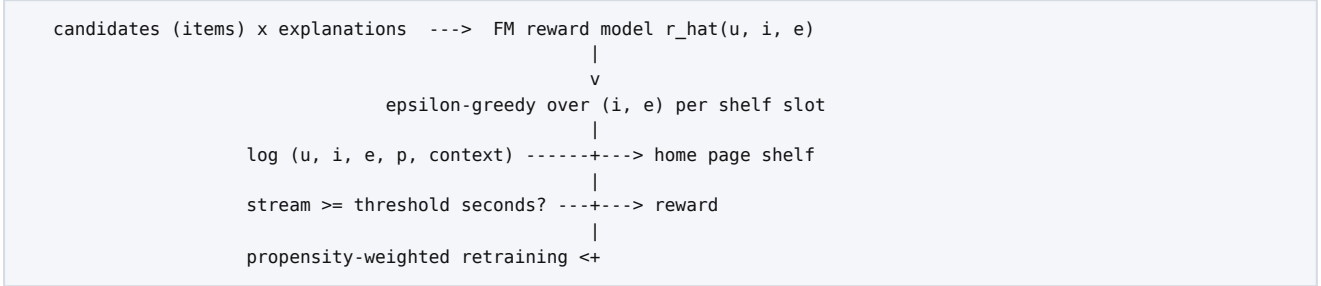
Design pattern



Logging and offline evaluation



BaRT: Bandits for Recommendations as Treatments



Architecture

```
 user event sequence (watched videos, context)
             |
             v
    +-----------------+
    |   RNN state s_t |  (shared user-state encoder)
    +--------+--------+
             |
     +-------+---------------------------+
     |                                   |
     v                                   v
softmax pi_theta(a|s)            softmax beta_hat(a|s)
over millions of videos          (behavior policy head;
(the policy we deploy)            gradient blocked into
     |                            the shared encoder)
     |                                   |
     +------> importance weight w = pi_theta / beta_hat
                    |
                    v
     top-K correction lambda_K = K * (1 - pi_theta)^(K-1)
                    |
                    v
     REINFORCE gradient: sum_t  w * lambda_K * R_t * grad log pi_theta(a_t|s_t)
     (with weight capping and normalized importance sampling)
```

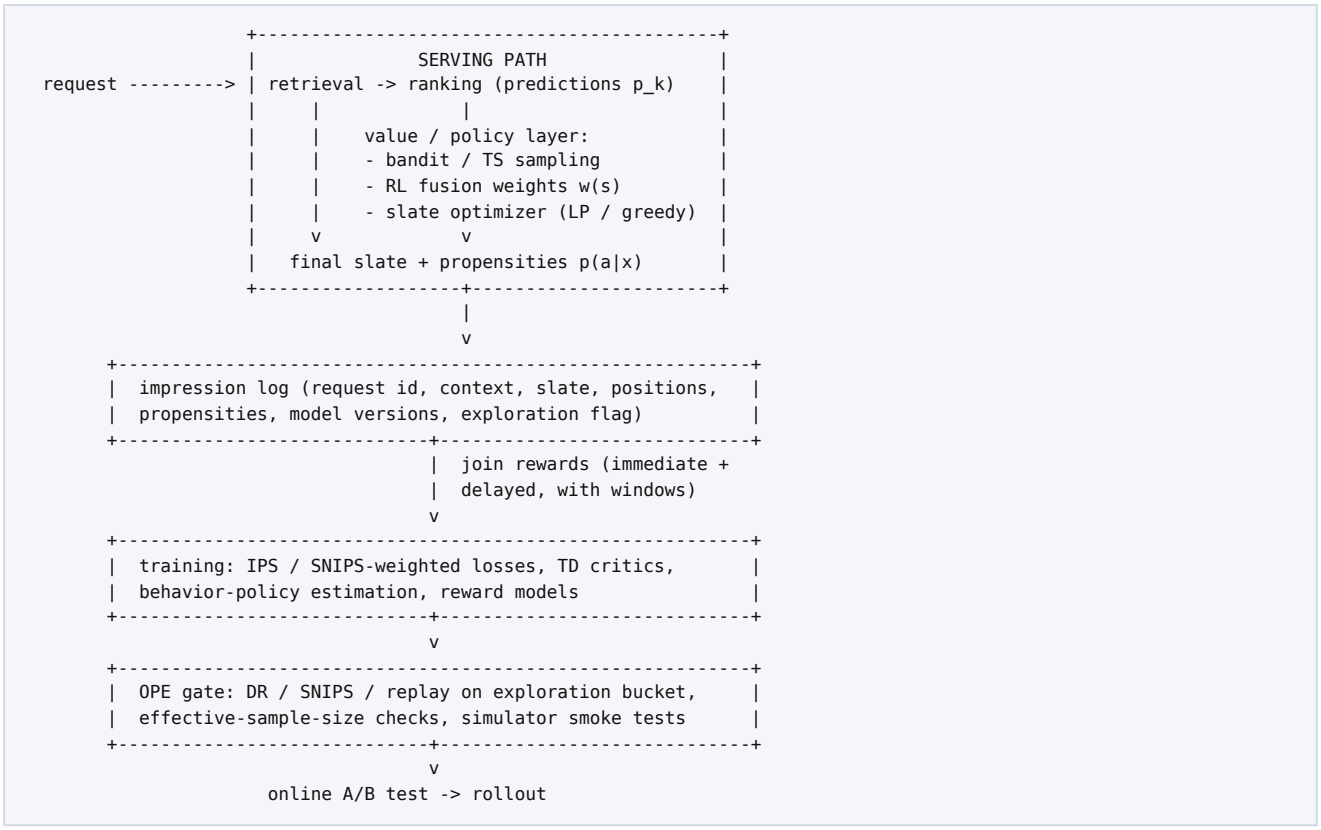
Slate optimization

```
 offline logs (s, slate A, clicked i, r, s')
          |
          +--> choice model P(i | s, A)   (trained as a classifier)
          |
          +--> Q_bar(s, i) via SARSA/TD on (s, i, r, s', i')
                          |
 serving:  candidates from retrieval (hundreds)
                          |
          score each i by P-weighted Q_bar
                          |
          slate = LP or greedy top-k  ---> user
```

Kuaishou: reinforcing user retention at billion scale (WWW 2023)

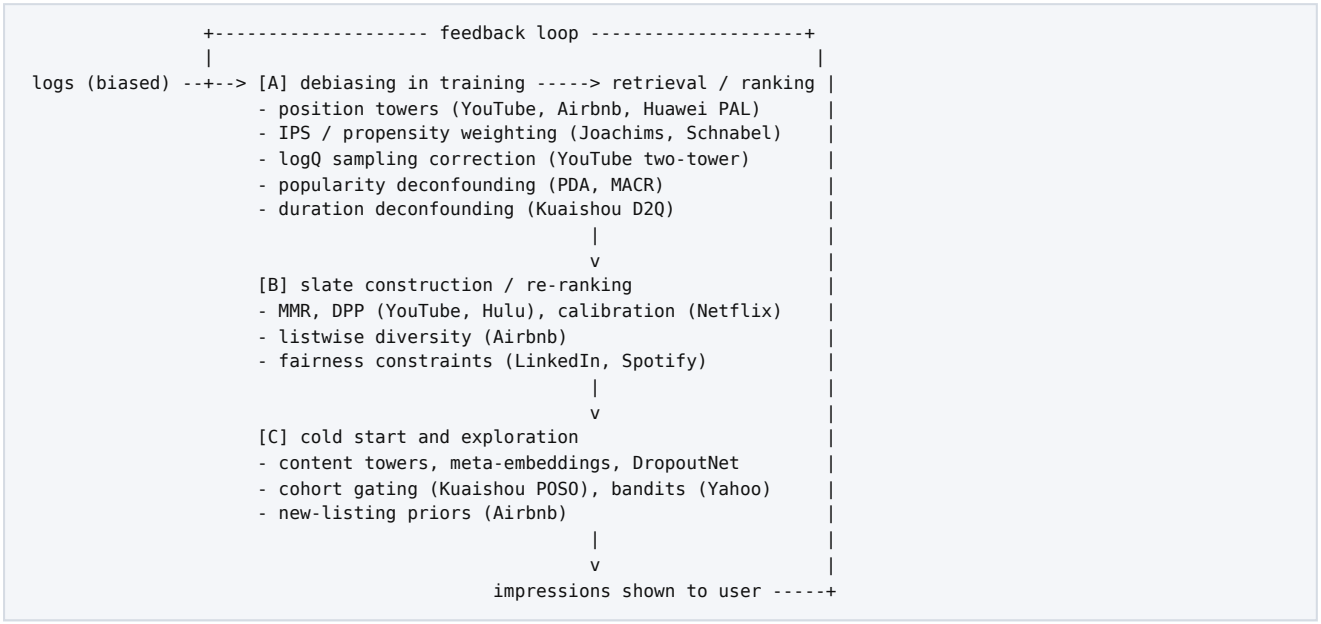
```
request t:  user state s_t (history, profile, time since last visit)
               |
               v
        actor pi(s_t) --> weights w_t = (w_watch, w_like, w_follow, ...)
               |
               v
   existing ranker: score(i) = sum_k w_t,k * prediction_k(i)   ---> feed
               |
 session ends; next visit after gap g  ---> reward r = f(g) (+ shaped terms)
               |
        critic V(s) trained by TD; actor updated by critic
```

Reference architecture



14. Debiasing, Diversity, Fairness and Cold Start: The Corrective Layers of Production Recommenders

Overview



Architecture.

```
user / context / candidate features          position, device (+ other bias feats)
                |                                        |
        +---------------+                        +--------------+
        | shared hidden |                        | shallow tower|
        |  layers       |                        | (small MLP / |
        +---------------+                        |  linear)     |
                |                                +--------------+
        +---------------+                               |
        | MMoE experts  |                               |
        | + task gates  |                               |
        +---------------+                               |
          |         |                                   |
  engagement     satisfaction                           |
  task towers    task towers                            |
          |                                             |
      logit_main  ------------------ (+) --------------- logit_bias
                                      |
                                  sigmoid ----> training loss (click etc.)

 Serving: the position feature is treated as missing, so logit_bias collapses
          to a constant and ranking uses the main towers only.
```

14.3.5 Duration bias in watch time: Kuaishou D2Q

```
     duration --------> exposure (system shows long videos more)
        |                 |
        v                 v
              watch time (label)

 D2Q: split training data into duration buckets
      -> within each bucket, convert watch time into a quantile label
      -> fit one model (shared parameters) predicting the within-bucket quantile
      -> serve on quantiles: compares "how well this video holds attention
         relative to similar-length videos"
```

Architecture.

```
candidate generators --> pointwise ranker (q_i = predicted utility)
                                  |
                                  v  top-N ranked candidates (N ~ a few hundred, approx.)
                  +-------------------------------+
                  | DPP re-ranker                 |
                  |  kernel:                      |
                  |  L_ii = q_i^2                 |
                  |  L_ij = alpha*q_i*q_j*        |
                  |        exp(-D_ij / (2 s^2))   |
                  |  D_ij from video embeddings / |
                  |  topical distance             |
                  |  greedy MAP, window by window |
                  +-------------------------------+
                                  |
                                  v
                 feed served in windows of k items; next window
                 chosen conditioned on items already selected
```

14.6 Cold Start

```
           +------------------+      +------------------------+
new item -->| content encoders  |----> | ID embedding warm-up   |--> ranker
(meta,     | (text, image,     |      | (meta-embedding [25],   |
 media)    |  audio, creator)  |      |  DropoutNet [24])       |
           +------------------+      +------------------------+
                    |                            ^
                    v                            |
          exploration traffic (bandits [32], boosted slots, priors [5])
                    |
                    v
          real-time training (minutes) -> ID embedding converges

new user --> onboarding signals + context (geo, device, referrer)
         --> cohort-specific gating (POSO [26]) --> popular-within-cohort fallback
```

14.7.2 LinkedIn Recruiter: fairness-aware re-ranking at scale

```
query --> candidate retrieval --> ML ranker (relevance scores)
                                      |
                                      v
          desired distribution p_q(attribute) = distribution among candidates
          matching the query (e.g., qualified pool), computed per query
                                      |
                                      v
          fairness re-ranker: per-attribute queues sorted by score;
          fill ranks greedily so that every prefix k has at least
          floor(p_a * k) and at most ceil(p_a * k) of each attribute a
          (DetGreedy / DetCons / DetRelaxed variants)
                                      |
                                      v
                           top-k shown to recruiter
```

14.8 Putting It Together: A Reference Corrective Stack

```
[Training data]
  impressions + engagements + position + device + example age + duration
    |
    |-- reweight: fixed examples per user [12]; IPS for explicit ratings [13]
    v
[Retrieval: two-tower]
  in-batch softmax with logQ correction via streaming frequency estimator [11]
  ID dropout so content towers handle cold items [24]
  meta-embedding initialisation for new IDs [25]
    |
    v  ~ thousands of candidates
[Ranker: multitask DNN / MMoE]
  main towers (engagement, satisfaction)
  + shallow position/device tower, additive logit, position dropout [1][5]
  duration-bucketed quantile targets for watch time [27]
  cohort gates for new users [26]
  example_age = 0 and position = missing at serving [12][1]
    |
    v  ~ hundreds of candidates
[Re-ranking / slate construction]
  fast greedy DPP or MMR within sliding windows [2][3][10]
  calibration to user's genre / creator-popularity mix [4]
  fairness prefix constraints where policy requires [21]
  exploration slots for cold items/creators (bandit bonus) [32]
    |
    v
[Serving + logging]  log true positions, exploration flags, propensities
                     (needed by every debiasing method above)
```

15. Re-ranking, Blending and Whole-Page Optimization

1.1 Canonical last-mile architecture

```
heavy ranker output: ~10^1-10^3 candidates, each with p_click, p_like, p_hide, ...
       |
       v
+--------------------+   value model: score = sum w_i * p_i (+ offsets)
| 1. score fusion    |   (Ch. 4: Meta value model, X weights, Kuaishou ensemble sort)
+---------+----------+
          v
+--------------------+   hard filters: integrity, blocked/muted, seen, legal,
| 2. eligibility     |   inventory, geo, age, dedup by item / cluster
+---------+----------+
          v
+--------------------+   context-aware rescoring: DLCM / PRM / SetRank / PEAR
| 3. listwise rerank |   or generator + evaluator (Seq2Slate, GRN, PIER, NAR4Rec)
+---------+----------+
          v
+--------------------+   MMR / DPP / author decay / calibration / cold-start lift
| 4. diversification |
+---------+----------+
          v
+--------------------+   interleave content types and sources, insert ads at
| 5. blending / slots|   allowed positions, modules (WTF, stories, carousels)
+---------+----------+
          v
+--------------------+   rows / shelves / modules chosen and ordered;
| 6. page assembly   |   cross-row dedup; device templates
+---------+----------+
          v
    response + impression log (slate, positions, propensities)
```

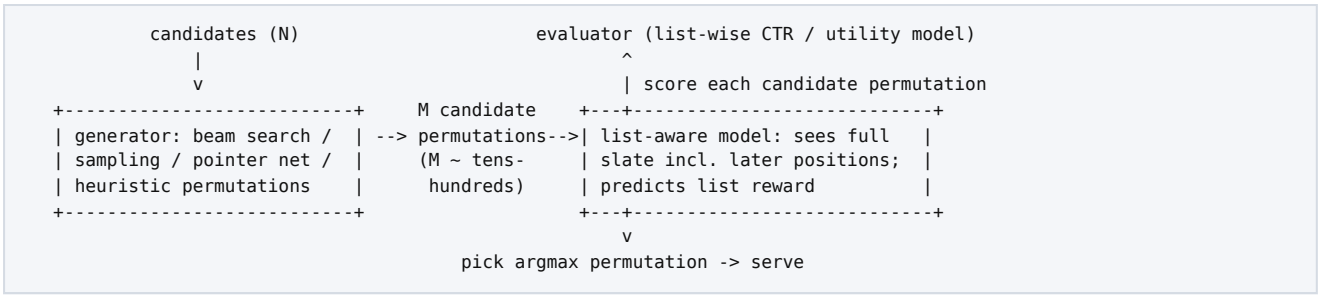
2.3 Multiplicative decays and lifts (X / Twitter)

```
weighted sum of ~25 heads  -> negative-score offset (keeps scores > 0)
   -> + exploration bonus
   -> x author-diversity multiplier: (1 - floor) * decay^k + floor
          (k = number of higher-ranked posts already chosen from that author)
   -> x out-of-network factor (< 1 for unfollowed accounts)
   -> cold-start lift (new author raised to at least a target rank)
   -> DPP reorder (vm-ranker) -> visibility filters -> timeline
```

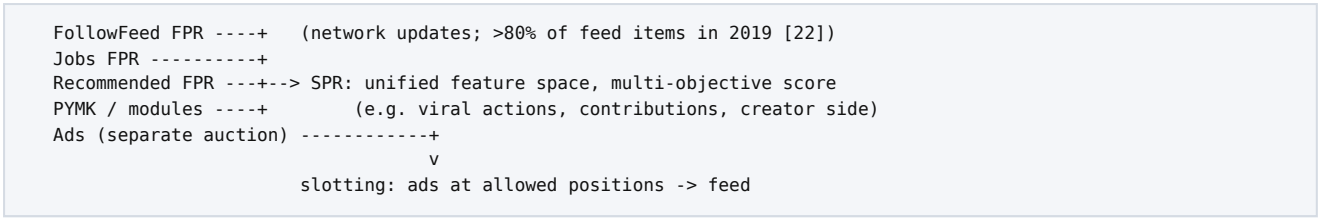
Architecture.

```
initial list from ranker: items i_1 .. i_n  (n = list length, tens of items)
      |
      v
per item input  x_i = [ item feature vector  ||  PV_i ]  +  PE_i
      |               PV_i = personalized vector: hidden layer of a pre-trained
      |                      CTR model taking (user, item) -> user-specific item rep
      |               PE_i = learnable position embedding (initial rank)
      v
linear projection to d_model
      v
N x Transformer encoder blocks (multi-head self-attention + FFN)
      |   every item attends to every other item -> O(n^2) but n is small
      v
linear + softmax over the list  -> re-ranked scores
      v
sort -> final list
```

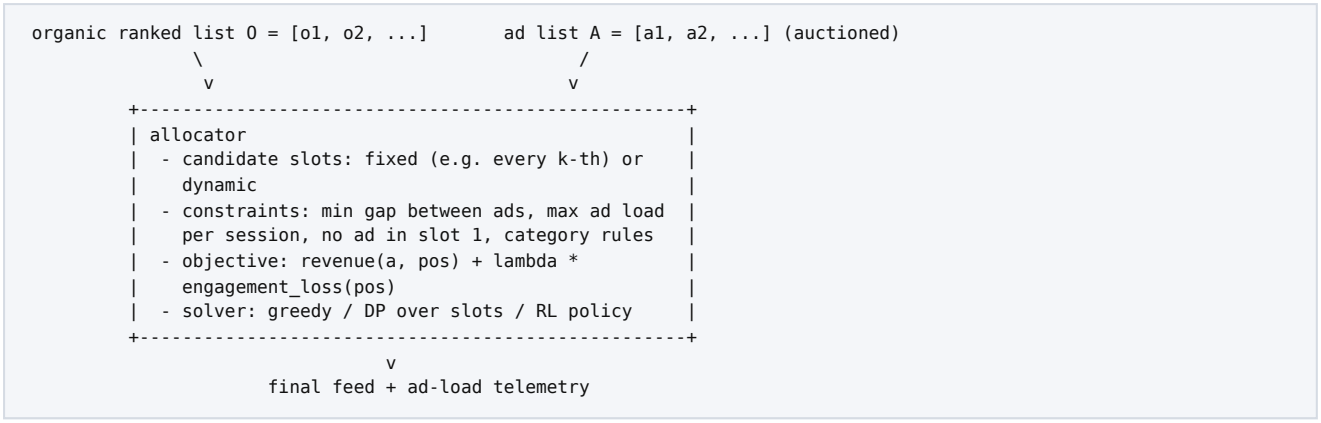
4.2 Generator-evaluator: Exact-K, PRS, GRN



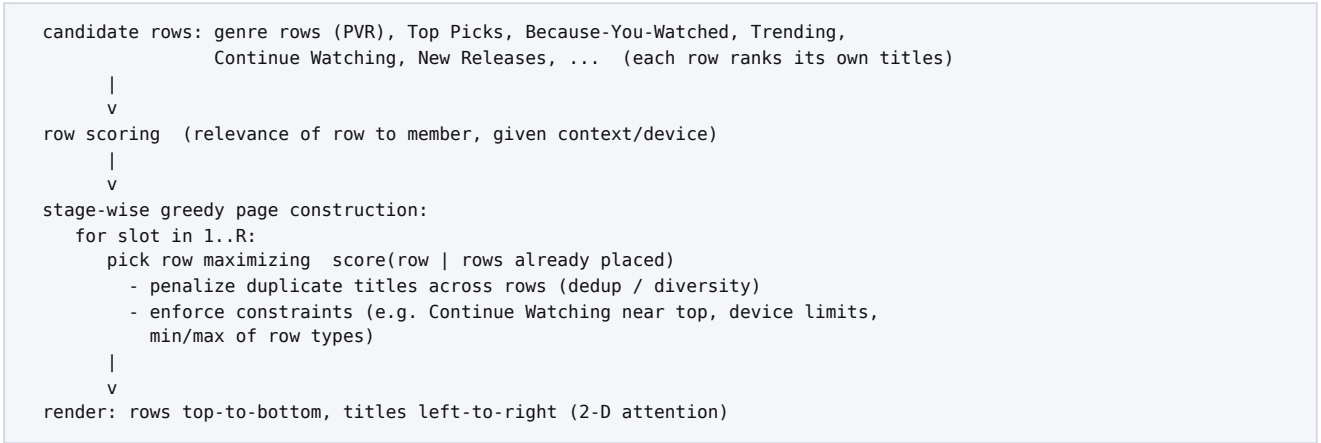
6.1 LinkedIn: first-pass rankers and a second-pass ranker



6.5 Ads and organic: the allocation problem



7.1 Netflix: page generation



8.2 Design patterns

```
config store (versioned, A/B-able)
     |  rules as data: {id, scope(surface, country), type(filter|cap|boost|slot),
     |                  params, priority, experiment_id}
     v
+-------------------------------------------------------------------------+
| rule engine in the mixer                                                |
|  1. hard filters (set operations; cheap; run early AND re-run at end)   |
|  2. score adjustments (multipliers/offsets; order-preserving sign care) |
|  3. sequential constraints (greedy fill: author cap, gap rules, DPP)    |
|  4. slot constraints (ads, modules, pinned rows)                        |
|  5. backfill if constraints leave holes (fallback sources)              |
+-------------------------------------------------------------------------+
     |
     v  log: which rule touched which item (for debugging and OPE)
```

16. Ads Recommendation Systems: pCTR/pCVR Prediction, Auctions, Pacing, Calibration and the Meta, Google, Pinterest, Snap and Twitter Ads Stacks

Overview

```
ad request (user, context, placement)
     |
     v
+-----------------+   eligibility: targeting, budget left, policy, frequency caps
| ad index / ACL  |   10^6 .. 10^7+ eligible ads (Meta: "tens of millions") [10][11]
+--------+--------+
         v
+-----------------+   two-tower / learned index (Andromeda) / GPU brute force
| retrieval       |   -> 10^3 .. 10^4 candidates
+--------+--------+
         v
+-----------------+   cheap model, often distilled from the heavy ranker
| light ranking   |   -> 10^2 candidates
+--------+--------+
         v
+-----------------+   multi-task DNN: pCTR, pCVR (per conversion type), quality,
| heavy ranking   |   negative feedback; calibrated per task
+--------+--------+
         v
+-----------------+   score = bid(or auto-bid) x p(action) + quality terms
| auction         |   allocation + pricing (GSP-like, VCG-like), reserve prices
+--------+--------+
         v
+-----------------+   ad load, spacing, slot choice vs organic items
| allocation      |
+--------+--------+
         v
impression --> click/conversion logs --> joiner --> online trainer (minutes-hours)
         ^                                                  |
         +-------------- pacing / bid controller <----------+ (spend telemetry)
```

3.1 Practical Lessons from Predicting Clicks on Ads at Facebook (ADKDD 2014)

```
raw features --> boosted trees T1..TK (retrained ~daily)
                   | leaf ids (one-hot)
                   v
             logistic regression (online SGD, near real time)
                   |
                   v
         p (downsampled space) --> q = p / (p + (1 - p) / w) --> auction
```

3.3 Lattice: consolidating the ads model fleet (2023, paper 2025)

```
before Lattice                          after Lattice
--------------------------------        ------------------------------------------
model(FB Feed, CTR)                     +--------------------------------------+
model(FB Feed, purchase CVR)            | Lattice: one multi-domain multi-task  |
model(IG Reels, CTR)          ==>       | model; domain-aware sparse experts;   |
model(IG Stories, install)              | heads per objective; Pareto balancing |
... (N surfaces x M objectives)         +-------------------+------------------+
                                                            | distillation
                                             lean serving models per stage/surface
```

Architecture [10].

```
user + context features
       |
       v
+-------------------------------------------------------------+
| Andromeda retrieval DNN on Grace Hopper (unified CPU-GPU mem) |
|                                                               |
|   level 0:  root clusters   --score--> keep top nodes         |
|   level 1:  sub-clusters    --score--> keep top nodes         |
|   level L:  individual ads  --score--> top ~10^3 ads          |
|   (hierarchy learned jointly with the scoring network)        |
+----------------------------+--------------------------------+
                             v
               ranking (Lattice / GEM-distilled models) -> auction
```

3.5 GEM: the Generative Ads Recommendation Model (November 2025)

```
          +-----------------------------------------------+
          | GEM (offline, LLM-scale, thousands of GPUs)    |
          | inputs: long user ad/organic sequences +       |
          |         non-sequence features across surfaces  |
          +-----------------------+-----------------------+
                                  | distillation / representation transfer
       +--------------------------+--------------------------+
       v                          v                          v
 retrieval models           light rankers               heavy rankers (Lattice)
 (Andromeda)                (per surface)               per surface / objective
       \_____________________________|_____________________________/
                                     v
                          auction + allocation
```

4.3 Sample-selection bias: ESMM

```
impression features --> shared embeddings
                            |                 |
                        CTR tower         CVR tower
                            |                 |
                          pCTR ------(x)---- pCVR
                            |                 |
                   loss on click       pCTCVR: loss on click & convert
```

7.1 GSP, VCG and why ads use "GSP-like" auctions

```
rank by  score_i = bid_i x pCTR_i (x pCVR_i) + quality_i         (expected value)
slot k goes to the k-th highest score
GSP price per click for winner k:
    cpc_k = (score_{k+1} - quality_k) / pCTR_k     (+ minimum increment, >= reserve)
=> winner k pays just enough to beat k+1, so pCTR errors move both rank AND price
```

throttling (participate in a fraction of eligible auctions) and **bid modification**

```
advertiser goal (budget B, target CPA)          spend telemetry (per minute)
          |                                                |
          v                                                v
+-------------------+   multiplier lambda_t   +--------------------------+
| bid controller /  |------------------------>| auction: bid_t = lambda_t |
| RL policy         |<------------------------|  x pCVR x target CPA      |
+-------------------+  spend, conversions,    +--------------------------+
    state: remaining budget, time of day, pace error
```

9. Ad load and allocation against organic content

```
organic ranked list O = [o1, o2, ...]      auctioned ad list A = [a1, a2, ...]
               \                                  /
                v                                v
       +-------------------------------------------------+
       | allocator                                       |
       |  slots: fixed (every k-th) or dynamic per user  |
       |  constraints: min gap, max ad load per session, |
       |               no ad in slot 1, category rules   |
       |  objective: revenue(a, pos) + lambda x          |
       |             engagement_loss(pos)                |
       +-------------------------------------------------+
```

17. Google & YouTube Case Studies: From Co-Visitation Graphs to Semantic IDs and LLM-Native Recommenders

Overview

```
Google / YouTube recommender lineage (simplified)

2007 Google News  --- MinHash + PLSI + co-visitation (MapReduce/Bigtable)
2010 YouTube      --- co-visitation related graph, seed expansion, heuristic rank
2016 Google Play  --- Wide & Deep (memorization + generalization)
2016 YouTube      --- DNN candidate gen (softmax -> ANN) + DNN ranker (watch time)
2018 MMoE / DPP   --- multi-gate experts; windowed DPP feed diversification
2019 YouTube      --- two-tower + logQ | MMoE + shallow tower | REINFORCE | SlateQ
2020 Google       --- DCN-V2, mixed negative sampling, ScaNN
2021 YouTube      --- "valued watch time" from surveys; self-supervised towers
2023 Google       --- TIGER: semantic IDs + seq2seq generative retrieval
2024 YouTube      --- semantic IDs replace video IDs in ranking embeddings
2025 YouTube      --- PLUM: LLM continued pre-training on SIDs, generative retrieval
```

Architecture (Das et al., WWW 2007)

```
Google News CF (2007)

click logs --MapReduce--> MinHash user clusters ---+
           --MapReduce--> PLSI (EM) user classes ---+--> Bigtable stats
live clicks --> co-visitation counts (online) ------+         |
                                                              v
request(user) --> candidate stories --> score = w1*minhash + w2*plsi + w3*covisit
```

Architecture (Davidson et al., RecSys 2010)

```
YouTube 2010 (batch-computed, served from Bigtable)

watch logs (24h sessions) --MapReduce--> co-visitation counts c_ij
                                      --> related(v_i) = top-N by r(v_i, v_j)
                                             r = c_ij / f(v_i, v_j)
user seeds S = {watched, liked, favorited, playlisted videos}
         |
         v
candidates C_n(S) = union of related-graph neighbours up to n hops
         |
         v
rank by linear mix of: video quality | user specificity | diversification
         |
         v
top ~4-60 recs (pre-computed several times a day; served with explanation)
```

Architecture

```
     wide (memorization)                    deep (generalization)
sparse ids + cross products         sparse ids -> embeddings (~32-d each)
           |                                   + dense features
           |                                      |
           |                                MLP (ReLU layers, e.g. 1024-512-256)
           |                                      |
           +----------- sum of logits -----------+
                            |
                        sigmoid -> P(app acquisition)

serving funnel: query -> retrieval (ML + rules, ~O(100) apps) -> Wide & Deep rank
```

Architecture

```
YouTube 2016

corpus (millions) ---> CANDIDATE GENERATION ---> hundreds ---> RANKING ---> dozens
                       user history + context       |          rich features,
other candidate  ----> -> MLP tower -> user vec u    |          impression data,
sources (earlier       train: sampled softmax        |          weighted LR on
work, co-vis, etc.)    serve: ANN top-N by <u, v_j>  +--------> watch time
                                                                     |
                                                      A/B test on watch time/user
```

Design

```
 query tower  q_i = f(user_i, context_i)          batch i = 1..B
 item tower   v_j = g(video_j features)
 logits       s_ij = <q_i, v_j> / tau  -  log p_j        (logQ correction)
 loss         weighted softmax CE over in-batch candidates (weight = reward)
 p_j          streaming estimate: hashed arrays of "last step seen" and
              moving average of the gap between occurrences; p_j ~ 1 / gap
```

Architecture

```
YouTube Watch Next ranker (2019)

query video + candidate + user + context features
               |
         shared hidden layer (cheaper than feeding raw inputs to every expert)
               |
     +-------- MMoE: N experts --------+
     |   gate_k(x) = softmax(W_k x)    |    (one gate per objective)
     +--+--------+--------+-----------+
        |        |        |
  engagement  engagement  satisfaction ... task towers
    (click)   (watch)     (like, dismiss, rating)
        |        |        |
        +--------+--------+--> logits
                            + shallow-tower bias logit (training only)
position, device ---> SHALLOW TOWER ------^   (position dropout ~10%, approx.)
               |
serving: position feature missing -> debiased per-task predictions
         -> weighted combination (weights tuned by online experiments) -> rank
```

Top-K Off-Policy Correction for a REINFORCE Recommender (WSDM 2019)

```
user event sequence (watches, context) --> RNN state s_t
                                                |
              +-------------------------------+-------------------+
              v                                                   v
     pi_theta(a|s): softmax over               beta_hat(a|s): behavior-policy head
     millions of videos (deployed)             (gradient blocked into shared state)
              |                                                   |
              +---------- importance weight  pi / beta_hat ------+
                                  |
               lambda_K = K * (1 - pi(a|s))^(K-1)     (top-K correction)
                                  |
         REINFORCE update: sum_t  w_t * lambda_K * R_t * grad log pi(a_t|s_t)
         + weight capping, normalized IS, TRPO-style constraint
```

Architecture

```
TIGER (Rajput et al., Google, 2023)

item text/metadata -> Sentence-T5 (768-d, approx.) -> RQ-VAE
    level 1 code c1 (256 options)  -> residual r1
    level 2 code c2 (256 options)  -> residual r2
    level 3 code c3 (256 options)  [+ collision token c4]
Semantic ID(item) = (c1, c2, c3, c4)

user history: SID(i1) SID(i2) ... SID(in)  (+ user id token, hashed)
         |
  Transformer encoder-decoder (T5-style, small)
         |
  beam search decodes the next item's SID tuple -> lookup -> items
```

PLUM: Pre-trained Language Models for industrial-scale generative recommendation (2025)

```
PLUM (Google / YouTube, 2025)

1. Item tokenization: Semantic IDs v2
   multimodal content embeddings (+ co-engagement signal) -> RQ-style
   multi-level codes; tuned for coverage and low collision at YouTube scale
2. Continued pre-training (CPT) of a pre-trained Gemini-family LLM
   on a mixture of: user watch sequences as SID tokens, SIDs aligned with
   text (titles, topics), so the model "speaks" SIDs and language
3. Task fine-tuning: generative retrieval
   input: user context + history (SIDs + text) -> decode SIDs of next videos
4. Serving: beam decoding of SIDs -> videos, as a retrieval source
   feeding the existing ranking stack
```

18. Netflix Case Study: From the Netflix Prize to a Foundation Model for Personalization

3.2 The blueprint

```
                  +-------------------------- OFFLINE --------------------------+
 member events ---> | Hadoop / Hive / Pig: data prep, model training, batch scoring |
 (plays, ratings,   | -> trained models + precomputed results                      |
  browsing, search) +--------------+------------------------+----------------------+
      |                            | models                 | results
      |                            v                        v
      |               +---------------- NEARLINE -----------------+
      +-------------> | event-driven compute (e.g. on play event:  |
      |  (pub/sub)    | recompute user factors / scores),          |
      |               | results written to Cassandra / EVCache /   |
      |               | MySQL                                      |
      |               +--------------------+---------------------+
      |                                    | cached results / features
      v                                    v
 +------------------------------ ONLINE -------------------------------+
 | request (member, device, context) -> fetch precomputed + live signals |
 | -> light model eval / filtering / context adaptation -> page          |
 | strict SLA; FALLBACK to precomputed or unpersonalized results         |
 +-----------------------------------------------------------------------+
```

4.2 Approaches described by Netflix

```
candidate rows (hundreds to thousands of possible row "themes"):
   genre rows (PVR) | Top Picks (Top-N) | Because You Watched (sims)
   Trending | Continue Watching | New Releases | My List | ...
      |      each row independently ranks its own titles
      v
row relevance model  score(row | member, context, device)
      |
      v
stage-wise greedy assembly, for slot k = 1..R:
     choose row maximizing  score(row | rows[1..k-1])
       - dedup titles already shown above (esp. in first visible columns)
       - diversity across row types and genres
       - constraints: e.g. Continue Watching high, row-type min/max,
         device-specific row counts and widths
      |
      v
artwork + evidence selection per (member, title)   -> Section 5
      |
      v
rendered page (top-m rows x top-n columns are what is actually seen)
```

5.3 Logging and offline evaluation

```
page request -> for each title shown:
   artwork policy (contextual model + exploration)
      -> chosen image_id, propensity p
      -> impression log (member, title, image_id, p, context)
      -> member sees image -> quality play?  -> reward log
offline:
   replay on randomized-exploration logs: keep events where the new
   policy picks the logged image; metric = take fraction (plays/impressions)
online:
   A/B test confirms replay wins
```

7.1 The measurement funnel

```
idea / new model
   |
   v
OFFLINE  ranking metrics on held-out plays; replay / OPE on randomized logs
   |      (screening only: offline wins often do not transfer [4][18])
   v
INTERLEAVING (stage 1 online)  team-draft, hours attributed to each ranker;
   |      approx. 100x fewer members than A/B for the same power [10]
   v
A/B TEST (stage 2 online)  ABlaze cells; OEC = retention,
   |      sensitive proxy = streaming hours; proxy rewards validated here
   v
launch -> long-term monitoring; quasi-experiments where no randomization
```

8. Consolidation: from many models to few

```
BEFORE                                  AFTER
canvas A -> model A (features A)        canvas A --+
canvas B -> model B (features B)        canvas B --+--> one multi-task model
canvas C -> model C (features C)        canvas C --+    input: (member, context,
search   -> model S (features S)        search   --+     task/canvas id, query?)
N pipelines, N on-calls                 1 pipeline, shared features, shared infra
```

9.2 Design (Chapters 5 and 6 summaries of [16], plus secondary sources [25][26])

```
member event streams (plays, browses, searches, thumbs, My List, ...)
     |
     v
interaction tokenizer: merge adjacent related events; attach attributes
     |  (sequence of tokens per member, lifetime history)
     v
+-------------------- FOUNDATION MODEL (offline, large GPU training) ----+
| item emb = gate(age) * ID emb + (1-gate) * metadata emb                 |
| causal Transformer, sparse attention, sliding-window training           |
| heads: next item | multi-token future items | genre / language aux      |
+-----------+-----------------------+------------------------+----------+
            |                       |                        |
            v                       v                        v
     (1) direct use           (2) embeddings             (3) fine-tune /
     next-title scores        member + title vectors     subgraph in
     as features / ranker     -> embedding store          downstream models
            |                       |                        |
            +------------> rows, search, notifications, page gen <------+
```

10.2 Maestro

```
Maestro (schedule / event trigger)
   -> Metaflow flow: fetch facts (Axion) -> build features (shared encoders)
      -> train (GPU, @resources) -> evaluate offline (replay/ranking metrics)
      -> publish model / embeddings (embedding store, aligned space)
   -> online services pick up new versions -> interleaving -> A/B (ABlaze)
```

19. Meta Case Study: From News Feed Ranking to Andromeda, Lattice, GEM and Generative Recommenders

```
2014   GBDT + LR ads CTR (online joiner, NE metric)                      [16]
2018   MSI ("meaningful social interactions") Feed value model           [3]
2019   DLRM open-sourced; Instagram Explore 3-pass distillation funnel    [9][5]
2020   Instagram constrained exploration; FB Search embedding retrieval   [6][37]
2021   Feed ranking explainer; survey-weighted objectives; Neo/ZionEX     [1][2][12]
2022   DHEN; DSI data-ingestion paper; TorchRec (RecSys)                  [17][15][13]
2023   Lattice (ads model consolidation); Explore scaled to 4 stages     [22][7]
2024   HSTU generative recs; Wukong; InterFormer; Andromeda   [30][19][20][25]
2025   GEM ads foundation model; Lattice paper; DHEN for CVR              [27][23][18]
2026   GEM trained at LLM scale (MFU 20-25%); ULTRA-HSTU                  [28][34]
```

1.2 Architecture: multi-pass ranking on "predictors"

```
inventory: all eligible posts for the viewer (> 1,000 on average)
      |
      v
pass 0: integrity processes + eligibility (remove violating / ineligible content)
      |
      v
pass 1: lightweight model narrows inventory to ~500 most relevant posts
      |
      v
pass 2: heavy multitask neural networks predict P(action_i) per post
        (like, comment, share, hide, "worth your time" survey answer, ...)
        -> combined into one value score per post
      |
      v
pass 3: contextual re-ranking (e.g. content-type diversity, avoid runs of
        similar posts) -> final ordered Feed
                                                    (after Meta 2021 [1][4])
```

Ranking funnel.

```
seed accounts --ig2vec KNN--> candidate accounts --> their media (+ other sources)
          |
          v
 ~500 candidates
          |   pass 1: distillation model (minimal features), trained to mimic the
          |           combined output of the later stages
          v
 ~150 candidates
          |   pass 2: lightweight neural network
          v
 ~50 candidates
          |   pass 3: full deep neural network (all dense features)
          v
 ~25 shown first (after integrity and diversity rules)
                               (after Instagram 2019 [5]; counts approx.)
```

2.3 Explore 2023: four stages and "caching everywhere"

```
+--------------+  <=1,500  +------------------+  top 100  +-----------------+  +--------+
| Retrieval    |---------->| 1st-stage rank   |---------->| 2nd-stage rank  |->| Final  |
| heuristics,  |           | two-tower,       |           | MTML network +  |  | rerank |
| item-CF, PPR,|           | cached towers,   |           | linear value    |  | integ.,|
| two-tower ANN|           | distilled from   |           | model (BO-tuned)|  | divers.|
| (weighted)   |           | 2nd stage        |           |                 |  |        |
+--------------+           +------------------+           +-----------------+  +--------+
  item tower: offline daily embeddings | user tower: online, fresh features
  some users' heavy-stage results precomputed off-peak
                                 (after Meta 2023 [7] and Explore system card [8])
```

3.1 DLRM (2019)

```
          dense features            sparse features (multi-hot IDs)
               |                       |        |        |
          +---------+             +------+ +------+ +------+
          | bottom  |             | EMB  | | EMB  | | EMB  |   <- almost all params,
          |  MLP    |             | T1   | | T2   | | Tk   |      model-parallel
          +---------+             +------+ +------+ +------+
               |                      | pool   | pool   | pool
               +----------+-----------+--------+--------+
                          |  pairwise interaction (dot products / concat)
                     +---------+
                     | top MLP |   <- data-parallel
                     +---------+
                          |
                      P(click)
```

3.3 TorchRec, FBGEMM and the data pipeline

```
serving --> Scribe --> ETL / label join --> Hive (DWRF) on Tectonic
                                                  |
                        DPP CPU fleet (read cols, decode, transform)
                                                  |  tensors
                                                  v
                     GPU trainers (TorchRec: sharded EMB + DP MLP, FBGEMM TBE)
```

4.3 Meta Lattice (2023 blog, 2025 paper)

```
before Lattice                          after Lattice
---------------------------------       ------------------------------------------
model(FB Feed, CTR)                     +--------------------------------------+
model(FB Feed, purchase CVR)            | Lattice: one multi-domain multi-task  |
model(IG Reels, CTR)          ==>       | model; domain-aware sparse experts;   |
model(IG Stories, install)              | heads per objective; Pareto balancing |
... (N surfaces x M objectives)         +------------------+-------------------+
                                                           | distillation
                                             lean serving models per stage/surface
```

5.2 Architecture

```
user + context features
       |
       v
+-----------------------------------------------------------------+
| Andromeda retrieval DNN on Grace Hopper (unified CPU-GPU mem)  |
|   level 0:  root clusters   --score--> keep top nodes          |
|   level 1:  sub-clusters    --score--> keep top nodes          |
|   level L:  individual ads  --score--> top ~10^3 ads           |
|   (hierarchy learned jointly with the scoring network)         |
+-----------------------------+-------------------------------+
                              v
         ranking (Lattice / GEM-distilled models) --> auction --> feed insertion
```

6.2 Role in the stack: central teacher, not online scorer

```
          +-------------------------------------------------+
          | GEM (offline, LLM-scale, thousands of GPUs)    |
          | long user ad/organic sequences + non-sequence  |
          | features across surfaces and objectives         |
          +-----------------------+-------------------------+
                                  | distillation / representation transfer
       +--------------------------+--------------------------+
       v                          v                          v
 retrieval models           light rankers              heavy rankers
 (Andromeda)                (per surface)              (Lattice family)
       \_____________________________|_____________________________/
                                     v
                     auction + allocation into feeds
```

7.3 The HSTU block

```
  X --> Linear --> SiLU --> split --> U, V, Q, K
                                       |
         SiLU(Q K^T + rab(position, time)) * V     <- no softmax; relative pos+time bias
                                       |
                        LayerNorm(.) * U           <- gating replaces the wide FFN
                                       |
                             Linear --> + X (residual)
```

20. TikTok/ByteDance and Kuaishou: Real-Time Short-Video Recommendation, from Monolith to OneRec

Online training loop.

```
client actions (play, like, skip...) --> Kafka: action log --+
serving-time feature log ------------> Kafka: feature log --+
                                                            |
                        Flink online joiner (join by request key;
                        in-memory cache + on-disk KV for late actions;
                        negative sampling)
                                                            |
                     +--------------------------------------+-----------+
                     v                                                  v
          Kafka: training examples                          HDFS (batch dumps)
                     |                                                  |
        online training workers <---------- same model code -----> batch training
                     |
        training PS (Cuckoo hash tables, admission + expiry)
                     |  sparse params: incremental sync of touched IDs (minutes)
                     |  dense params: synced far less often
                     v
        serving PS --> online inference for the feed
```

20.5.2 RankMixer (2025) and TokenMixer-Large (2026)

```
 raw features --> group/project --> T tokens x D
    [ token mixing (heads shuffled across tokens) -> add & norm
      per-token FFN / sparse MoE (separate weights per token) -> add & norm ] x L
 --> mean pool --> multi-task heads (finish, like, play time, ...)
```

20.5.4 The ByteDance pattern

```
      +-------------------- Monolith-style online training ------------------+
      | Kafka actions + feature logs -> Flink joiner -> streaming trainer    |
      | Cuckoo-hash sparse PS  --(minute-level incremental sync)--> serving  |
      +-----------------------------------------------------------------------+
 request --> retrieval (two-tower / Deep Retrieval paths / multi-channel)
         --> pre-rank --> rank: LONGER (10k seq, KV cache) + RankMixer tokens
         --> value-model fusion (P_like*V_like + E_playtime*V_playtime + ...)
         --> diversity rules (no same creator / same sound back to back)
         --> For You feed
          (composite, synthesised from [1][4][6][11][12]; not an official diagram)
```

20.6.1 RecFlow and the six-stage funnel

```
corpus (10^8+ videos)
   | multi-channel retrieval: i2i, u2i two-tower, KuaiFormer, ...
   v
~8,000 merged candidates
   | pre-ranking (lightweight)
   v
~500
   | cascade rank -> full rank (multi-task: pLTR, pWTR, pFTR, watch time, ...)
   v
tens
   | ensemble sort (pXTR fusion) + re-rank (listwise, diversity, business rules)
   v
~6-10 exposed                                  (after [16][23][24]; approximate)
```

20.7.1 TWIN (KDD 2023)

```
per behaviour j:  K_j = W * inherent_j        (precomputed, cached per item)
                  b_j = w . cross_j            (cheap, user-specific)
per candidate c:  s_j = Q(c) . K_j / sqrt(d) + b_j
CP-GSU: top-100 by s_j over ~10^4 behaviours -> ESU: multi-head target attention
```

20.9.1 OneRec V1 (February 2025 paper, June 2025 technical report)

```
static feats -----+
short seq (20) ---+--> Encoder (Transformer, RMSNorm) --> 405 context tokens
pos-feedback(256)-+                                              | cross-attn
lifelong (1e5) -> hier. k-means (2000) -> QFormer (128) -------+
                                                                 v
               [BOS] s1 s2 s3 --> Decoder (causal self-attn + MoE FFN)
                                                                 | beam search
                                                                 v
               candidate sessions --> reward system:
                   P-Score (SIM-based, one tower per objective: CTR, LTR, VTR, ...)
                   + format reward (legal SIDs only) + SIR (viral down-weighting)
                                                                 |
               IPA / DPO (V1 paper)  ->  ECPO = early-clipped GRPO (tech report)
```

21. Twitter/X Case Study: From WTF and GraphJet to the 2023 the-algorithm and the 2026 Phoenix Transformer

Architecture

```
          follow graph (in memory, Cassovary)
                       |
  user u --> personalized PageRank --> circle of trust (hubs, top-k)
                       |
          hubs  ==follow edges==>  authorities
                       |
                  SALSA iterations
                       |
     authorities ranked --> "Who to Follow" candidates --> filters --> UI
```

Production descendants

```
Event Bus / Kafka (fav, RT, follow, client events)
          |
     recos-injector  (filter, combine, per-service topic)
          |
  +--------+---------------------+
  |                              |
 UTEG (GraphJet, 24-48h window)  other GraphJet recos services
  |
 input: viewer's RealGraph-weighted followees
 traverse user->tweet edges, aggregate engagement weight
  |
 top tweets with social proof "<followee> liked"
```

21.3.1 SimClusters (KDD 2020)

```
follow graph A (consumers x producers)
       |  cosine sim over followers, threshold
       v
producer-producer graph --MH community detection (k~145K)--> KnownFor V (20M x 145K)
       |
  U = A . V  ---> InterestedIn (consumer, long-term)
       |                        \
       |                         +--> producer embeddings (multi-community)
  fav stream (Heron) --> tweet vec += InterestedIn(faver)   [real-time]
       |
SimClusters ANN: cluster -> top tweets index; user vec -> candidate tweets
```

21.4.2 Architecture

```
client request (For You)
       |
 HOME MIXER (Scala, Product Mixer framework)
 ForYouProductPipelineConfig
  +-- ForYouScoredTweetsMixerPipeline  (mix tweets + ads + WTF users)
  |     +-- ScoredTweetsRecommendationPipeline
  |     |     candidate pipelines (fetch caps from ScoredTweetsParam):
  |     |       InNetwork (Earlybird)   <= 600
  |     |       TweetMixer / CR-Mixer   <= 400   (SimClusters, TwHIN ANN, ...)
  |     |       UTEG (GraphJet)         <= 300
  |     |       FRS (FutureGraph)       <= 100
  |     |     feature hydration (~6000 features)
  |     |     ScoringPipeline: heavy ranker (MaskNet, Navi)
  |     |     heuristics: author diversity, in/out balance, fatigue, dedup
  |     |     visibility filtering
  |     +-- ConversationService pipeline  (reverse-chron backup on failure)
  |     +-- Ads pipeline
  |     +-- WhoToFollow pipeline (FRS)
  +-- marshal -> timeline (modules, social context, cursors) -> client
```

21.4.6 The heavy ranker (MaskNet)

```
 ~6000 features (aggregates, graph, embeddings, content)
       |  double-norm-log featurization
       v
 +-----------+-----------+-----------+-----------+
 | MaskBlock | MaskBlock | MaskBlock | MaskBlock |   parallel
 | x*mask(x) | x*mask(x) | x*mask(x) | x*mask(x) |   (1024 each)
 +-----------+-----------+-----------+-----------+
       | concat (4096) -> MLP 2048 (shared)
       v
 10 task heads: MLP 256-128-1, sigmoid  -> P(engagement_i)
       |
 score = sum_i w_i * P_i   (weights from config)
```

21.6.2 Request path

```
FOR YOU REQUEST
  |
HOME MIXER (Rust) -- Post Pipeline (PhoenixCandidatePipeline)
  1 query hydration: user action sequence (recent engagements = main model input),
    following list, blocks/mutes, muted keywords, seen/served posts, topics
  2 candidate sources (parallel)
      in-network : Thunder (in-memory recent posts of followees)   cap 1200
      out-of-net : Phoenix retrieval (two-tower ANN)               cap 1000
                   SimClusters (cluster similarity, <=48h)
                   Popular posts (shared list)                      cap 500
  3 candidate hydration: text/media, author + labels, quoted post, lang, counts
  4 pre-scoring filters (dedup, 48h age, self, OON replies/RTs, seen/served,
    muted kw, block/mute, subscription, election filter, holdouts ...)
  5 scoring: PhoenixScorer -> RankingScorer (weighted sum, author decay,
    OON discount, new-author boost) -> VMRanker (DPP rerank)
  6 selection: TopKScoreSelector
  7 post-selection: VFCandidateHydrator -> VFFilter -> AncillaryVFFilter
    -> DedupConversationFilter
  |
Blending Pipeline (ForYouCandidatePipeline): ads blender, WTF, prompts
  |
side effects (after response): record served posts, refresh cache, log events
```

21.6.5 Phoenix ranking (transformer with candidate isolation)

```
 keys ->      | user | history (S) | candidates (C) |
 query user   |  v   |   v v v v   |   x x x x      |
 query hist   |  v   |   v v v v   |   x x x x      |
 query cand_1 |  v   |   v v v v   |   v x x x      |  <- self only
 query cand_2 |  v   |   v v v v   |   x v x x      |
 ...
 => score(c) depends only on (user, history, c): batch-invariant, cacheable
```

21.7 Visibility and Trust & Safety in 2026

```
CONTENT UNDERSTANDING (async)            LABELING RULES             SERVE
grox/ (LLM + classifiers: spam, adult,   scarecrow/ (event rules,   visibility-filtering/
  violent media, text/image embeddings)    botmaker rule engine)      per (post, viewer):
media-model-proxy/, clip/, pnsfwmedia/   abuse-enforcement-svc/       ALLOW | NOTICE |
agatha/ (blocks+reports vs favs)           (+ abuse-ledger appeals)   INTERSTITIAL | DROP
bdsm/ (behaviour sequences)              safety-label-user-agg/       |
user-cred-v2/ (PageRank cred)                        |                VFFilter,
              \_____________ labels -> storage _/                    AncillaryVFFilter
```

22. Pinterest Case Study: From Pixie Random Walks to PinFM Foundation Models

1. The home feed funnel at a glance

```
 user request
     |
     v
+---------------------- CANDIDATE GENERATION (recall) ----------------------+
| Pixie walks | learned two-tower EBR (Manas) | conditional / multi-emb EBR  |
| token indexes | PinRec generative retrieval | explore/exploit sources      |
+-----------------------------------+-----------------------------------+
                                    | merge + dedup (thousands)
                                    v
                  Light-Weight Scorer (two-tower pre-ranking)
                                    | hundreds
                                    v
      Pinnability ranker (wide&deep + DCN-style crossing + MTL heads)
        <- TransAct / TransAct V2 sequence modules, PinFM, PinnerFormer
                                    |
                                    v
        Multi-Objective Blender (utility weights; PRL-PUTS per request)
                                    |
                                    v
                        home feed grid  (after [25][26][33])
```

Architecture

```
user actions (saves, clicks)        offline graph build (daily)
        |                                   |
        v                                   v
query pins Q = {q1..qk}, weights      pruned Pin<->Board graph (~120 GB)
        |                                   |
+------ Pixie server (full replica, everything in RAM) ---------------+
| per q: step budget ~ weight(q) scaled by degree(q)                  |
|   walk pin -> board -> pin ... biased by user features,             |
|   restart with prob alpha; count visits V_q[p]                      |
|   early-stop when n_p pins have >= n_v visits                       |
| combine: V[p] = (sum_q sqrt(V_q[p]))^2   (multi-hit boost)          |
+-----------------------------+-------------------------------------+
                              v
                  top-K pins -> ranking -> surface
```

Architecture and scale

```
 TRAINING (producer-consumer)
  CPU producers: sample (query, positive) pairs from co-engagement
                 -> random-walk neighbourhoods (top-T by visit count)
                 -> gather visual + text features            --+
  GPU consumer:  2 conv layers, importance pooling,             |
                 max-margin loss, random + curriculum negatives <+
 INFERENCE (MapReduce, no redundant computation)
  layer-1 embeddings for all nodes -> join -> layer-2 -> Pin embeddings
                 -> ANN index -> Related Pins / Homefeed candidates
```

Architecture

```
user's recent actions (pin, action type, surface, time features)
    |  each pin: 256-d PinSage embedding (frozen) + learned feature embs
    v
causal Transformer encoder
    v
L2-normalised user embedding, recomputed daily for every user
    |
    +--> ANN retrieval against pin embeddings
    +--> feature in Homefeed ranking and in ads ranking models
```

8.2 TransAct: real-time actions inside the ranker (KDD 2023)

```
real-time action stream                                  candidate pin
        |                                                       |
latest ~100 actions: PinSage emb + action type + timestamp    |
        |                                                       |
early fusion: concat each action with candidate embedding <--+
        v
small Transformer encoder
        v
output compression (first K tokens + max pooling) -> flatten
        v
concat with PinnerFormer + other features -> DCN v2 crossing -> MLP
        v
multi-task heads (repin, click, hide, ...) -> utility
```

8.3 TransAct V2: lifelong sequences (2025)

```
lifelong seq (~16k, int8)    real-time seq     impression seq
           \                       |                /
            +-- candidate-anchored NN search (dot product in PinSage
                space); most recent real-time actions always kept --+
                                    |
                        short selected subsequence
                                    v
                single-kernel Transformer (Triton)
                                    v
      ranker heads  +  auxiliary Next Action Loss
```

8.4 PinFM: a foundation model for user activity (RecSys 2025)

```
user history --> context Transformer (once per request) --> cached K,V
                                                                |
candidates c1..c1000 --> crossing attention over cached K,V -+--> features
                                                                  -> ranker
```

10. GPU serving for ranking

```
CPU era                  GPU era (2022+)            kernel era (2025+)
tree/linear + MLP  -->   batched DNN, CUDA graphs -->  fused Triton kernels,
per-candidate CPU        fused embeddings,            int8/int4 sequences,
cost ~ model size        GPU preprocessing            KV dedup (DCAT)
```

11. Pinterest ads ranking

```
ad candidates (offline/online ANN, targeting)
         v
MTMD lightweight ranking (multi-task, multi-domain)
         v
engagement model: user seq Transformer + PinnerFormer + features
                  -> DCNv2 / ML-DCN crossing -> pCTR, long-CTR heads
         v
auction + blending with organic feed
```

23. LinkedIn Case Study: Feed, PYMK, Jobs, LiRank, LiGNN, 360Brew and the ML Platform Beneath Them

23.1 The shape of LinkedIn's recommendation problems

```
members (QPS = member count x interaction frequency)
      |
      v
+-------------------+   large K (item pool >> member pool, e.g. posts),
|  Retrieval layer  |   high recall, two-tower / GPU exhaustive / LLM dual-encoder
+-------------------+
      |  hundreds to ~2,000 candidates
      v
+-------------------+   small K, high precision; DCN-style rankers (LiRank),
|   Ranking layer   |   or a text foundation ranker (360Brew)
+-------------------+
      |
      v
re-ranking / blending / fairness / ads slotting  -->  member-facing list
```

23.2.2 GLMix and GAME (Photon ML)

```
GLMix / GAME (coordinate descent with offsets)
-----------------------------------------------
round r:
  1. fixed effect  f(x)       <- fit on ALL rows, offset = sum(random effects)
  2. per-member    g_m(x)     <- fit per member m on its rows, offset = f + h
  3. per-job       h_j(x)     <- fit per job j on its rows,  offset = f + g
repeat until convergence
serve: logit = f(x) + g_member(x) + h_job(x)
```

23.3.2 Venice: the derived-data online store

```
Hadoop/Spark offline features --batch push--\
                                             >--> Venice store (per region,
Kafka -> Samza nearline features --stream---/     active-active, CRDT merge)
                                                      |   read compute:
Feathr defs (anchors, derived, registry)              |   dot / cosine
   |-- offline: PIT join onto impressions --> training data
   `-- online: batch-get features <---------- ranking services
                       (thin <10ms | fast <2ms | Da Vinci in-process <1ms)
```

23.4.1 Architecture (2019)

```
 FollowFeed FPR ----+   (network updates; >80% of feed items, 2019)
 Jobs FPR ----------+
 Recommended FPR ---+--> SPR: unified features, multi-objective score
 PYMK / modules ----+        sum_k w_k * P(action_k) (+ creator-side terms)
 Ads (separate auction) -------------+
                                     v
                    slotting (ads in allowed positions) -> Feed
```

23.4.4 Feed retrieval in the LLM era (2024-2026)

```
2019:  FPR_1 ... FPR_n (per-inventory)  -->  SPR (GLMix/DNN)  --> blend
2024:  + LiNR GPU out-of-network retriever; LiRank deep SPR; LiGNN embeddings
2025:  LLM (LLaMA-3) dual-encoder retrieval, ~2k cands, quantized numeric tokens
2026:  unified LLM retrieval + sequential generative ranker (approx., secondary)
       GPU "retrieval as ranking" + SilverTorch tensor ANN for pre-ranking
```

23.5.2 Architecture and key techniques

```
  sparse IDs --> embedding tables (compressed vocab, quantized)
  dense feats ----------------------------+
  member history seq --> Transformer -----+--> concat
                                         |
            +----------------------------+-----------------+
            |                            |                 |
       Residual DCN               Dense Gating          wide MLP
    (low-rank, attention,              |                   |
     residual cross layers)            |                   |
            +-------------+----------+--------------------+
                          v
                multi-task heads (Feed actions / job apply / ad click)
                          v
                jointly trained isotonic calibration layer
                          v
               calibrated P(action)  -->  value model / auction
```

23.6.1 LiNR: model-based neural retrieval on GPUs (2024)

```
 model binary = { query tower weights, item matrix [N x d], item attrs }
 request:  q    = tower(member)                      (GPU)
           mask = prefilter(attrs)                  (GPU, exact)
           s    = q . items^T over masked rows      (dense / quantized)
           topk(s) --> ranking
```

23.6.2 LiGNN: production GNNs (KDD 2024)

```
heterogeneous graph (member/job/post/ad/company/skill, typed + timestamped edges)
    |  densify low-degree nodes (cold start)
    v
temporal multi-hop sampling --> GNN (ID emb + content feats, long-term losses)
    |  7x faster: adaptive sampling, batch grouping/slicing, shm queue
    v
embeddings (batch + nearline) --> Venice / ANN --> Feed, Jobs, PYMK, Ads models
```

23.7 People You May Know (PYMK)

```
 member m
   |-- 2-hop graph walk (friends-of-friends, weighted by edge strength)
   |-- co-attribute joins (same company/school/time overlap)
   |-- address-book / imported contacts (where consented)
   |-- GNN embedding ANN (LiGNN-style; covers cold-start members)
   v
 candidate union (dedupe, filter existing connections, blocked, pending)
   v
 ranker: P(invite) x P(accept | invite)  (reciprocity), plus growth objectives
   v
 diversification + frequency caps --> PYMK module (also slotted into Feed)
```

23.9.2 Formulation

```
Instruction:  member profile + jobs + past interactions (applied / viewed /
              dismissed / no interaction); predict apply/view/dismiss for the
              "Question" job. (May include steering, e.g. "focus on skills,
              location, years of experience"; "30% weight to profile match,
              70% to historical activity")
Member Profile:  current position, company, location
Past job interaction data:
   applied: [Title, Location, Country, Company, Description] ...
   viewed:  [ ... ]
Question:  Will the member apply to the following job: [ ... ]
Answer:    The member will apply
```

24. Amazon and E-commerce: From Item-to-Item Collaborative Filtering to Query-Aware Retrieval

1.2 The algorithm

```
user history / cart / page context
       |
       v
+---------------------+     lookup per item      +--------------------------+
| seed items (k)      | ------------------------> | similar-items table      |
| purchased, rated,   |                           | item -> [(item, score)]  |
| carted, viewed      | <------------------------ | precomputed offline      |
+---------------------+   top-n lists per seed    +--------------------------+
       |
       v
merge + aggregate scores, drop owned/seen items, apply business filters
       |
       v
"Customers who bought this also bought", "Your Recommendations", cart recs
```

2.3 P-Companion: diversified complementary recommendation (CIKM 2020)

```
query item q
   |
   v
[product embedding (content + GNN over behaviour graph)]
   |
   +--> [complementary type transition]  type(q) -> {t1, t2, ..., tk}
   |                                       (diverse set of complement types)
   v
for each ti: [type-conditioned item embedding] --> ANN within type ti
   |
   v
merge per-type top items -> diversified "frequently bought together" list
```

3.1 Semantic Product Search (KDD 2019)

```
query text ---tok--> [uni|bi|char-3gram|hashed OOV] --embed--> avg pool --+
                                                                          |
                       shared embedding table                            cos
                                                                          |
product text -tok--> [uni|bi|char-3gram|hashed OOV] --embed--> avg pool --+
(title, attributes)
                    offline: embed catalog -> ANN index; online: embed query -> kNN
```

3.3 Query-aware recommendation as a design pattern

```
          query q, user u, context (page, cart, locale, device)
                    |
    +---------------+-----------------+----------------------+
    |                                 |                      |
lexical retrieval              semantic two-tower       item-to-item / complement
(inverted index)               (query emb -> ANN)       expansion from top results
    |                                 |                      |
    +---------------+-----------------+----------------------+
                    v
      candidate union, filters (stock, delivery, region, policy)
                    v
      ranker: f(q, u, item, context) with ESCI-style relevance + behaviour labels
                    v
      slotting: exact matches first; substitutes; complements / "related" widgets
```

4.3 Architecture of a Personalize-based deployment

```
clickstream / orders ----batch (S3 import)-----> datasets --> solution version (train)
        |                                                          |
        +--real-time PutEvents--> event tracker ------------+     |
                                                            v     v
app request (user, item, ctx) --> campaign / recommender endpoint
                                   - candidate scoring (recipe model)
                                   - filters (stock, owned, category)
                                   - promotions / exploration slots
                                  <-- ranked item ids (+ scores)
batch inference jobs --> S3 --> email, push, offline segments
```

6.2 Personalized embedding-based recommendations (2021)

```
new listing --> item tower (title tokens, category, aspects) --> item emb --> ANN index
                                                                                ^
user events (recent N) --> user tower (sequence encoder) --> user emb --kNN--+
                                                                              |
                                          candidates --> downstream ranker --> module
```

7.2 Unified Embedding (KDD 2023)

```
query + user features --> query/user tower ----------------------------+
                                                                       | cosine
listing --> [transformer text emb] + [term emb] + [graph emb] -> fuse --+
                                    |
                     ANN index (+ quality-aware scoring) --> union with lexical
                                                          --> learning-to-rank
```

10. A reference architecture for e-commerce recommendation

```
                       +--------------------------------------------+
 events (view, cart,   |  streaming layer: session features,        |
 purchase, search) --->|  real-time user embedding, cart state      |
                       +--------------------------------------------+
                                    |
 request(user, page, query?, cart)  v
 ----------------------------> candidate generation (union)
         - item-to-item tables (co-view: substitutes; co-buy: complements)
         - two-tower ANN (content-rich item tower; sequence user tower)
         - query-conditioned semantic + lexical retrieval
         - complement-type expansion (type -> items)
         - popularity / trending / exploration slots
                                    |
                                    v
              business filters: stock, delivery region, store availability,
              already-owned, policy, merchant/seller constraints
                                    |
                                    v
              ranker: multi-task (click, add-to-cart, purchase, return?)
              features: user x item x query x context; price, deal, rating
                                    |
                                    v
              re-rank: diversity (by product type), dedupe variants,
              promotions, explanations ("because you bought X")
                                    |
                                    v
              widgets: "Frequently bought together", "Customers also viewed",
              "Buy it again", "Recommended for you", search results
```

25. Spotify and Audio Recommendation: Discover Weekly, BaRT, Podcasts, AI DJ, LLM Playlists, Deezer Flow, Pandora, and the Music Streaming Stack

1.3 Discover Weekly as a pipeline

```
 play logs, playlist co-occurrence (user-curated playlists as "documents")
       |                          |
       v                          v
 implicit MF / word2vec-style   playlist-track co-occurrence
 track & user vectors           (tracks that appear together)
       |                          |
       +-----------+--------------+
                   v
       candidate tracks near the user's taste vector(s)
       (ANN search: Annoy, later Voyager)
                   |
  filters: not already heard, artist diversity, explicit-content
  settings, cold-start audio-model candidates
                   |
                   v
        ranking + sequencing into a 30-track weekly playlist
                   |
                   v
          batch generation for every user, written to storage,
          served on Monday (precomputed, not real-time)
```

1.4 Content-based audio models for cold start

```
audio (30s clips) -> mel-spectrogram -> conv layers (time-pooled)
                                            |
                                            v
                            predicted latent vector  ~=  CF vector (from ALS/LMF)
                                            |
          loss: MSE / weighted-MSE to the CF factors of tracks with enough plays
          use: embed tracks that have no plays -> same retrieval space
```

2.2 BaRT: Bandits for Recommendations as Treatments

```
         user context (time, device, history embeddings)
                        |
    candidate cards  x  candidate explanations ("Because you listened to ...",
    (playlists,          "Your top mixes", "Popular podcasts", ...)
     albums, shows)
                        |
                        v
         reward model  P(stream >= ~30s | user, card, explanation, ctx)
         (factorization machine with explanation interactions)
                        |
              epsilon-greedy policy  ---> log (action, propensity, reward)
                        |                          |
                        v                          v
                 rendered shelf            off-policy evaluation and
                                           retraining (IPS-weighted)
```

3.2 Generalized user representations (2024)

```
Stage 1: content & interaction encoders           Stage 2: user aggregation
----------------------------------------          -------------------------
audio / metadata / CF  -> item embeddings  ---->  user listening history (items,
(per-entity pretrained)                           weighted by recency/engagement)
                                                         |
                                                         v
                                                autoencoder -> compact user vector
                                                (refreshed frequently, stored in a
                                                 feature/embedding store)
                                                         |
                        +---------------+----------------+---------------+
                        v               v                v               v
                  Home ranking    search ranking   podcast recs    ads / other
                  (downstream models consume the general vector as a feature)
```

4.3 2T-HGNN for audiobooks (WWW 2024)

```
heterogeneous graph:  audiobooks -- podcasts -- (co-listened / co-saved edges)
                      nodes carry LLM-based text embeddings of title/description
                             |
                             v
            HGNN (GraphSAGE-style message passing) -> item embeddings
                             |
user features: music + podcast + audiobook  -->  user tower
history, demographics                            |
                             +-------------------+
                             v
             two-tower (2T) model: user vec . item vec
                             |
                ANN retrieval -> downstream ranking on Home / search
```

5.1 Spotify AI DJ (2023)

```
user history -> recommender -> "segments" (theme + 4-8 tracks)  [precomputable]
                                    |
         editor-curated facts/ ----> LLM script generation (short, templated,
         style guides                checked/filtered before use)
                                    |
                              TTS voice model (Sonantic-derived)
                                    |
              client: [voice intro][track][track]...[voice transition][track]...
              user taps DJ button -> skip whole segment -> new theme
```

5.2 AI Playlist (2024) and prompt-based playlist generation

```
prompt "upbeat 80s synthpop for running"
    |
    v
LLM (fine-tuned seq2seq) --decode--> semantic IDs  e.g. <c12><c7><c3>  (per track)
    |                                    |  (IDs from clustering CF embeddings)
    |                                    v
    |                        ID -> track lookup (no fuzzy title matching)
    v
personalization re-rank with user vector, dedupe, artist diversity, sequencing
```

6.2 Semi-personalized cold start (KDD 2021)

```
existing users: embeddings u from MF on listening data
        |
        v
k-means over warm users  ->  K clusters (each with precomputed top tracks)
        |
new user registers: demographics (country, age), onboarding answers
(selected artists/genres), first-day interactions
        |
        v
feedforward network predicts the user's embedding u_hat (trained on warm users:
input = their registration-time features, target = their eventual embedding)
        |
        v
assign u_hat to nearest cluster -> serve that cluster's precomputed recs
        (semi-personalized: cheap, cacheable, robust to noisy u_hat)
```

7.1 Music Genome Project

```
seed (artist/song) -> gene vector (expert-annotated, ~hundreds of attributes)
         |
         v
candidate songs by weighted gene-space similarity
         |
+ collaborative signals from thumbs up/down across listeners (added over time)
+ per-station adaptation: thumbs on this station reweight genes/candidates
         |
         v
ensemble ranker -> next song (licensing constraints: DMCA radio rules limit
                  skips and how often an artist/album can repeat)
```

10.2 Batch versus real-time split

```
 BATCH (daily/weekly)                          REAL-TIME (per request / per track)
 --------------------                          -----------------------------------
 MF / embedding training                       session context, last N tracks, skips
 audio & text content embeddings               user vector refresh (frequent)
 ANN index builds (Voyager/Annoy)              shelf/card ranking on Home (BaRT-like)
 Discover Weekly / Release Radar / Daylists*   radio / autoplay / DJ next-segment
 LLM explanation text precompute               bandit exploration + propensity logging
```

26. Marketplace and Travel Recommenders: Airbnb, Booking.com, Uber Eats, DoorDash and Others

26.2.1 Overall architecture

```
guest query (place, dates, guests, filters)
       |
       v
+--------------------------+     availability, price, geo filters
| Retrieval                |---> (inverted index; since 2025 also
|  - geo + availability    |      two-tower EBR with IVF clusters as filters [8][9])
+--------------------------+
       |  candidates (order of thousands, approximate)
       v
+--------------------------+     features: listing, query, guest, real-time
| First-pass / base ranker |     session features from embeddings [1]
|  DNN (pairwise/listwise) |     (journey ranker: multi-task base module [5])
+--------------------------+
       |
       v
+--------------------------+
| Second pass / context    |     diversity given listings above [6],
| twiddlers, business rules|     cancellation / rejection penalties [5]
+--------------------------+
       |
       v
 ranked results  ---> logs (impressions, clicks, bookings) ---> training
                       \--> interleaving experiments [10]
```

26.2.4 Architecture, bias and cold start (KDD 2020)

```
  query + guest features            listing features (+ position at train time)
          |                                    |
    [ query tower ]                     [ listing tower ]
          |                                    |
      "ideal listing"  ----- distance -----  listing vector
                               |
                     score -> pairwise loss
  at serving: position feature := 0 ; new-listing engagement := neighbour estimate
```

26.5.1 Graph learning for dish and restaurant recommendations (2019)

```
orders (per city) -> weighted bipartite graphs (user-dish, user-restaurant)
     -> GraphSAGE (sampled 2-hop, max-margin + low-rank positives), daily
     -> user / dish / restaurant embeddings
     -> similarity features ---> homefeed ranker ---> carousels and store list
```

26.5.3 Eats search pipeline: hydration is the hidden cost

```
eater request
   |
   v
retrieval paths (lexical, embedding/ANN, geo)  --> tens of thousands of candidates
   |                                                  (before the fix)
   v
light hydration (ranking features only)
   |
   v
first-pass ranker (high recall, low latency)
   |
   v
second-pass ranker (multi-objective)
   |
   v
full hydration (presentation data) only for shown results
```

26.6.3 Feature and prediction infrastructure

```
events (orders, views, ETAs) --Kafka--> Riviera (Flink, real-time features) --+
warehouse tables -----------> Fabricator (batch, declarative)  ----------------+
                                                                               v
                                                          online feature store (Redis)
                                                                               |
homepage request --> candidate stores (geo, open now) --> Sibyl prediction -----+
                                                              (store ranking + UCB)
                                                                     |
                                                                     v
                                                  carousels + store list on homepage
```

27. The Chinese Ecosystem: Alibaba, Tencent, Meituan, Baidu, Huawei, JD, Xiaohongshu and the Architecture of Hyperscale Commerce and Content

Overview

```
         THE CHINESE "FULL-LINK" RECOMMENDATION / ADS FUNNEL
  (Alimama's own description: recall -> pre-rank -> rank -> exposure
               -> click -> conversion -> repurchase [12])

corpus 10^7 - 10^9 items
  |  multi-channel recall: item-CF, EGES i2i, MIND multi-interest ANN,
  |  TDM tree beam search, RALM look-alike, graph walks (Graph-Learn / embedx)
  v  ~10^3 - 10^4 candidates
pre-rank ("cu pai")  COLD: small DNN with cross features, compute-aware
  v  ~10^2 - 10^3
rank ("jing pai")    DIN / DIEN / SIM / ETA / BST target attention;
  |                  ESMM / PLE / AITM multi-task heads (CTR, CVR, CTCVR, ...)
  v  ~10^1 - 10^2
re-rank              PRM / PEAR / PIER / generator-evaluator permutations
  v
exposure -> click -> conversion -> repurchase (labels flow back via streaming)
```

1.1 DIN: Deep Interest Network (KDD 2018)

```
candidate ad emb --------------------------+
behavior_1 emb --[act unit(b1, ad)]-- w1 --+
behavior_2 emb --[act unit(b2, ad)]-- w2 --+--> sum_i w_i * b_i  (candidate-specific
...                                        |                      user interest)
behavior_T emb --[act unit(bT, ad)]-- wT --+
       concat(user interest, profile, ad, context) --> MLP (Dice) --> pCTR
```

1.3 Lifelong sequences: MIMN/UIC, SIM, ETA

```
UIC (event-driven)                     SIM (request-driven, two-stage)
behavior event --> update memory       request(user, candidate c)
                   M_u in UIC            |  GSU: UBT lookup [user][cat(c)]
request --> read M_u (O(1)) --> MIMN     |       (22 TB index, hard search)
                                         v  top-K behaviors
                                       ESU: multi-head target attn(c, top-K)
                                         v
                                       concat with short-term DIN/DIEN -> MLP
```

1.4 ESMM: Entire Space Multi-Task Model (SIGIR 2018)

```
loss = sum_i CE(y_i, pCTR(x_i)) + sum_i CE(y_i AND z_i, pCTR(x_i) * pCVR(x_i))

 impression x --> [shared embedding layer] --+--> CTR tower --> pCTR ----+
                                             |                           (x)--> pCTCVR
                                             +--> CVR tower --> pCVR ----+
```

1.6 TDM and JTM: learned tree indexes for retrieval

```
offline (XDL)                                  online (TDM Serving)
item embs --k-means--> tree T_k                request(user behaviors)
    ^                    |                       |
    |      samples: ancestors(+), same-level(-)   v
    +---- train DNN M_k <-+                    level 1: score children of root (Blaze)
         (repeat until stable)                  keep top-K --> expand children
                                                ... level L: top-K leaves = items
export: tree.pb + dense/sparse/item_emb  --->   + business filters in Index layer
```

2.1 PLE: Progressive Layered Extraction (RecSys 2020 best long paper)

```
level 1:  [VTR experts]   [shared experts]   [VCR experts]
             gate_A          gate_shared         gate_B
level 2:  [VTR experts]   [shared experts]   [VCR experts]
             gate_A                             gate_B
          tower_A -> VTR (effective view, AUC)   tower_B -> VCR (completion, MSE)
```

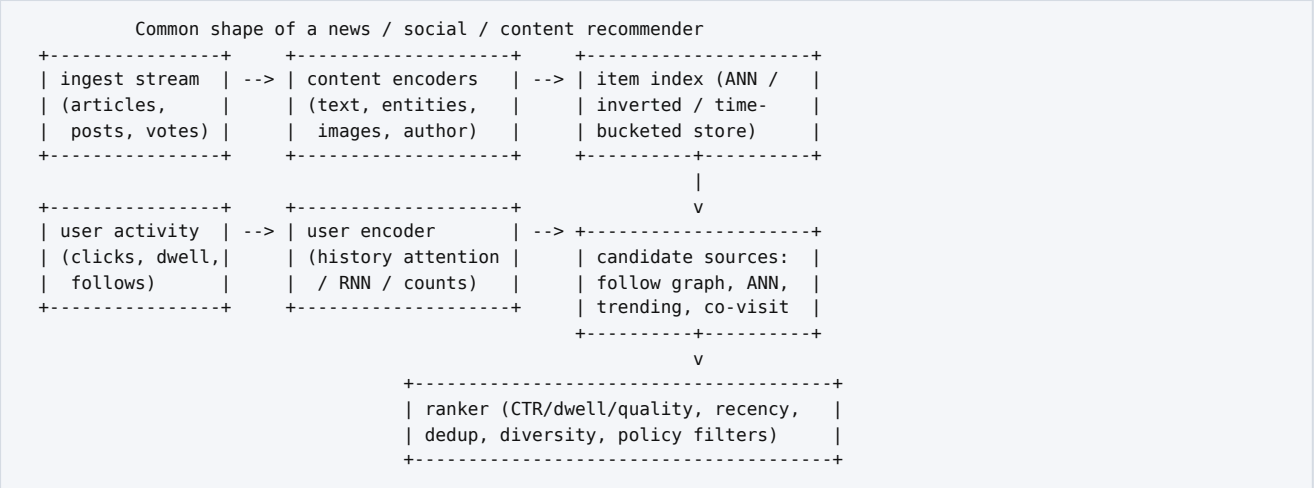
7. Architecture patterns characteristic of Chinese hyperscale

```
       ALGORITHM-SYSTEM CO-DESIGN: WHERE THE WORK MOVES

 request-time compute  -->  moved to ...                 example
 --------------------      -------------------------    ----------------------
 attend over history        event-time memory update     UIC / MIMN [6]
 attend over 54K items      offline category index       SIM UBT, 22 TB [7]
 pick top-k behaviors       bit ops on SimHash codes     ETA [8], SDIM [59]
 embed images per sample    parameter-server compute     AMS / DICM [24]
 traverse graph n hops      write-time pre-sampling      DGS, 20 ms P99 [28]
 score 10^7 items           beam search on learned tree  TDM Serving [19]
 score m candidates         one pass, masked attention   MTGR [57]
```

28. News, Social and Content Platforms: From Click Co-visitation to Open Feed Protocols

Overview



Architecture and models



28.3.2 The model family: DKN, NPA, NAML, LSTUR, NRMS

```
title words (<=30) -> GloVe 300d -> multi-head self-attn (20 heads x 20 dim)
                   -> additive attention (hidden 200) -> news vector r_i
clicked news r_1..r_50 -> multi-head self-attn -> additive attention -> user u
candidate news r_c  -> score = <u, r_c>
training: 1 positive + 4 sampled negatives from same impression, softmax CE
```

"Hot" ranking for posts

```
score   = ups - downs
order   = log10(max(|score|, 1))
sign    = +1 / 0 / -1 by sign of score
seconds = epoch_seconds(post_time) - 1134028003
hot     = round(sign * order + seconds / 45000, 7)
```

Architecture

```
  user client                PDS                     Feed Generator (did:web)
 +-----------+  getFeed   +------+  getFeedSkeleton  +----------------------+
 | app view  | ---------> | PDS  | ----------------> | index built from the |
 |           |            |      |  (JWT signed by   | firehose; returns    |
 |           | <--------- |      |   user's key)     | [post URIs + reason] |
 +-----------+  hydrated  +------+ <---------------- +----------^-----------+
                posts       hydrate: post content,              |
                            author, like/repost counts   com.atproto.sync.
                                                         subscribeRepos (firehose)
```

Trending: a chi-square-like spike detector

```
expected = accounts sharing the link one day earlier (min 1)
observed = accounts sharing it now
score    = 0                          if observed < 5 or expected > observed
         = (observed - expected)^2 / expected   otherwise
max_score kept with cooldown 2 days; decayed = max_score * 0.5^(age / 8h)
trend if decayed >= 1
```

What is verifiable

```
Snap-style retrieval stack (composite of [24]-[26] and Chapter 2)
+----------------+   +-------------------+   +-------------------------+
| social graph + |-->| GiGL: subgraph    |-->| user/item/node embeds    |
| engagement     |   | sampling, GNN     |   | -> ANN index (EBR)       |
+----------------+   +-------------------+   +------------+------------+
+----------------+   +-------------------+                v
| item text/LLM  |-->| semantic IDs      |-->  generative retrieval (GRID)
+----------------+   +-------------------+                v
                                          multi-source candidates -> ranker
```

29. LLMs in Recommender Systems (2023-2026): Rankers, Features, Agents, Distillation and Production Deployments

29.1 A taxonomy: where and how the LLM is used

```
                    WHERE THE LLM SITS (2023-2026)
+-------------------------------------------------------------------------+
| offline / nearline                          | online (per request)      |
|                                             |                           |
|  [item text, reviews, images]               |  user query / dialogue    |
|        |                                    |        |                  |
|        v                                    |        v                  |
|  LLM knowledge gen (KAR, LLMRec)            |  LLM agent / intent parser |
|  LLM embeddings (RLMRec, NoteLLM, HLLM,     |  (InteRecAgent, AI Playlist) |
|   LinkedIn LLaMA-3 dual encoder)            |        |  tool calls      |
|  semantic-ID tokenizer (TIGER, PLUM)        |        v                  |
|  LLM teacher labels (DLLM2Rec, SLMRec)      |  retrieval -> rank -> rerank |
|        |                                    |   ^          ^        ^    |
|        +--> features/IDs/students ----------+---+          |        |    |
|                                             |  LLM ranker (360Brew, LlamaRec)|
|                                             |  LLM rerank / explain (DJ)  |
+-------------------------------------------------------------------------+
  cost amortised over items or users          cost paid per request x candidate
```

29.3 LLM-generated features, knowledge and embeddings (the most deployed pattern)

```
 OFFLINE / NEARLINE (batch, cacheable)                    ONLINE (unchanged latency)
 item text, reviews, images, user history summaries
         |
         v
 +----------------------+    +------------------+    +---------------------------+
 | LLM (frozen, API or  |--->| text encoder /   |--->| feature store: dense vecs, |
 | fine-tuned): generate |    | adapter (small)  |    | tags, semantic IDs         |
 | reasoning, attributes |    +------------------+    +-------------+-------------+
 | profiles, labels      |                                          |
 +----------------------+                                          v
                                                  conventional CTR / two-tower model
                                                  (DIN, DCN, SASRec ...) at full QPS
```

29.3.5 LinkedIn: LLaMA-3 dual-encoder retrieval for the Feed (2025)

```
member profile + recent activity text --> LLaMA-3 (fine-tuned) --pool--> member vec
                                                                              |
post text (nearline, on create/update) --> LLaMA-3 (shared or twin) --pool--> post vec
                                                                              |
                        ANN / GPU exhaustive search over hundreds of millions  v
                        ---------------------------------------> ~2,000 candidates
                                                                 -> Feed ranker
```

29.4.1 YouTube PLUM (2025)

```
1. Item tokenization: Semantic IDs v2 (multimodal content emb + co-engagement)
      -> RQ-style multi-level codes, tuned for coverage and low collision
2. Continued pre-training (CPT) of a Gemini-family LLM on:
      user watch sequences as SID tokens + SID<->text alignment (titles, topics)
3. Task fine-tuning for generative retrieval:
      user context + history (SIDs + text) -> decode SIDs of next videos
4. Serving: beam decoding of SIDs -> videos, as one retrieval source
      feeding the existing ranking stack
```

29.5 LLM as the ranker in production: LinkedIn 360Brew (2025)

```
Instruction: predict apply/view/dismiss for the Question job (optional steering)
Member Profile: position, company, location
Past job interactions:  applied: [title, location, company, description] ...
                        viewed:  [...]   dismissed: [...]
Question: Will the member apply to: [job text]
Answer:   "The member will apply"   <-- score = logit(answer tokens)
```

29.6.2 InteRecAgent (Microsoft RecAI): LLM brain, recommender tools

```
user: "something like Hollow Knight but co-op, under $20"
  |
  v
LLM planner --(1 plan call)--> [SQL: price<20 AND tag~co-op] -> candidate bus
                               [item-sim: Hollow Knight]     -> candidate bus
                               [rank: preference(user profile)] -> top-k
  |                                                              |
  +<------------------------ results (IDs -> titles) ---------------+
  v
LLM writes the answer + explanation; profile memory updated (prefer/unwanted)
```

29.7 Knowledge distillation from LLMs

```
                  TEACHER (offline, expensive)
    LLM ranker / 360Brew-like / LLM-scale FM      <- trained on text + interactions
        |  soft labels, rankings, rationales, embeddings
        v
 +-------------+   +-------------------+   +----------------------+
 | small LM    |   | SASRec / two-tower|   | CTR model features   |
 | (1-3B)      |   | student           |   | (LLM-derived dense)  |
 +-------------+   +-------------------+   +----------------------+
        \_____________________ online serving at full QPS _____________/
```

29.10.1 Where the cost lands

```
cost per day  ~=  (#requests) x (#candidates scored by LLM) x (tokens per candidate)
                  x (cost per token)                          <- per-request LLM ranker
              vs. (#items + #users refreshed) x (tokens) x (cost per token)
                                                              <- offline feature/embedding
```

30. Scaling Laws and Hardware for Recommenders: Compute, Embeddings, Accelerators and Cost

Overview

```
            parameters (log)                     dense FLOPs / example (log)
                ^                                    ^
  2019-2022     |  ######## embeddings               |  #  small MLPs
  "sparse era"  |  #        dense                    |     (memory-bound)
                +----------------------->            +-------------------->
  2024-2026     |  ######   embeddings (still huge)  |  ########  attention /
  "dense era"   |  ####     dense (HSTU, GEM, OneRec) |  ########  token mixing
                +------------------------>           +-------------------->
```

3.1 Wukong (Meta, ICML 2024)

```
embeddings X0 (n x d)
     |
+----v-------------------------------------------+
| Wukong layer i                                  |
|   FMB: X_{i-1} X_{i-1}^T -> flatten -> MLP --+  |
|   LCB: W * X_{i-1} (linear recombine) -------+--> concat -> LN + residual
+-------------------------------------------------+
     |  (repeat L times: orders 1 .. 2^L)
     v
top MLP -> task heads
```

4.1 The sparse side: sharding, caching, offload

```
sparse half (model-parallel)               dense half (data-parallel)
+-------------------------------+          +---------------------------+
| ID features -> sharded tables  |  all-to-  | interaction + MLP/attn     |
| (TW/RW/CW, HBM + host offload)  |==> all ==>| replicated on every GPU    |
| bandwidth- & capacity-bound    |  pooled   | FLOP-bound, all-reduce     |
+-------------------------------+   embs    +---------------------------+
         ^ scale: memory, NICs                    ^ scale: tensor cores
```

5. Google TPU SparseCore: embeddings in silicon

```
TPU v4 chip
+-------------------------------------------------------------+
|  TensorCores (MXU systolic arrays) <-- dense MLP / attention |
|  SparseCores (dataflow tiles)      <-- embedding gather,     |
|        |                                 pool, scatter-add   |
|        +--> ICI (3D torus, OCS-reconfigurable) all-to-all     |
|  HBM: shard of every large embedding table                   |
+-------------------------------------------------------------+
  x 4,096 chips per pod (optically switched)
```

6.3 Andromeda and the mixed fleet

```
Meta recsys fleet (as publicly described, 2024-2026)
  training: GPU clusters (ZionEX lineage -> large GPU clusters for GEM)
  retrieval: Andromeda DNN on Grace Hopper (unified memory) [18]
  ranking:  Lattice / GEM-distilled students on GPUs and MTIA [17][15][21]
  teacher:  GEM, LLM-scale, thousands of GPUs, MFU 20-25% [15][16]
```

8. Power, cost and datacenter footprint

```
cost per served recommendation  ~=  (training $ / requests served over model life)
                                  + (serving FLOPs x $/FLOP at achieved MFU)
                                  + (embedding bytes x $/GB-month of the memory tier)
                                  + (network all-to-all at training scale)
levers: MFU up | precision down | distill teacher->student | consolidate models |
        tier embeddings (SRAM/HBM -> DRAM/LPDDR -> SSD) | cache hot rows
```

9. Putting it together: a reference scaling stack (2026)

```
                  +-----------------------------------------+
 offline teacher  | Foundation recsys model (GEM / HSTU-scale) |  GPU cluster,
 (LLM-scale)      | long sequences, MoE, FP8, MFU 20-45%       |  power-law scaling
                  +--------------------+--------------------+
                                       | distillation, embeddings
           +---------------------------+-----------------------------+
           v                           v                             v
 retrieval (GPU + unified mem)   light ranking (CPU/GPU)       heavy ranking (GPU/ASIC)
 learned index, sub-linear       cheap towers, caching          RankMixer/Wukong-style,
 (Andromeda-like)                                               M-FALCON KV sharing
           \___________________________|_____________________________/
                                       v
            sparse substrate: sharded embeddings (TorchRec/TPU SparseCore),
            tiered memory (HBM -> DRAM/LPDDR -> SSD), INT8/INT4 rows,
            incremental checkpoints, row admission/eviction
```

31. Search Meets Recommendation: Learning-to-Rank, Personalized and Semantic Search, Hybrid Retrieval and Unified Models

31.1 The common skeleton

```
 explicit query (text, filters)     implicit context (user, session, page, geo)
           \                                  /
            v                                v
 +-------------------- QUERY UNDERSTANDING --------------------+
 | normalize, spell-correct, segment, classify (category,      |
 | intent, locale), extract attributes, rewrite / expand,      |
 | encode -> dense vector and/or sparse terms                  |
 +-------------------------------------------------------------+
            |                |                    |
            v                v                    v
 lexical / boolean     dense EBR (ANN)      behavior / graph sources
 (inverted index,      (two-tower, IVF,     (co-click, item-to-item,
  BM25, filters)        HNSW, SPANN)         history, trending)
            \                |                    /
             v               v                   v
 +--------- MERGE: union / RRF / quotas / dedup (10^3 - 10^4) -----+
                             |
                             v
 first-pass ranker (GBDT/LambdaMART or small DNN, in-index or near data)
                             |
                             v
 second-pass ranker (deep multi-task model, personalization, cross-features)
                             |
                             v
 re-rank: diversity, business rules, ads blending -> page
                             |
                             v
 logs: impressions, clicks, add-to-cart, bookings -> labels, propensities
```

31.2.3 LambdaMART

```
per query q, documents d1..dn with graded labels l_i
     |
     v
current ensemble F_m(x) --> scores s_i
     |
     v
for each pair (i, j) with l_i > l_j:
      rho_ij   = 1 / (1 + exp(sigma * (s_i - s_j)))
      lambda_ij = sigma * |delta NDCG(i <-> j)| * rho_ij
      lambda_i += lambda_ij ;  lambda_j -= lambda_ij
     |
     v
fit regression tree m+1 to {(x_i, lambda_i)}
leaf value = sum(lambda) / sum(second derivative)   (Newton step)
     |
     v
F_{m+1} = F_m + eta * tree_{m+1}
```

31.4.1 Facebook Search: EBR inside the inverted index (KDD 2020)

```
query text + searcher ctx --> query tower --> q (e.g. 128-d)
                                              |
     Unicorn:  (nn q :radius r)  AND  (term:location ...)  OR  (lexical terms)
                                              |
index:  term = IVF centroid id ; payload = PQ(residual)
                                              |
hybrid candidate set --> ranking (embedding similarity also a ranking feature)
```

31.4.3 Etsy Unified Embedding (KDD 2023)

```
query + user recent behavior --> query/user tower -----------+
                                                             | cosine
listing text (transformer) --+                               |
listing tokens (term emb)  --+--> product tower --> [v ; w * quality] --> ANN
query-listing graph emb    --+                               |
                                                             v
                                lexical retrieval  --union-->  ranker
```

31.4.5 Airbnb: from embeddings-as-features to EBR

```
guest query (place, dates, guests, filters) + guest context
       |
       v
retrieval: geo + availability filters  AND  IVF cluster ids of query vector [17]
       |  (order of thousands of listings, approximate)
       v
base ranker: two-tower / multi-task journey DNN, pairwise loss [14][15]
       |
       v
second pass: diversity given listings above [16], twiddlers [15]
       |
       v
interleaving for fast screening (~50x sensitivity vs A/B, approximate) [19]
```

31.5.3 Score fusion

```
 query --> [text() OR nearestNeighbor()]  on each content node
                |
          first-phase: cheap expression (e.g. bm25 + closeness) on all hits
                |
          second-phase: GBDT / small NN on top-N per node
                |
  container: global-phase over merged top-K: RRF or normalize, cross-encoder
                |
             results
```

31.7 Unified search and recommendation

```
L1 shared infra      same ANN platform, feature store, ranking service
                     (Manas, Vespa, Unicorn)
L2 shared space      one embedding space for queries, users and items
                     (OmniSearchSage, Etsy, Facebook unified embedding)
L3 joint behavior    search history feeds recommendation and rec history feeds
                     search (UniSAR, Bing short/long-term)
L4 one model         a single generative model serves both tasks
                     (Spotify GR study, Amazon SynerGen, Kuaishou OneSearch)
```

31.7.3 L4: one generative model for both tasks

```
 OneSearch-style unified generative search (simplified)

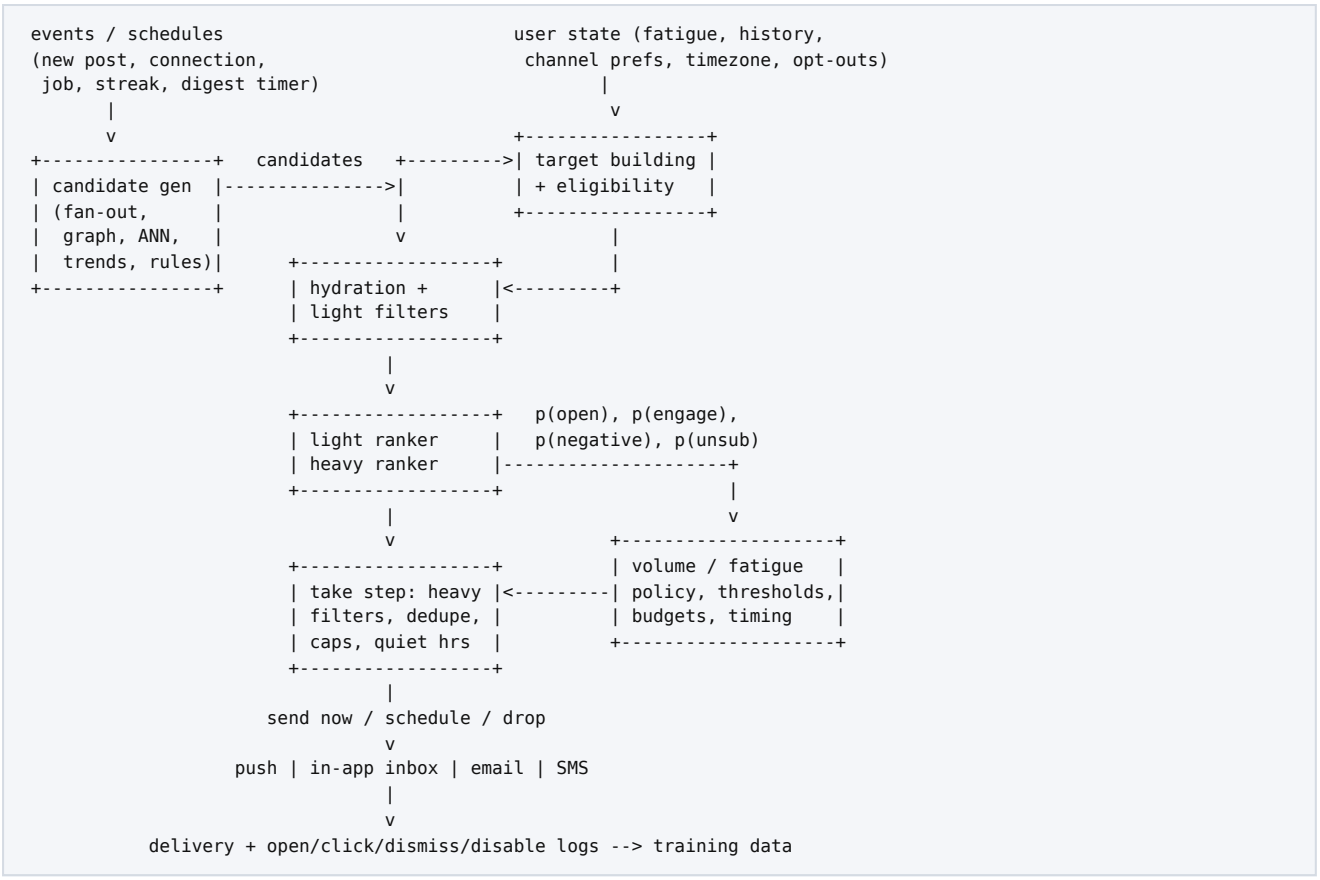
 query text --> keyword-aware encoding --+
 user behavior (search + rec history) --+--> encoder
                                         |
                                         v
                               decoder (beam search over
                               valid semantic-ID prefixes)
                                         |
            SID tokens:  [RQ1 4096][RQ2 1024][RQ3 512][OPQ 256][OPQ 256]
                                         |
                                         v
                   items  --> optional light re-rank --> results
 training: SFT on (query, user) -> purchased/clicked item SIDs,
           then preference / reward alignment
```

31.8 Reference architecture: a unified discovery stack

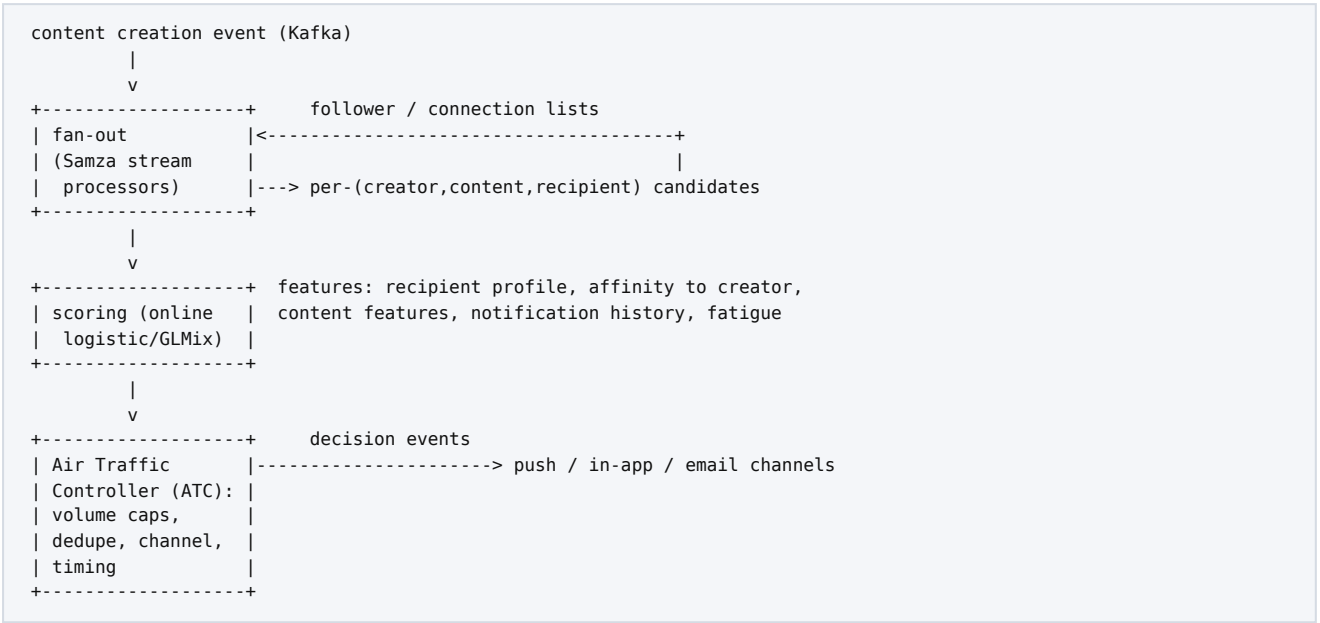
```
                    +------------------- shared feature store -------------------+
                    | user long/short-term seqs (search + rec), item stats,      |
                    | query stats, real-time session events                      |
                    +------------------------------------------------------------+
request(surface, query?, user, ctx)
      |
      v
query understanding (only if query): spell, classify, attributes, rewrite, encode
      |
      v
retrieval portfolio (per surface quotas)
 - lexical / learned sparse (search only; also relevance guardrail)
 - unified two-tower ANN (query+user tower; empty query for feed)
 - item-to-item / graph (related items; expansion from top search hits)
 - generative SID decoding (optional, constrained)
      |
      v
merge: RRF or learned blend; dedup; filter (availability, policy)
      |
      v
first pass (in-index or near data): GBDT LambdaMART / small DNN
      |
      v
second pass: multi-task DNN (click, purchase/booking, negative outcomes),
             surface-specific heads, position debiasing
      |
      v
re-rank: diversity, ads, business rules  --> page
      |
      v
logs + interleaving + A/B
```

32. Notification, Push and Email Recommendation: Volume, Fatigue and Timing

32.1 The canonical notification recommendation pipeline



32.4.1 Concourse: near-real-time activity-based notifications



Design

```
per-user features (activity level, history, past response)
                |
                v
+-----------------------------------------+
| volume response model:                  |
| predict E[weekly activity | k sends/week] |
| and p(unsubscribe | k sends/week)       |
+-----------------------------------------+
                |
                v
+-----------------------------------------+
| optimizer: choose k_u per user maximizing |
| sum_u activity_u(k_u)                   |
| s.t. unsubscribe / total-volume budget  |
+-----------------------------------------+
                |  weekly budget k_u
                v
+-----------------------------------------+
| per-notification ranker picks the best  |
| k_u candidates; scheduler spreads them  |
+-----------------------------------------+
```

33. Content Understanding and Integrity in Recommender Systems: Multimodal Item Embeddings, Integrity Filters, Borderline Demotion, Spam and the Policy Layer

33.1 Where content understanding and integrity sit in the stack

```
ITEM INGEST (async, per upload)                          SERVING (per request)
+-----------------------------------------+
| raw item: image, video frames, audio,   |
| text, OCR, ASR, author, metadata        |
+--------------------+--------------------+
                     |
         +-----------v-----------+
         | content encoders      |  CLIP-like image/text, video MLLM,
         | (multimodal)          |  audio CNN, text LLM
         +-----+-----------+-----+
               |           |
    dense embedding     semantic ID (RQ codes)
               |           |
  +------------v--+   +----v---------------+     +-----------------------------+
  | integrity     |   | ANN / SID index    |---->| retrieval (cold + warm)     |
  | classifiers,  |   | (gated by labels)  |     +--------------+--------------+
  | LLM labelers, |   +--------------------+                    |
  | hash/near-dup |                                             v
  +------+--------+                                  eligibility pre-filter
         |  labels: violating / borderline /        (blocks, mutes, hard labels)
         |  spam / adult / quality score                       |
         v                                                     v
  +----------------+                                 ranking (MTL heads incl.
  | label store    |------------------------------> negative-feedback heads)
  +----------------+                                           |
         |                                                     v
         +-----------------------------------------> value model: weights,
                                                      demotion multipliers
                                                               |
                                                               v
                                                     re-rank + visibility filter
                                                     (drop / interstitial / label)
                                                               |
                                                               v
                                                          final page
```

33.2.2 Pinterest: unified visual embeddings

```
training sets (per product task)     shared CNN backbone      unified embedding
+----------------------------+
| Lens (camera-to-Pin)       |--+
| Shop-the-Look (product)    |--+-->  image -> CNN -> FC -> emb --> proxy softmax
| Flashlight (in-image crops)|--+                                  per task (training)
+----------------------------+                   |
                                                 v
                                    serving: one (binarized) embedding per image
                                    used by every visual search product
```

33.2.4 Kuaishou OneRec: multimodal LLM item tokenizer

```
title, tags, ASR, OCR, cover, frames
          |
   miniCPM-V-8B  ->  1,280 tokens  -> QFormer -> 4 vectors
          |                                      |
 item-pair contrastive (CF signal)     RQ-KMeans, 3 levels
 + title generation (aux)                       |
                                      semantic ID (c1, c2, c3)
                                                |
                      generative recommender decodes SIDs per session
```

33.3.1 Meta Feed: integrity first

```
inventory (> 1,000 posts)
   |
pass 0: integrity processes + eligibility   <- violating / ineligible content removed
   |
pass 1: lightweight model -> ~500 posts
   |
pass 2: multitask NNs -> P(like), P(comment), P(hide), P("worth your time"), ...
        -> one value per post
   |
pass 3: contextual re-rank (diversity, avoid runs of similar posts)
```

33.3.3 X / Twitter: visibility filtering as a separate rule engine

```
CONTENT UNDERSTANDING (async)          LABELING RULES              SERVE
grox/  LLM + classifiers: spam,        scarecrow/ event rules,     visibility-filtering/
       adult, violent media,           botmaker rule engine        per (post, viewer):
       text/image embeddings           abuse-enforcement-svc/        ALLOW | NOTICE |
media-model-proxy/, clip/,              (+ abuse-ledger appeals)     INTERSTITIAL | DROP
  pnsfwmedia/                          safety-label-user-agg/            |
agatha/ (blocks+reports vs favs)               |                     VFFilter,
bdsm/   (behaviour sequences)                  |                     AncillaryVFFilter
user-cred-v2/ (PageRank credibility)           |
             \____________ labels -> storage ___/
```

33.4.2 YouTube: raters, borderline classifiers and authoritative boosts

```
human raters (public guidelines)        survey responses (1-5 stars)
          |                                        |
 borderline / authoritativeness             satisfaction model
 classifiers (score every video)             (predict for every watch)
          |                                        |
          v                                        v
candidate scoring:  value = f(engagement preds, satisfaction pred)
                    x demotion for borderline  x boost for authoritative
          |
 recommendation surfaces (Home, Up Next)  -- search and subscriptions treated separately
```

33.7.2 A reference policy layer

```
                +-----------------------------------------------------------+
scored          |  POLICY LAYER (config + rules, versioned, owned by policy) |
candidates -->  |                                                           |
(p_engage,      |  1. hard eligibility: blocks, mutes, age gates, legal,    |
 p_negative,    |     jurisdiction (e.g. election filters)                  |
 labels)        |  2. value model: sum w_i * p_i, large negative weights    |
                |     on report / block / see-less; survey-predicted value  |
                |  3. demotions: borderline x a, low quality x b, OON spam  |
                |     ineligible; boosts: authoritative x c                 |
                |  4. diversity and fatigue: author decay, DPP              |
                |  5. final visibility filter on top-K (+ ancillary items)  |
                |  6. logging: every drop/demotion with reason, holdouts,   |
                |     exploration share, per-account transparency report    |
                +-----------------------------+-----------------------------+
                                              |
                                              v
                                         final page
```

34. Open-Source Frameworks, Tooling, Benchmarks and Public Datasets

```
             The open-source recsys stack (typical composition)
+-------------------------------------------------------------------------------+
| Orchestration / compute     Metaflow (flows, versioning)   Ray (Data/Train/Serve)|
+-------------------------------------------------------------------------------+
| Features                    Feast: offline store <-> registry <-> online store   |
+-------------------------------------------------------------------------------+
| Training                    TorchRec+FBGEMM | TFRS (+TFRA) | Merlin HugeCTR/Models|
|                             RecBole | MS Recommenders | LightFM | implicit       |
+-------------------------------------------------------------------------------+
| Retrieval index             FAISS | ScaNN | Milvus | Weaviate | Qdrant | Vespa   |
+-------------------------------------------------------------------------------+
| Serving / ranking           Triton (Merlin Systems) | Vespa phased ranking |     |
|                             TorchRec inference (C++/TorchScript)               |
+-------------------------------------------------------------------------------+
| Benchmarks / data           MovieLens Criteo Avazu Amazon KuaiRand MIND Yambda |
|                             BARS / FuxiCTR / RecBole protocols                 |
+-------------------------------------------------------------------------------+
```

34.1.1 TorchRec and FBGEMM (Meta / PyTorch)

```
      TorchRec sparse/dense hybrid step (per batch, conceptual)
GPU0..GPUn each hold: replicated dense MLPs (DDP) + a SHARD of embedding tables
1. input KJT (ids per feature) --all-to-all--> owner shard of each table/row range
2. owner shard: fused table-batched lookup (FBGEMM TBE) -> pooled embeddings
3. pooled embeddings --all-to-all--> back to the sample's GPU
4. dense forward/backward (interaction + MLPs), allreduce dense grads
5. embedding grads --all-to-all--> owner shard; fused optimizer update in-kernel
Pipeline: step 1 of batch i+1 overlaps steps 2-5 of batch i
```

34.1.3 NVIDIA Merlin (NVTabular, HugeCTR, Models, Transformers4Rec, Systems)

```
request(user_id)
   |
QueryFeast(user features) --> PredictTensorflow(query tower) --> QueryFaiss(top-100)
   |                                                                 |
   +--> FilterCandidates(remove already-seen) <--------------------+
            |
        QueryFeast(item features) -> UnrollFeatures -> PredictTensorflow(ranker)
            |
        SoftmaxSampling(temperature, top-10)  --> response
Ensemble(...).export(dir);  tritonserver --model-repository=dir
```

34.3.2 ScaNN (Google Research)

```
query q
  |
  v
[1] partition (tree, num_leaves ~ sqrt(n)) -> pick num_leaves_to_search leaves
  |
  v
[2] score candidates with asymmetric hashing (AH), anisotropic quantization,
    dimensions_per_block = 2   (or brute force, optionally int8-quantized)
  |
  v
[3] rescore top k' (reordering_num_neighbors > k) with exact / int8 dot product
  |
  v
top-k
```

34.3.4 Vespa

```
 query ---> container (stateless)
              | scatter
    +---------+---------+
    v         v         v
 content   content   content       per node: retrieve (WAND / HNSW / filters)
 node 1    node 2    node N          -> first-phase (cheap expr, all hits)
    |         |         |            -> second-phase (ML model, top rerank-count)
    +---------+---------+
              | gather top hits
              v
       global-phase rerank (e.g. ONNX cross-model on GPU) -> response
```

34.4.1 Feast

```
 event log (user_id, item_id, ts, label) --+
                                           |  point-in-time join
 feature tables (entity, ts, values) -----+--> training set (no leakage)
           |
           | materialize(-incremental)
           v
 online store (Redis/Dynamo/...) <-- get_online_features(entity rows) <-- ranker
```

34.6 Benchmarking practice

```
 Reproducible offline benchmark loop
 raw logs -> frozen preprocessing (versioned ID, e.g. Avazu_x4) -> temporal split
        -> library (RecBole / FuxiCTR / TorchRec) with fixed config + seeds
        -> full-ranking metrics (nDCG@k, Recall@k, AUC/logloss) + latency/memory
        -> compare vs simple baselines (popularity, ItemKNN, iALS, SAR)
        -> only then: online A/B (Chapter 12)
```

35. The Recommender System Design Interview Playbook: Feeds, PYMK, Ads, E-commerce and Notifications

35.1.2 The canonical architecture to draw first

```
              OFFLINE                              ONLINE (per request)
+-------------------------------------+   +-------------------------------------+
| logs -> label join -> training data |   | request(user, context)              |
|   |                                 |   |   |                                 |
|   v                                 |   |   v                                 |
| train retrieval (two-tower, graph)  |   | candidate generation (parallel):    |
| train rankers (MTL DNN)             |   |   ANN(user emb) | graph | trending  |
|   |                                 |   |   | follows | rules  -> 10^3-10^4 |
|   v                                 |   |   v                                 |
| item embeddings -> ANN index -------+-->| filter: seen, blocked, policy       |
| batch features -> feature store ----+-->| light ranker            -> 10^2-10^3|
| stream features (Kafka/Flink) ------+-->| heavy ranker (MTL)      -> scores   |
+-------------------------------------+   | value fn + re-rank (diversity, biz) |
              ^                           |   -> top K (10^1)                   |
              |                           +-----------------|-------------------+
              +--- impression / engagement logs <----------+
```

Architecture

```
user_id + context
     |
     v
+------------ CANDIDATE GENERATION (parallel, ~1-2K total) ------------+
| two-tower ANN (recent watches) | co-watch / item-to-item | subs      |
| trending / fresh (explore)     | search-history sources  | multi-int |
+-----------------------------------------------------------------------+
     |
     v   filter: watched, blocked, region/age policy, dedupe
light ranker (distilled two-tower or small DNN)  ->  ~100-500
     |
     v
heavy ranker: multi-task DNN (MMoE) -> p(click), E[watch time], p(like), p(share),
              p(dislike), p(survey satisfied); shallow position-bias tower
     |
     v
value function  sum_i w_i * pred_i  ->  re-rank: diversity (DPP / MMR), freshness,
                                        creator caps, policy  ->  top ~10-20
```

Architecture: Facebook's four passes

```
inventory: all eligible posts for viewer (> 1,000 on average)
  -> pass 0: integrity + eligibility filters
  -> pass 1: lightweight model -> ~500 posts
  -> pass 2: multitask NNs predict P(like), P(comment), P(share), P(hide),
             P("worth your time" survey) ... -> one value score per post
  -> pass 3: contextual re-ranking (content-type diversity, no runs of similar posts)
```

Architecture

```
OFFLINE (daily / hourly)                          ONLINE
graph snapshot at t ----------------------+      request(user)
  -> FoF expansion (2-hop) ~10^5-10^6     |        -> read precomputed PYMK list
  -> + EBR: GNN embeddings -> ANN         |        -> filter: already connected,
  -> + co-attributes: school, company,    |           blocked, pending, privacy
       contacts, location                 |        -> light re-score with fresh
  -> ranker: p(send) * p(accept)          |           signals (recent views)
  -> top ~100 per user -> KV store -------+------> -> diversity, frequency caps
```

Architecture

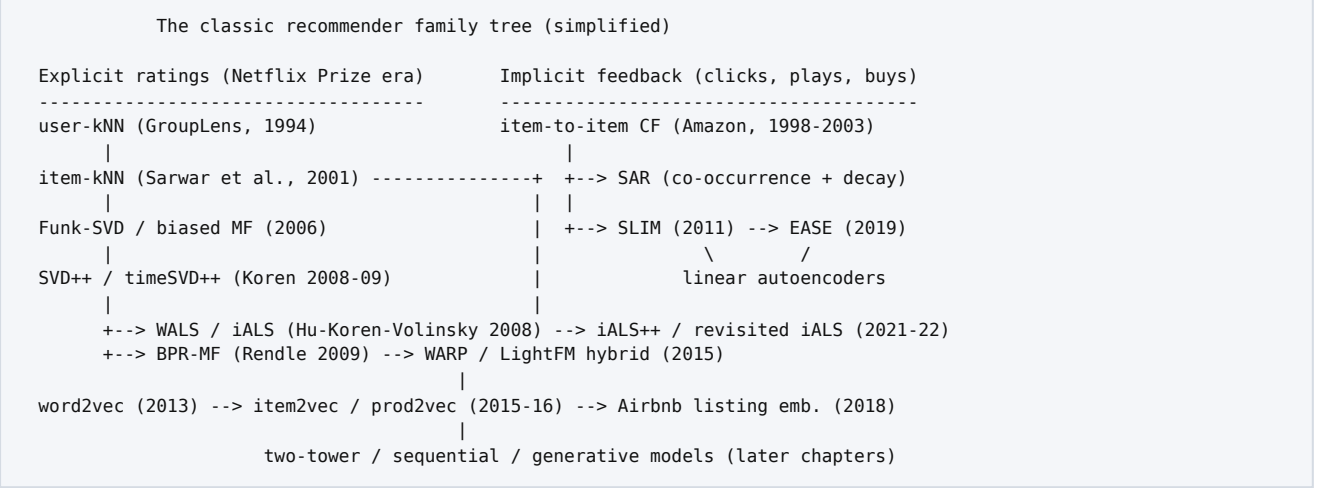
```
ad request (user, placement)
  -> eligibility: targeting (age, geo, interests), budget/pacing, policy
  -> retrieval: tens of millions of eligible ads -> a few thousand      (Andromeda-like)
  -> light ranker -> hundreds
  -> heavy ranker: pCTR, pCVR, p(hide) heads; calibration layer
  -> auction: rank by bid * pCTR * pCVR + quality; price (GSP/VCG-style)
  -> blend with organic feed (ad load, spacing)
logs -> online joiner (impression + click within window) -> streaming trainer
```

Architecture (from X's open-sourced pushservice)

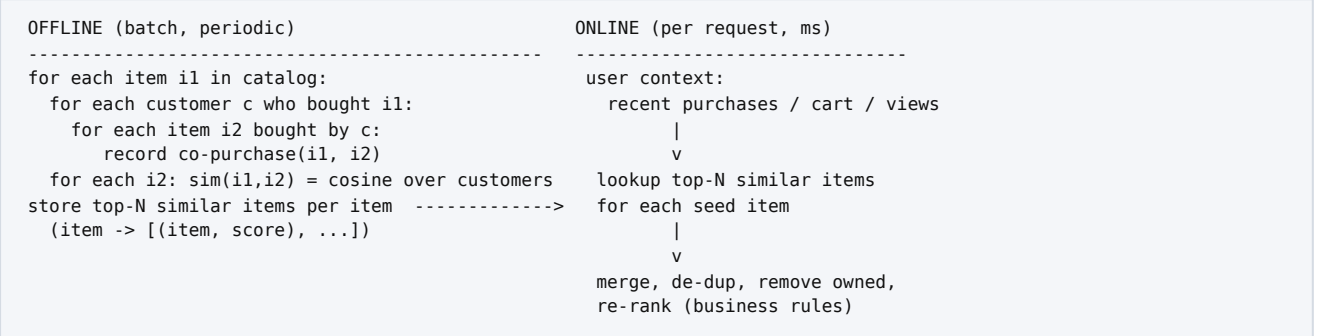
```
user id -> Target (eligibility, settings) -> query candidate sources
  -> hydrate candidates -> light filters
  -> LIGHT RANKER (pre-selects from the large pool)
  -> HEAVY RANKER (multi-task: p(open), p(engage), negative-feedback risk)
  -> take step: run expensive predicates (quality, fatigue) one candidate at a time
     until ONE passes -> send push + in-app notification
```

36. Classic Algorithms Foundations: Neighborhoods, Factorization, Linear Autoencoders, Embeddings, and Why Baselines Still Win

Overview



1.2 Item-to-item CF (Amazon)



3.3 The ALS trick that makes it scale

```
x_u = (Y^T C^u Y + lambda I)^(-1) Y^T C^u p(u)
    where  Y^T C^u Y = Y^T Y  +  Y^T (C^u - I) Y
                       ^^^^^     ^^^^^^^^^^^^^^^^
                  precompute     only n_u nonzero rows (items u touched)
                  once, O(f^2 n)
```

5.2 EASE (Steck, WWW 2019)

```
min_B ||X - X B||_F^2 + lambda ||B||_F^2   s.t. diag(B) = 0

P   = (X^T X + lambda I)^(-1)                # item-item, m x m
B_ij = - P_ij / P_jj   for i != j ,  B_jj = 0
scores(u) = x_u B                            # dense item-item "autoencoder"
```

5.2 EASE (Steck, WWW 2019)

```
 Linear item-item models: what each one learns

              similarity source           constraints            training cost
item-kNN      cosine / Jaccard heuristic  top-k sparsify         O(nnz * avg deg)
SAR           co-occurrence + rescale     time-decayed A         O(nnz + m^2)
RP3beta       3-step random walk          popularity penalty     sparse mat-mults
SLIM          learned (elastic net)       W >= 0, diag 0         m regressions
EASE          learned (ridge)             diag 0                 one m x m inverse
```

6.3 Airbnb listing embeddings (KDD 2018 best paper)

```
Airbnb-style session embedding pipeline
---------------------------------------
 click logs --> sessionize (30-min gap) --> filter (dwell >= threshold)
      |
      v
 SGNS training (window, negatives = random + same-market, global ctx = booking)
      |
      v
 listing vectors (d=32) --> ANN index --> "similar listings" candidates
      |
      +--> real-time features: sim(candidate, recently clicked / skipped)
                                --> fed to GBDT/NN search ranker
```

10. Reference architecture: classic models in a modern pipeline

```
                           +-----------------------------+
  interaction logs --------> | batch training (daily)     |
  (views, plays, buys)       |  - item-kNN / RP3beta      |----> item->topN table (KV)
                             |  - iALS / BPR (Spark/GPU)  |----> item & user factors
                             |  - EASE (if m small)       |----> B matrix / top-N
                             |  - item2vec on sessions    |----> item vectors
                             +-------------+--------------+
                                           |
                                 ANN index build (Faiss/ScaNN/Annoy)
                                           |
request --> context (recent items, user id) v
        --> candidate generation: union of
              [kNN lookups of recent items] + [ANN on user factor]
              + [content / trending for cold users]          (~100s-1000s)
        --> feature join (incl. CF scores, embedding sims as features)
        --> learned ranker (GBDT or DNN)                          (~10s)
        --> business rules, diversity, dedup  --> response
```

37. Real-Time and Session-Based Recommendation: Streaming Updates, Online Learning, Nearline Pipelines

and In-Session Intent

37.1.1 Offline, nearline, online: the Netflix blueprint (2013)

```
member events (plays, ratings, browsing, search)
       |
       +--> OFFLINE  (Hadoop/Hive/Pig): training, batch scoring; no data or
       |             complexity limit, but stale and context-blind
       |
       +--> NEARLINE (event-driven, no request waiting): e.g. on a play event
       |             recompute member factors and scores; results written to
       |             Cassandra / EVCache / MySQL
       |
       v
    ONLINE (per request, strict SLA): fetch precomputed + live signals,
           light model eval, filtering, context; FALLBACK to precomputed
           or unpersonalized results on timeout
```

37.3.1 Twitter's aggregation framework: one definition, two engines

```
          client events (Kafka client-event topics)
               |                          |
  daily DataRecord dumps (HDFS)        Storm/Heron (summingbird)
               |                          |
  batch aggregates -> Manhattan KV     real-time aggregates -> memcached
               \_____________________   ___________________/
                                    \ /
            Home Mixer / pushservice feature hydration
```

37.4.1 GraphJet (Twitter, VLDB 2016)

```
Kafka (fav, RT, follow, client events)
       |
 recos-injector (filter, combine, per-service topic)
       |
 UTEG / GraphJet (24-48 h window, all in RAM, full replica per server)
       |  input: viewer's weighted followees
       v
 tweets ranked by followee engagement, with social proof
```

37.5.1 Pinterest TransAct (KDD 2023)

```
real-time action stream                                  candidate Pin
        |                                                      |
latest ~100 actions: PinSage emb + action type + timestamp     |
        |                                                      |
early fusion: concat each action with candidate embedding <---+
        v
small Transformer encoder
        v
compress (first K tokens + max pooling) -> flatten
        v
concat with PinnerFormer + other features -> DCN v2 -> MLP
        v
multi-task heads (repin, click, hide, ...) -> utility
```

Architecture.

```
client actions --> Kafka: actions ---+
serving-time feature log --> Kafka --+
                                     v
          Flink online joiner (join on request key; in-memory cache +
          on-disk KV for late actions; negative sampling)
                 |                                     |
         Kafka: training examples              HDFS (batch dumps)
                 |                                     |
         online training workers <--- same code ---> batch training
                 |
         training PS: Cuckoo-hash tables (frequency admission + expiry)
                 |  sparse: incremental sync of touched IDs (minute-level)
                 |  dense : synced far less often
                 v
         serving PS -> online inference
```

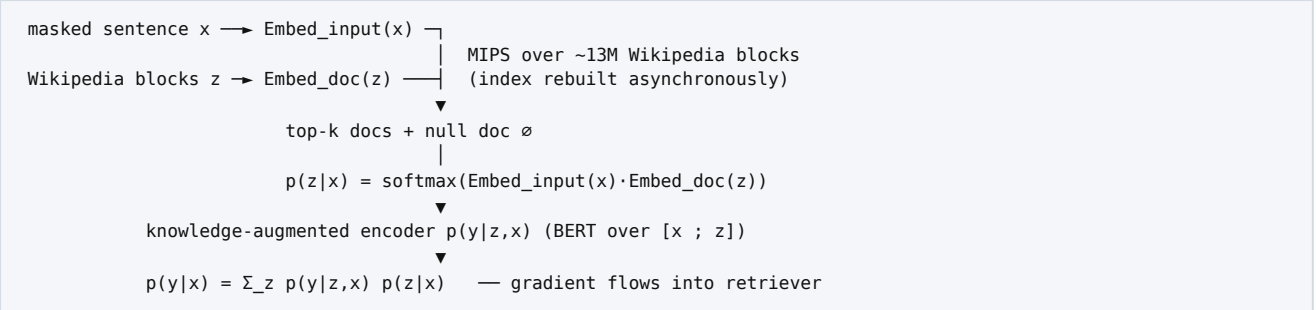
37.8 TikTok and the short-video case

```
    +---------------- short-video real-time stack (composite) ----------------+
    | client swipe/watch events -> Kafka                                       |
    |   |-> Flink features (counters, last-N actions) -> KV  -> ranker input  |
    |   |-> fresh-item / graph indexes (incremental insert)  -> retrieval     |
    |   '-> Flink joiner(serving feature log) -> online trainer                |
    |          sparse rows: minute-level incremental push -> serving PS        |
    |          dense weights: hourly or slower                                 |
    | request: lifelong history (cached, may lag) + last-N actions (fresh)     |
    +--------------------------------------------------------------------------+
```

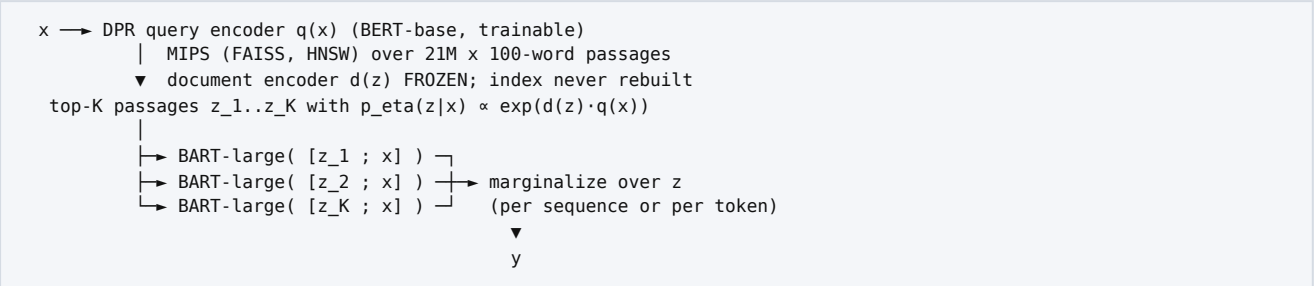
2. RAG — Architecture Diagrams

01. RAG Foundations and Taxonomy: From REALM and RAG to the Modular Reference Architecture

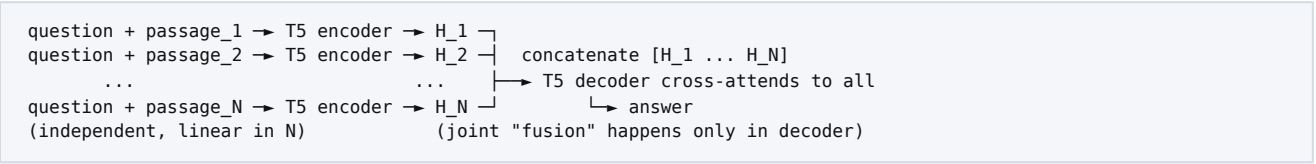
Architecture



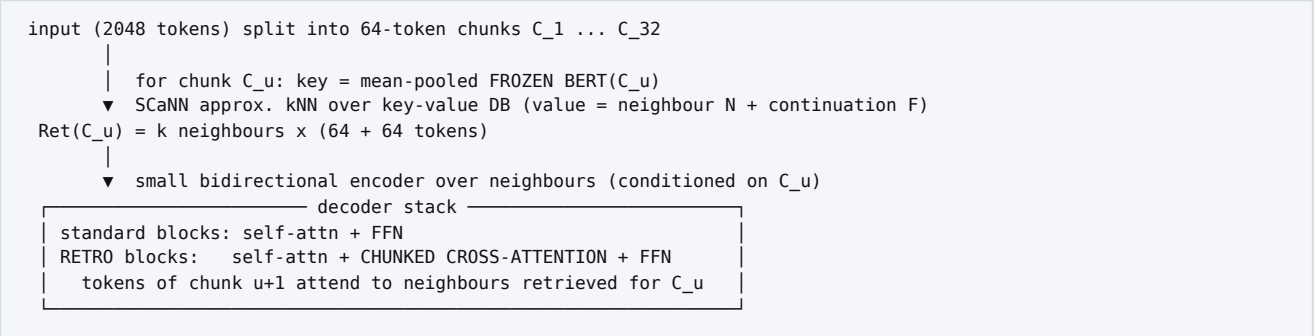
Architecture



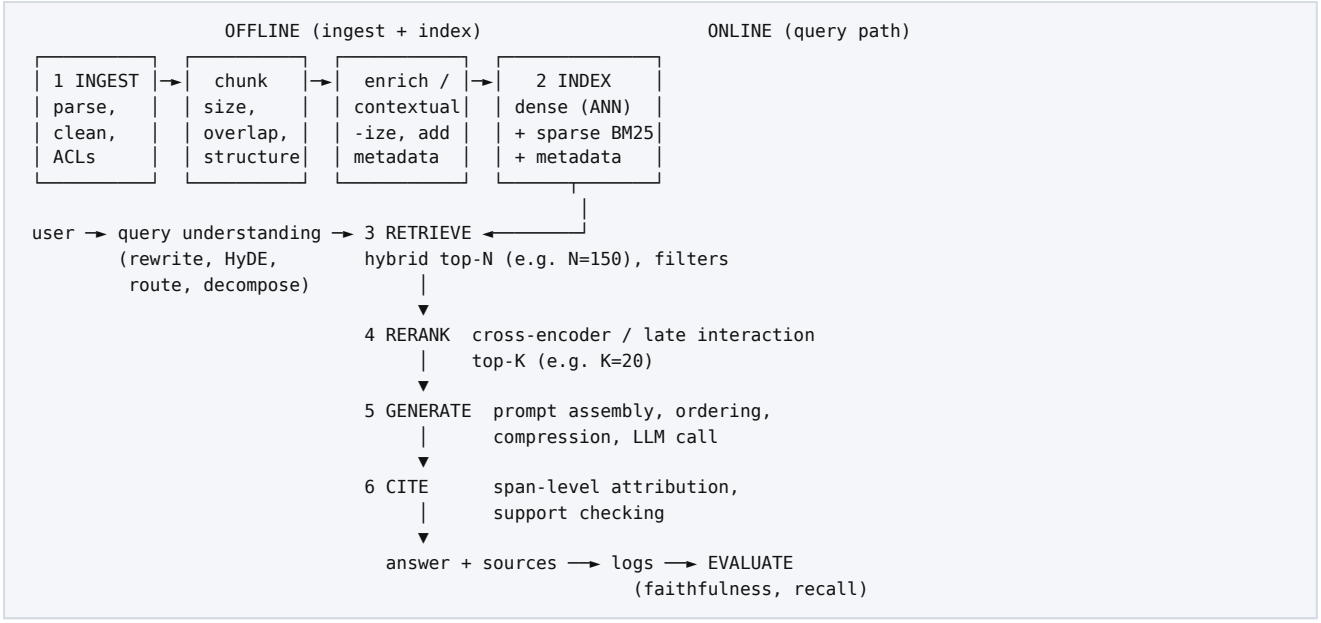
Architecture



Architecture

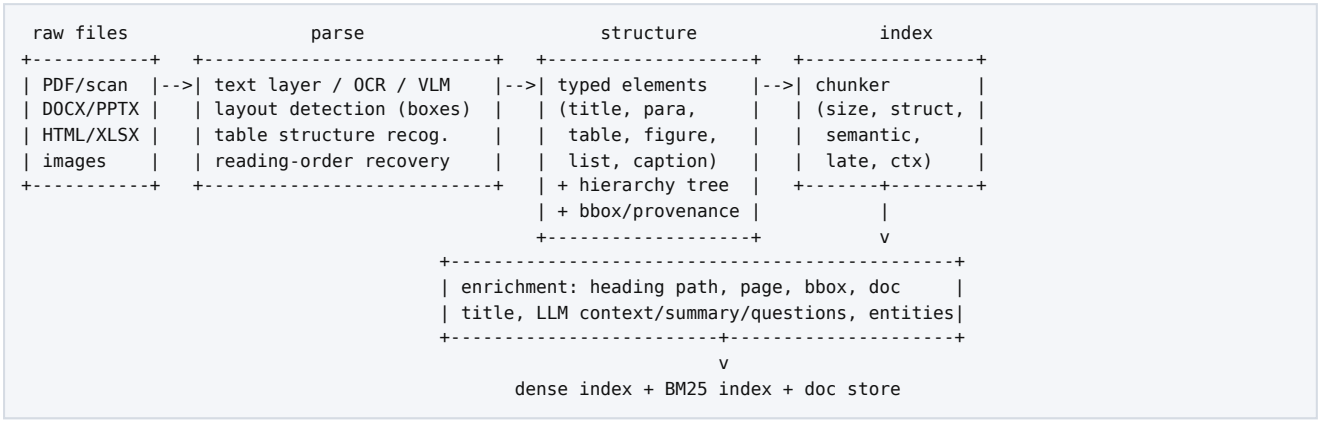


10. The end-to-end reference architecture

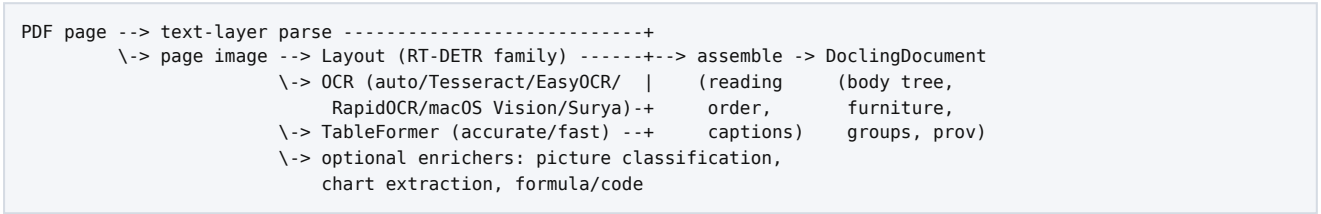


02. Document Parsing and Chunking: From Pixels to Retrievable Units

Overview



Docling (IBM Research)



Reducto



Hierarchical and multi-granularity chunking

```
             doc
      +--------+--------+
   [2048]            [2048]           parent nodes (returned to LLM)
   /  |  \           /  |  \
 [512][512][512]  [512][512][512]     intermediate
  /|\
[128][128][128] ...                   leaf nodes (embedded / searched)

AutoMergingRetriever: if most children of a parent are hit -> return parent
```

Anthropic Contextual Retrieval

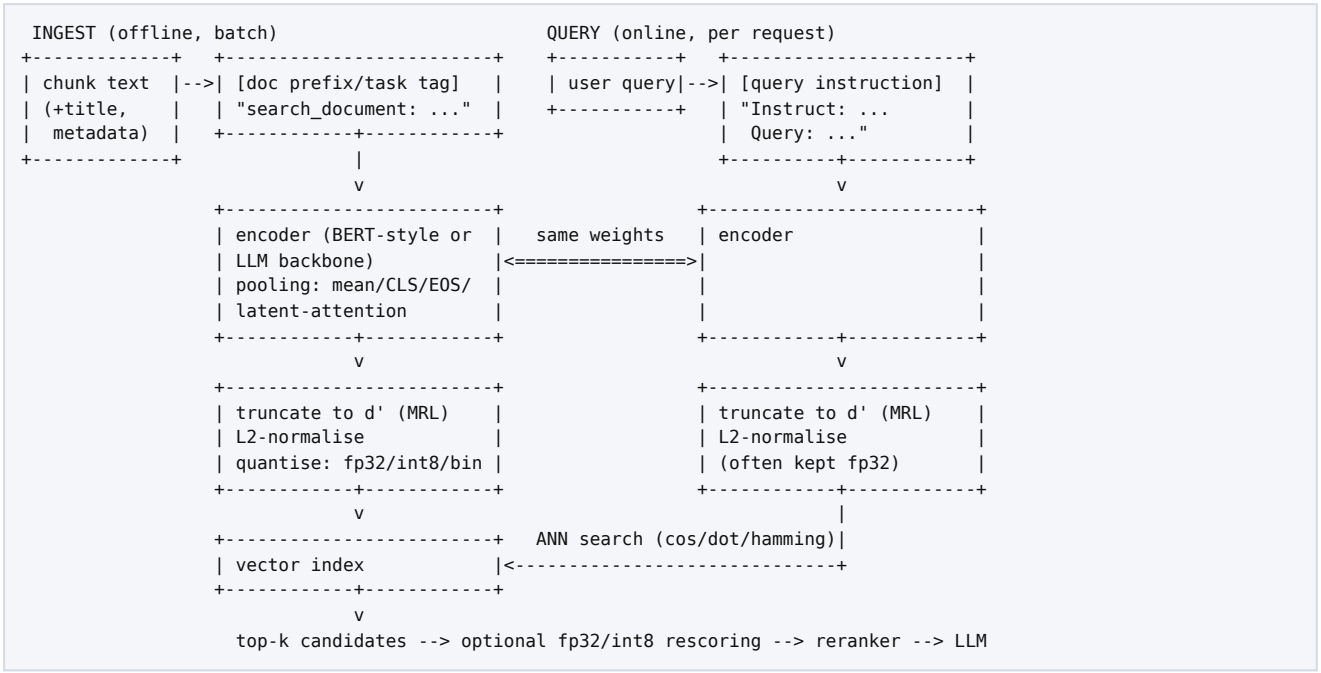
```
for doc in corpus:
   cache(doc)                                # prompt caching: write once
   for chunk in doc.chunks:                  # process sequentially to hit cache
       ctx = LLM("<document>{doc}</document> <chunk>{chunk}</chunk> situate it")
       text = ctx + "\n" + chunk
       dense.add(embed(text)); bm25.add(text)
query -> dense top-150 + bm25 top-150 -> rank fusion -> rerank -> top-20 -> LLM
```

Enrichment pipeline pattern.

```
chunk --> +heading path, page, bbox, doc title          (free, from parser)
      --> +entities, dates, doc type, ACL tags          (NER/rules; for filtering)
      --> +situating context / questions / summary      (LLM; for matching)
      --> embed( context + heading path + chunk )       (dense)
      --> bm25 ( context + chunk )                      (sparse)
      --> store( raw chunk, metadata, parent_id, prov )  (for generation + citation)
```

03. Embedding Models for Retrieval: Commercial APIs, Open Families, Compression and Adaptation

Overview



1.3 Voyage AI (voyage-3, voyage-3-large, voyage-4 family)

```
shared-space asymmetric deployment (voyage-4 style)
docs (once, offline) --> voyage-4-large  --+
                                           +--> same vector space --> index
queries (per request) --> voyage-4-lite ---+        (MRL d', int8/binary)
```

2.1 BAAI BGE: from C-Pack to BGE-M3 and LLM embedders

```
                      +--> [CLS] -> dense vector (1024)      --- s_dense
text -> XLM-R-style   |
        encoder (8192) +--> per-token weights -> sparse lexical --- s_sparse
                      |
                      +--> per-token vectors -> multi-vector  --- s_colbert
hybrid score = w1*s_dense + w2*s_sparse + w3*s_colbert   (example weights 0.4/0.2/0.4)
```

2.3 Alibaba GTE and Qwen3-Embedding

```
Qwen3 LLM (0.6B/4B/8B, causal attention)
         |
         v
Stage 1: weakly-supervised pre-training on ~150M SYNTHETIC pairs
         (generated by Qwen3-32B: persona from Persona Hub x query type x
          difficulty x language, documents from Qwen3 pre-training corpus)
         |
         v
Stage 2: supervised fine-tuning on ~7M labelled pairs + ~12M filtered
         synthetic pairs (keep if cosine sim > 0.7); InfoNCE with hard and
         in-batch negatives, false-negative masking
         |
         v
Stage 3: model merging of SFT checkpoints (slerp)
         |
         v
embedding = hidden state at final [EOS] token, MRL-capable, 32K context
```

3.3 Scalar (int8) and binary quantization

```
query --embed fp32--> q
  |
  +--> binary(q) --Hamming top-40--> [binary index: ~5.2 GB, in RAM]
                                          |
                                          v
           load int8 vectors of the 40 candidates  [int8 store: ~47.5 GB, on disk]
                                          |
                      rescore: dot(q_fp32, doc_int8) --> top-10 --> reranker/LLM
total ~5.2 GB RAM + ~52 GB disk  vs  ~200 GB RAM + ~200 GB disk for pure fp32
```

Reference pipeline.

```
domain corpus chunks
       |
       v
(1) synthetic queries: LLM writes 1-3 questions per chunk      [13][48][51]
    (or T5 query generator, as in GPL)                         [47]
       |
       v
(2) hard-negative mining: retrieve top-k with base embedder/BM25,
    drop the positive and near-duplicates (false-negative risk)
       |
       v
(3) optional pseudo-labels: cross-encoder scores (q, pos, neg) --> margins (GPL)
       |
       v
(4) train: MultipleNegativesRankingLoss / Cached-MNRL (big effective batch)
           or MarginMSE (distillation), wrapped in MatryoshkaLoss
       |
       v
(5) merge: average fine-tuned weights with base weights (soup/slerp)  [10][25][51]
       |
       v
(6) evaluate on held-out REAL queries (NDCG@10, recall@k) vs base + reranker
       |
       v
(7) re-embed corpus (or train query-side adapter only to avoid re-indexing)
```

04. Vector Databases and ANN Indexing: From In-Memory Graphs to Object-Storage-Native Engines

Overview

```
                the generic vector-database data path (2024-2026 designs)
 writes                                                                  queries
   |                                                                        |
   v                                                                        v
+--------+   +-----------------+   +------------------+   +-----------------------+
| router |-->| WAL / log       |-->| index builder /  |-->| immutable index files |
| (auth, |   | (object storage,|   | compactor        |   | (object storage)      |
| quota) |   |  Kafka, quorum) |   | (cluster, quant.,|   +-----------+-----------+
+--------+   +--------+--------+   |  graph build)    |               | lazy load
                      |            +------------------+               v
                      |  tail unindexed suffix          +-----------------------------+
                      +-------------------------------->| query node: RAM/SSD cache,  |
                         ("freshness layer")            | ANN + filter + merge + top-k|
                                                        +-----------------------------+
```

4.1.4 DiskANN / Vamana: graphs that live on SSD

```
query --> entry point (near global centroid)
  loop:
    expand frontier using PQ codes held in RAM  (cheap approximate distances)
    pick best beam_width candidates
    read their SSD blocks (full vector + adjacency list in one I/O)
    compute exact distances, update candidate list
  until candidate list converges --> top-k
knobs: max_degree, search_list_size, beam_width_ratio,
       pq_code_budget_gb_ratio, search_cache_budget_gb_ratio
```

4.2.4 Cost-based planners: the convergent design

```
          estimate |allowed| from inverted / payload / bitmap index
                                |
        +-----------------------+------------------------+
        | tiny                  | medium                 | large (>~ 40-95%)
        v                       v                        v
  exact scan of allowed IDs  graph walk w/ allow-list   plain ANN + post-filter
  (payload index, bitmaps)   (+ACORN / extra edges)     (filter rarely bites)
        |                       |                        |
        +---------- fallback: if ANN returns < k, rerun exactly -------+
```

4.4.1 Pinecone serverless

```
writers --commit--> blob storage + log (sequences all mutations)
                        |                         |
                tail log |                         | tail log
                        v                         v
        +-------------------------+     +-----------------------+
        | index builder           |     | freshness layer       |
        | geometric partitioning  |     | compact index over data|
        | (coarse->fine hierarchy)|     | newer than manifest's  |
        | immutable segments ->   |     | max indexed version    |
        | blob; update manifest   |     +-----------+------------+
        +-----------+-------------+                 |
                    | segments (lazy load + cache)  |
                    v                               v
        +-----------------------------------------------------------+
        | query executors: load only relevant partitions of the  |
        | target namespace, merge with freshness results         |
        +-----------------------------------------------------------+
```

4.4.2 turbopuffer

```
client --API--> stateless ./tpuf query nodes (multi-tenant, Rust)
                |  route by namespace for cache locality; any node can serve any ns
                v
        +----------------------+   miss   +-----------------------------------+
        | memory + NVMe cache  |--------->| object storage, prefix per         |
        +----------------------+          | namespace:                         |
                                          |  /wal/  one file per group commit  |
                                          |  /index/ centroids.bin,            |
                                          |          clusters-N.bin,           |
                                          |          exact filter + BM25 idx   |
                                          +-----------------------------------+
writes: buffer -> group commit (<= 1 WAL entry/s/ns) -> CAS -> async indexer
reads (strong): indexed data + scan of unindexed WAL tail; ~10 ms consistency check
```

4.4.3 Milvus (Zilliz)

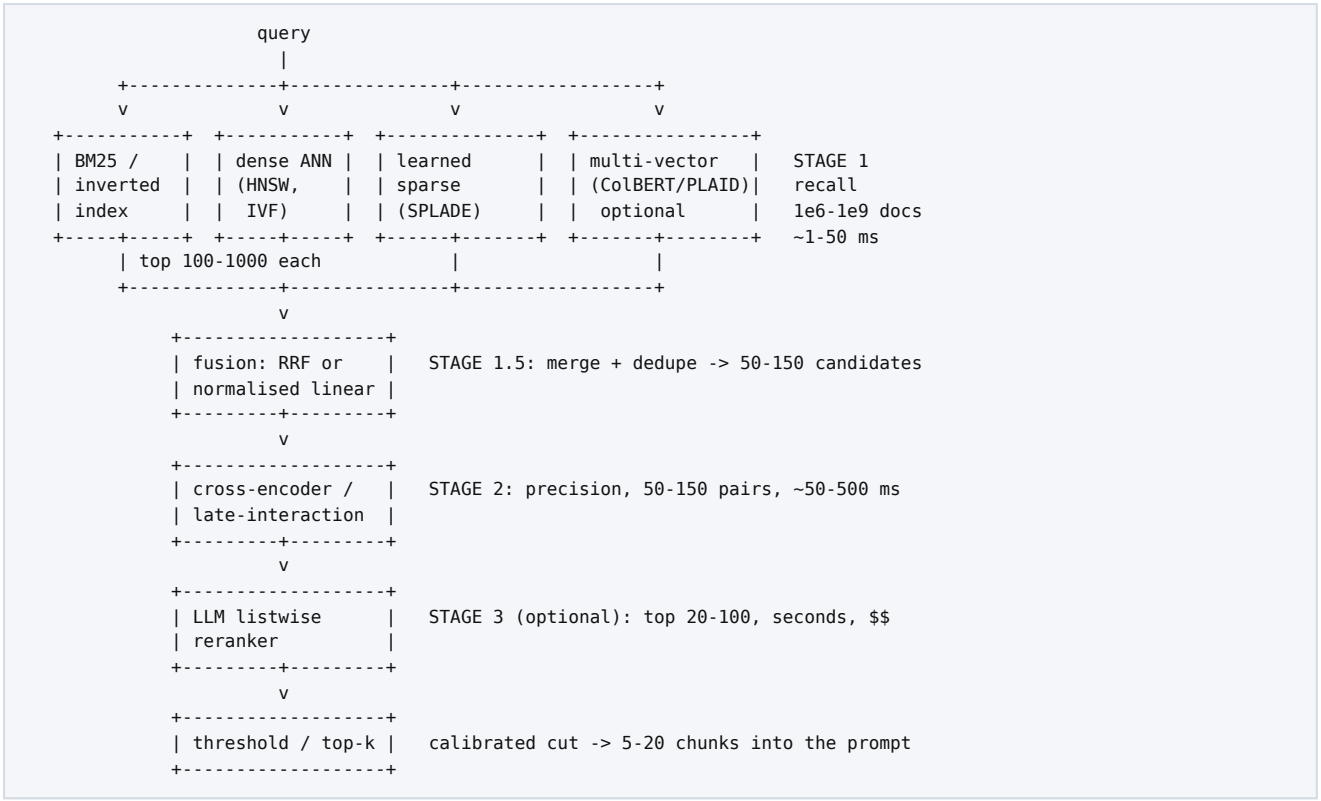
```
SDK --> [ Access layer: stateless proxies (MPP reduce of partial results) ]
                |
        [ Coordinator: exactly one active; DDL/DCL, TSO/time-tick,
          WAL<->streaming-node binding, query views, compaction/index tasks ]
                |
  +-------------+-----------------+-------------------------+
  | Streaming node                | Query node              | Data node
  | shard "mini-brain": WAL       | loads sealed segments   | compaction,
  | consumption, growing-segment  | from object storage,    | index build
  | search, growing -> sealed     | serves historical data  |
  +-------------+-----------------+-------------------------+
                |
  [ Storage: etcd (meta) | WAL (Woodpecker/Kafka/Pulsar) | object storage (S3/MinIO) ]
```

4.4.6 Chroma

```
client -> Gateway (auth, quota, rate limit, logical plan)
            | writes                          | reads: rendezvous hash(collection_id)
            v                                 v
        Log (WAL) ------------------->  Query executor (mem + SSD-cached indexes,
            |      consistent read        consults log tail for freshness)
            v
        Compactor: log -> vector/FTS/metadata indexes -> object storage;
                   registers new index versions in System DB (SQL catalog)
```

05. Hybrid Search and Reranking: Fusion, Learned Sparse, Late Interaction and Multi-Stage Cascades

Overview

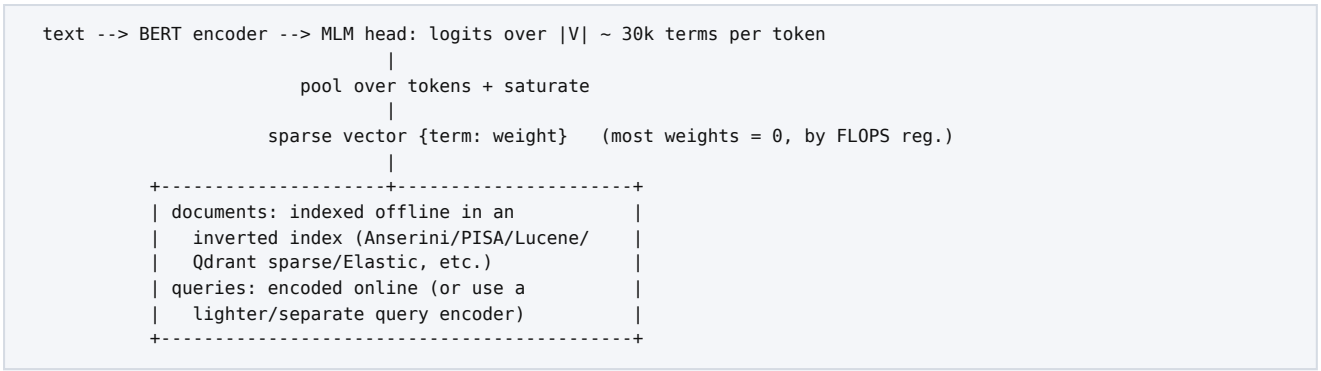


1.4 Where fusion runs

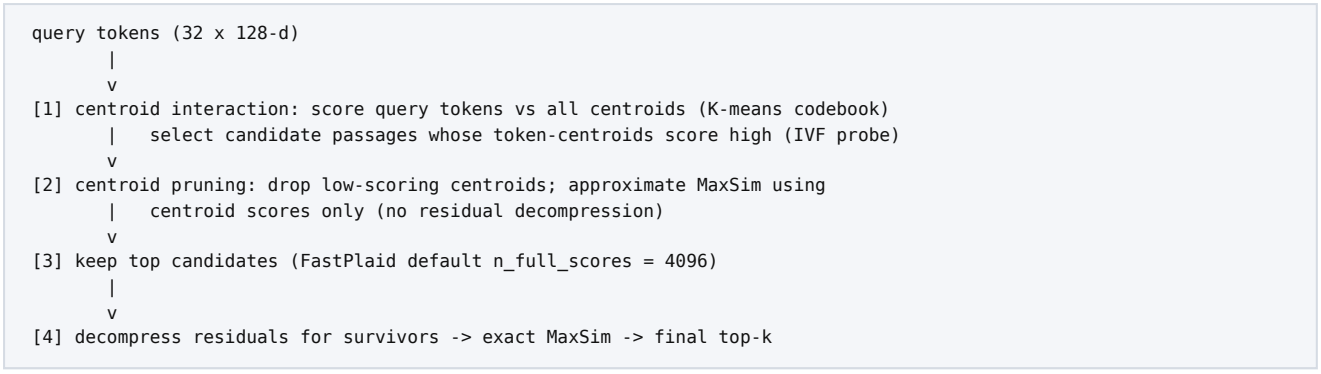
```
Azure AI Search                        Elasticsearch / OpenSearch
---------------                        --------------------------
text query --> BM25 (maxTextRecallSize   retriever tree:
                default 1000)              rrf | linear
vector q(s) -> HNSW (k each)                 |-- standard (BM25)
       \          /                          |-- knn (dense)
        RRF (k=60, per-query weights)        |-- sparse (ELSER/SPLADE-like)
          |  top 50 by default              text_similarity_reranker
        semantic ranker (top 50 only)          wraps any retriever,
          |  @search.rerankerScore 0..4        rank_window_size default 10
        response

Vespa                                  Weaviate
-----                                  --------
content nodes: first-phase (all hits)  hybrid(query, alpha)
               second-phase (100/node)    -> BM25F + vector in parallel
container:     global-phase (100 total)   -> relativeScoreFusion (default)
               RRF / normalize_linear        or rankedFusion
               ONNX cross-encoder         -> optional reranker module
```

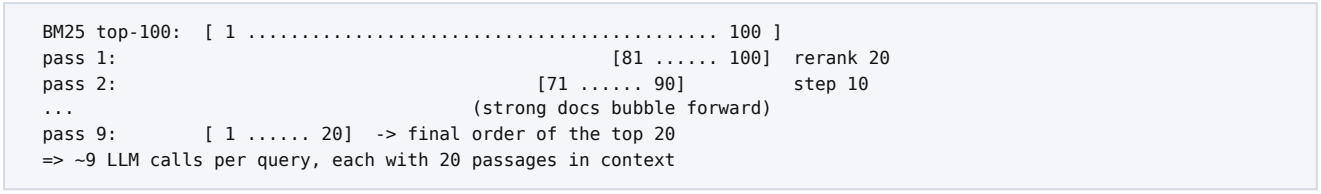
2.3 SPLADE architecture



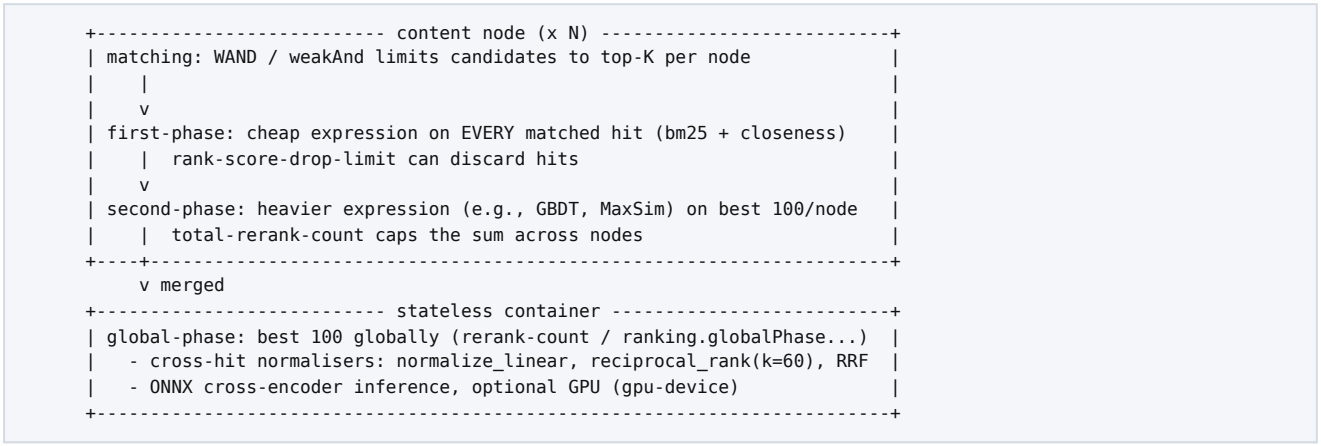
3.3 PLAID



5.1 RankGPT: listwise permutation generation

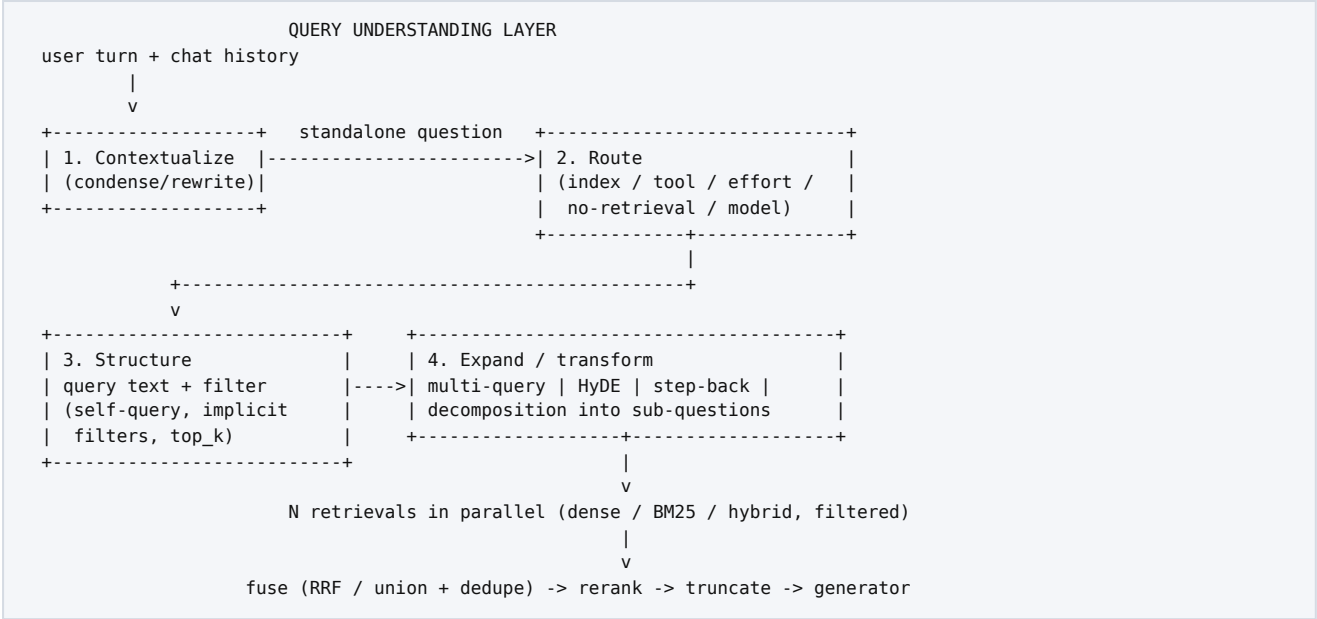


7.2 Vespa's phased ranking as a reference design

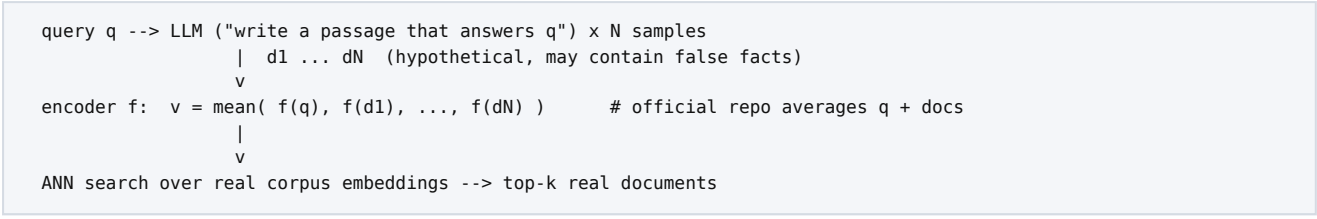


06. Query Understanding and Transformation: HyDE, Multi-Query, RAG-Fusion, Step-Back, Decomposition, Conversational Rewriting, Routing and Self-Query

Overview



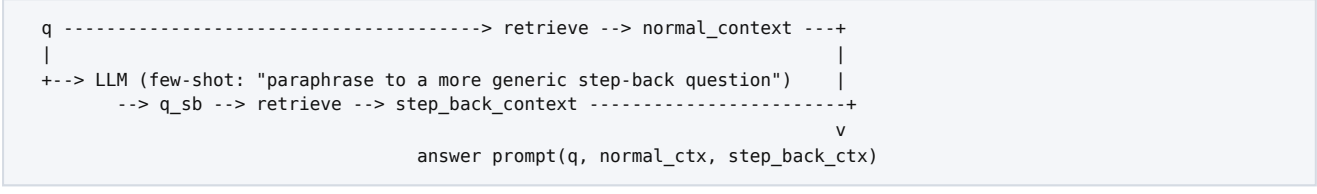
2.1 HyDE (Hypothetical Document Embeddings)



3.2 RAG-Fusion: multi-query plus Reciprocal Rank Fusion



4. Step-back prompting



5.2 Framework patterns: answer individually vs. answer recursively

```
PARALLEL ("answer individually")         SEQUENTIAL ("answer recursively")
q -> LLM -> [s1, s2, s3]                 q -> LLM -> [s1, s2, s3]
      |      |     |                      s1 -> retrieve -> a1
     ret    ret   ret                     s2 + (s1,a1) -> retrieve -> a2
      |      |     |                      s3 + (s1,a1),(s2,a2) -> retrieve -> a3
     a1     a2    a3                     answer(q) = a3 (or synthesize)
      \      |     /
 synthesize(q, Q/A pairs)
```

5.4 Managed decomposition: Azure AI Search agentic retrieval

```
retrieve(query, conversation history, reasoningEffort)
     |
     v
[query planning LLM]  (skipped at "minimal")
     |  focused subqueries + knowledge-source selection
     v
subqueries run in parallel over knowledge sources
(keyword | vector | hybrid; indexed indexes or remote sources)
     |  each subquery semantically reranked (L2)
     v
[medium only] semantic classifier (L3) judges sufficiency
     |  insufficient -> ONE follow-up iteration with revised plan
     v
merged grounding content (+ optional references, activity log, answer synthesis)
```

7.1 Condense-question and history-aware retrievers

```
history H = [(u1,a1), ..., (u_{t-1}, a_{t-1})],  follow-up u_t
      |
      +-- H empty? --yes--> retrieve(u_t)
      |
      no
      v
LLM: rewrite(H, u_t) -> standalone q_t  -->  retrieve(q_t) --> answer(H, u_t, docs)
                                             (answer step still sees the raw turn)
```

8.2 Semantic (embedding) routing

```
offline: route r -> utterances u_r1..u_rn -> encoder -> utterance index
online:  q -> encoder -> sim(q, u_ri) aggregated per route
         best route score >= threshold_r ?  -> route name (static)
                                            -> route + LLM-extracted args (dynamic)
         else -> None (fallback pipeline / general LLM)
```

9.1 LangChain SelfQueryRetriever

```
q --> LLM(query-constructor prompt: schema + attribute descriptions
         + allowed comparators/operators + few-shot examples)
   --> {"query": "teenager love",
        "filter": "and(or(eq(artist,'Taylor Swift'), eq(artist,'Katy Perry')),
                       lt(length,180), eq(genre,'pop'))",
        "limit": optional int}
   --> parse into StructuredQuery IR (Comparison / Operation / limit)
   --> store-specific Translator (Pinecone, Qdrant, Chroma, Weaviate, Milvus,
       Elasticsearch, OpenSearch, PGVector, MongoDB Atlas, Neo4j, Redis, ...)
   --> vectorstore.search(query, filter=native_filter, k=limit)
```

07. GraphRAG and Knowledge Graphs: Community Summaries, Graph Memory, and When Graphs Beat Vectors

Overview

```
 index-time LLM cost  ------------------------------------------------>  high
 |                                                                          |
 vector RAG     LazyGraphRAG   FastGraphRAG    HippoRAG/LightRAG    GraphRAG (standard)
 (embed only)   (NLP concept   (NLP graph +    (LLM triples or      (LLM entities +
                 graph, no     LLM community    entity/relation      relationships +
                 summaries)    reports)         profiles)            summaries + claims
                                                                     + community reports)
 query-time cost, global questions: high <-------------------------> low (pre-summarized)
```

1.2 Indexing pipeline

```
documents
   |
   v
[1] Compose TextUnits ---- chunk (default 1,200 tokens); unit of extraction + provenance
   |
   v
[2] Document processing -- link documents <-> TextUnits; export Documents table
   |
   v
[3] Graph extraction ----- LLM per TextUnit -> entities (title,type,description)
   |                       + relationships (source,target,description)
   |                       merge by (title,type) / (source,target)
   |                       LLM summarizes merged description lists
   |                       optional: claim extraction -> Covariates (off by default)
   v
[4] Graph augmentation --- hierarchical Leiden community detection, recursive until
   |                       communities reach a size threshold -> multi-level hierarchy
   v
[5] Community summarization -- LLM report per community (overview, key entities,
   |                            relationships, claims), then a shorter summary
   v
[6] Text embeddings ------ TextUnit text, entity descriptions, report content
                           -> vector store (LanceDB default; Azure AI Search)
outputs: Documents, TextUnits, Entities, Relationships, Communities,
         Community Reports, (Covariates)
```

1.3 Query modes

```
         query
           |
  +--------+---------+---------+          level-k community reports
  v        v         v         v          (batched into context-size chunks)
[map]    [map]     [map]     [map]   --> rated points {text, score}
  \        |         |        /
   +-------+----+----+-------+
                v
       filter/sort by score, pack to token budget
                v
            [reduce] --> final answer (+ report/source references)
```

1.9 LazyGraphRAG: deferring all LLM work to query time

```
INDEX (no LLM)                          QUERY (budgeted LLM work)
--------------                          -----------------------------------------
docs -> chunks                          query -> LLM: 3-5 subqueries, expanded query,
   -> noun-phrase concepts                       refine with matching graph concepts
   -> co-occurrence concept graph       per subquery:
   -> graph-statistics optimization        best-first: rank chunks by embedding sim,
   -> hierarchical communities                        rank communities by top-k chunks
   -> chunk embeddings                     breadth-first: LLM sentence-level relevance
                                                      test on top-k untested chunks of
                                                      communities in rank order
                                           iterative deepening: recurse into
                                                      sub-communities after z empty
                                                      communities in a row
                                           stop: no relevant communities left or
                                                 relevance-test budget / q reached
                                        map: concept subgraph -> group chunks by
                                             community -> LLM extracts claims
                                             -> rank, filter to context window
                                        reduce: LLM answers expanded query from claims
```

2.2 Architecture

```
INDEX                                       QUERY
-----                                       -----
chunks --LLM--> entities + relations        query --LLM (keyword role)-->
          (name, type, description;            low-level keywords  (entities)
           relation keywords/description)      high-level keywords (themes)
   |                                               |                |
   v                                               v                v
dedupe/merge into graph  <------------------  match entity       match relation
   |                                          vectors (local)    vectors (global)
   +--> KV store: entity/relation profiles           \              /
   +--> vector store: entity, relation, chunk         +--> 1-hop neighbourhood +
   +--> graph store: nodes/edges                      |    source chunks
   +--> doc-status store                              v
                                              (mix mode: + naive chunk vectors)
                                                      v
                                              [optional reranker] -> LLM answer
```

3.1 HippoRAG (NeurIPS 2024)

```
OFFLINE INDEXING                            ONLINE RETRIEVAL
passages --LLM OpenIE--> (s, p, o) triples  query --LLM NER--> query entities
                     |                                          |
                     v                                retrieval encoder: nearest KG nodes
          schemaless KG (noun-phrase nodes)                     |
                     +                                          v
   retrieval encoder (ColBERTv2/Contriever):        seed PPR at those nodes (node
   synonymy edges between similar nodes             specificity weighting)
   (example scripts: cosine >= 0.8)                             |
                     |                                          v
          node -> passage incidence matrix          PPR mass over nodes -> aggregate to
                                                    passages -> rank -> LLM reader
```

4.1 Retrievers (neo4j-graphrag-python)

```
question --embed--> vector index on :Chunk(embedding) --top-k (node, score)-->
   retrieval_query:
     MATCH (node)<-[:PART_OF_CHUNK]-(e:__Entity__)-[r]-(nbr:__Entity__)
     RETURN node.text, collect(e.name + ' ' + type(r) + ' ' + nbr.name)
   --> context strings --> RagTemplate(context, question, examples) --> LLM
```

4.2 KG construction pipeline

```
           LEXICAL LAYER                          DOMAIN LAYER
(:Document)<-[:PART_OF_DOCUMENT]-(:Chunk)-[:NEXT_CHUNK]->(:Chunk)
                                    ^
                                    | [:PART_OF_CHUNK]
                              (:__Entity__:Person)-[:WORKS_AT]->(:__Entity__:Org)
```

Extraction granularity.

```
docs -> parse -> chunk (+lexical graph: Document/Chunk/NEXT_CHUNK)
     -> [sample] -> LLM schema proposal / prompt auto-tune
     -> per-chunk LLM extraction (structured output, schema-constrained)
     -> validate + prune (schema patterns, direction, empties)
     -> entity resolution (exact -> fuzzy/semantic -> LLM adjudication)
     -> write graph (batched) + embed nodes/relations/chunks
     -> optional: community detection (Leiden) + LLM reports
     -> optional: temporal validity on edges (valid_from / invalidated_at)
```

7.3 Decision heuristics

```
Is the question about the WHOLE corpus (themes, trends, "what are the main...")?
   yes -> community-hierarchy method: LazyGraphRAG (cheap index) or GraphRAG global /
          dynamic selection (precomputed reports, repeated queries on a stable corpus)
   no  -> Does the answer need >= 2 hops through shared entities?
            yes -> PPR graph (HippoRAG 2 / fast-graphrag) or vector entry + Cypher
                   expansion (Neo4j); DRIFT if a GraphRAG index already exists
            no  -> Is the domain inherently relational/typed with exact constraints
                   (org charts, tickets, products, regulations)?
                     yes -> property graph + Text2Cypher / curated KG (KAG, Neo4j)
                     no  -> hybrid vector + BM25 + reranker (chapter 05); add a graph
                            only when evaluations show multi-hop failures
Do facts change over time and must history be queryable? -> temporal KG (Graphiti)
```

08. Agentic and Iterative RAG: Self-RAG, CRAG, FLARE, IRCoT, Adaptive-RAG, ReAct, RL-Trained Search Agents and Agentic Search

Overview

```
  FIXED PIPELINE                                                    AUTONOMOUS AGENT
  |---------------|----------------|----------------|----------------|---------------|
  Naive RAG     IRCoT / DSP      FLARE / DRAGIN   Self-RAG / CRAG   ReAct, Search-R1,
  (1 retrieval) (loop fixed by   (retrieval fires (model or small   Claude Code,
                 the program)     on uncertainty)  critic grades    multi-agent research
                                                   and corrects)    (model owns the loop)
  control: code ------------------------------------------------------> control: policy
  cost:    1 LLM call + 1 search ----------------------------> 10s-100s of calls/searches
```

2.2 IRCoT: chain-of-thought as the query generator

```
  question q ---> BM25 (Elasticsearch) ---> paragraphs P0
       |
       v
  +-------------------------------------------------------------+
  | REASON: LLM(q, P_all, CoT so far) -> next CoT sentence s_i      |
  |   if s_i contains "answer is" or i == max_steps: stop          |
  | RETRIEVE: BM25(s_i) -> top-K paragraphs, add to P_all          |
  +-------------------------------------------------------------+
       |  (loop)
       v
  READER: answer from q + collected paragraphs (or read off CoT)
```

3.1 FLARE (Forward-Looking Active REtrieval)

```
context y_<t ---> LLM drafts next sentence s'_t (no retrieval)
                       |
       all token probs >= theta ?  ---- yes ---> accept s'_t, continue
                       | no
                       v
      query = s'_t (the tentative sentence itself)
                       |
              retrieve (BM25 over Wikipedia / Bing)
                       |
      regenerate s_t conditioned on retrieved docs ---> continue
```

4. Self-reflective retrieval: Self-RAG

```
x, y_<t --> [Retrieve?] --no--> generate y_t --> [ISUSE]
                | yes (or P(yes) > threshold delta)
                v
        Contriever-MS MARCO: top-K passages d1..dK
                |
   +------------+------------+-------------+
   v            v            v             v      (parallel, one per passage)
 d1 -> [ISREL] y_t^1 [ISSUP] [ISUSE]   ...   dK -> ... y_t^K
                |
   score_k = log p(y_t^k) + sum_G w_G * s_G(critique tokens)
                |
   segment-level beam search (keep top-B) --> next segment
```

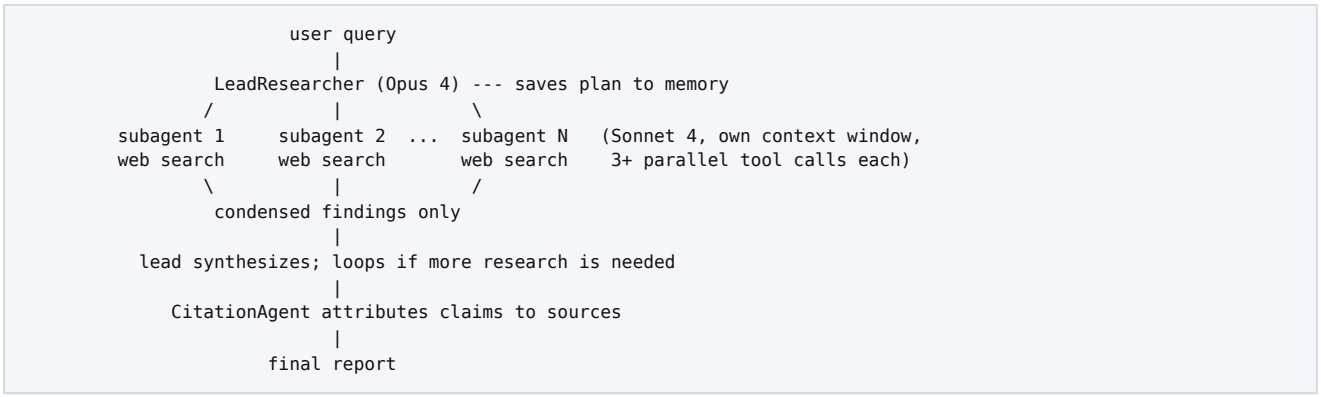
Architecture.

```
query x --> retriever --> ~10 docs
                |
   T5-large retrieval evaluator scores each (x, d_i) pair
                |
   +------------+--------------------+-------------------------+
   | any score > upper threshold      | all scores < lower      | otherwise
   v                                  v                         v
CORRECT                          INCORRECT                 AMBIGUOUS
knowledge refinement:            discard retrieval;        both: refined internal
decompose docs -> strips,        ChatGPT rewrites x to     + web external
score strips, drop irrelevant,   keyword query -> web      knowledge
recompose in order               search API (prefers
                                 Wikipedia pages) ->
                                 fetch pages -> refine
                \________________________________|_____________________________/
                                                 v
                                              generator
```

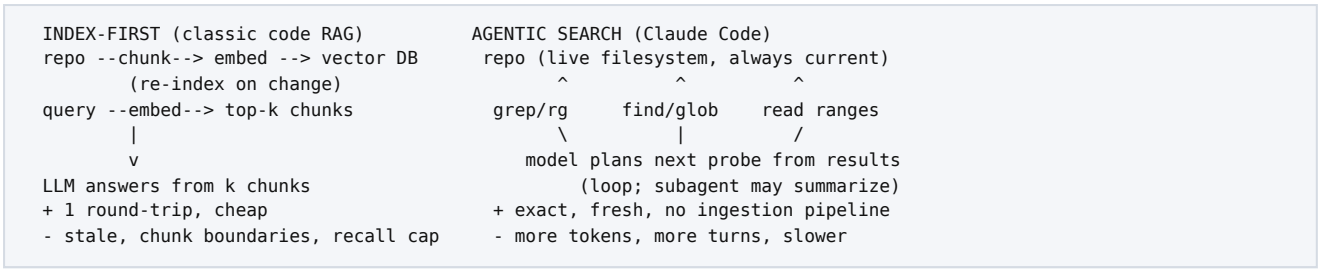
7. ReAct-style retrieval agents

```
Thought: I need the castle David Gregory inherited.
Action:  search[David Gregory (physician)]
Obs:     "... inherited Kinnairdy Castle in 1664 ..."
Thought: Now find how many storeys Kinnairdy Castle has.
Action:  search[Kinnairdy Castle]
Obs:     "... a tower house of five storeys ..."
Action:  finish[five]
```

10. Multi-agent research: Anthropic's Research system

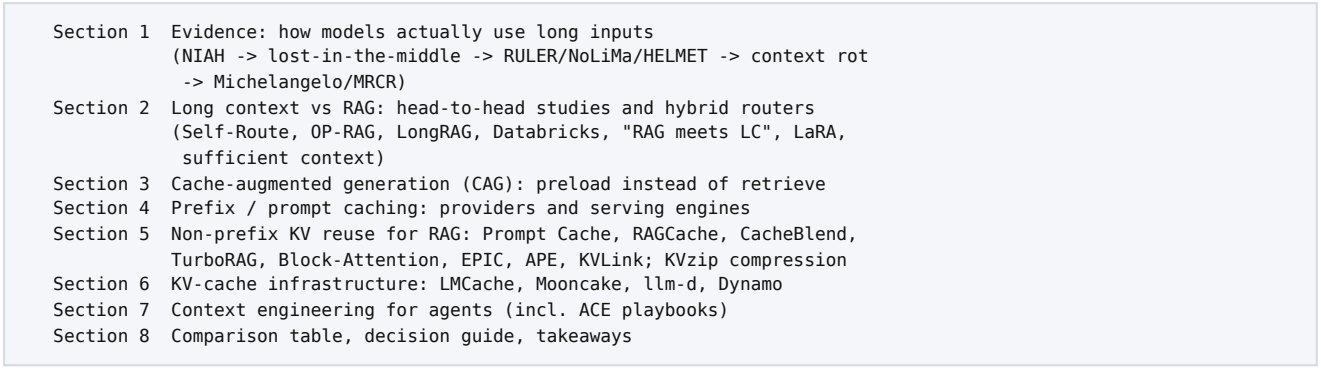


11.1 Claude Code's design

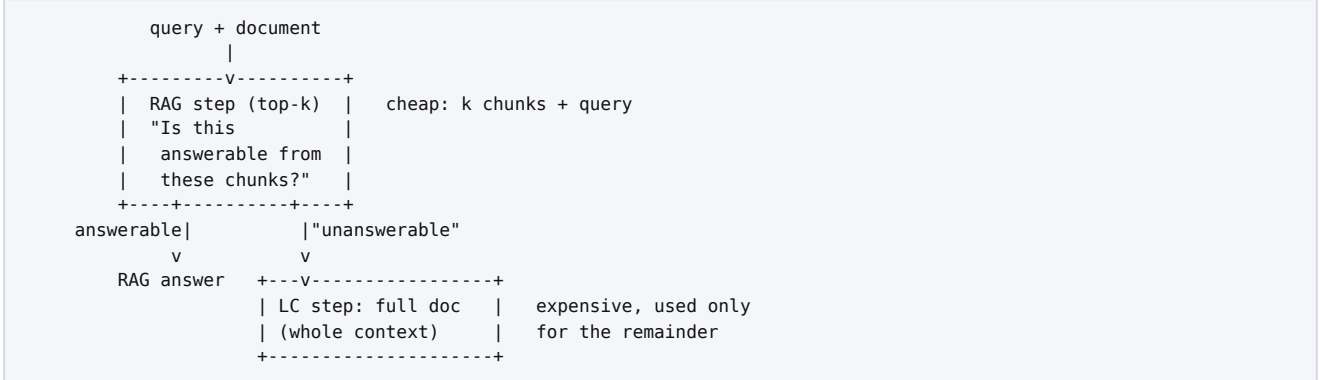


09. Long Context vs RAG: Cache-Augmented Generation, Prompt and KV Caching, Context Rot, and Context Engineering

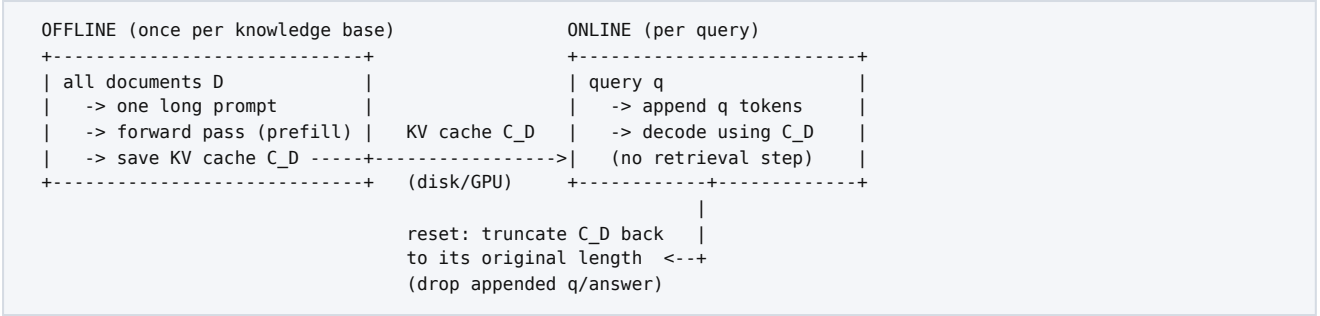
Overview



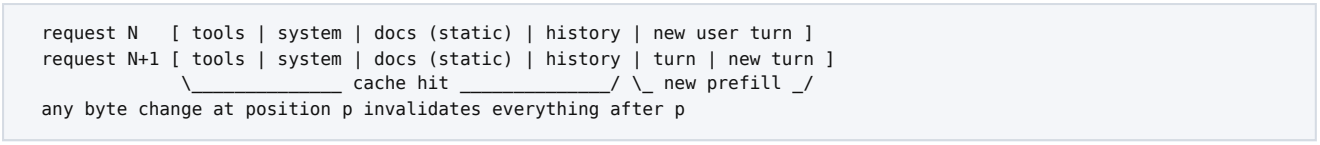
Architecture.



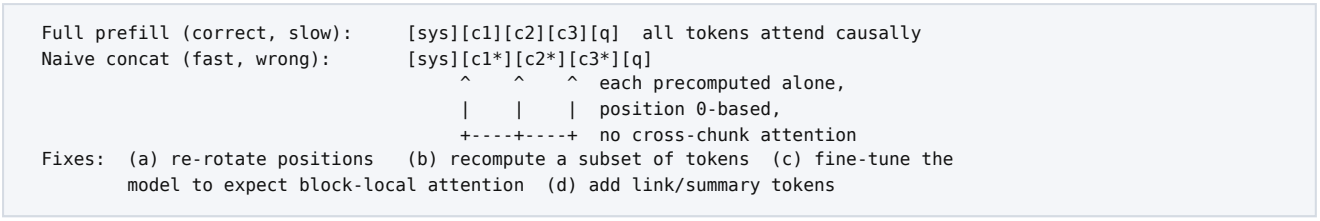
3.1 Design



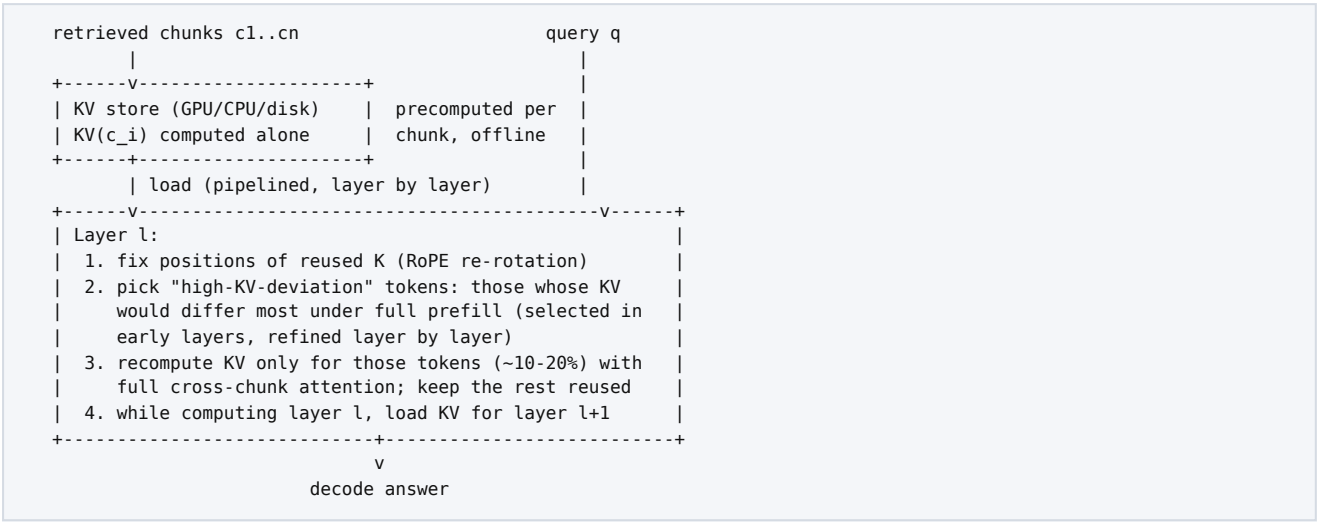
4.1 The mechanism



5.1 The problem



Architecture.



6. KV-cache infrastructure

```
                +------------------------------+
 requests ----->| cache-aware router           |  hashes prefix blocks, looks up
                | (llm-d / Dynamo / gateway)   |  global KV index, picks worker
                +---+----------+----------+---+
                    |          |          |
               +----v---+  +---v----+  +--v-----+     prefill / decode pools
               | vLLM / |  | vLLM / |  | SGLang |     (disaggregated or not)
               | worker |  | worker |  | worker |
               +----+---+  +---+----+  +---+----+
                    |  KV connector (LMCache / NIXL / Mooncake TE)
               +----v-----------------------------------------+
               | tiered KV store: HBM -> CPU DRAM -> SSD -> S3  |
               +------------------------------------------------+
```

7.4 Context-engineering reference architecture

```
+------------------------------- context window (per step) -------------------+
| [tools: fixed set, masked]  [system prompt: stable]   <- cached prefix       |
| [pinned knowledge / CAG corpus (optional)]            <- cached prefix       |
| [compacted summary of older turns]                    <- re-cached on compact|
| [recent turns verbatim] [recent tool results]                                |
| [recitation: current plan / todo]  [new observation]  <- uncached tail       |
+--------------------------------+---------------------------------------------+
          ^ select (JIT retrieval) |  write (notes, files)    | isolate
          |                        v                          v
+---------+--------+   +--------------------+   +-----------------------------+
| RAG / search /   |   | memory dir, files, |   | sub-agents with clean       |
| code grep tools  |   | todo.md, NOTES.md  |   | windows -> condensed results|
+------------------+   +--------------------+   +-----------------------------+
```

8.2 Decision guide

```
Is the knowledge static and < ~effective context length of your model?
   |-- yes --> Is it queried repeatedly within the cache TTL?
   |             |-- yes --> CAG / provider prompt caching (pin it in the prefix)
   |             |-- no  --> long context per query, or small RAG
   |-- no  --> RAG.
               Are the same chunks hot across many queries, and is TTFT critical?
                 |-- yes, can fine-tune --> TurboRAG / Block-Attention
                 |-- yes, stock model   --> CacheBlend via LMCache (or EPIC/APE)
                 |-- no                 --> standard RAG, cache only the static prefix
Hard global / multi-hop questions?  --> route escalations to long context (Self-Route)
Agent loop?  --> context engineering: stable prefix, JIT retrieval, compaction,
                 notes, sub-agents; track cache hit rate
```

10. Multimodal RAG: Visual Document Retrieval, Cross-Modal Embeddings, and Video and Audio Retrieval

Overview

```
                     MULTIMODAL RAG: FOUR RETRIEVAL FAMILIES
 source artefact                index representation                      generator
+----------------+   (1) parse/OCR/caption/ASR -> text chunks (+KG)   +--------------+
| PDF page       |-------------------------------------------------->|              |
| slide, chart   |   (2) render -> VLM -> 1 dense vector per page      |   VLM / LLM  |
| table, figure  |-------------------------------------------------->|  (page img,  |
| screenshot     |   (3) render -> VLM -> ~700-1,030 patch vectors     |   frames,    |
| video, audio   |-------------------------------------------------->|  transcript, |
|                |   (4) any modality -> shared omni-modal space       |   captions)  |
+----------------+-------------------------------------------------->+--------------+
```

Architecture

```
OFFLINE (per page)                                ONLINE (per query)
PDF -> rasterise -> 448x448 image                 "which year has highest outlay?"
          |                                                 |
  SigLIP-So400m (32x32 = 1,024 patches)             Gemma tokenizer + LM
          |                                                 |
  Gemma-2B contextualisation (+ prompt)             linear proj -> n_q x 128
          |                                                 |
  linear proj -> 1,030 x 128 (bf16/fp32)                    |
          |                                                 v
  multi-vector store ---------------------->  MaxSim(Q,D) = sum_i max_j <q_i, d_j>
                                                            |
                                            top-k page IDs -> page images -> VLM
```

10.5 Serving Late-Interaction Visual Retrieval at Scale

```
cost ~ (#pages scored) x (#vectors/page) x (bytes/vector) x (#query tokens)
         |                    |                  |
         |                    |                  +-- scalar int8 (4x), binary (32x),
         |                    |                      PLAID residuals [20][27][32]
         |                    |
         |                    +-- token pooling: hierarchical (3x), row/col mean (32x),
         |                        k-means [2][18][26]
         +-- multi-stage: MUVERA / pooled / dense prefetch -> full MaxSim rerank
             on 100-1,000 candidates [19][22][24]
```

Production patterns: Vespa, Weaviate and NVIDIA

```
PDF/slide decks -> rasterise (1,500-2,500 px) -> page store (BLOB/object storage)
                        |
                        v
             Col* / DSE embedder (GPU)  -> [pooled|FDE|binary] ANN index
                                        -> full-precision multivectors (rerank store)
query -> query encoder -> stage 1: ANN prefetch (100-1,000) -> stage 2: MaxSim rerank
      -> optional VL reranker / LLM-agent selection (top 5-10)
      -> VLM generator(page images + query) -> answer + page-image citations + heatmaps
```

RAG-Anything (HKUDS, 2025)

```
PDF/DOCX/PPTX --MinerU--> {text | image | table | equation} + hierarchy
     text  ---------------------------------> chunks + entities --+
     image --VLM caption + spatial relations-> entity nodes ------+--> multimodal KG
     table --structured interpretation-------> entity nodes ------+    + vector index
     eqn   --LaTeX parse + concept mapping---> entity nodes ------+
query -> vector search + graph traversal -> modality-aware rerank -> LLM/VLM answer
```

VideoRAG (HKUDS, 2025; KDD 2026): graph plus segment embeddings

```
video --split 30 s--> segment_i
   |-- audio  -> Whisper ASR ------------------+
   |-- 5 frames -> VLM caption ----------------+--> text chunks -> entity/relation KG
   |-- frames  -> ImageBind 1,024-d -> visual vector index           + text vectors
query --> {KG/entity search, chunk search, visual search} -> top segments
      --> re-sample 15 frames -> fine captions w/ timestamps -> LLM/VLM answer
```

11. RAG Evaluation: Metrics, Judges, Benchmarks, Golden Datasets and Production Monitoring

Overview

```
            reference answer / nuggets / gold doc IDs  (golden dataset)
                  |                |                          |
                  v                v                          v
query -> [retriever] -> top-k -> [reranker] -> context -> [generator] -> answer+cites
  |           |                            |          |              |              |
  |   IR metrics: Recall@k, nDCG,  context precision  |   faithfulness/groundedness  |
  |   MRR, MAP (needs qrels / IDs) context recall     |   (answer vs context, NLI    |
  |                                (claims of ref vs  |    or LLM judge)             |
  |                                 context)          |                              |
  +---------------- answer relevancy (answer vs query) --------------------------+
                   answer correctness (answer vs reference / nuggets)
                   citation recall / precision (each sentence vs cited passages)
                   refusal / negative rejection (when evidence is absent)
```

3.2 Architecture

```
  sample = {user_input, retrieved_contexts, response, reference?, reference_contexts?}
                                |
            +-------------------+--------------------+
            v                   v                    v
    [claim extractor LLM]   [question generator]  [chunk relevance judge]
       response -> claims     response -> N qs     (chunk, ref|resp) -> 0/1
            |                   |                    |
    [NLI verifier: LLM or   [embed + cosine vs     [rank-aware AP over v_k]
     HHEM T5 classifier]     user_input]               |
            v                   v                    v
         faithfulness       answer relevancy      context precision / recall
```

5. ARES: trained judges plus prediction-powered inference

```
 corpus docs ----> [LLM few-shot query/answer generator] ----> synthetic (q, passage, a)
                                                                + generated negatives
                                                                        |
                                                                        v
                                                  [fine-tune classifier judges, one per
                                                   dimension: ctx-rel, faithful, ans-rel]
                                                                        |
  RAG system outputs (unlabeled, large) ----------------------------->  | predictions
  human-labelled validation set (small, ~50 to several hundred) --------+
                                                                        v
                                    [Prediction-Powered Inference: correct judge bias using
                                     labelled set; output score + 95% confidence interval]
```

8. Citation accuracy

```
answer sentences s_1..s_n, each with citation set C_i
  for each s_i:
     recall_i = NLI( concat(passages in C_i) => s_i )            # 1 or 0
     for each c in C_i:
        irrelevant(c) = not NLI(c => s_i) and NLI(C_i \ {c} => s_i)
  citation_recall    = mean_i recall_i
  citation_precision = 1 - (#irrelevant citations / #citations)
```

9. Nugget-based evaluation and the TREC RAG Track

```
 topics --> [retrieval runs over MS MARCO v2.1 segments] --> pooled passages
                                                                  |
                                        [UMBRELA: LLM relevance labels 0..3]
                                                                  |
                       passages with judgment >= 2 --> [AutoNuggetizer: create nuggets]
                                                                  |  label vital / okay
 RAG answers (with [n] citations) --> [assign: support / partial / not] <-+
                                                                  |
                       vital_score, all_score, strict_* (full=1.0, partial=0.5)
```

12. Building golden datasets

```
  1. SEED             2. SYNTHESIZE              3. CURATE             4. FREEZE
+----------------+  +-------------------------+  +-------------------+  +---------------+
| prod query logs|  | KG over corpus (ragas): |  | SME review: keep, |  | versioned set |
| SME-written qs |->| nodes, NER, similarity  |->| fix, drop; label  |->| + slices      |
| support tickets|  | edges -> scenarios ->   |  | gold doc IDs,     |  | + judge cal.  |
| failure cases  |  | single/multi-hop,       |  | reference answer, |  | subset (PPI)  |
+----------------+  | specific/abstract qs    |  | nuggets (vital?)  |  +-------+-------+
                    +-------------------------+  +-------------------+          |
         ^                                                                      |
         +------------------ 5. GROW: add every production failure -------------+
```

13. Production monitoring

```
user --> app (instrumented: OpenTelemetry / OpenInference spans)
           | retrieve span: query, chunk ids, scores, latency
           | rerank span
           | llm span: prompt, context, answer, citations, tokens, cost
           v
    [trace store] --(sample 100%)--> cheap checks: HHEM/MiniCheck groundedness,
           |                         citation-NLI, refusal detector, PII, latency/cost
           |--(sample 1-10%)-------> LLM judges: context relevance, answer relevance,
           |                         completeness (pinned judge + prompt version)
           |--> user feedback (thumbs, edits, escalations)
           v
    dashboards + alerts (per slice: tenant, intent, doc source, language)
           |
           +--> annotation / review queue --> golden dataset (Section 12) --> CI gate
```

12. Enterprise RAG in Production: Case Studies from Engineering Blogs

A reference architecture distilled from the case studies

```
                      +---------------------- offline / async ----------------------+
sources (wikis, docs, |  connectors -> parse/normalize -> enrich (LLM summaries,    |
tickets, chats, CRM,  |  keywords, FAQs) -> chunk -> embed -> index (lexical+vector |
code, policies, DBs)  |  + ACL metadata + graph edges) -> eval sets from prod logs  |
                      +--------------------------------+---------------------------+
                                                       |
user --> [input guard] --> [router / intent] --> [query rewrite / decompose]
                                  |                          |
                                  |          [retrieve: hybrid + ACL filter + rerank]
                                  |                          |
                                  v                          v
                              [tools / APIs / SQL]   [context assembly + citations]
                                  \                          /
                                   +-----> [LLM generate] <+
                                                  |
                       [output guard: grounding, policy, PII, format]
                                                  |
                       pass -> answer + sources   fail -> retry / human handoff
                                                  |
                       logs -> LLM judge + human review -> fixes to KB/prompts/index
```

1.2 Microsoft 365 Copilot: grounding through Microsoft Graph and the Semantic Index

```
user prompt (Word/Teams/Outlook/...)
       |
       v
[Copilot orchestration] --pre-processing ("grounding")--> Microsoft Graph
       |                      (search: Semantic Index + keyword;   (mail, files,
       |                       only content the user can view)      chats, calendar)
       v
modified prompt + grounding data --> [LLM (Azure OpenAI, in service boundary)]
       |
       v
[post-processing: responsible-AI checks, security/compliance/privacy review,
 app command generation]
       |
       v
response + citations to the app
```

1.5 Slack AI: RAG inside the trust boundary

```
Slack user --> Slack AI service --(user's ACL)--> Slack search / message store
                    |                                    |
                    |<----------- relevant messages -----+
                    v
       [escrow VPC in Slack's AWS: hosted LLM]  (provider: no data access)
                    |
                    v
    summary / answer (ephemeral) --> user
```

2.1 DoorDash: Dasher support with a two-tier guardrail and an LLM judge

```
Dasher chat --> [conversation summarizer: extract core issue]
                         |
                         v
             [retrieve: similar resolved cases + KB articles]
                         |
                         v
             [LLM: draft response from retrieved context]
                         |
                         v
    +--------------[LLM Guardrail]---------------+
    | tier 1: cheap semantic-similarity check     |
    |   (response vs retrieved KB content)        |
    | tier 2: LLM evaluator, only if tier 1 flags |
    |   (grounding, coherence, policy compliance) |
    +-------------------+-------------------------+
           pass         |          fail
            v           |            v
        send reply      |   retry / escalate to human agent
                        v
    [LLM Judge offline: retrieval correctness, response accuracy,
     grammar, coherence to context, relevance] + human review
                        |
                        v
    [quality pipeline: fix KB articles, retrieval, prompts; regression tests]
```

2.2 LinkedIn: a knowledge graph over support tickets

```
historical tickets
  |  parse each ticket into a tree (rule-based + LLM)
  v
intra-ticket trees: ticket -> sections (summary, description, priority, resolution)
  |  connect tickets: explicit links (clone/related) + implicit (embedding similarity)
  v
inter-ticket graph  --(section embeddings)--> vector index
-------------------------------------------------------------------------------
query: [LLM extracts entities + intents]
         -> [embedding search finds candidate tickets]
         -> [LLM translates question to graph query (Cypher-like) over sub-graphs]
         -> [retrieved sub-graphs -> LLM answer]
```

2.5 Uber Genie and Enhanced Agentic RAG

```
sources: Engwiki (internal wiki), internal Stack Overflow, engineering docs
    |  Spark ETL: scrape + chunk
    v
embeddings (OpenAI embedding models via Uber's Michelangelo gateway)
    |
    v
SIA "Search In Action" (Uber's in-house vector DB)  <-- refreshed on schedule
    |
Slack question --> Knowledge Service --> retrieve top chunks --> LLM (via gateway)
    |                                                            |
    |<----------------- answer + source links -----------------+
    v
user feedback buttons (resolved / helpful / not relevant) --> metrics dashboard
```

Architecture.

```
user query + page context (job post, profile, feed post)
    |
    v
[Routing: small/fast LLM picks a specialized "agent"]  (job assessment, knowledge, ...)
    |
    v
[Retrieval (recall-oriented): agent decides which internal APIs ("skills") to call:
  people search, job data, company data, external search; plus embedding-based
  retrieval of few-shot examples]
    |
    v
[Generation (precision-oriented): larger LLM composes answer from noisy context]
    |
    v
streamed response (all stages stream, to cut time to first token)
```

3.4 Ramp: RAG as constrained classification

```
business profile (name, description, website text, ...)
      |  embed
      v
[retrieve top-k candidate NAICS codes by embedding similarity]
      |
      v
[LLM picks from candidates only, constrained output; optional second pass
 at a finer level of the hierarchy]
      |
      v
validated code (must be in candidate set) + explanation
```

13. Code Retrieval and RAG: Codebase Indexes, Repo Maps, and Agentic Search in Coding Assistants

Overview

```
                    CODE CONTEXT ARCHITECTURES (2023-2026)

 latency budget:   < 1 s (completion)      seconds (chat)        minutes (agent)
                   |                       |                     |
 INDEX-FIRST       Copilot/Cursor/Augment embeddings + sync ---> exposed as a tool
 LEXICAL ENGINE    Cody: Jaccard windows   Cody: BM25 over Sourcegraph search
 STRUCTURAL        tree-sitter intent      Aider repo map        LocAgent / RepoGraph
 AGENTIC SEARCH                                                  Claude Code, SWE-agent,
                                                                 SWE-grep subagents
                   <------------- convergence: hybrid toolboxes ------------->
```

13.2 Cursor: Merkle-tree sync, chunk embeddings, and teammate index reuse

```
CLIENT (IDE)                                  CURSOR BACKEND
files --> Merkle tree (SHA-256 per file,
          folder = H(children))  -- root/subtree hashes -->  server Merkle tree
                  | diverging branches only                      |
changed files --> syntactic chunks --> embed (async) -----> chunk-embedding cache
                                                            (keyed by chunk content)
                                                                    |
query --> embed ---------------------------------------> vector DB (obfuscated
                                                            paths + line ranges)
local read of returned line ranges <------ top-k ids ----------+
(code text is read from the client's own disk)
```

13.3 GitHub Copilot: a remote code-search index reconciled with the local diff

```
query --+--> embed query (started eagerly)
        |
        +--> GitHub semantic code search @ indexed commit --> candidate chunks
        |                                                       (exclude diffed files)
        +--> local diff vs indexed commit
                |-- > 2,000 files or > 70% changed --> skip remote, warn
                |-- small diff --> local embeddings over changed files (15 s cap)
        merge --> ignore-filter --> sort by distance --> [optional reranker]
             --> top (token_budget / 250) chunks into the prompt
```

13.4 Sourcegraph Cody: from embeddings to search-engine retrieval and layered long context

```
user query --> tokenize + cleanse --> Sourcegraph search over selected repos
                                        (up to 10 repos; adapted BM25 + learned signals)
                                                  | ranked file snippets
IDE local context (open files) ---------------+ |
                                              v v
                                  global ranking across context providers
                                              |
                             take first N snippets (N depends on snippet length)
                                              |
                         prompt = prefix + user input + context --> LLM
```

13.6 Windsurf and Cognition: SWE-grep, an RL-trained fast-context subagent

```
main agent (frontier model)            Fast Context subagent (SWE-grep)
"where is X handled?" ------------->  turn 1: 8 parallel {grep, glob, read}
                                       turn 2: 8 parallel ...
                                       turn 3: 8 parallel ...
                                       turn 4: answer = [(file, line ranges)]
reads only cited ranges <-----------  reward in training: F0.5(files, lines)
```

13.8 Aider: the PageRank-ranked repository map

```
repo files --tree-sitter tags--> defs/refs per file (disk cache by mtime)
      |
      v
file graph: referencer --(ident, w = mul * sqrt(#refs))--> definer
      |      mul: x10 mentioned, x10 long-name, x0.1 private, x0.1 common, x50 in-chat
      v
personalized PageRank (chat files / mentioned names get personalization)
      v
ranked (file, symbol) defs --binary search on count--> map within --map-tokens
      v
prompt: repo map + files the user or LLM added to the chat
```

13.10 SWE-bench retrieval findings: what the evaluations say

```
Evidence map (who found what)
repo size / query type      | finding                              | source
----------------------------+--------------------------------------+-----------
small repo, issue text      | grep/find sufficient; agent persists | [18][25]
large repo (>1k files)      | +semantic search: +2.6% retention    | [2]
any repo, latency-critical  | RL fast parallel grep subagent       | [19]
issue -> function           | trained embed + rerank > agents/cost | [35]
identifiers shared w/ query | BM25 ~ best dense (RepoEval)         | [39]
NL issue -> code            | dense >> BM25 (SWE-bench NDCG)       | [39]
```

14. Answer Engines and Web-Grounded LLMs: Perplexity, Google AI Overviews and AI Mode, Bing/Copilot, ChatGPT Search, Search APIs for Agents, Deep Research and Citation Systems

Overview

```
                  THE ANSWER-ENGINE DESIGN SPACE
latency      ~1 s            ~5-30 s                 minutes to tens of minutes
            |---------------|----------------------|-------------------------|
pattern     single-shot     agentic search         deep research
            grounding       (search inside the     (plan, fan out, read
            (1 query,       reasoning loop,        hundreds of pages,
            top results)    a few to ~10 calls)    write a cited report)
examples    AI Overviews,   ChatGPT search,        OpenAI / Gemini / Claude
            Bing grounding, Claude web search,     Research, Perplexity
            Sonar-style     AI Mode, Perplexity    deep research, Exa /
            answers         Pro search             Parallel / Tavily research
```

2. Reference architecture

```
user query (+ conversation, location, locale)
     |
     v
[1] TRIGGER / CLASSIFY ---- no search needed --> answer from parametric knowledge
     |  (is the answer current? niche? risky?)
     v
[2] QUERY GENERATION: rewrite to standalone; generate 1..N queries ("fan-out")
     |
     v
[3] RETRIEVAL: own index | search API | SERP provider     (filters: domain, recency,
     |                                                     locale, safe-search)
     v
[4] CONTENT EXTRACTION: snippet | query-biased highlights | full page | PDF | live crawl
     |
     v
[5] CONTEXT ASSEMBLY under a token budget (dedupe, per-URL caps, threshold filtering,
     |   code-based filtering)
     |
     +--> [6] AGENTIC LOOP? read results -> reflect on gaps -> new queries -> [3]
     v
[7] GENERATION with inline citations (markers, spans, or post-hoc citation pass)
     |
     v
[8] CITATION RENDERING + compliance (show sources, show queries, provider widgets)
```

3.2 Google: AI Overviews, AI Mode and Deep Search

```
  "best e-bike for a steep 10 km commute in the rain, under $2k"
                               |
                   Gemini (custom for Search)
                   decomposes the information need
      +-----------+-----------+-----------+------------+-------------+
      v           v           v           v            v             v
  "e-bike hill  "waterproof "e-bike     "e-bike      shopping      reviews /
   torque"       rating      range 10km  under 2000"  graph /       forums
                 IPX"        hills"                   prices        (filtered)
      +-----------+-----------+-----+-----+------------+-------------+
                                  v
              merge, dedupe, rank evidence; reason over gaps
                                  v
             answer with inline links  (Deep Search: same pattern,
                                        hundreds of queries, a report)
```

7. Deep research agents

```
 user request
     |
     v
 [clarify / rewrite]  -- ChatGPT: intermediate model asks questions; API: skipped;
     |                   Gemini: optional collaborative planning; Azure: gpt-4o scoper
     v
 [plan]  lead agent / research model writes a plan (persisted to memory)
     |
     +------------+------------+------------+     parallel workers / search threads
     v            v            v            v
  search ->    search ->    search ->    MCP / file   each: search, fetch, read,
  read ->      read ->      read ->      search ->    reflect, refine query
  compress     compress     compress     compress     (own context window)
     +------------+------------+------------+
     |
     v
 [synthesize] lead agent decides: enough? -> else spawn more work
     |
     v
 [cite]  inline citations from the model, or a separate citation pass
     |
     v
 report (+ charts, structured output); background mode + webhooks / polling
```

15. RAG Frameworks and Managed Services: From Orchestration Libraries to Retrieval-as-a-Service

15.1 A reference model: who owns which layer?

```
+-------------------------------------------------------------------------------+
| L5 ORCHESTRATION  agent loop / graph / pipeline; tools; memory; HITL; retries  |
|    LangGraph, Haystack Pipeline+Agent, LlamaIndex Workflows, AI SDK agents,    |
|    Agent Framework, DSPy modules (+ optimizer that compiles prompts/weights)   |
+-------------------------------------------------------------------------------+
| L4 SYNTHESIS      prompt assembly, grounded generation, citations, guardrails  |
|    Anthropic Citations, Bedrock RetrieveAndGenerate, Azure answer synthesis,   |
|    OpenAI file_search annotations, Vertex grounding / check-grounding          |
+-------------------------------------------------------------------------------+
| L3 RETRIEVAL      query rewrite/decompose, hybrid search, fusion, rerank,      |
|                   metadata filters, thresholds                                |
+-------------------------------------------------------------------------------+
| L2 INDEX          vector store / inverted index / graph / SQL; ACLs           |
+-------------------------------------------------------------------------------+
| L1 INGEST         connectors, parsing/OCR/layout, chunking, enrichment, embed  |
+-------------------------------------------------------------------------------+

 Frameworks: thin over L1-L3 (adapters), thick at L5.
 Managed services: thick at L1-L4, increasingly reaching into L5 (agentic retrieval).
 Anthropic Citations: L4 only (bring your own retrieval).
```

15.2.1 LangChain, LangGraph and Deep Agents

```
user question
     |
     v
+-----------+   vector search tool   +--------------+
|orchestrator|--------------------->| vector store |
|  (agent)   |<--- chunk ids -------+--------------+
+-----+------+
      | writes chunks as files (offload)
      v
+-------------+   parallel    +-----------+  +-----------+
| agent FS    |-------------->| subagent 1|  | subagent n|  read + summarise
+-------------+               +-----+-----+  +-----+-----+
                                    \             /
                                     v           v
                                orchestrator synthesises answer + citations
```

15.2.6 Vercel AI SDK

```
Next.js useChat UI --POST--> /api/chat route --streamText(tools)--> LLM
                                    |                                  |
                                    |   tool: getInformation(q)  <---+
                                    v
                    embed(q) -> SQL: 1 - cosineDistance(...) > 0.5
                                    |   ORDER BY sim DESC LIMIT 4
                                    v
         Postgres + pgvector: embeddings(1536-d, HNSW, vector_cosine_ops)
                                    ^
    tool: addResource(content) -> chunk on '.' -> embedMany -> INSERT
```

Architecture and data flow.

```
S3 / SharePoint / Salesforce / Confluence / web          Redshift (structured)
         |                                                       |
         v  StartIngestionJob (one job at a time per KB)         v
parse: default | BEDROCK_FOUNDATION_MODEL | BEDROCK_DATA_AUTOMATION
         |                                          metadata/schema ingestion
chunk: FIXED_SIZE | HIERARCHICAL | SEMANTIC | NONE | Lambda custom
         |                                                       |
embed (e.g. Titan Text v2, Nova Multimodal Embeddings)           |
         v                                                       v
vector store: OpenSearch Serverless | Aurora | Pinecone |   NL -> SQL generation
  Redis | MongoDB Atlas | Neptune Analytics (GraphRAG) |
  S3 Vectors | Kendra GenAI index (managed retriever)
         |
Retrieve / RetrieveAndGenerate
  overrideSearchType HYBRID|SEMANTIC, implicitFilterConfiguration,
  QUERY_DECOMPOSITION, rerankingConfiguration (amazon.rerank-v1:0,
  cohere.rerank-v3-5:0), Guardrails contextual grounding
```

15.3.2 Google: Vertex AI RAG Engine and Vertex AI Search

```
GCS / Drive / Slack / Jira / web --import_files(chunking cfg)--> RagCorpus
                                                                  |
    +--------------------------+--------------------------------+
    | backend: managed store | Vector Search | Pinecone | Weaviate |
    |          Feature Store | Vertex AI Search engine            |
    +--------------------------+--------------------------------+
                               |
 Gemini generate_content(tools=[Retrieval(VertexRagStore(top_k, threshold,
                               llm_ranker))])  or  retrieve_contexts -> any LLM
```

15.3.3 Azure AI Search: from hybrid + semantic ranker to agentic retrieval

```
query + chat history
       |
       v
+---------------- knowledge base (retrievalReasoningEffort) ----------------+
| minimal: no LLM | low: 1 planning pass | medium: + semantic classifier    |
|                 |                      |   -> 1 follow-up iteration       |
| auto: light pass first, escalate up to medium only if grounding is weak   |
+------------------+-----------------------------+----------------------+
                   | parallel subqueries           |
      +------------+------------+-----------------+-------------+
      v            v            v                 v             v
  search index   Blob/OneLake  SharePoint(remote)  Web    MCP server, Work IQ,
  (hybrid+L2)    (indexed)     (user-delegated)          Fabric, Azure SQL
      \____________|____________|___________________|_____________/
                                v
      merged grounding docs + references (+ activity log, + answer synthesis)
```

16. RAG Security and Governance: Permission-Aware Retrieval, Injection, Poisoning, Privacy and Audit

Overview

```
                    THREAT MAP OF A RAG PIPELINE

 writers of     +-----------+   +----------+   +---------+   +-----------------+
 source docs -->| connectors|-->| parse /  |-->| embed + |-->| index / vector  |
 (incl. the     | + ACL sync|   | chunk    |   | metadata|   | store (+ ACLs)  |
 attacker)      +-----------+   +----------+   +---------+   +-----------------+
    ^  POISONING (16.4)   ^ stale ACLs (16.1)      ^ INVERSION (16.5)   |
    |  HIDDEN INSTR. (16.3)                        | cross-tenant (16.2)|
                                                                        v
 user --query--> [authn] --> [query + identity] --> retrieve --> top-k chunks
                                |  ^                                    |
                                |  +-- authz filter / ReBAC check ----+
                                v                                       v
                          prompt assembly: system + user + UNTRUSTED CONTEXT
                                |
                                v
                             LLM --> tools / links / images / email (EXFIL)
                                |
                                v
                   answer + citations --> audit log / traces (PII in logs)
```

Design space: where the authorization check happens

```
 (A) PRE-FILTER (filter pushed into the ANN / BM25 query)
     identity --> resolve groups / authorized IDs --> vector_search(q, filter=ACL)
     + top-k is always k authorized hits     - needs ACL metadata on every chunk
     + nothing unauthorized leaves the index - large ID lists, stale on ACL change

 (B) POST-FILTER (retrieve broadly, then check each hit)
     vector_search(q, k = k * m) --> CheckBulk(user, hits) --> keep allowed --> top-k
     + uses the live authz system (fresh)    - recall loss if most hits are denied
     + no ACL copy in the index              - extra latency per query

 (C) SOURCE-DELEGATED (federated / query-time call to the source system)
     query --> connectors search *as the user* (OAuth on-behalf-of) --> merge
     + exact source semantics                - slow, rate-limited, no global ranking
```

3. Detection classifiers.

```
          CaMeL-style isolation applied to RAG (simplified)

 user query --> P-LLM (trusted) --> program:  docs = search(q)
                                              facts = Q_LLM.extract(docs, schema)
                                              reply(format(facts))
                                  |
                                  v
                 interpreter: every value tagged {source, readers}
                 policy: values derived from docs may NOT choose recipients,
                         URLs or tool names; send() requires readers ⊇ recipient
                                  |
          Q-LLM (no tools) <------+ sees untrusted docs, returns typed values only
```

16.5.3 PII redaction pipelines

```
 source --> [1 ingest-time detect + transform] --> index  (minimize what is stored)
 query  --> [2 query-time scan / pseudonymize]  --> LLM    (protect 3rd-party APIs)
 LLM    --> [3 output scan / deanonymize]       --> user   (stop leakage)
 traces --> [4 log redaction / retention]       --> SIEM   (logs are a datastore)
```

16.5.4 Embedding inversion attacks

```
  Vec2Text loop
  e* (stolen vector) --> hypothesizer --> x0 --> embed(x0)=e0
       ^                                                  |
       +---- corrector(e*, x_t, e_t) --> x_{t+1} <-----+   (repeat ~20 steps, beam)
```

16.8 A reference architecture for a governed RAG service

```
INGESTION PLANE                                              (write path = trust path)
+-------------------------------------------------------------------------------+
| connector (least-privilege, per-source)                                       |
|   -> fetch content + ACLs + labels + version                                  |
|   -> parse (hidden-text normalization) -> injection classifier (chunked 512t)  |
|   -> PII detect/transform (hard-redact / label / vault)                       |
|   -> chunk -> project ACL + label + source trust onto EVERY chunk             |
|   -> embed (private model) -> encrypt vectors w/ tenant key -> tenant shard    |
|   -> emit provenance record (doc id, version, hash, ACL snapshot id)          |
+-------------------------------------------------------------------------------+
QUERY PLANE
+-------------------------------------------------------------------------------+
| authn (OIDC) -> principal + groups from token (never from request body)       |
| -> tenant routing -> retrieve with ACL pre-filter -> post-check high-sens docs |
|    via ReBAC -> robust aggregation / cluster filter for low-trust sources     |
| -> prompt assembly: spotlighted untrusted context, minimal user-data mixing   |
| -> LLM (injection-hardened) -> output scan (PII, verbatim-overlap, URLs)      |
| -> egress policy: no auto-fetched images, link allow-list, confirm side-effects|
+-------------------------------------------------------------------------------+
GOVERNANCE PLANE
+-------------------------------------------------------------------------------+
| OTel GenAI traces (refs not content) -> immutable audit store (per tenant)    |
| revocation SLO monitors | CI red-team suites | erasure cascade (doc->chunk->   |
| vector->cache->trace) | label-based model routing (internal vs external LLM)  |
+-------------------------------------------------------------------------------+
```

17. Fine-tuning vs RAG and Training the Retrieval Stack

17.1.7 Synthesis

```
              Does the model need NEW FACTS, or a NEW SKILL?
                              |
          +-------------------+---------------------+
          | facts (change often / need              | skill (format, reading,
          | citations / access control)             | domain jargon, ranking)
          v                                          v
  RAG + better retrieval                     fine-tune a component
  (train retriever/reranker,                   |
   sec. 17.3-17.6)                             +-- generator reads docs badly
          |                                    |      -> RAFT / RA-DIT LM-ft
          | facts static, corpus small,        +-- retriever misses domain intent
          | latency-critical, offline?         |      -> synthetic queries + negatives
          v                                    +-- ranking quality / latency
  consider synthetic CPT (EntiGraph)           |      -> distil LLM -> cross-encoder
  AND keep RAG on top [10]                     +-- retriever != what the LM needs
                                                      -> LM-supervised retrieval (LSR)
```

17.2.1 RAFT: Retrieval-Augmented Fine-Tuning (UC Berkeley, 2024)

```
P fraction of examples:      Q + D* + D_1 ... D_{k-1}  -->  A* (CoT with quotes)
(1-P) fraction of examples:  Q + D_1 ... D_k (no oracle) -->  A*

A* = reasoning that quotes the oracle verbatim between
     ##begin_quote## ... ##end_quote## markers, then the final answer
default: 1 oracle + 4 distractors; CoT generated by GPT-4-1106
```

17.2.2 RA-DIT: Retrieval-Augmented Dual Instruction Tuning (Meta FAIR, ICLR 2024)

```
           +---------------------- LM-ft ----------------------+
instruction|  for each (x, y) and each of top-3 chunks c:      |
dataset    |    minimise -log p_LM(y | "Background: c" + x)    |
(20 sets)  |  one training instance per chunk                  |
           +---------------------------------------------------+
           +---------------------- R-ft -----------------------+
           |  LSR target: p_LSR(c|x,y) ~ exp(p_LM(y|c+x)/tau)  |
           |  minimise KL(p_R(c|x) || p_LSR(c|x,y)) over top-10 |
           |  update QUERY encoder only (index stays valid)    |
           +---------------------------------------------------+
inference: top-10 chunks -> 10 parallel prompts -> weighted ensemble
```

17.2.3 ChatQA (NVIDIA, 2024): context-enhanced instruction tuning plus a tuned conversational retriever

```
Stage 1: general SFT (about 128K instruction + dialog samples)
         -> conversational, but weak at context-grounded QA
Stage 2: context-enhanced instruction tuning on a blend of
         - HumanAnnotatedConvQA (about 7k dialogues)
         - SyntheticConvQA (about 7k dialogues from GPT-3.5-Turbo)
         - single-turn context QA (DROP, NarrativeQA, ...)
         - a slice of stage-1 data (to keep instruction following)
         - about 1.5k turns annotated "unanswerable"
Retriever: dense retriever fine-tuned on multi-turn conversational QA,
           taking dialogue history + current turn as the query
```

17.3.4 Reinforcement learning on retrieval outcomes (2025)

```
 question --> policy LLM --(query)--> frozen search engine / retriever
                  ^                           |
                  |                      top-k docs
                  |                           v
            PPO / GRPO update <-- reward: answer EM (Search-R1)
                                  or recall / nDCG (DeepRetrieval)
```

17.4.1 InPars v1 and v2 (NeuralMind / Zeta Alpha, 2022-2023)

```
corpus (BEIR dataset) --sample 100k docs--> GPT-J 6B, 3-shot MS MARCO "gbq" prompt,
                                            greedy, 1 query/doc   (~30 h on one A100)
       |
       v
monoT5-3B (MS MARCO-tuned) scores all 100k (q, d) pairs  (~1.5 h on TPU v3-8)
       |  keep top 10k as positives
       v
negatives: 1 random doc from BM25 top-1000 for each synthetic query
       |
       v
fine-tune monoT5-3B (already MS MARCO-tuned) 1 epoch on 10k pos + 10k neg
(Adafactor, lr 1e-3, batch 64+64; < 10 min on TPU v3-8)  -> one reranker per dataset
       |
serve: BM25 top-1000 (Pyserini, k1=0.9, b=0.4) -> InPars-v2 monoT5-3B rerank
```

17.5.1 ANCE (Microsoft, 2020): global negatives from an asynchronously refreshed index

```
Trainer (GPUs) --------- checkpoint k ---------> Inferencer (GPUs)
   ^                                              re-encode entire corpus
   |                                              rebuild ANN index (Faiss)
   +---- ANCE negatives = top-ranked docs <-------+  (swap in when done)
         from the (slightly stale) index
```

17.6 Distillation into retrievers and rerankers

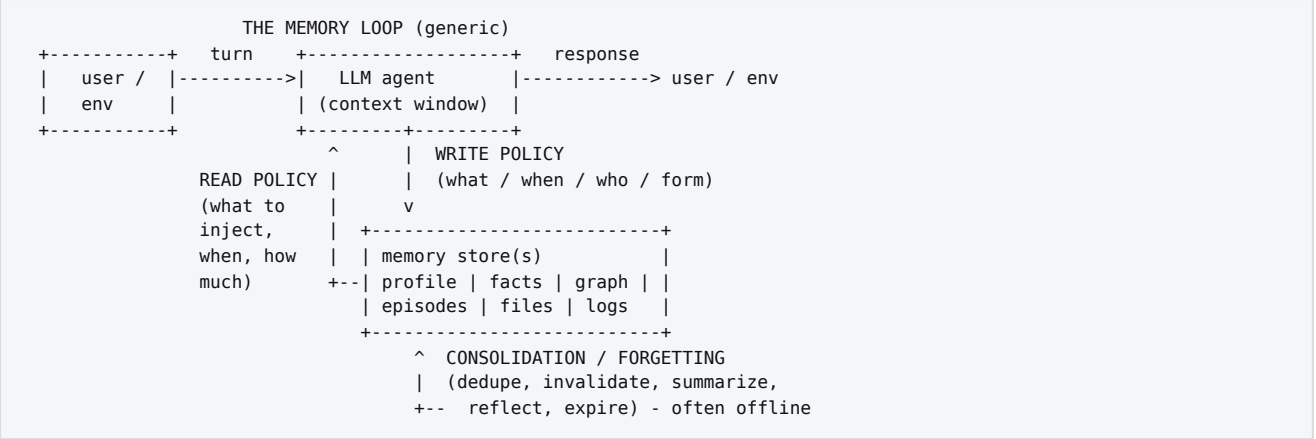
```
 GPT-4 / R1-class LLM  --permutations / reasoning traces-->  7B-32B LLM reranker
        |                                                    (RankZephyr, Rank1)
        | permutation distillation (RankNet)                        |
        v                                                           v
 cross-encoder (DeBERTa / monoT5 / ModernBERT)  <--- scores --- (teacher for)
        |
        | MarginMSE / KL over top-k (GPL, ColBERTv2)
        v
 late-interaction (ColBERTv2)  /  bi-encoder (GPL student, E5)
        |
        | query-side-only or adapters to avoid re-indexing
        v
 production first stage
```

17.7 End-to-end domain adaptation playbook

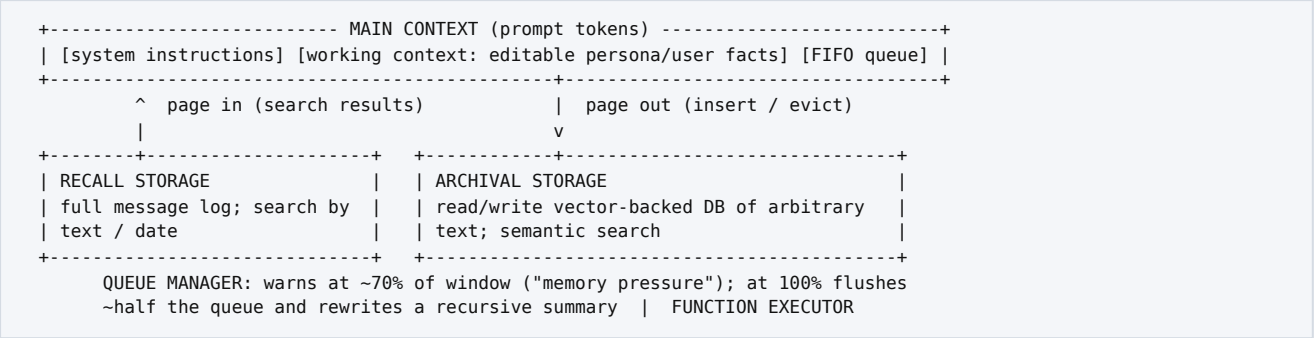
```
0. Baseline: BM25 + off-the-shelf embedder + off-the-shelf reranker; build an eval set
   of REAL user queries with graded labels (LLM-judged + human spot checks)
1. Synthetic queries per chunk
     - generic LLM prompt  -> fast, MS-MARCO-like intents
     - few-shot task prompt (Promptagator) -> intent-specific
     - filter: reranker score (InPars-v2) or round-trip top-1 (Promptagator)
2. Mine negatives
     - dense top-k with rank window + positive-aware margin (NV-Retriever)
     - denoise with cross-encoder (RocketQA) or keep soft scores (GPL)
3. Train reranker first (cheapest big win; no re-index) -- BCE / listwise
     - optionally distil an LLM ranker's permutations (RankGPT) into it
4. Train retriever
     - query-side only / adapter if re-embedding is expensive (Atlas, RA-DIT)
     - else full fine-tune with MarginMSE from the step-3 reranker (GPL)
     - optionally LM-supervised (LSR) if a target generator is fixed (REPLUG, RA-DIT)
     - merge with base weights to limit forgetting (Qwen3 / Gemini model soups)
5. (G) If the generator misreads context: RAFT-style SFT with oracle + distractors,
     CoT with verbatim quotes, P < 100%; or RA-DIT LM-ft over top-k chunks
6. (G) If facts must live in weights (offline / latency): synthetic CPT (EntiGraph),
     fact-level coverage, early stopping; keep RAG on top
7. Re-evaluate end-to-end; track hallucination rate, not only retrieval metrics
```

18. Long-Term Memory Systems for LLMs and Agents: MemGPT/Letta, Mem0, Zep/Graphiti, ChatGPT and Claude Memory, LangMem, and the Write/Read Policies Behind Them

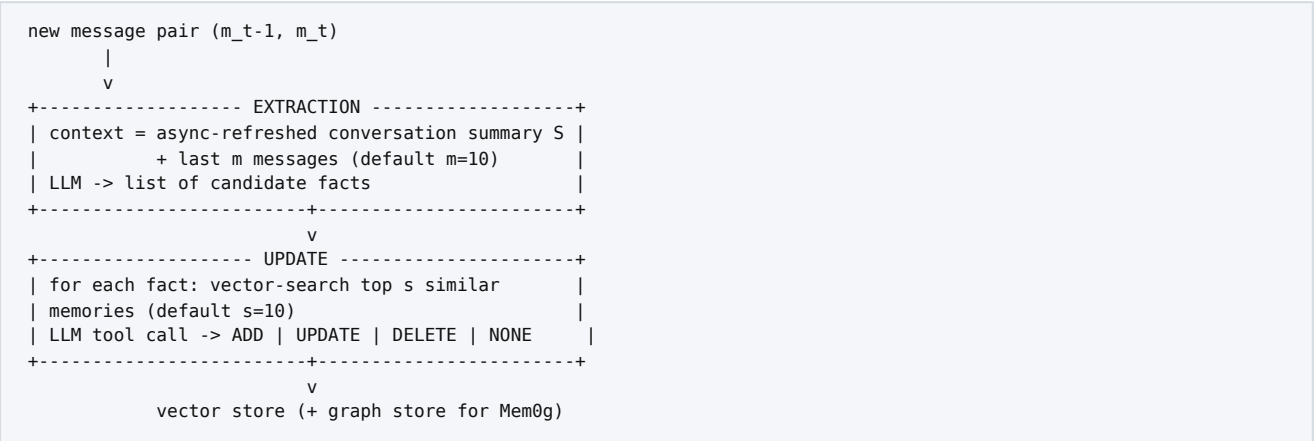
Overview



2.2 Architecture



3.2 Original architecture (2025 paper, v1/v2 library)



4.2 Architecture

```
 EPISODES (raw messages / JSON / text, each with a reference timestamp)
    |  ingest: entity extraction (window of recent messages) -> entity resolution
    |          (embedding + full-text candidate match, LLM dedupe) -> fact (edge)
    |          extraction + dedupe -> temporal extraction -> contradiction check
    v
+-------------------------------------------------------------------------+
|  Episode subgraph   : raw episodes; every derived fact links back here |
|  Semantic subgraph  : entity nodes (evolving summaries) + fact edges   |
|                       edge = (fact text, embedding, valid_at,          |
|                       invalid_at, created_at, expired_at, episodes[])  |
|  Community subgraph : clusters of entities with summaries              |
+-------------------------------------------------------------------------+
    |  retrieve: search (cosine | BM25 | breadth-first traversal)
    |            -> rerank (RRF | MMR | node distance | episode mentions |
    |               cross-encoder) -> context constructor (facts + dates)
    v
 prompt context string
```

5. LangMem and LangGraph Memory: A Toolkit View

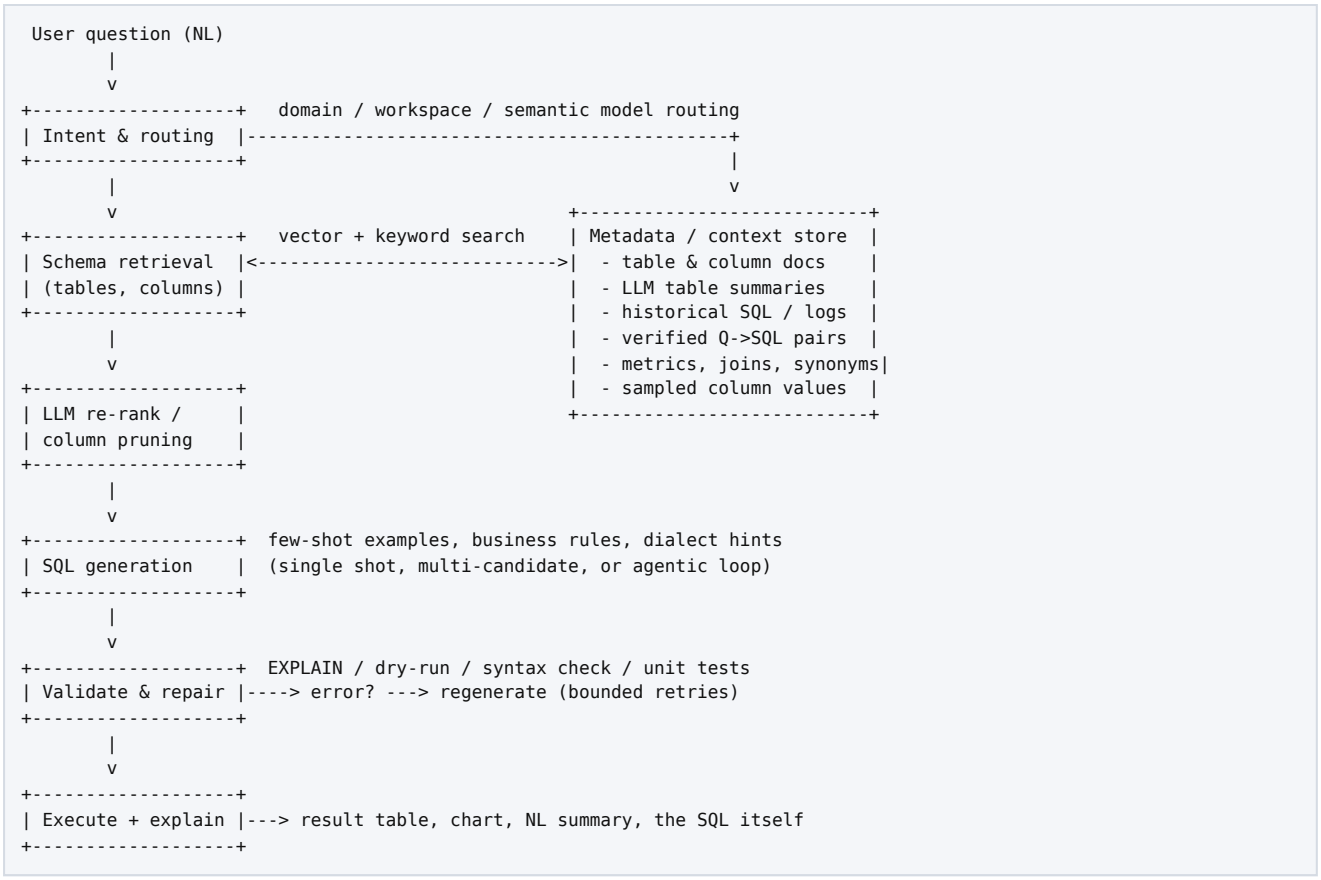
```
        HOT PATH (conscious)                BACKGROUND (subconscious)
user -> agent --tool: manage_memory--> store <-- memory manager (after turn /
        agent --tool: search_memory--> store      after inactivity; reflects,
        (+latency, +tool-choice burden)           consolidates; no added latency)
```

6.2 Reverse-engineered architecture

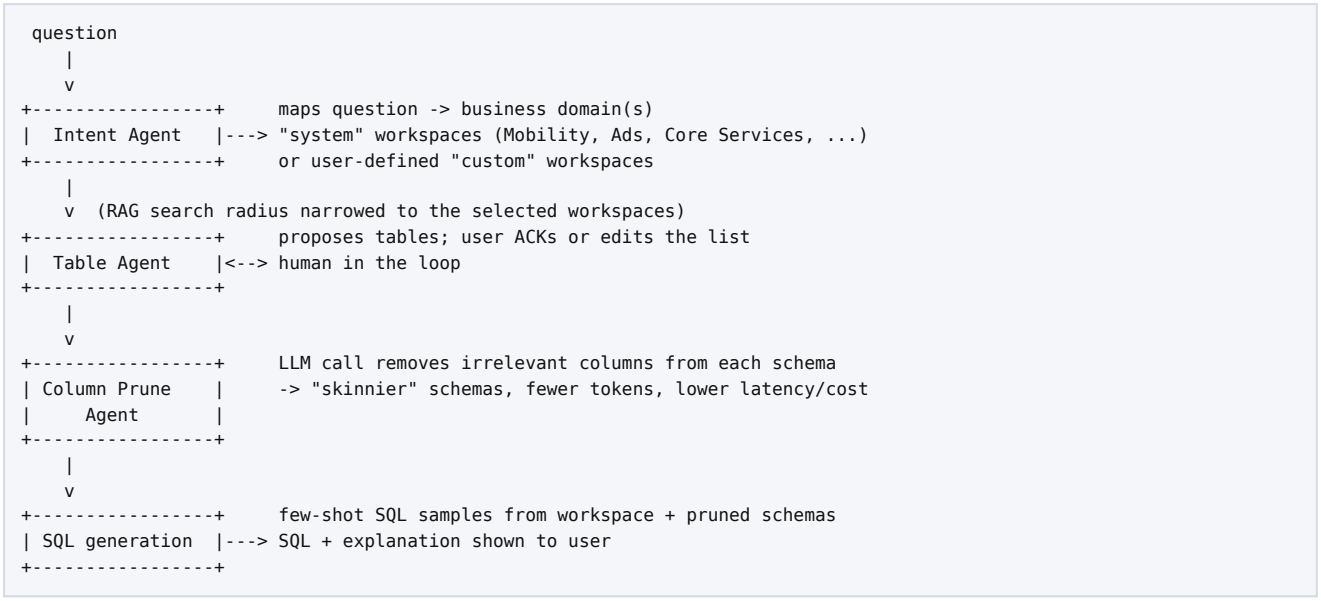
```
[system + developer instructions]
[session metadata]        device/env info; injected per session, not persisted
[user memory]             explicit long-term facts (saved memories)
[recent conversations]    lightweight summaries of recent chats
[current session window]  sliding window of this conversation's messages
```

19. Structured-Data RAG: Text-to-SQL Architectures, Semantic Layers, Table QA and Schema Retrieval

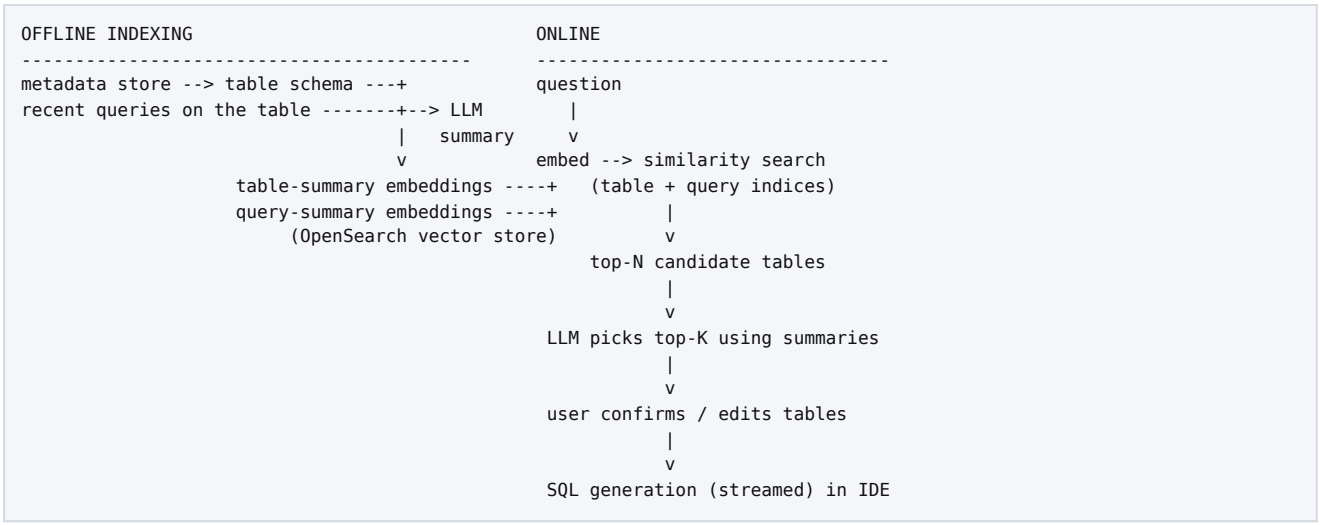
19.1 The canonical structured-data RAG pipeline



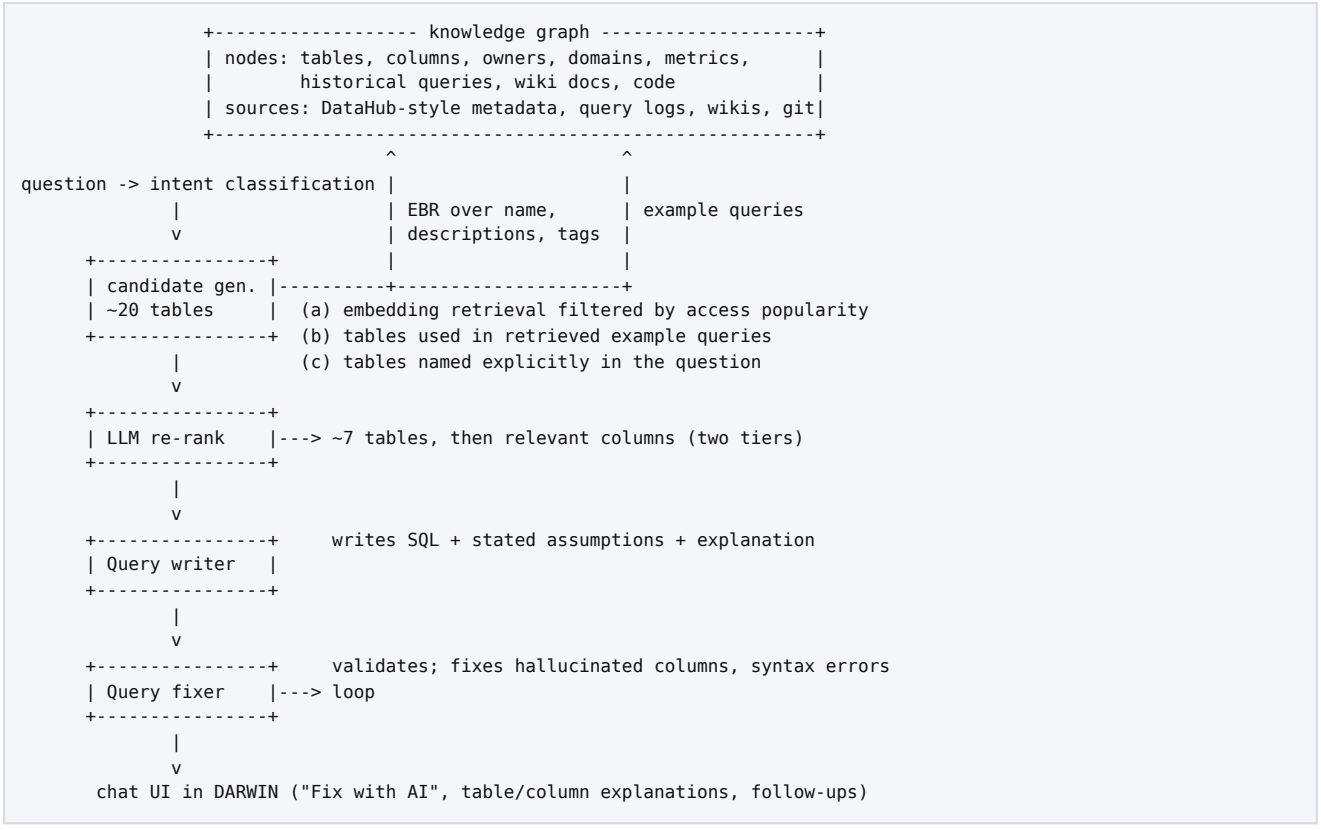
Architecture



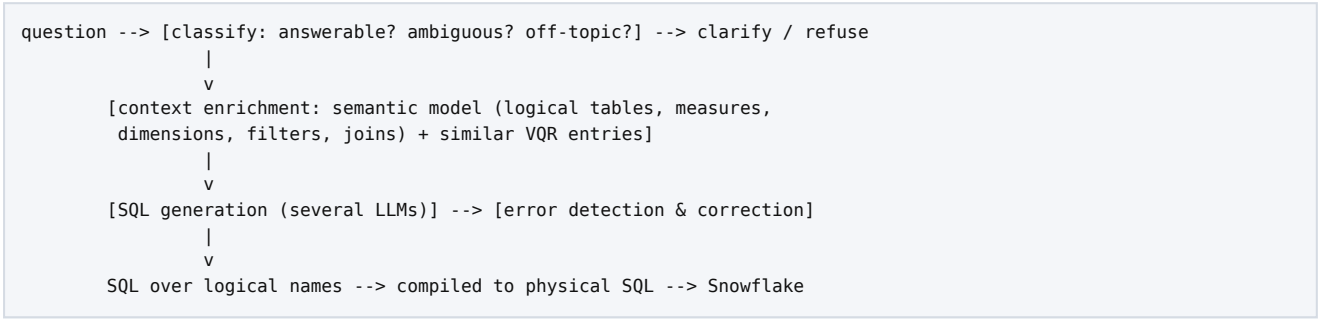
Architecture (2024 design)



Architecture



Agentic architecture



Architecture

```
                  +--------------- Genie space / agent -----------------+
question --------> | planner -> knowledge search (semantic indices over   |
                   |            UC assets, knowledge store, examples)      |
                   |        -> SQL generation (multiple LLMs / sub-agents) |
                   |        -> parallel reasoning paths + judge/aggregate  |
                   |        -> trusted-asset match? -> "trusted" label     |
                   +-------------------------------------------------------+
                                 |
                                 v
              Databricks SQL warehouse -> table/chart + explanation
              (Agent mode: decompose -> many SQL queries -> report)
```

CHESS (2024)

```
Information Retriever --> Schema Selector --> Candidate Generator --> Unit Tester
 (keywords, LSH over      (prunes hundreds     (writes + revises      (NL unit tests
  DB values, vector DB     of columns to         SQL using execution    pick best
  over catalog docs)       usually < 10)         feedback)              candidate)
```

19.11 Table QA without SQL: TableRAG and Chain-of-Table

```
TableRAG                                    Chain-of-Table
---------                                   --------------
question -> expand -> schema queries        table T0, question
                   -> cell queries            | plan op1 (e.g. f_select_row)
schema retrieval  (column embeddings)         v
cell retrieval    (distinct col=value pairs,  T1 -> plan op2 (f_group_by) -> T2
                   capped by encoding budget)   | ... until [E] (end)
        |                                       v
        v                                   final table -> LLM answer
compact prompt -> program-aided LM (e.g. Python/pandas) -> answer
```

20. Production RAG Operations: Ingestion, Freshness, Migrations, Latency, Cost, Observability, Caching and Debugging

1.1 The canonical ingestion pipeline

```
  sources (wiki, drive, tickets, code, DB)
        |  connectors: full crawl + webhooks / CDC / polling
        v
  +--------------+   +------------+   +-----------+   +------------------+
  | fetch + ACLs |-->| parse /    |-->| chunk +   |-->| change detection |
  | + metadata   |   | normalize  |   | enrich    |   | (hash per doc /  |
  +--------------+   +------------+   +-----------+   |  per chunk)      |
                                                      +--------+---------+
                                    unchanged: skip <---------+
                                                              | changed / new
                                                              v
                              +-------------------+    +----------------+
                              | embed (batch GPU  |--->| upsert vectors |
                              | or API, cached by |    | + lexical idx  |
                              | content hash)     |    | + metadata     |
                              +-------------------+    +-------+--------+
                                                               |
                        record manager / docstore  <-----------+
                        (source_id -> {chunk ids, hashes, ts, model ver})
                                                               |
                                     cleanup: delete stale ids  v
```

1.3 Case study: Notion's dual-path pipeline and Page State

```
 page edits --> Kafka --> online consumers --+
                                             |      +-------------------+
 historical pages --> Spark batch -----------+----> | embed + upsert to |
                                                    | vector DB (gen N) |
 routing table: workspace_id -> generation  ------> +-------------------+
```

1.4 Case study: Cursor's Merkle-tree sync and index reuse

```
client                                  server
+-----------------+   root hash      +----------------------+
| Merkle tree of  |----------------->| stored tree per      |
| file hashes     |<-- diff paths ---| namespace            |
+--------+--------+                  +----------+-----------+
         | changed files only                   |
         v                                      v
  chunk (AST) -> hash -> [embedding cache hit?] -> embed -> upsert
```

4.2 Blue-green and named vectors (Qdrant pattern)

```
           writes ---------------+-------------------+
                                 v                   v
                         +--------------+     +--------------+
  queries --> alias --> | coll_v1      |     | coll_v2      | <-- backfill
              "docs"    | (model A)    |     | (model B)    |     worker
                        +--------------+     +--------------+
  cutover: point alias "docs" -> coll_v2 ; keep v1 for rollback window
```

5.1 Where the time goes

```
client
  |-- auth/ACL lookup ............ 5-50 ms   (often underestimated)
  |-- query rewrite (LLM, opt.) .. 100-500 ms
  |-- query embedding ............ 5-50 ms
  |-- ANN + BM25 (parallel) ...... 5-100 ms  (cold object-store: 100s of ms)
  |-- rerank (cross-encoder) ..... 30-400 ms (depends on candidates x length)
  |-- prompt assembly + fetch .... 10-80 ms
  |-- LLM prefill  -> TTFT ....... dominant for long contexts
  '-- LLM decode ................. dominates total wall time
```

7.3 KV caching of retrieved knowledge

```
retrieved: [c7][c2][c9]  ---> KV store (GPU/CPU/SSD) lookup per chunk
                                  |hit       |hit       |miss
                                  v          v          v
                             [KV c7]    [KV c2]   prefill c9
                                   \        |        /
                               selective recompute of ~few % tokens
                                            v
                                       decode answer
```

8.2 Standards: OpenTelemetry GenAI and OpenInference

```
app (LangChain / LlamaIndex / custom)
  | OTel SDK + GenAI / OpenInference instrumentation
  v
OTel collector --> trace store (Phoenix / Langfuse / APM vendor)
                     |
                     +--> sampled traces --> offline evaluators (faithfulness,
                     |                       context precision, citation check)
                     +--> dashboards: p95 per span, cache hit %, $/query,
                                      empty-retrieval %, index lag
```

9.2 Triage flow

```
bad answer reported
  |
  v
1. Replay trace. Is the correct source doc in the corpus?  --no--> FP1 / ingestion
  |yes                                                            (connector, parser,
  v                                                              filter, ACL?)
2. Is the current version indexed? compare src hash vs index --stale--> freshness
  |yes                                                                  playbook
  v
3. Was it in retriever top-N (pre-rerank)? --no--> FP2: query rewrite, hybrid,
  |yes                                            chunking, model version mix?
  v
4. Survived rerank / context assembly?  --no--> FP3: reranker, top-k, token budget
  |yes
  v
5. In prompt but answer wrong?  --yes--> FP4-FP7: prompt, model, format, citations
  |
  v
6. Was a cache involved? semantic hit / stale prompt cache --> cache playbook
```

21. RAG System Design Interview Playbook: Enterprise Q&A, Support Bots, Code Assistants, and Web Answer Engines

1.3 The canonical component diagram

```
OFFLINE / ASYNC INGESTION PATH
 sources --> connectors/crawlers --> parse & clean --> chunk --> enrich
 (wiki, tickets,  (incremental,       (OCR, tables,    (struct-   (title, path,
  code, web)       webhooks, ACLs)     dedupe)          aware)     context blurb)
                                                                   |
          +------------------+-----------------+----------------+
          v                  v                 v
     lexical index       vector index       metadata / ACL store
     (BM25 / inverted)   (HNSW / IVF / BQ)  (doc->groups, freshness)

ONLINE ANSWER PATH
 user --> auth --> query understanding --> hybrid retrieve --> fuse (RRF)
          |        (rewrite, condense,     (BM25 + dense,     |
          |         route, decompose)       ACL pre-filter)   v
          |                                                 rerank (cross-enc.)
          |                                                      |
          +--> conversation memory          context assembly (top-k, order,
                                            dedupe, token budget)
                                                   |
                                                   v
                       LLM generation with citations --> guardrails --> user
                                                   |
                      logs, feedback, LLM-judge <-+--> eval sets, dashboards
```

2.4 Traffic and tokens

```
avg_QPS   = DAU x queries_per_user_per_day / 86,400
peak_QPS  ~ 3-10 x avg_QPS (state assumption)
input_tok = system_prompt + history + k_chunks x chunk_tokens + question
LLM_cost  = (input_tok x p_in + output_tok x p_out) x queries
```

2.5 Latency budget (interactive chat, illustrative)

```
auth + ACL lookup           10-30 ms
query rewrite (small LLM)   100-300 ms   (skip when query is self-contained)
hybrid retrieval            20-100 ms    (BM25 + ANN in parallel)
rerank top-50..100          50-200 ms    (cross-encoder, GPU)
LLM time-to-first-token     300-1000 ms  (depends on model and prompt size)
guardrail check             50-500 ms    (cheap similarity first, LLM judge if needed)
```

3.3 Architecture

```
+-----------+   webhooks / polling   +-------------------+
| SaaS apps |----------------------->| connector workers |--+
| (wiki,    |   (per-source rate     | fetch content +   |  |
|  drive,   |    limits, backoff)    | ACLs + activity   |  |
|  tickets) |                        +-------------------+  |
+-----------+                                               v
                     +-----------------------------------------------+
                     | normalize -> chunk -> embed -> index          |
                     | per-chunk fields: tenant, acl_principals[],   |
                     | source, updated_at, author, doc_type          |
                     +--------------------+--------------------------+
                                          |
        +---------------+------------------+--------------+
        v               v                                 v
  inverted index   vector index (tenant-sharded)   knowledge graph
  (BM25)           with ACL filter fields          (people, docs, teams,
                                                    activity edges)
ONLINE:
user -> SSO -> expand principals (user + groups, cached) -> query rewrite
     -> parallel BM25 + ANN with filter acl_principals IN user_principals
     -> RRF -> rerank (+authority/recency/graph features) -> top 5-10
     -> LLM answer with inline citations -> post-check: cited doc still visible?
```

4.4 Architecture

```
customer msg --> session store (history, account ctx, locale)
      |
      v
intent + condense (small LLM): "core issue" + entities (order id, product)
      |
      +--> policy router: sensitive intent? (fraud, legal, safety) --> human
      |
      v
retrieval: help-center (hybrid) + resolved-ticket KG / similar cases
      |                                  + tool calls (order status API)
      v
rerank --> grounded generation (answer + citations + proposed action)
      |
      v
guardrail tier 1: similarity(answer, retrieved evidence) >= tau ?
guardrail tier 2: LLM judge for policy / grounding (only if tier 1 uncertain)
      |pass                                |fail or timeout
      v                                    v
  send to user                       escalate to agent with summary
      |
      v
offline: LLM-judge sampling, CSAT, resolution, repeat-contact, reopen rate
```

5.4 Architecture

```
IDE plugin                                  server
+-----------------------------+             +-------------------------------+
| cursor ctx (prefix/suffix)  |  completion | low-latency code LLM (FIM)    |
| open-tab snippet matcher    |------------>| prompt cache                  |
| local symbol index (LSP)    |  <~ hundreds |                              |
| Merkle tree of workspace    |  of ms total +-------------------------------+
+-------------+---------------+
              | diff sync (changed hashes only)
              v
+-------------------------------------------------------------+
| chunk by syntax (functions/classes) -> embed (async, cached   |
| by content hash) -> per-repo/branch vector index + code search |
| + symbol/code graph                                          |
+-----------------------------+-------------------------------+
                              |  chat / agent queries
                              v
retrievers: keyword(+rewrite) | embeddings | graph (defs/refs) | open files
        -> rank (recsys-style) -> pack under token budget -> chat LLM
        -> answer/edit; agent may iterate (grep, read file, run tests)
```

6.3 Reference architecture: Perplexity on Vespa

```
 crawler fleet --> fetch queue (priority by freshness/authority)
      |                       ^
      v                       | recrawl signals (news, change rate)
 parse + boilerplate removal + spam/quality classifiers
      |
 split into sections/spans --> per-span: text, embedding, signals
      |
      v
 distributed index (sharded by doc id), lexical + vector + attributes
-------------------------------------------------------------------------
ONLINE (~hundreds of ms budget for retrieval):
query --> classify (navigational / fresh / complex) --> rewrite / decompose
      --> fan-out to all shards: hybrid retrieval of spans
      --> stage-1 cheap rank (per shard) --> merge top-N
      --> stage-2 ML ranker (many signals) --> stage-3 cross-encoder (top-K)
      --> LLM synthesize answer with numbered citations (streamed)
      --> complex queries: iterate (search -> read -> search) as agent
```

22. Master Index of RAG Engineering Blogs, Talks and Guides

22.1 How to use this index

```
                      WHO PUBLISHED IT
          +-------------+-------------+-------------+-------------+
          | Product co. | Cloud/LLM   | Infra vendor| Practitioner|
          | eng. blogs  | providers   | (vector DB, | essays,     |
          | (Uber,      | (Anthropic, | frameworks) | courses,    |
          | DoorDash..) | MSR, AWS..) |             | talks       |
  +-------+-------------+-------------+-------------+-------------+
W | Ingest| Cursor,     | Contextual  | Jina late   | Applied LLMs|
H | /chunk| GitHub      | Retrieval   | chunking,   |             |
A |       |             |             | Vespa ColPali             |
T +-------+-------------+-------------+-------------+-------------+
  | Retr- | Pinterest,  | GraphRAG,   | Weaviate    | Eugene Yan, |
I | ieve  | Ramp,Harvey | LazyGraphRAG| hybrid,     | Hamel       |
T |       |             |             | Pinecone rr |             |
  +-------+-------------+-------------+-------------+-------------+
C | Gen / | DoorDash    | Citations,  | RAGAS, HF   | J. Liu      |
O | guard | guardrails  | BenchmarkQED| cookbook    | course      |
V | / eval|             |             |             |             |
E +-------+-------------+-------------+-------------+-------------+
R | Agent | Uber EAg-RAG| Multi-agent | LangChain   | Chip Huyen  |
S | -ic   |             | research,   | decon. RAG  |             |
  |       |             | ctx. eng.   |             |             |
  +-------+-------------+-------------+-------------+-------------+
```

22.2.1 Uber: Genie and Enhanced Agentic RAG

```
Slack question
     |
     v
+-----------------+    +--------------------+    +---------------------+
| Query optimizer |--->| Source identifier  |--->| Hybrid retrieval    |
| (agent)         |    | (agent: which docs)|    | vector + BM25 top-k |
+-----------------+    +--------------------+    +----------+----------+
                                                            |
                    +---------------------------+           v
answer + cites <----| Generator (LLM)           |<---| Post-processor     |
                    +---------------------------+    | (context refine)   |
                                                     +--------------------+
Offline: PDF/GDoc loaders -> LLM formatting (tables, headings) -> chunks -> index
```

22.2.7 Code assistants: GitHub Copilot, Cursor, Sourcegraph

```
     Cursor-style incremental code index
local repo --hash--> Merkle tree --diff vs server--> changed files only
                                                          |
                                          syntactic (AST) chunking
                                                          |
                        content-hash cache hit? --yes--> reuse vector
                                  | no
                                  v
                          embed -> vector namespace (per repo / team)
```

22.2.9 Text-to-SQL and data agents: Uber QueryGPT, Swiggy Hermes

```
question --> domain router --> [workspace / charter]
                                  |  table summaries + example SQL (vector index)
                                  v
                        candidate tables (top-N) --> LLM re-select (top-K)
                                  |                    |
                                  +------> user confirms --+--> SQL gen --> run/validate
```

22.3.2 Microsoft Research: the GraphRAG family

```
GraphRAG (eager)                        LazyGraphRAG (deferred)
docs -> LLM entity/rel extraction       docs -> noun-phrase extraction (NLP)
     -> graph -> Leiden communities          -> concept graph -> communities
     -> LLM community summaries        query -> LLM expands to 3-5 subqueries
query -> map over summaries                   -> best-first chunk ranking
      -> reduce -> answer                     -> LLM relevance tests (budget)
                                              -> map claims -> reduce answer
cost paid at index time                 cost paid at query time, tunable
```

23. Canonical RAG and Retrieval Paper Reading List: 100+ Papers (2017-2026) by Theme, with Venue, Link and System-Design Takeaway

Overview

```
                      OFFLINE (indexing)
docs --> parse --> chunk/propositions --> embed ------> ANN index (HNSW/DiskANN/IVF)
           |        (Dense X, late      (DPR, E5,     (HNSW, FAISS, ScaNN, SPANN)
           |         chunking, RAPTOR,   ColBERT,
           |         contextual retr.)   SPLADE)  \--> sparse index (BM25/SPLADE)
           \--> entity/relation extraction --> graph + communities (GraphRAG, HippoRAG)

                      ONLINE (query time)
query --> rewrite/expand --> retrieve (hybrid) --> rerank --> compress --> generate
          (HyDE, Query2doc,   (RRF fusion)     (monoT5,   (RECOMP,     (FiD, RAG,
           step-back)                           RankGPT)   LLMLingua)   Self-RAG)
   ^                                                                        |
   |____ iterate / route / abstain (IRCoT, FLARE, CRAG, Search-R1, CoRAG) __|

serving: KV reuse (RAGCache, CacheBlend), drafting (Speculative RAG)
evaluation: BEIR, MTEB, KILT, RAGAS, ARES, RGB, BRIGHT, FRAMES, TREC RAG
```

2. Retrievers: dense, sparse, late-interaction and general embedders

```
single-vector bi-encoder     late interaction (ColBERT)      learned sparse (SPLADE)
q -> [v]  d -> [v]           q -> [v1..vn] d -> [u1..um]     q,d -> |V|-dim sparse
score = <v_q, v_d>           score = sum_i max_j <vi,uj>     score = <w_q, w_d>
1 vector / passage           1 vector / token (compressed)   inverted index, BM25-like
cheapest, weakest OOD        best quality/storage trade-off  exact-match + expansion
```

6. Indexing granularity, chunking and hierarchy

```
granularity spectrum
proposition --- sentence --- chunk (100-800 tok) --- section --- doc --- summary tree
(Dense X)                   (+context prefix:                   (LongRAG) (RAPTOR)
                             Anthropic contextual retrieval;
                             +late chunking: embed whole doc,
                             pool per chunk)
```

7. Iterative, adaptive and corrective retrieval

```
ReAct / IRCoT style loop                  Self-RAG / CRAG style gates
+---------+  thought   +----------+       retrieve? --no--> generate
|  LLM    |----------->| search   |          | yes
| reason  |<-----------| tool     |       retrieve -> grade docs
+---------+  evidence  +----------+          | correct    | ambiguous | incorrect
    | stop when answer is supported       refine     refine+web    web search
    v                                        \_____________|____________/
 answer (+ citations)                              generate -> critique (IsSup)
```

8. RL-trained search agents and deep research (2025-2026)

```
rollout:  <think> ... </think> <search> q1 </search> <information> d1..dk </information>
          <think> ... </think> <search> q2 </search> <information> ...  </information>
          <answer> a </answer>
reward:   rule-based outcome match(a, gold)       (no process reward)
trick:    mask retrieved tokens from the policy-gradient loss (not model-generated)
infra:    retriever served as an HTTP tool (e5 + Wikipedia in Search-R1's quick start)
```

9. Graph and structured RAG

```
GraphRAG (Microsoft)                          HippoRAG
chunks -> LLM entity/relation extraction      chunks -> OpenIE triples -> KG
      -> KG -> hierarchical Leiden communities  query -> entities -> seed nodes
      -> LLM community summaries (C0..C3)       Personalized PageRank over KG
global query: map over summaries -> reduce      -> passage scores -> top-k passages
cost: LLM pass over the whole corpus at index   cost: OpenIE pass, cheap queries
```

11. Compression, caching and serving efficiency

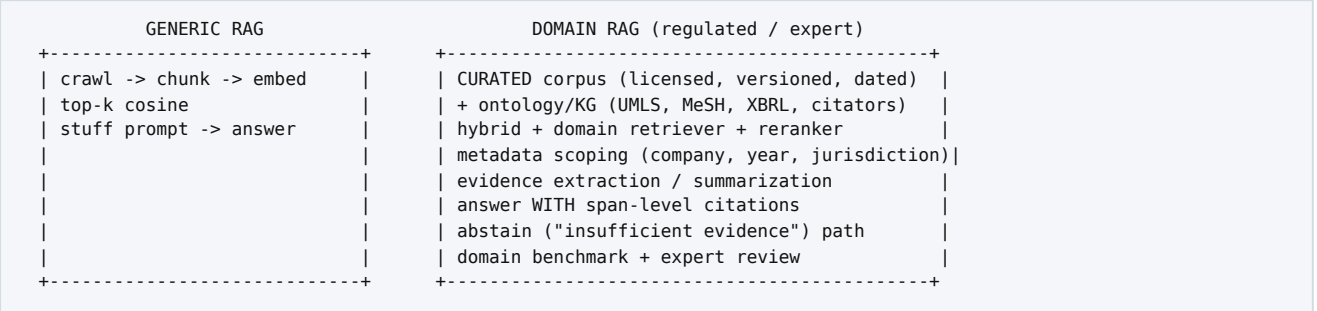
```
standard prefix cache:  [sys][docA][docB][q]  hit only if identical prefix
RAGCache:               knowledge tree of doc KV states across GPU/host memory,
                        prefix-aware eviction, overlap retrieval with prefill
CacheBlend:             reuse non-prefix chunk KV caches, recompute a small subset
                        of tokens to restore cross-attention; overlap with KV load
```

A suggested reading order (two weeks)

```
week 1  (core)                          week 2  (frontier)
d1  DrQA [1], DPR [4], RAG [5]          d6  Self-RAG [53], CRAG [54], Adaptive-RAG [55]
d2  FiD [6], RETRO [8], In-Ctx RALM[10] d7  Search-o1 [58], Search-R1 [59], CoRAG [57]
d3  ColBERTv2 [14], SPLADE [17],        d8  GraphRAG [67], LazyGraphRAG [70],
    HNSW [27], DiskANN [31]                 HippoRAG 2 [73]
d4  monoT5 [34], RankGPT [35],          d9  RAGCache [88], CacheBlend [89],
    HyDE [37], contextual retr. [46]        Speculative RAG [56], ColPali [92]
d5  Lost in Middle [77], Power of       d10 BEIR [96], BRIGHT [98], RAGAS [100],
    Noise [78], Sufficient Ctx [83]         ARES [101], Seven Failure Points [107]
```

24. Domain-Specific RAG for Regulated and Expert Fields: Medicine, Law, Finance and Science

Overview



1.1 MedRAG and the MIRAGE benchmark

```
                question (options optionally hidden)
                              |
          +---------+---------+----------+----------+
          |         |         |          |          |
        BM25   Contriever  SPECTER    MedCPT     (any subset)
          |         |         |          |
          +----+----+---------+----------+
               | rank fusion across retrievers
               v
 MedCorp snippets: PubMed | StatPearls | Textbooks | Wikipedia
               |
               v  top-k snippets (k swept, e.g. up to 32)
       CoT prompt -> LLM (GPT-4, GPT-3.5, Mixtral, Llama 2, Meditron, PMC-LLaMA)
               |
               v
       option letter + rationale
```

1.2 i-MedRAG: iterative follow-up queries

```
Q --> LLM: generate n_queries follow-ups --> retrieve+answer each --> append Q/A log
 ^                                                                      |
 +----------------------------- repeat n_rounds ----------------------+
                                      |
                                      v
                         final answer conditioned on Q/A log
```

1.4 Clinical-guideline RAG: a recurring pattern

```
 guideline PDFs (versioned, dated, issuing body)
       |
       v
 structure-aware parsing: recommendation units, strength-of-evidence grade,
 population/intervention tags, tables & flowcharts -> text
       |
       v
 index: hybrid (BM25 + dense) + metadata {body, version, date, population}
       |
 query --> concept normalization (UMLS CUIs / MeSH) --> filtered retrieval
       |
       v
 answer that quotes recommendation text + grade + guideline/version citation
       |
       v
 abstain / escalate if no recommendation covers the population
```

1.5 Ontology- and graph-aware medical retrieval: UMLS, MeSH, SPOKE

```
 Tier 1 (top)    : private user data (e.g., MIMIC-IV notes) -> LLM-extracted graph
                         |  "trinity" links
 Tier 2 (middle) : medical books + papers (MedC-K, PubMed-linked S2ORC) -> graph
                         |
 Tier 3 (bottom) : medical dictionary (UMLS) -- already graph-structured
 Retrieval walks top -> down so every patient-level claim can be grounded
 in literature and finally in a controlled-vocabulary definition.
```

2.2 LegalBench and LegalBench-RAG

```
query: "Does the NDA allow the receiving party to share with consultants?"
    |
    v
retriever under test --> returned spans [(file, start, end), ...]
    |
    v
overlap with gold spans at CHARACTER granularity
    -> precision@k = chars(returned ∩ gold) / chars(returned)
    -> recall@k    = chars(returned ∩ gold) / chars(gold)
```

2.3 Commercial legal assistants: Lexis+ AI, Westlaw / CoCounsel

```
user question (+ jurisdiction, practice area)
      |
      v
query understanding -> legal-issue decomposition
      |
      v
retrieval over curated corpus: cases, statutes, regulations,
secondary sources (treatises, practice notes) + editorial metadata
(headnotes, key numbers / topic taxonomies, citator status)
      |
      v
citator filter: drop / flag overruled or negatively treated authority
      |
      v
LLM drafts memo-style answer with linked citations
      |
      v
lawyer verifies (UI links each citation to the source document)
```

3.1 FinanceBench

```
 Configuration                          Retrieval scope        Outcome (qualitative)
 -----------------------------------    --------------------   -------------------------
 closed book                            none                   mostly wrong / refuse
 shared vector store (all filings)      corpus-wide top-k      81% wrong/refuse (GPT-4T)
 single-document vector store           the right filing       better
 long context (whole filing in prompt)  entire document        better; slow, limited
 oracle (gold evidence page)            perfect                upper bound
```

3.2 SEC-filing pipeline design

```
EDGAR (full-text search / submissions / XBRL APIs)
      |
      +--> XBRL facts (us-gaap tags, units, periods) --> SQL/columnar store
      |                                                       ^
      +--> filing HTML/PDF --> section parser (Item 1A Risk, Item 7 MD&A, Notes)
                  |                table extractor (keep row/col headers + units)
                  v
            chunks with metadata {CIK, ticker, form, fiscal_year, period, section}
                  |
query --> entity+period resolver (ticker, "FY23", "last quarter") --> HARD filter
      --> router: numeric? -> XBRL/SQL tool  |  narrative? -> hybrid retrieval
      --> calculator / code tool for ratios, growth, sums
      --> answer with citation (form, period, section, page) + computed trace
```

4.1 PaperQA and PaperQA2 (FutureHouse)

```
                     +------------------ agent LLM (ToolSelector) ------------------+
question ----------->  |  chooses tools in any order, within budget/timeouts           |
                     +---------------------------------------------------------------+
        |                          |                          |                  |
        v                          v                          v                  v
 [paper_search]           [gather_evidence]          [citation traversal]  [gen_answer]
 LLM writes keyword       embed query, top-k         follow refs/citers    best summaries
 queries -> metadata      chunks (evidence_k=10)     of good papers        (<=5 sources)
 providers (Semantic      -> per-chunk, query-       (internal versions; -> cited answer
 Scholar, Crossref,       specific SUMMARY + score    not in OSS repo)   or "insufficient
 Unpaywall, OpenAlex)     -> LLM re-score/select                           information"
 -> fetch PDFs, chunk,       = "RCS": rerank +
    embed, add to state      contextual summarization
```

Architecture.

```
query
  |
  v
bi-encoder retriever (Contriever continued-pretrained on peS2o)
  |  over OpenScholar Datastore: ~45M open-access papers, ~236M passage embeddings
  |  (+ optional Semantic Scholar API and web search)
  v
cross-encoder reranker (trained from BGE-reranker-large)
  |
  v
generator: OpenScholar-8B (Llama 3.1 8B, ~13k instruction examples)
  |
  v
SELF-FEEDBACK LOOP: model critiques its draft ("missing evidence on X",
  "citation unsupported") -> issues new retrievals -> revises
  |
  v
citation verification pass -> final answer with per-sentence citations
```

4.3 Ai2 ScholarQA

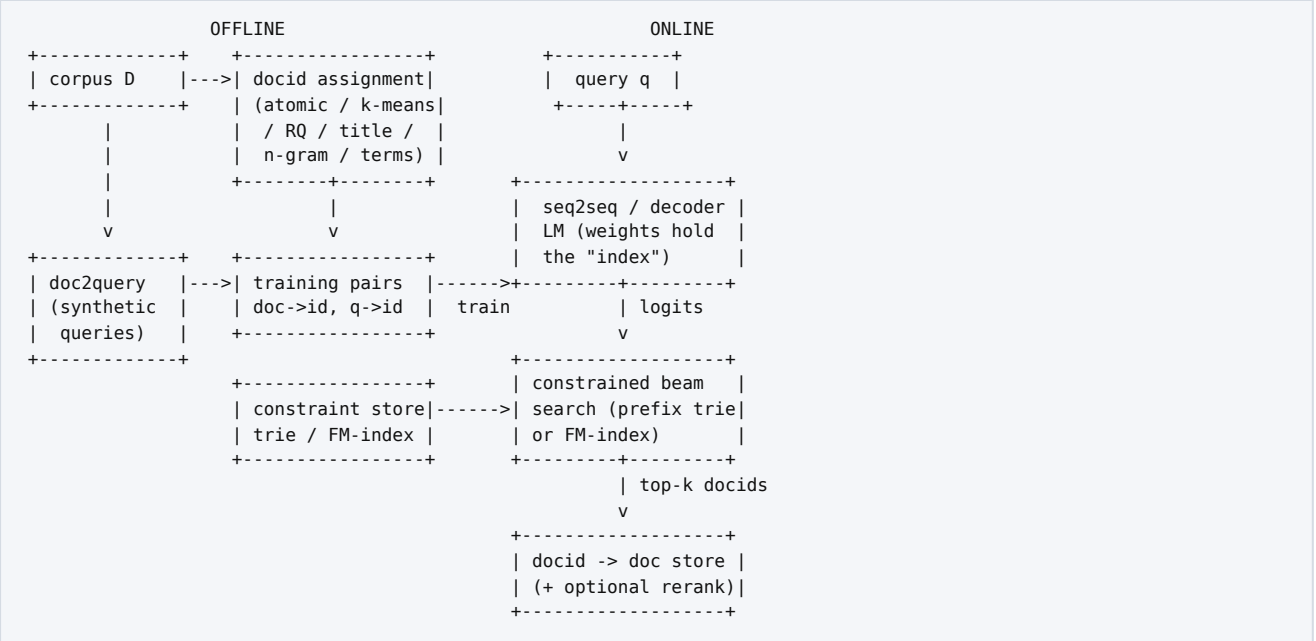
```
1. Query decomposition (LLM): rewrite query; extract filters (year, venue,
   field of study); produce keyword-search variant
2. Retrieval: Semantic Scholar snippet search over ~11M+ full texts
   and ~100M+ abstracts; keyword search adds abstracts
3. Rerank: cross-encoder (mxbai-rerank-large-v1); aggregate passages to PAPER level
4. Quote extraction (LLM): pull exact supporting quotes per paper
5. Outline + clustering (LLM): plan sections, choose format (paragraph vs list),
   assign quotes to sections
6. Section generation (LLM): write each section from its quotes + prior sections,
   inline citations to quotes
7. Literature-review tables for list-type sections (ArxivDIGESTables method)
```

5.2 Ontology-aware retrieval

```
raw query --> NER --> normalize to IDs ------------------------------+
                      medicine: UMLS CUI / MeSH descriptor / RxNorm   |
                      law:      citation parser, key-number/topic     |
                      finance:  ticker->CIK, XBRL us-gaap concept     |
                      science:  S2 paper IDs, field-of-study, DOIs    |
                                                                      v
     expansion (synonyms, children)  +  hard filters  +  graph neighbors
                                     |
                                     v
                  hybrid retrieval (BM25 on expanded terms + dense)
```

25. Generative Retrieval and Model-Based Indexing: DSI, NCI, SEAL, TIGER, CorpusBrain and Their Hybrids

1.2 Canonical architecture



3.1 Architecture

```
doc --BERT--> embedding --hierarchical k-means--> semantic docid  (e.g. 3-7-1-4)
doc --docT5query--> synthetic queries --+
real queries --------------------------+--> T5 encoder
                                             |
                                  PAWA decoder: prefix-aware,
                                  weight-adaptive output layer
                                  (same digit has different
                                  embedding at each position/prefix)
                                             |
                                 consistency regularization
                                 (two dropout passes must agree)
                                             |
                                 constrained beam search -> docids
```

4.2 Architecture

```
query --> BART-large --> constrained decoding over FM-index
                              |
            FM-index (compressed suffix array over all passages):
            - next-token lookup: which tokens can follow prefix p?
            - occurrence count: how many times does p appear?
            - locate: which documents contain p?
                              |
              generated n-grams {g1, g2, ...} with LM scores
                              |
         document score = aggregate over n-grams it contains
         (LM prob x corpus-frequency-based weight; intersective
          scoring combines multiple non-overlapping n-grams)
                              |
                       ranked passages
```

5.2 Architecture

```
  item content (title, brand, category, description)
            |
     sentence encoder (e.g. Sentence-T5)  -> dense item vector x
            |
     RQ-VAE: level 1 codebook -> c1, residual r1 = x - e(c1)
             level 2 codebook -> c2, residual r2 = r1 - e(c2)
             level 3 codebook -> c3
             (+ extra token to disambiguate collisions)
            |
     semantic ID = (c1, c2, c3, c4)   e.g. (12, 201, 7, 0)
            |
  user history: [SID(item_1)] [SID(item_2)] ... [SID(item_n)]
            |
     Transformer encoder-decoder  --> autoregressively decode SID(item_n+1)
            |
     constrained to valid SIDs (prefix trie)  -> next items
```

6.2 Architecture and pre-training

```
  Wikipedia page P  --sample sentence / lead para / anchor ctx-->  pseudo-query
                                                                        |
                                    BART-large  <---- pretrain: pseudo-query -> title(P)
                                        |
                    fine-tune (few or many labels) on KILT tasks
                                        |
           query --> constrained beam search over KILT title trie --> pages
```

7.3 Closing the gap: RIPOR and PAG (UMass, 2024)

```
 phase 1: train T5 as a dense encoder (relevance-based)
 phase 2: residual-quantize its doc vectors -> RQ docids (relevance-aware prefixes)
 phase 3: seq2seq training with prefix-oriented ranking loss:
          at every prefix length, rank relevant docid prefixes above negatives
          mined from constrained beam search, + dense-retriever training data
```

9.1 GR as a coarse router over a vector index

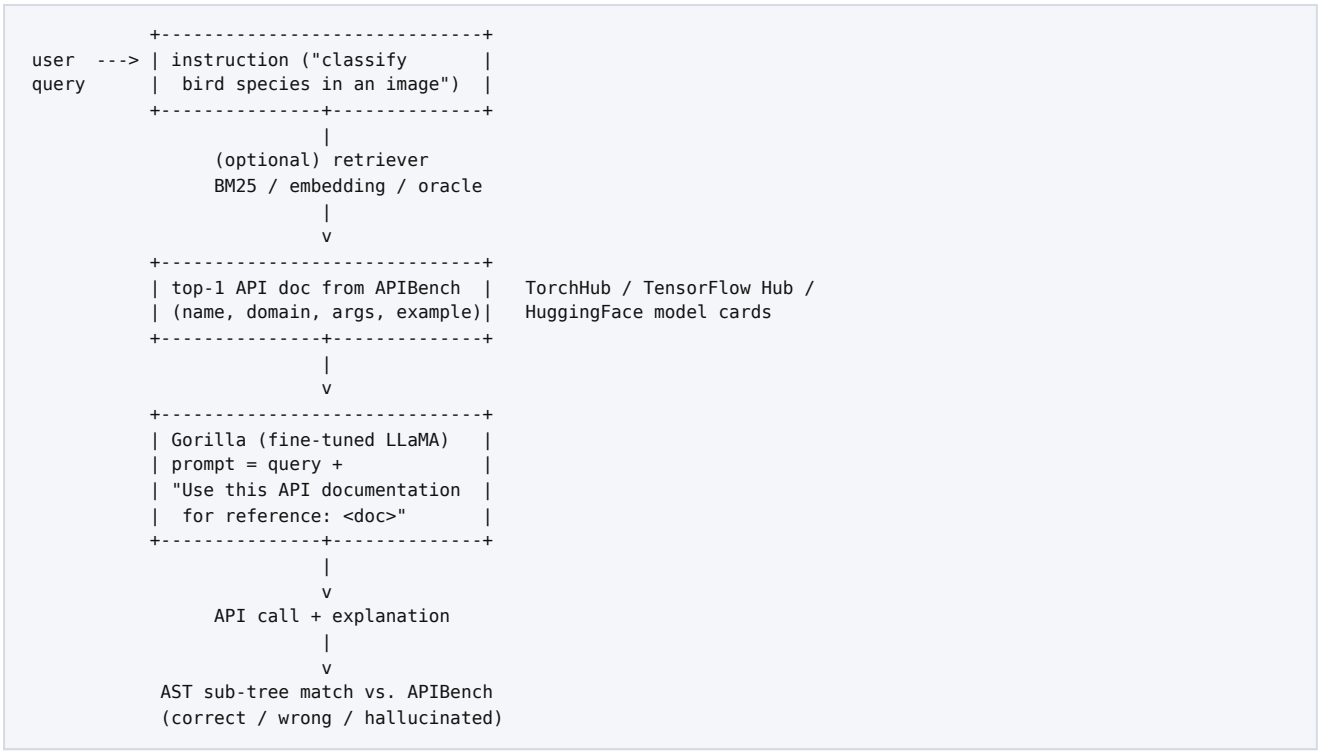
```
query --> small seq2seq --> beam over coarse codes (c1,c2) --> top clusters
                                                                  |
query --> dense encoder ----------------------------------------> ANN within clusters
                                                                  |
                                                        merged candidates --> reranker
```

9.3 GR plus LLM rerankers (2024-2026)

```
                +--> dense ANN (bi-encoder) -----------+
query -----------+                                     +--> fuse (RRF) --> LLM reranker
                +--> generative retriever (beam k) ----+          |
                |    (titles / RQ codes / n-grams)                v
                +--> BM25 / SPLADE --------------------+     top-n ctx --> LLM answer
```

26. Tool Retrieval: RAG over APIs, Function Catalogs and MCP Servers

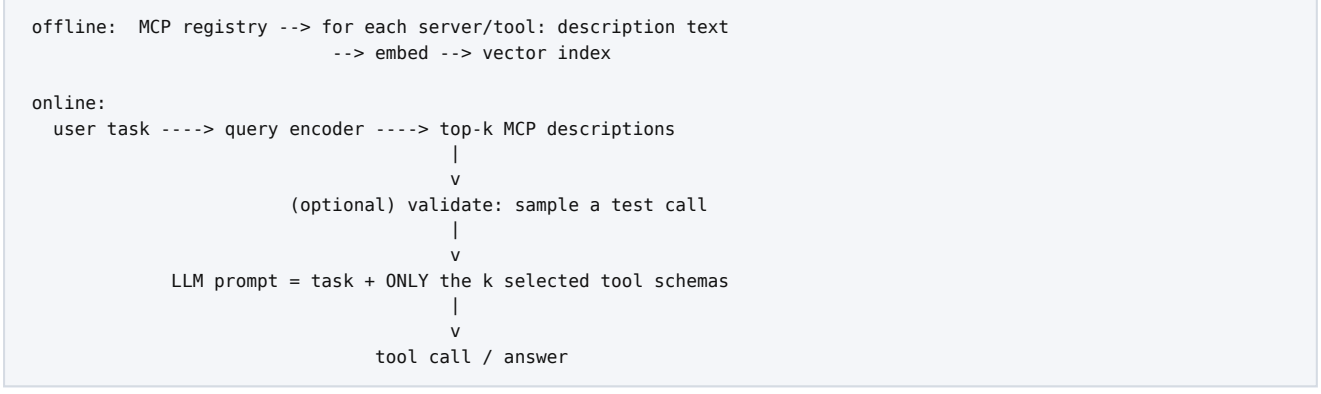
Architecture



Architecture



Architecture



5.1 Tool search tool and `defer_loading`

```
request.tools = [ tool_search_tool_{regex|bm25}_20251119,      <- always loaded
                  hot_tool_1 .. hot_tool_k,                     <- defer_loading: false
                  tool_1 .. tool_N (N <= 10,000)                <- defer_loading: true ]

+-------------------- Anthropic API (server side) ---------------------+
|  system-prompt prefix = search tool + hot tools   (cacheable)        |
|                                                                      |
|  Claude --server_tool_use(pattern|query, limit)--> search over       |
|          names, descriptions, arg names, arg descriptions            |
|       <--tool_search_tool_result: [tool_reference x <=5 default]--   |
|                                                                      |
|  API expands each tool_reference inline into the full definition     |
|  (prefix untouched -> prompt cache preserved)                        |
|                                                                      |
|  Claude --tool_use(discovered_tool)--> client executes, returns      |
+----------------------------------------------------------------------+
```

Retriever choice.

```
            +------------------- catalog build (offline) -------------------+
MCP servers->| normalize schema -> tool doc text -> +synthetic queries        |
OpenAPI    ->| server summary  -> embed(server), embed(tool), BM25 index     |
functions  ->| dedupe / merge near-duplicate tools                           |
             +------------------------------+-------------------------------+
                                            |
             +-------------- runtime (per turn) ----------------------------+
             | context: system + hot tools + search tool  (cached prefix)    |
             | model -> search(query) -> hybrid top-50 -> rerank -> top-5    |
             |       <- tool_reference x5 (expanded inline, not in prefix)   |
             | model -> tool_use -> result -> may search again               |
             | log: query, returned ids, chosen id, success -> eval set      |
             +----------------------------------------------------------------+
```

27. Multilingual and Cross-Lingual RAG: Benchmarks, Embeddings, Language Bias, Tokenization and Sovereign Deployments

Overview

```
           THE MULTILINGUAL RAG DESIGN SPACE
+-----------+   +-------------------+   +-----------------+   +-------------+
| Query     |   | Retrieval         |   | Context         |   | Generation  |
| (lang Lq) |-->| - translate q?    |-->| - which langs?  |-->| - prompt    |
| script,   |   | - shared space?   |   | - translate     |   |   language  |
| code-mix  |   | - per-lang index? |   |   passages?     |   | - answer in |
+-----------+   | - hybrid lexical  |   | - token budget  |   |   Lq?       |
                +-------------------+   +-----------------+   | - citations |
                        ^                                     +-------------+
                        |
         +--------------------------------+
         | Corpus (langs L1..Ln)          |
         | script-aware chunking,         |
         | per-language analyzers,        |
         | optional doc translation       |
         +--------------------------------+
```

Architectural choices: translate versus shared multilingual space

```
(A) TRANSLATE-THEN-RETRIEVE (query translation, "English pivot")
    q(Lq) --MT--> q(en) --> [English index] --> passages(en) --> LLM --> answer(Lq)

(B) DOCUMENT TRANSLATION (translate corpus offline)
    docs(L1..Ln) --MT offline--> docs(en) --> [English index]
    q(Lq) --MT--> q(en) --------------^      (keep originals for citation)

(C) SHARED MULTILINGUAL SPACE
    q(Lq) --enc--> v_q --+
                         +--> [one index over docs(L1..Ln) encoded by same model]
    docs(Li) --enc--> v_d+       --> passages(mixed langs) --> LLM --> answer(Lq)

(D) DISTILLED NEURAL CLIR (ColBERT-X / Translate-Distill)
    teacher: MT + English cross-encoder scores (q_en, d_en)
    student: multilingual late-interaction encoder scores (q_en, d_Li) directly
```

(C) Shared multilingual embedding space

```
        BGE-M3 ONE-PASS, THREE-SIGNAL RETRIEVAL
 text --> XLM-R-style encoder (8192 ctx) --> hidden states H
            |                 |                       |
      [CLS] -> dense      token weights -> sparse   all tokens -> multi-vec
      1024-d ANN index    inverted index           late interaction (rerank)
            \_________________|_____________________/
                   weighted score fusion --> cross-encoder rerank
```

Debiasing the "English is best" result (Park et al., 2026)

```
        DELTA-STYLE QUERY FUSION (monolingual alignment)
 q(ko) --+-- as is ----------------> search(ko docs)  --+
         +-- MT -> q(en) ------------> search(en docs)  --+--> fuse/rerank --> LLM
         +-- MT -> q(ja), q(zh) ... -> search(ja,zh docs)-+
         (each language partition is queried in its own language)
```

Consequences specific to RAG

```
 Same 5 passages, same information, different language
 English    [#####                ]  ~1.0x tokens
 Mid-res    [##########           ]  ~2x   (approximate, varies by tokenizer)
 Low-res    [####################+]  up to ~4-15x  [24]
 -> fewer passages fit in the budget, higher prefill cost, slower TTFT
```

Script-aware chunking pipeline

```
raw doc --> Unicode NFC normalize --> language/script ID (per paragraph)
        --> per-script sentence segmenter
              CJK: punctuation rules     Thai: dictionary/model segmenter [27]
              Arabic/Persian: punctuation + ZWNJ-safe normalization
              Indic: danda + grapheme-cluster-safe boundaries
        --> pack sentences into chunks sized in TOKENS of the embedder,
            with per-language targets derived from the measured premium [24]
        --> attach title/heading prefix (BERGEN prefixes article title) [18]
        --> store: original text, lang tag, script tag, optional translation
```

Reference architecture: a sovereign multilingual RAG stack

```
                  REGION / JURISDICTION BOUNDARY (data residency)
 +-------------------------------------------------------------------------+
 |  Ingest                                                                 |
 |   docs --> NFC + lang/script ID --> script-aware chunker --> chunks     |
 |              |                                                (lang tags) |
 |              +--> optional MT to pivot (IndicTrans2 / NLLB) [18,28]      |
 |                                                                         |
 |  Index                                                                  |
 |   multilingual embedder (BGE-M3 / mE5 / regional embedder) [11,13,29]   |
 |   + per-language lexical or learned-sparse index                        |
 |   partitions by language for monolingual-alignment search [21]          |
 |                                                                         |
 |  Query                                                                  |
 |   q --> lang ID --> query variants per corpus language (MT, short) [21] |
 |     --> hybrid search per partition --> cross-lingual reranker          |
 |     --> token-budgeted packer (per-language premium) [24]               |
 |                                                                         |
 |  Generate                                                               |
 |   regional/open LLM (SEA-LION / Apertus / ...) [29,30]                  |
 |   prompt written in target language; answer + entities in Lq [18]      |
 |   abstain policy tested on NoMIRACL-style data [23]                     |
 |                                                                         |
 |  Evaluate                                                               |
 |   per-language: retrieval nDCG, char-3gram recall, correct-language     |
 |   rate, citation-language mix, hallucination/error rate [18,20,22,23]   |
 +-------------------------------------------------------------------------+
```

28. On-Device, Edge and Air-Gapped RAG: Small Models, Embedded Indexes and Private Compute

Overview

```
             ON-DEVICE / EDGE RAG REFERENCE STACK
 +-------------------------------------------------------------------+
 |  App / OS assistant (Siri, Recall, in-app chat, field tool)       |
 +-------------------------------------------------------------------+
 |  Orchestrator: query rewrite, tool routing, escalation policy     |
 +-----------------+-------------------------+-----------------------+
 |  SLM generator  |  Retriever              |  Escalation           |
 |  1-4B, 2-4 bit  |  lexical + vector (+KG) |  verifiable cloud     |
 |  ~4K-8K ctx     |  rerank optional        |  (PCC-like) or none   |
 +-----------------+-------------------------+-----------------------+
 |  Embedded index: SQLite/vec0, Lance files, ObjectBox HNSW,        |
 |  OS index (Spotlight / AppContentIndexer); encrypted at rest      |
 +-------------------------------------------------------------------+
 |  Background indexer: OCR, chunking, small embedder (NPU/CPU),     |
 |  scheduled when charging / idle; incremental updates              |
 +-------------------------------------------------------------------+
 |  Hardware: shared DRAM (8-16 GB), NPU/GPU, flash, battery, TPM/SE |
 +-------------------------------------------------------------------+
```

4.1 sqlite-vec

```
create virtual table vec_docs using vec0(
  doc_id integer primary key,
  user_id integer partition key,      -- shard per user / tenant
  embedding float[768],
  embedding_coarse bit[768]           -- 96 bytes per row
);
select doc_id, distance from vec_docs
 where embedding match :q and user_id = :u
 order by distance limit 10;
```

5.1 EdgeRAG (Seemakhupt, Liu, Khan, 2024)

```
          EdgeRAG retrieval path (after [1])
query --> embed --> compare to centroids (resident) --> top-n clusters
                                                          |
            +---------------------------------------------+
            v
  cluster large?  --yes-->  load precomputed embeddings from storage
      | no
      v
  in cache?       --yes-->  use cached embeddings
      | no
      v
  re-embed cluster chunks on device --> insert into cost-aware cache
            |
            v
  exact search within clusters --> top-k chunks --> SLM generator
```

5.2 MiniRAG (HKU Data Intelligence Lab, 2025)

```
     MiniRAG query flow (after [2][3])
query -> SLM: {entities<=5, answer_type_keywords}
           |                          |
           v                          v
  entity-name vector index      type pool filter
           |                          |
           +---> graph: k-hop (k=2) neighbors, path scoring
                        |
                        v
          chunk nodes linked to top paths + direct chunk hits
                        |
                        v
         compact context (~500 tok node ctx) -> SLM answer
```

6.2 Architecture

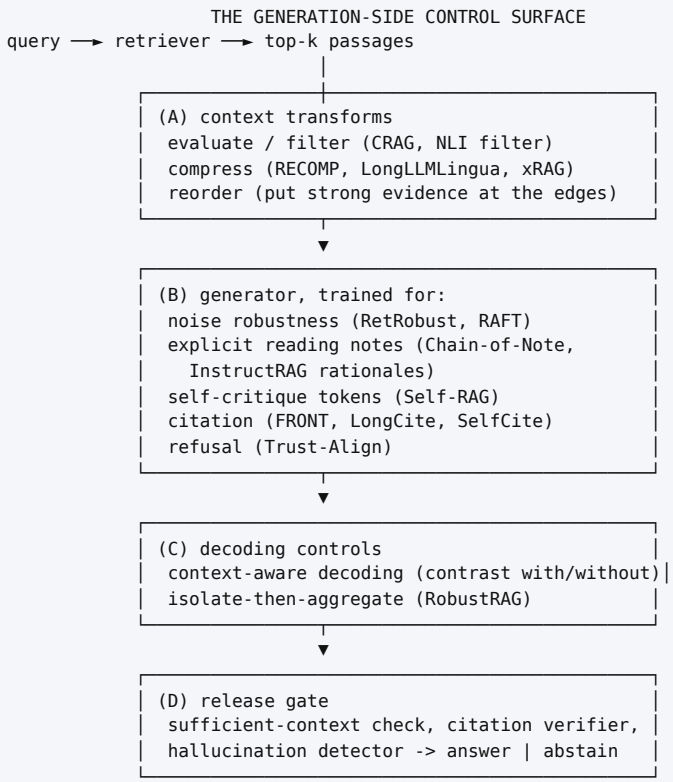
```
                       APPLE INTELLIGENCE (simplified)
 +------------------------------ device -------------------------------+
 | apps --donate--> Spotlight / Core Spotlight index (lexical+semantic) |
 | screen --> onscreen awareness        App Intents / "App Toolbox"     |
 |                 \                     /                              |
 |                  v                   v                               |
 |               system orchestrator (Siri)                              |
 |                 |               |                                     |
 |       on-device AFM (3B /   escalate? ---- yes ----+                  |
 |       20B sparse, adapters)                        |                  |
 +----------------------------------------------------|------------------+
                                                      | encrypted to keys
                                                      | of attested nodes,
                                                      v via 3rd-party relay
                        +------------- Private Cloud Compute -------------+
                        | stateless inference nodes (Apple silicon, and   |
                        | from 2026 NVIDIA GPUs in Google Cloud);         |
                        | no persistent storage of request data;          |
                        | published binaries + transparency log           |
                        +-------------------------------------------------+
```

7.2 Architecture

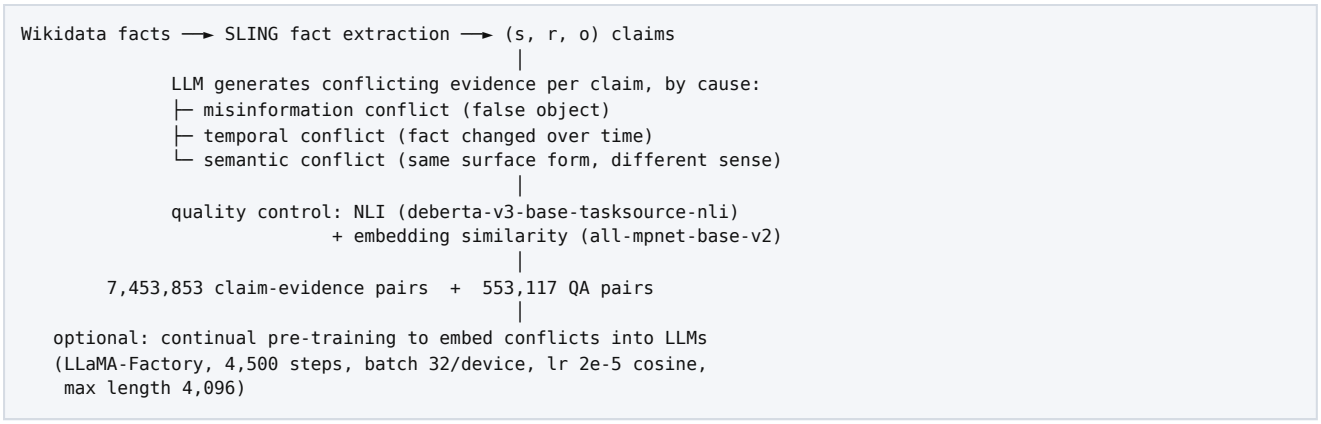
```
            RECALL PIPELINE (simplified, after [14][15][16])
screen change --> snapshot capture --> filters (apps, sites, private
                                          browsing, sensitive info, DRM)
                        |
                        v
          local OCR + semantic/visual embedding (NPU)
                        |
                        v
    encrypted snapshot store + vector index (local disk only)
                        |  unlocked just-in-time by Windows Hello ESS
                        v
user query (text/voice) --> semantic + lexical search --> timeline hits
                        --> relaunch via UserActivity deep link
```

29. The Generation Side of RAG: Knowledge Conflicts, Robustness, Compression, Attribution, and Abstention

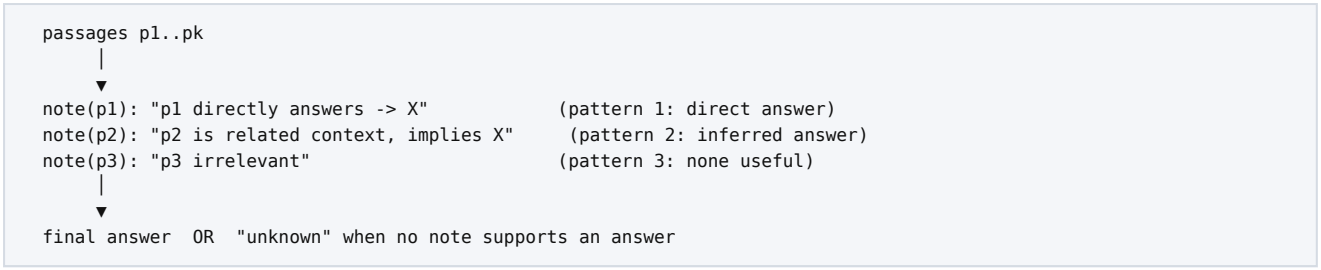
Overview



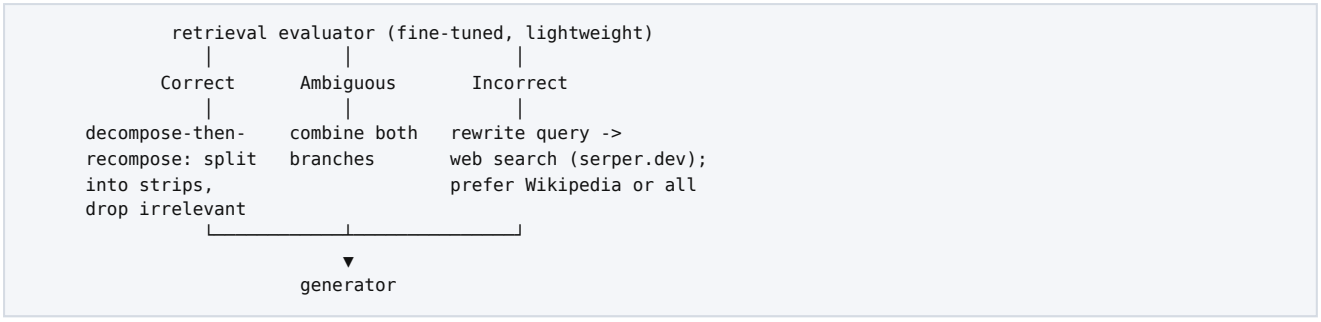
29.1.4 ConflictBank (NeurIPS 2024 Datasets and Benchmarks)



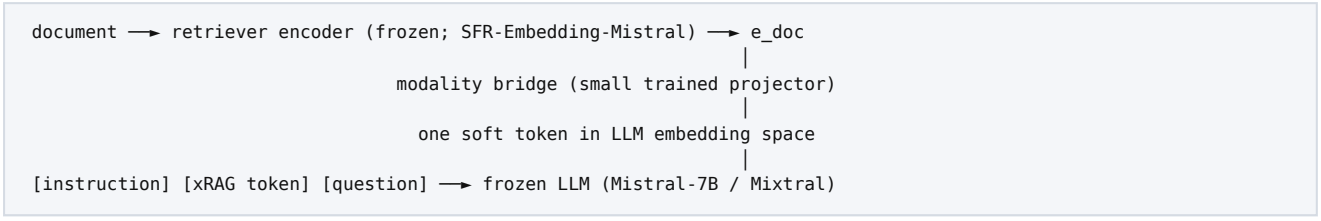
29.2.3 Chain-of-Note (Tencent AI Lab, 2023)



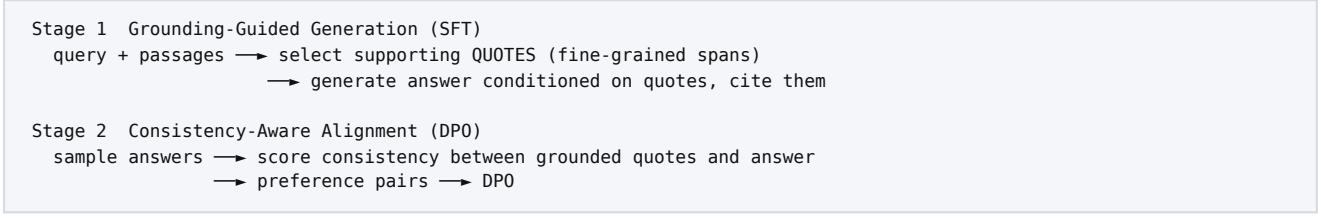
29.2.6 CRAG: Corrective RAG (2024)



29.3.4 xRAG (NeurIPS 2024): one-token context



29.4.2 FRONT: fine-grained grounded citations (HIT, ACL 2024 Findings)



29.6.2 Sufficient context (Google, ICLR 2025)

```
query + context ──► sufficiency autorater ──► s ∈ {sufficient, insufficient}
       │                                          │
       └──► generator ──► answer + confidence p   │
                                  │               │
                  selective policy: f(s, p) > τ ? answer : abstain
                  (τ tuned for a target coverage/accuracy point)
```

29.8 Reference Architecture: A Conflict-Aware, Attributed Generator

```
query
  │
  ├─► closed-book pass (small budget) ─► prior answer a0, logprob p0 ──────┐
  │                                                                        │
  └─► retrieve top-k ─► CRAG-style evaluator ──(incorrect)──► rewrite+web  │
                            │ (correct / ambiguous)                        │
                            ▼                                              │
            strip-level filter / LongLLMLingua (question-aware,            │
            dynamic ratio, reorder strongest -> edges)                     │
                            │                                              │
                            ▼                                              │
    generator fine-tuned with distractors + rationales + quote-first       │
    citations + grounded refusal (RAFT / InstructRAG / FRONT / Trust-Align)│
         optional: CAD (trusted sources only) or isolate-then-aggregate    │
         (untrusted / user-uploaded sources)                               │
                            │                                              │
                            ▼                                              │
    answer a1 + citations {claim, doc, span}                               │
                            │                                              │
    verifiers: exact-span check | NLI recall/precision | ablation reward   │
                            │                                              │
    conflict check: a0 != a1 and p0 high? ◄────────────────────────────────┘
                            │
    release gate: sufficiency x confidence > τ ? answer : abstain / escalate
```

30. Algorithm-System Co-Design for RAG Inference: Overlap, Reuse, Compression, and Placement

Overview

```
           Where RAG latency comes from (one request, non-iterative)

query --> [embed] --> [ANN search] --> [rerank] --> [prefill: Q + k docs] --> [decode]
            ~ms        10s ms to s      ~10s ms     dominant for long ctx   per token
                        (CPU/GPU/       (small LM)   O(L) FLOPs, O(L^2)     memory-bw
                         FPGA)                        attention               bound
  overlap targets:   PipeRAG, RaLMSpec, TeleRAG, Stream2LLM, RAGCache-DSP
  reuse targets:                                    RAGCache, CacheBlend, TurboRAG,
                                                    Block-Attn, CacheGen, LMCache
  shrink targets:                                   REFRAG, Speculative RAG
  placement targets: Chameleon, VectorLiteRAG, HedraRAG, RAGO (all stages)
```

Architecture

```
                  retrieved docs D (n docs)
                           |
          embed (instruction-aware encoder) + K-Means into k clusters
                           |
         sample 1 doc per cluster -> m subsets delta_1..delta_m (k docs each)
        /                  |                    \
  [Drafter 7B #1]   [Drafter 7B #2]  ...  [Drafter 7B #m]     (parallel endpoints)
    alpha_1, beta_1   alpha_2, beta_2        alpha_m, beta_m   (answer, rationale)
        \                  |                   /
         [Generalist verifier: one forward pass per draft, no docs in prompt]
          rho_j = rho_draft * rho_self-consistency * rho_self-reflection
                           |
                  answer = argmax_j rho_j
```

3.1 PipeRAG (Amazon; KDD 2025)

```
time ->
RETRO (serial):  [gen chunk i][retrieve i+1][gen chunk i+1][retrieve i+2] ...
PipeRAG:         [gen chunk i      ][gen chunk i+1    ][gen chunk i+2 ] ...
                 [retrieve(stale q_i - s)][retrieve(stale)] ...  nprobe sized to fit
```

4. Reusing KV caches of retrieved knowledge

```
prefix cache hit requires identical token prefix:
  req A: [sys][doc7][doc3][q_A]      req B: [sys][doc3][doc7][q_B]
         ^^^^^^^^^^^ hit for A only once [sys][doc7] cached; B misses on doc3
non-prefix reuse (CacheBlend/TurboRAG/Block-Attn): reuse doc3 and doc7 KV anywhere,
  then repair cross-attention (selective recompute) or train model to not need it
```

4.1 RAGCache (Peking University and ByteDance; ACM TOCS)

```
                    root (system prompt)
                   /        |         \
                doc12      doc3       doc40          <- GPU tier (hot)
               /    \        \
           doc3    doc8     doc12                    <- host tier (warm, swapped once)
           |
         doc8                                         path = ordered doc sequence
```

Architecture (Meta Superintelligence Labs, NUS and Rice; 2025)

```
retrieved passages --> split into k-token chunks
       |
  [light encoder: RoBERTa-base/large]  -> 1 embedding per chunk   (query-independent,
       |                                                           precomputable/cacheable)
  [2-layer MLP projection] -> decoder token-embedding space
       |
  RL policy: which chunks to EXPAND back to full tokens (selective compression)
       |
decoder input = [question tokens] + [chunk embeddings ... | expanded chunk tokens]
       |
  [LLaMA-2-7B decoder]  -> answer          input length shrinks ~k-fold
```

8.1 TTFT for one non-iterative request

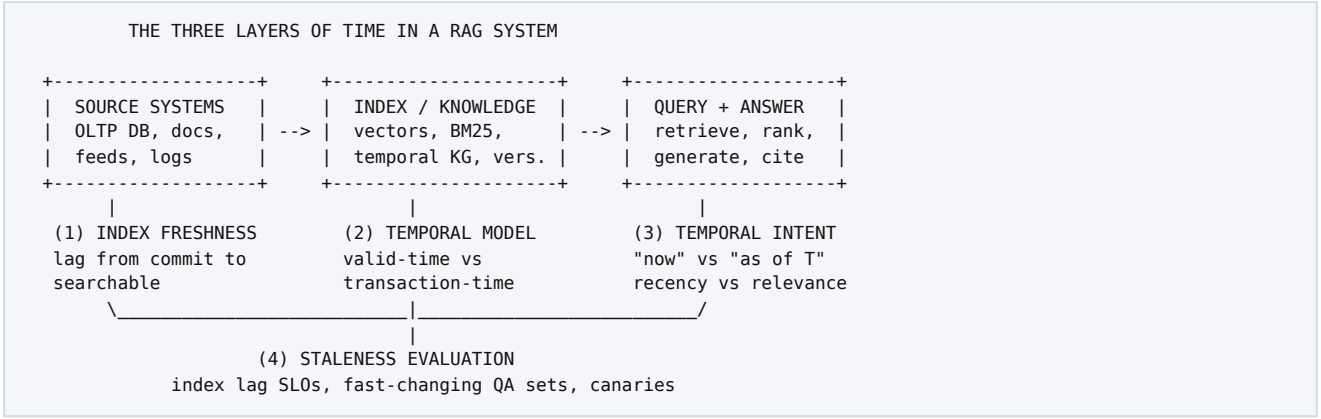
```
TTFT ~= T_embed(q)
      + T_search(index, nprobe, placement)          # CPU / GPU / FPGA
      + T_rerank(k)                                  # small model, many tokens
      + T_kv_load(bytes_hit / BW_tier)               # 0 if no reuse; PCIe/NVMe/net
      + T_prefill((1 - h) * L_ctx + r * h * L_ctx + L_q)
      + T_queue
      - T_overlap                                    # hidden by pipelining
where
  L_ctx   = retrieved tokens (k chunks x chunk length)
  h       = fraction of context tokens with reusable KV (cache hit ratio)
  r       = recompute ratio on hits (0 for exact prefix; ~0.15 CacheBlend; 0 TurboRAG)
  T_overlap = min(T_search, T_pre-retrieval-LLM) (TeleRAG), or speculative prefill
              on partial results (RAGCache), or streaming chunks (Stream2LLM)
```

8.3 Throughput and cost per query

```
QPS ~= min( QPS_search(cores or GPU share, batch_s),
            QPS_prefill(GPU FLOPs, effective tokens),
            QPS_decode(HBM bandwidth, batch_d limited by free KV memory) )
cost/query ~= sum_over_resources( price_per_hour / 3600 / QPS_at_that_resource )
```

31. Real-Time and Time-Aware RAG: Streaming Ingestion, Temporal Retrieval and Staleness Evaluation

Overview



2.1 The CDC pattern: Debezium + Kafka as the change log

```
 CDC -> STREAM -> EMBED -> VECTOR SINK (canonical streaming RAG ingestion)

+-----------+  WAL/binlog  +-----------+   topic per table   +----------------+
| Postgres  | -----------> | Debezium  | ------------------> | Kafka          |
| (OLTP)    |              | (Connect) |   c/u/d events      | (ordered log)  |
+-----------+              +-----------+                     +-------+--------+
                                                                     |
                      +----------------------------------------------+
                      v
           +------------------------+     +--------------------+
           | Stream processor       |     | Embedding service |
           | Flink / Pathway /      |<--->| (batched, cached, |
           | Bytewax: parse, chunk, |     |  rate-limited)    |
           | dedupe, key by doc_id  |     +--------------------+
           +-----------+------------+
                       | upsert(doc_id#chunk_i, vec, payload{ts,...})
                       | delete(doc_id#*) on tombstone
                       v
           +------------------------+       +----------------------+
           | Vector / hybrid index  | <---- | Query service (RAG)  |
           | Qdrant/Milvus/Mongo/ES |       | filters on ts fields |
           +------------------------+       +----------------------+
```

2.2 Flink-native RAG: Confluent Cloud streaming agents

```
   QUERIES-AS-STREAMS (Confluent quickstart Lab 2 shape)

documents --> documents_embed --> [Mongo Atlas sink] --> vector index
                                                            ^
queries ----> queries_embed ----> search_results ------------+ (vector search)
                                      |
                                      v
                           search_results_response  (LLM answer + snippets)
```

3.4 Temporal-intent gating

```
            TEMPORAL-INTENT-GATED RETRIEVAL

query --> [temporal intent classifier] --+-- "timeless"  --> pure relevance
            (LLM prompt or small model)  |
                                         +-- "current"   --> relevance x decay(now)
                                         |                   + filter valid_to=NULL
                                         |
                                         +-- "as of T"   --> valid_from<=T<valid_to
                                         |                   decay centered on T
                                         +-- "range"     --> filter ts in [T1,T2]
                                                             (logs, events)
```

4.1 Temporal knowledge graphs: Zep / Graphiti

```
      BI-TEMPORAL FACT EDGE (Graphiti-style)

 (Alice) --[ROLE: VP Sales]--> (Acme)
      valid_from = 2021-03     valid_to = 2024-07      <- world time
      created_at = 2021-03-04  invalidated_at = 2024-07-15  <- system time
      episode_ref = slack#88213 (provenance)

 (Alice) --[ROLE: CRO]-------> (Acme)
      valid_from = 2024-07     valid_to = NULL (current)

 "Who runs sales now?"      -> edges with valid_to = NULL
 "Who ran sales in 2022?"   -> edges with valid_from <= 2022 < valid_to
 "What did we believe on 2024-07-10?" -> filter on created/invalidated (system time)
```

6.2 Event-stream and log RAG: observability copilots

```
      OBSERVABILITY COPILOT: TOOL-MEDIATED TEMPORAL RETRIEVAL

 user: "why did p99 latency spike after 14:00?"
    |
    v
 +--------------+  1. alert rules / incidents   +---------------------+
 |  LLM agent   | ----------------------------> | Grafana / Alerting  |
 |  (planner)   |  2. PromQL range [13:30,15:00]| Prometheus / Mimir  |
 |              | ----------------------------> |                     |
 |              |  3. Loki patterns, same range | Loki                |
 |              | ----------------------------> |                     |
 |              |  4. runbook / postmortem RAG  | vector KB (static)  |
 +------+-------+ ----------------------------> +---------------------+
        |
        v   answer = live telemetry (exact time range) + static knowledge (semantic)
```

8. Reference architecture

```
          REFERENCE: TIME-AWARE STREAMING RAG

 SOURCES             INGEST                    STORES                QUERY
 -------             ------                    ------                -----
 OLTP DB --CDC--+                       +-> vector/hybrid idx --+
 Docs/Drive ----+--> stream processor ---+   (valid_from/to,      |   +----------+
 News feeds ----+    chunk, dedupe,      |    published_at,       +-->| intent   |
 Kafka topics --+    version, embed      |    indexed_at)         |   | gate     |
                     (batch+retry)       +-> temporal KG ---------+   +----+-----+
                          |              |   (bi-temporal edges)  |        |
                          |              +-> versioned snapshot --+   timeless/now/
                          v                  (time travel, eval)  |   as-of/range
                 lag + backlog metrics                            |        |
                          |                LIVE TELEMETRY ---------+   +---v------+
                          v                (Prom/Loki/SIEM via     |   | retrieve |
                 freshness SLO alerts       tools, time-bounded)   +-->| + decay  |
                          ^                                           | + filter |
                          |                                           +----+-----+
                 canary writer/prober <-------------------------------+    |
                                                                      +-----v-----+
                                                                      | generate  |
                                                                      | cite dates|
                                                                      +-----------+
```

32. Personalized and Conversational RAG: User Profiles, Per-User Scopes, Conversational Search, Voice Latency, Clarification and Shopping Assistants

Overview

```
             +-------------------- per-user scope (tenant / partition) -------+
             |  profile items   PTKB statements   memories   purchase history |
             +--------^-----------------^--------------^-------------^--------+
                      | profile retrieval (LaMP)      | (Ch. 18)    | recsys tools
user turn t --> [context resolver] --> rewritten query + selected profile facts
  ^              (CAsT rewrite /          |
  |               ConvDR encoder)         v
  |                              [shared corpus retrieval] --> rerank --> generate
  |                                       |                              |
  |   [clarify-or-answer policy] <--------+ (ambiguity / low confidence)    |
  +------------- clarifying question  <---+     cited answer / spoken reply <+
```

Architecture

```
 input x (e.g., article text)     user profile P_u = {d_1 ... d_n} (user's past docs)
        |                                  |
        +------> rank P_u by relevance to x: bm25 | contriever | recency
                                 |  (optional: prepend timestamps to items)
                                 v
                     top num_retrieved items --> prompt template --> LLM --> y_u
```

Reference baseline architecture (2023)

```
context + utterance --> t5-base-canard rewriter (automatic runs)
                            |      or resolved_utterance (manual runs)
PTKB --> cross-encoder ms-marco-MiniLM-L-6-v2 scores each (query, statement) pair
                            |  top num_ptkb statements appended to rewritten query
                            v
               BM25 (or QLD), optional RM3, k=100..1000 over ClueWeb22 subset
                            v
               cross-encoder rerank --> top num_psg passages (default 3)
                            v
          t5-base-finetuned-summarize-news ("summarize: ..."), max 250 tokens
                            v
         response + passage provenance + PTKB provenance (official JSON)
```

OpenAI Realtime: preambles, supervisors and async tools

```
user speech --> realtime model --(filler: "One moment...")--> audio out
                     | tool: getNextResponseFromSupervisor(relevantContext)
                     v
              text supervisor (gpt-4.1) --> lookupPolicyDocument / RAG / APIs
                     | message with citations   (~2 s in the README example)
                     v
              realtime model reads the answer aloud --> audio out
```

A clarify-or-retrieve policy

```
turn --> resolve context --> ambiguity / missing-slot detector (ClariQ-style 1..4)
             |                                  |
             | low need                         | high need and clarification budget left
             v                                  v
      personalize from profile        ask ONE targeted question
      (fill slots from PTKB/memory)    (facets from retrieval results or a bank)
             v                                  v
         retrieve + answer            user answer --> merge into query --> retrieve
```

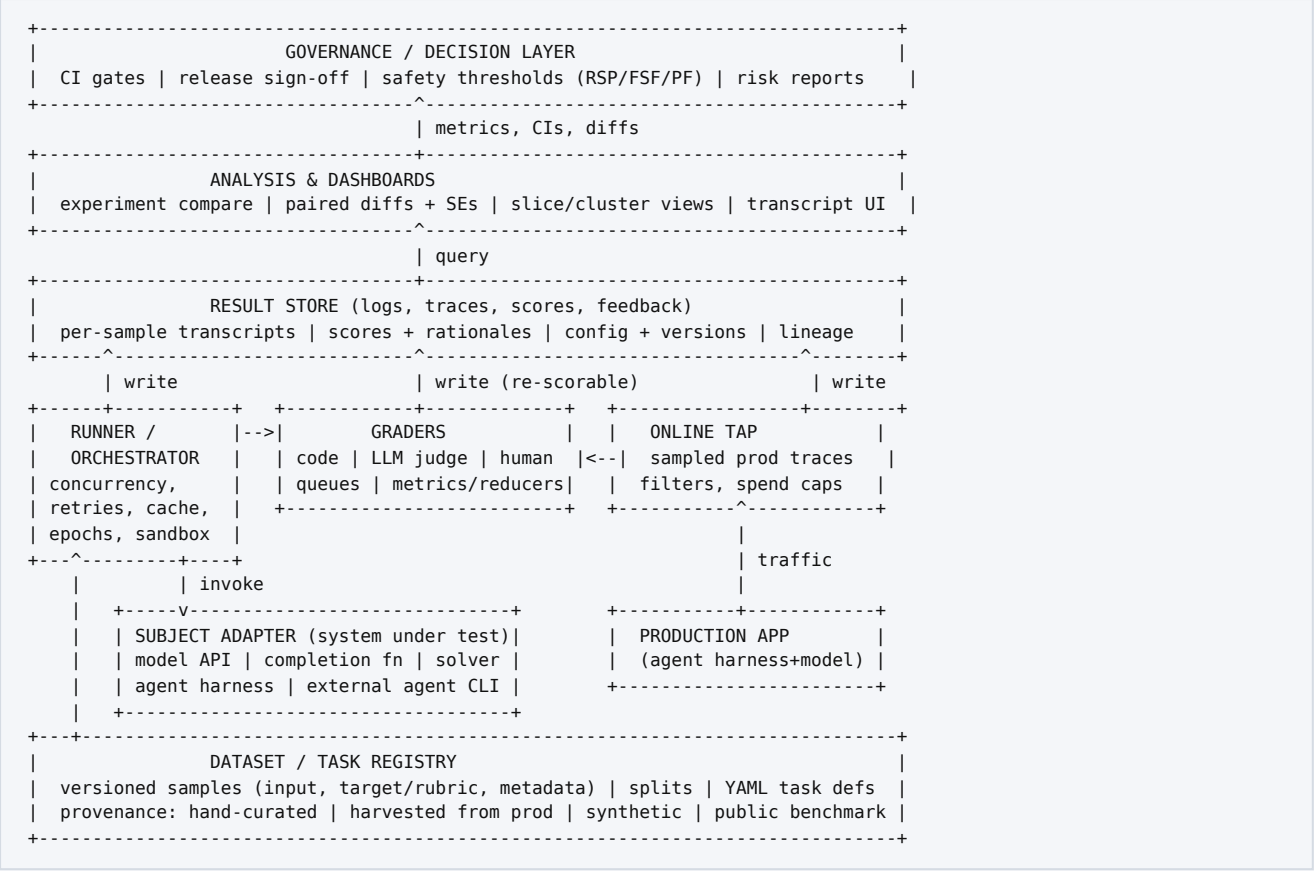
Tool-based conversational recommendation: InteRecAgent

```
user: "popular sports games like FIFA, not too pricey"
  LLM planner --> SQL tool (genre=sports, price<X) --> candidate set
              --> item-similarity tool (FIFA)      --> candidate set (merged)
              --> ranker(schema=preference, prefer=[...], unwanted=[...]) --> top-k
  LLM --> grounded recommendation with item facts from the profile table
```

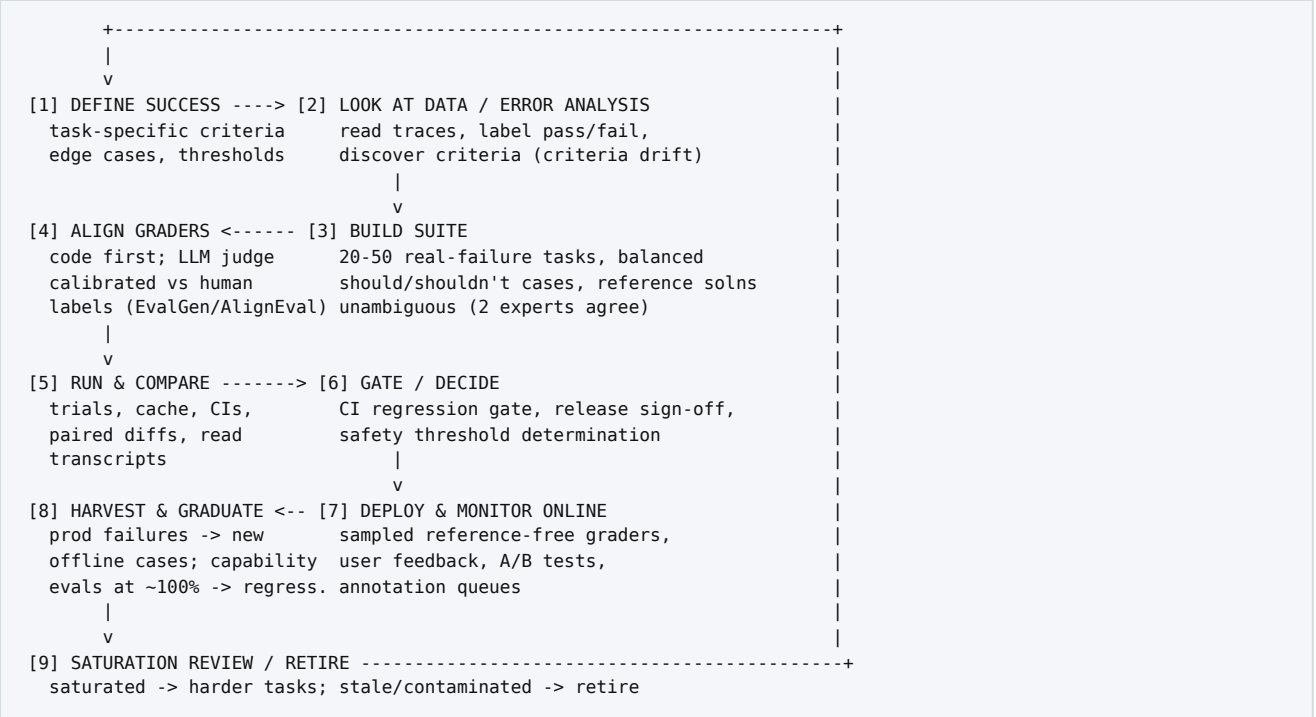
3. LLM Evaluation — Architecture Diagrams

01. Evaluation Foundations and Taxonomy: Purposes, Modes, and the Eval Stack

6. The eval stack: a reference architecture

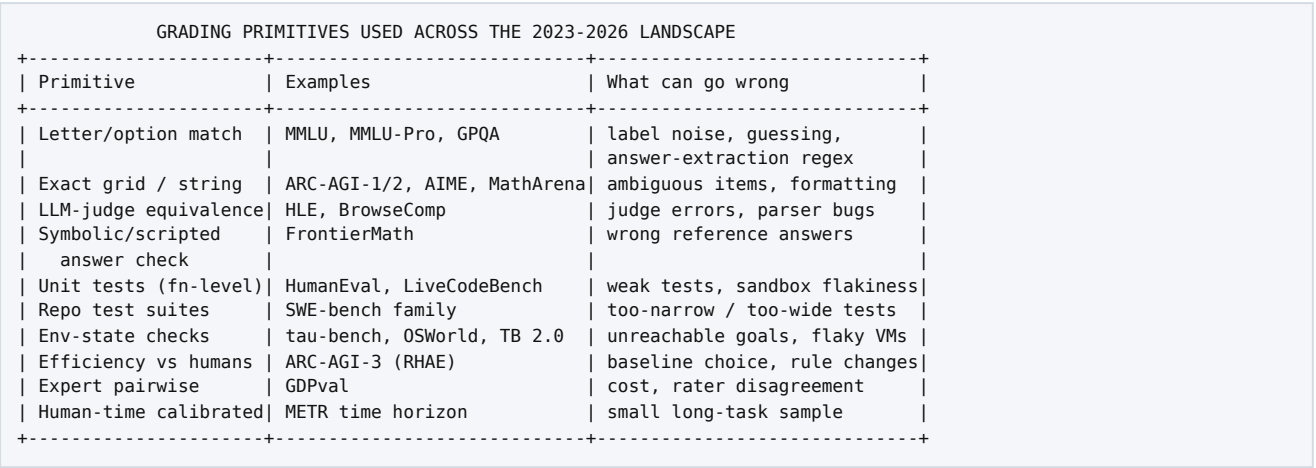


8. The eval lifecycle



02. The Benchmark Landscape Through 2026: How Frontier Benchmarks Are Built and Graded

Overview



Overview

```
author items --> adversarial filter --> expert review --> split
(humans,          (drop items frontier    (correctness,     |  public / semi-private /
 scraped repos,    models already solve)   ambiguity)       |  private / commercial
 contests)                                                  v
                                    harness (prompt, tools, container, VM, budget)
                                                            |
                                                            v
                  grader (regex | judge LLM | tests | state hash | humans)
                                                            |
                                                            v
           aggregate (acc, pass@k, pass^k, RHAE, win rate, logistic horizon)
                                                            |
                                                            v
           post-release audit --> errata / v2 / retirement  (the stage most
                                                              designs underbudget)
```

1.3 GPQA: "Google-proof" by construction

```
domain expert writes Q + 4 options + explanation
        |
        v
expert validator #1 answers, gives feedback --> writer revises
        |
        v
expert validator #2 answers (measures objectivity)
        |
        v
3 non-expert validators (PhDs in OTHER fields), unrestricted web access
        |
        v
subsets: Extended (all)  >  Main (448)  >  Diamond (198: experts agree,
                                             <=1 of 3 non-experts correct)
```

Construction pipeline.

```
~1,000 subject experts submit Q (MC or exact short answer, some multimodal)
        |
        v
LLM difficulty gate: exact-match Qs must stump several frontier LLMs;
MC Qs must be at or below chance across all LLMs   [10]
        |
        v
human expert review, two rounds (refine, then approve)
        |
        v
public set (2,500 Qs) + private held-out set (overfitting detection)
        |
        v
post-release community bug bounty --> errata; dataset finalised Apr 3, 2025
```

Architecture.

```
hand-built turn-based game (no instructions, no stated goal)
   levels L1..Ln, each with a target state
        |
 agent <-- frame (grid + available_actions + levels_completed) --+
   |                                                             |
   +-- action (ACTION1..ACTION7, reset) --------------------------+
        |
per-level action count A_ai  vs  human baseline A_h
        |
RHAE_level = min(cap, (A_h / A_ai)^2);  0 if A_ai > 5 * A_h
        |
average across levels, games, private environments
```

Architecture.

```
weekly contests (LeetCode / AtCoder / Codeforces)
      |  scrape statement + tests, tag release_date, difficulty
      v
release_v1 (400 probs, to Mar 2024) ... release_v6 (1055 probs, to Apr 2025)
      |
      v
scenarios: codegeneration | selfrepair | testoutputprediction | codeexecution
      |
      v
score only problems with release_date > model cutoff  (--start_date)
```

Architecture.

```
41 repos total
+-----------------------+   +------------------------+   +-----------------------+
| PUBLIC (731 tasks)    |   | HELD-OUT (858 tasks)   |   | COMMERCIAL (276)      |
| 11 copyleft (GPL)     |   | 12 separate repos,     |   | 18 proprietary startup|
| repos, released on HF |   | kept private to detect |   | repos; results public,|
|                       |   | future overfitting     |   | data never released   |
+-----------------------+   +------------------------+   +-----------------------+
              \_____________________ human-verified, augmented _________________/
                      (issue text + requirements + interface spec)
```

7.2 tau-bench, tau2-bench and tau3-bench: tool-agent-user loops graded by end state

```
+------------------+   natural language   +-----------------+   API calls
| user simulator   | <------------------> |  agent under    | -------------> +-------+
| (LLM + scenario, |                      |  test (policy   |                | DB /  |
|  optional tools) |                      |  doc + tools)   | <------------- | env   |
+------------------+                      +-----------------+   results      +-------+
                                                                                 |
 reward = DB_state_hash(final) == DB_state_hash(replay(gold_actions))           |
          x  all communicate_info strings present in agent messages    <---------+
```

7.5 OSWorld and OSWorld-Verified

```
VM snapshot --config scripts--> initial state --> agent loop:
    screenshot / a11y tree --> model --> pyautogui action (max_steps budget)
                                                |
evaluator script: read files/app state/URLs from VM --> compare to expected --> 0/1
```

Architecture.

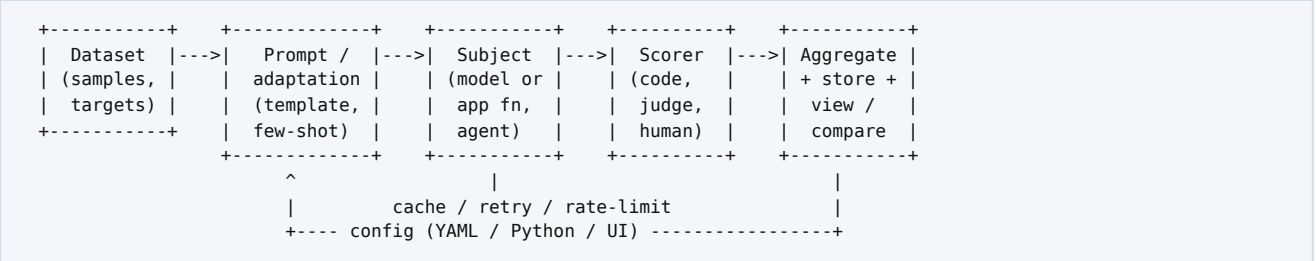
```
task suites with human baselines (HCAST, RE-Bench, SWAA, ...)
   each task: human_minutes (geo-mean of successful human baselines)
        |
agent runs --> score_binarized in {0,1}, weighted (diversity-adjusted)
        |
fit logistic:  P(success) = sigma( a - b * log2(human_minutes) )
        |
horizon_50 = human_minutes where P = 0.5   (also p80)
bootstrap over tasks/runs --> CI;  regress log(horizon) on release date
```

Grading.

```
request + reference files --> model deliverable  \
                          --> expert deliverable  >--> blinded occupational expert
                                                 /    ranks unlabeled outputs
                                                      (>1 hour per comparison)
metric: win-or-tie rate vs. human expert
automated grader (experimental, gold subset): 66% agreement with experts
vs. 71% human inter-rater agreement
```

03. Eval Harnesses and Platforms: The Architecture of Fourteen Systems

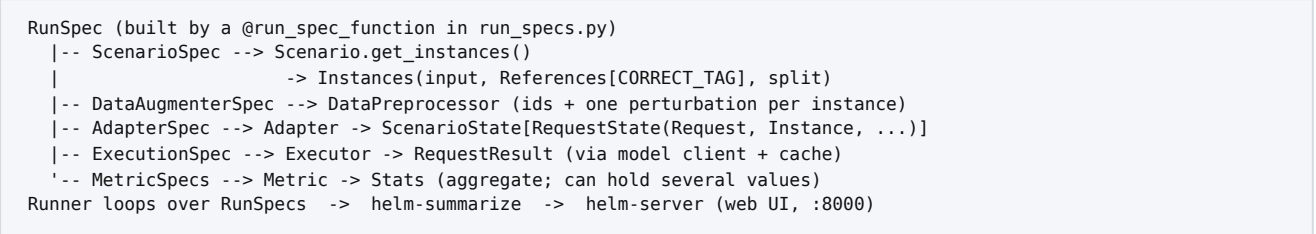
1. The common skeleton



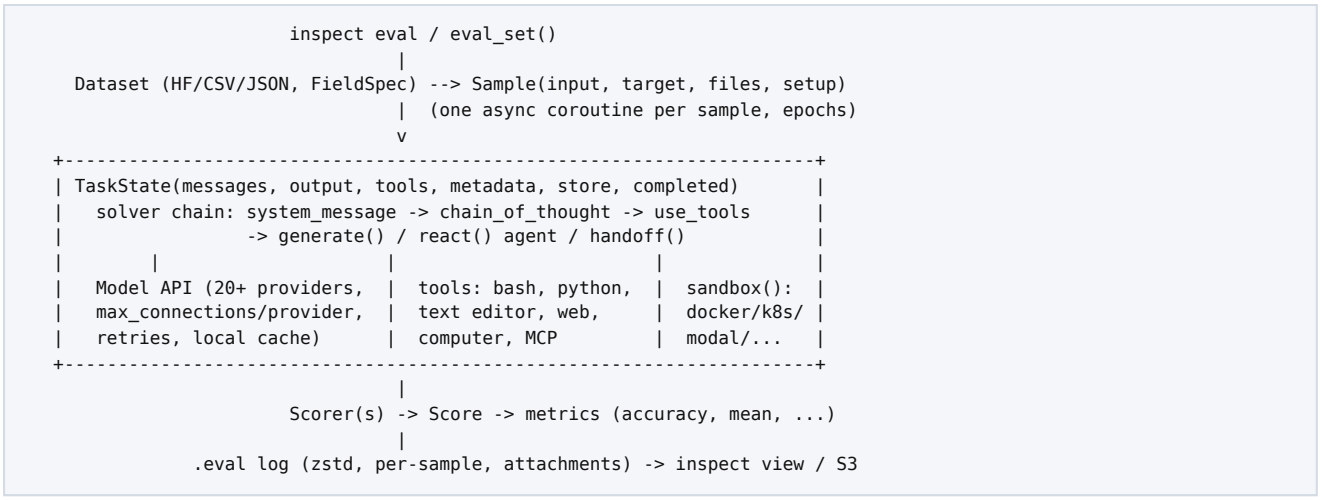
Architecture



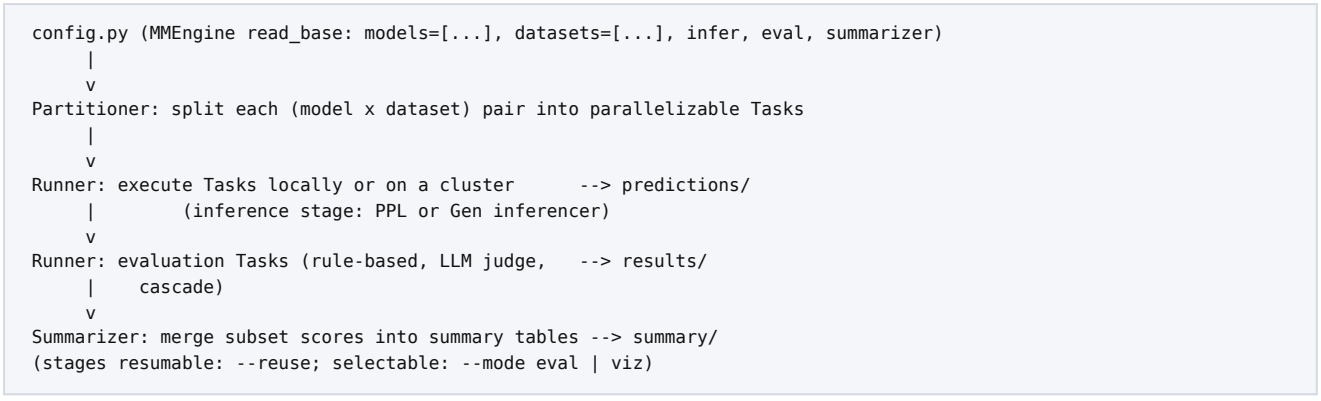
Architecture



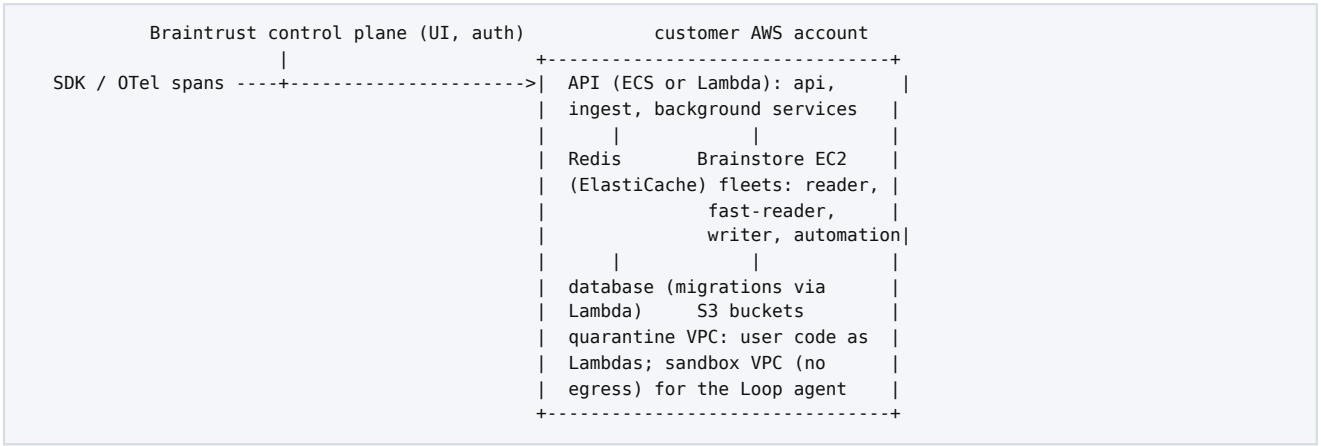
Architecture



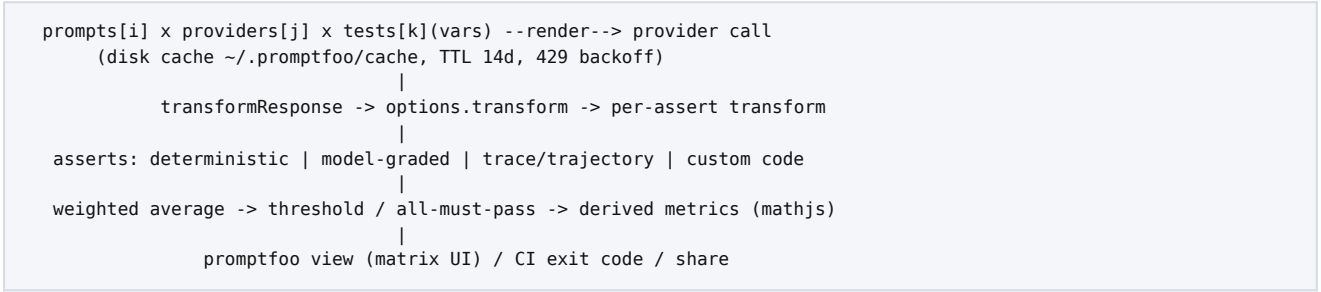
Problem and architecture



Architecture



Problem and architecture

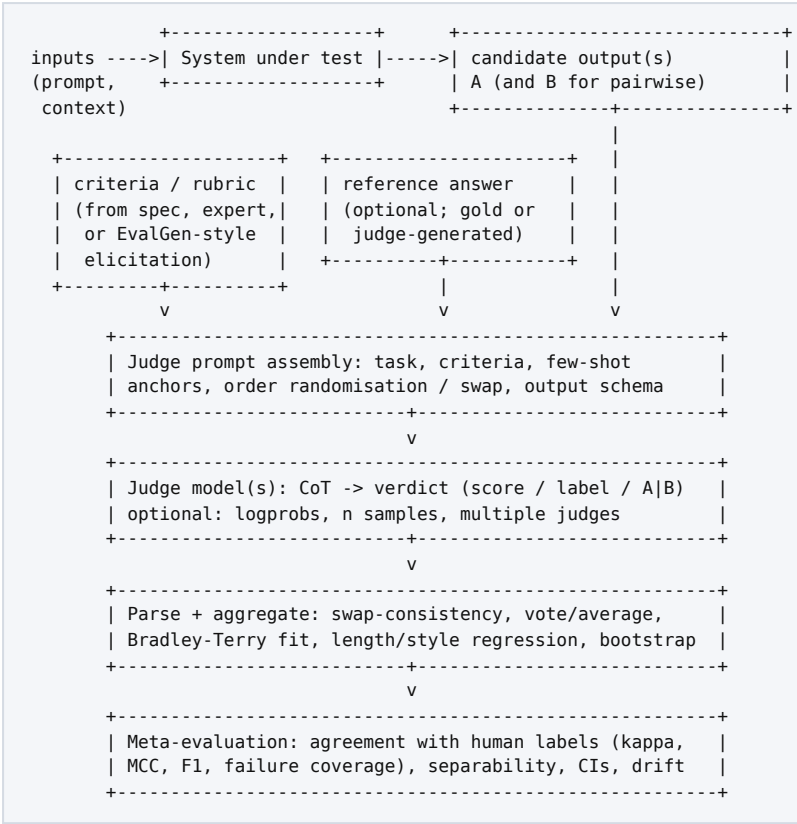


Problem and architecture

```
SDK batch --> Web (auth + schema check only)
                | write raw event --> S3 (durable, replayable)
                | enqueue reference --> Redis queue
                v
             Worker: tokenize, compute cost, merge events by object id
                |
                v
         ClickHouse (ReplacingMergeTree; wide, mostly immutable observations)
Prompt / auth endpoints: separate deployment + Redis read-through cache
SDKs: serve last-fetched prompt immediately, refresh in background
```

04. LLM-as-a-Judge: Protocols, Biases, Calibration, Ensembles, Fine-Tuned Judges and Judge Alignment

1. The canonical judge loop



3.2 G-Eval: form-filling with auto-CoT and probability-weighted scores

```
criteria text --> [LLM: generate evaluation steps (auto-CoT)] --> steps
                                                                    |
(source, output) + steps --> [LLM: fill form, score 1..N] --> token probs
                                                                    |
                       score = sum_i  p(score = i) * i   <--------+
```

3.3 Pairwise against an anchor: AlpacaEval and Arena-Hard

```
Chatbot Arena prompts --> embed + cluster into topics
        |
        v
LLM annotator scores each prompt on 7 criteria
(specificity, domain knowledge, complexity, problem-solving,
 creativity, technical accuracy, real-world application)
        |
        v
keep clusters with mean score >= 6/7 (250 clusters) --> sample 2 each
        |
        v
500 prompts x (model vs GPT-4-0314 baseline) x 2 orders = 1,000 judgments
        |
        v
judge (GPT-4-Turbo): CoT, writes own answer first, then 5-way label
A>>B, A>B, A~=B, B>A, B>>A
        |
        v
Bradley-Terry fit over all comparisons, 100x bootstrap -> score + 95% CI
```

3.4 Rubric grading at scale: HealthBench

```
conversation + rubric items [(criterion_i, points_i, tags_i)]
         |
         +--> for each item i:  grader(conversation, criterion_i)
         |        -> {"explanation": ..., "criteria_met": true|false}
         v
score = sum(points_i for met items) / sum(points_i where points_i > 0)
         |
         v
example scores -> mean clipped to [0,1] -> bootstrap std (1,000 resamples)
         -> per-tag (theme / axis) breakdowns
```

5.3 Selective judging with guarantees: Trust or Escalate

```
item --> [Mistral-7B judge] --conf >= t1--> accept verdict
             | abstain
             v
         [GPT-3.5 judge] ----conf >= t2--> accept verdict
             | abstain
             v
         [GPT-4-Turbo judge] -conf >= t3--> accept verdict
             | abstain
             v
         escalate to human
thresholds t_i calibrated on a held-out human-labelled split
(reference config: calibration set 500, alpha = 0.15, delta = 0.1)
```

8.1 EvalGen: "Who validates the validators?"

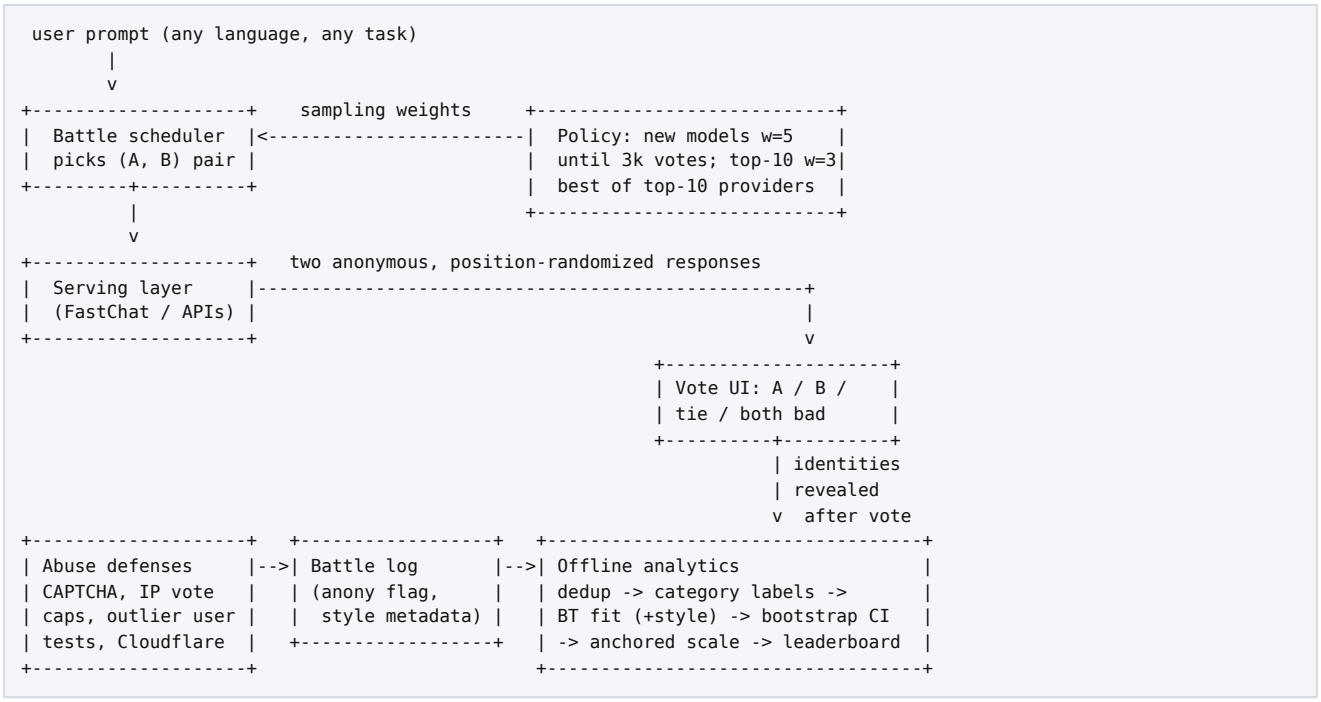
```
developer prompt + sample (vars, prompt, response) triples
         |
         v
[LLM] suggest evaluation criteria  --> developer edits / confirms
         |
         v
per criterion: [LLM AssertionEmitter] generate multiple candidate
evaluators (code assertions or LLM-judge prompts), streamed
         |
         v
Executor (background): run every candidate on every example;
update each example's grading priority from candidates' pass rates
         |
         v
developer grades examples thumbs up/down, choosing next example by
"priority" policy (or random) to get a mix of good and bad outputs
         |
         v
per candidate: confusion matrix vs. developer grades ->
  F1, MCC, Cohen's kappa, failure coverage, false-failure rate
         |
         v
select best candidate per criterion (sorted by MCC, then lower FFR)
-> report card: coverage of failures, false failures, chosen set
```

10. A reference architecture for a production judge service

```
            +------------------- offline (alignment) --------------------+
            |  expert labels + critiques (binary, principal expert)       |
            |        |                                                    |
            |        v                                                    |
            |  criteria elicitation (EvalGen-style) <--- criteria drift   |
            |        |                                                    |
            |        v                                                    |
            |  candidate judges (prompt variants / models / fine-tunes)   |
            |        |                                                    |
            |        v                                                    |
            |  meta-eval: MCC, kappa, failure coverage, FFR, swap-        |
            |  consistency, adversarial set (LLMBar/JudgeBench-like)      |
            |        |                                                    |
            |        v                                                    |
            |  optimise (few-shot, MemAlign/GEPA) -> freeze + version     |
            +--------+----------------------------------------------------+
                     |
online / CI          v
outputs --> per-criterion judges (isolated; rubric items; CoT -> JSON)
              |  pairwise? -> both orders; cross-family panel if comparative
              v
         aggregate: rubric points, vote/mean, Bradley-Terry + style
         covariates, clustered SEs / bootstrap CIs
              |
              +--> low confidence / abstain --> escalate (bigger judge, human)
              v
         dashboards + sampled human audit --> drift alarms --> back to offline
```

05. Human Evaluation and Arenas: Pairwise Preference Platforms, Their Statistics, Their Critics, and Expert Annotation Pipelines

1.2 Architecture and data flow



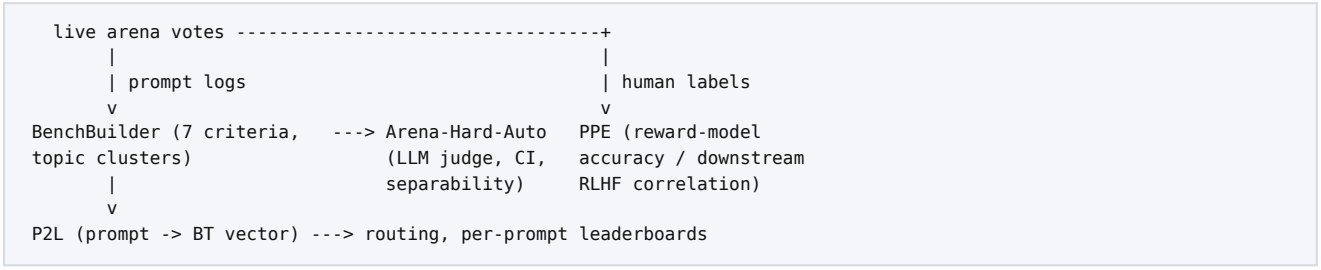
2.2 Bradley-Terry as logistic regression

```
For battle i, X_i in R^M has +1 at the left model, -1 at the right model.
Y_i = 1 if the left model wins, else 0.

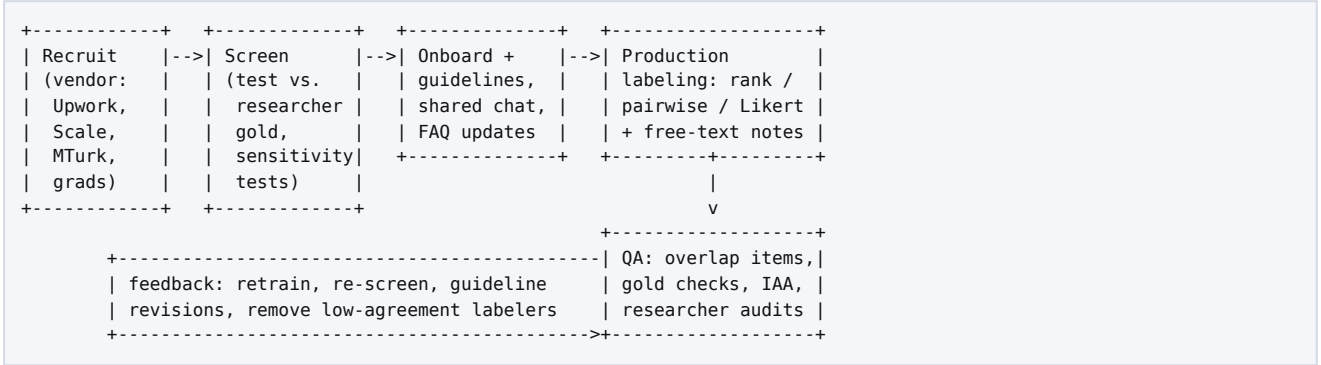
beta_hat = argmin_beta  (1/n) * sum_i BCE( sigmoid(X_i . beta), Y_i )

Reported Arena Score = 400 * beta (natural-log scale converted via log(10))
                     + 1000, then shifted so an anchor model has a fixed value.
```

6.3 Other derived products

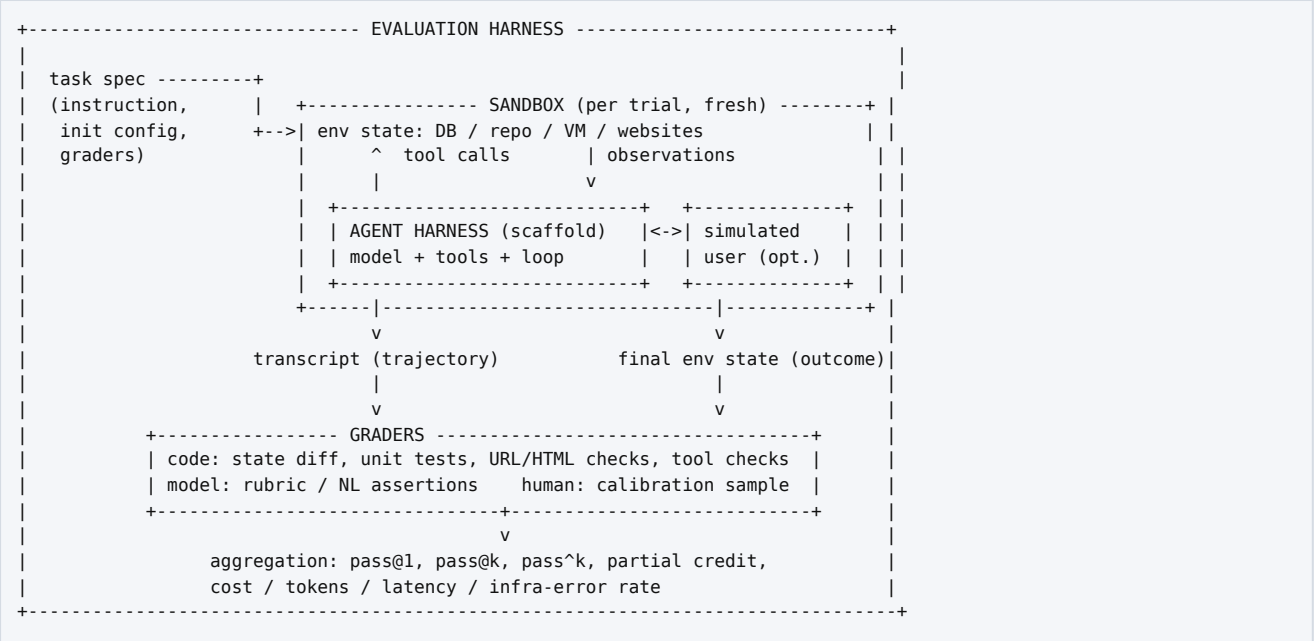


7. Expert annotation pipelines

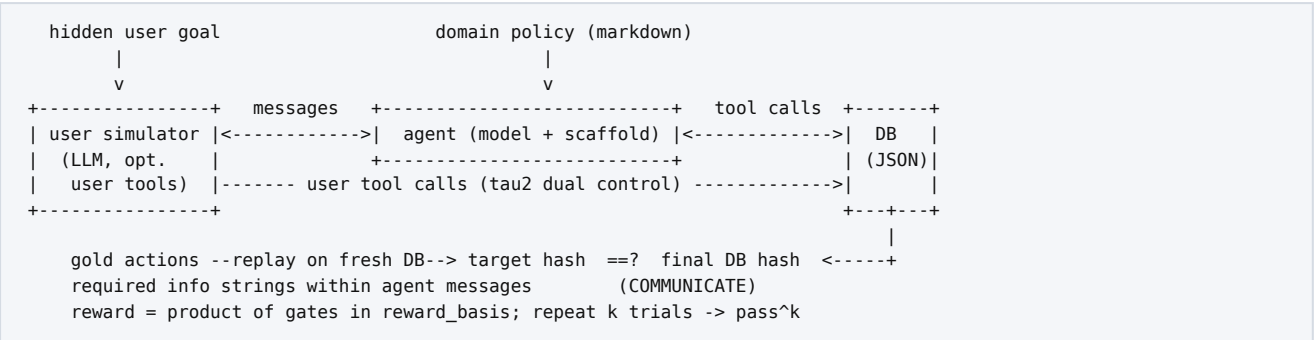


06. Agent Evaluation Architectures: Trajectories, Outcomes, Sandboxes and Reliability

1. Anatomy of an agent evaluation



Architecture



Architecture

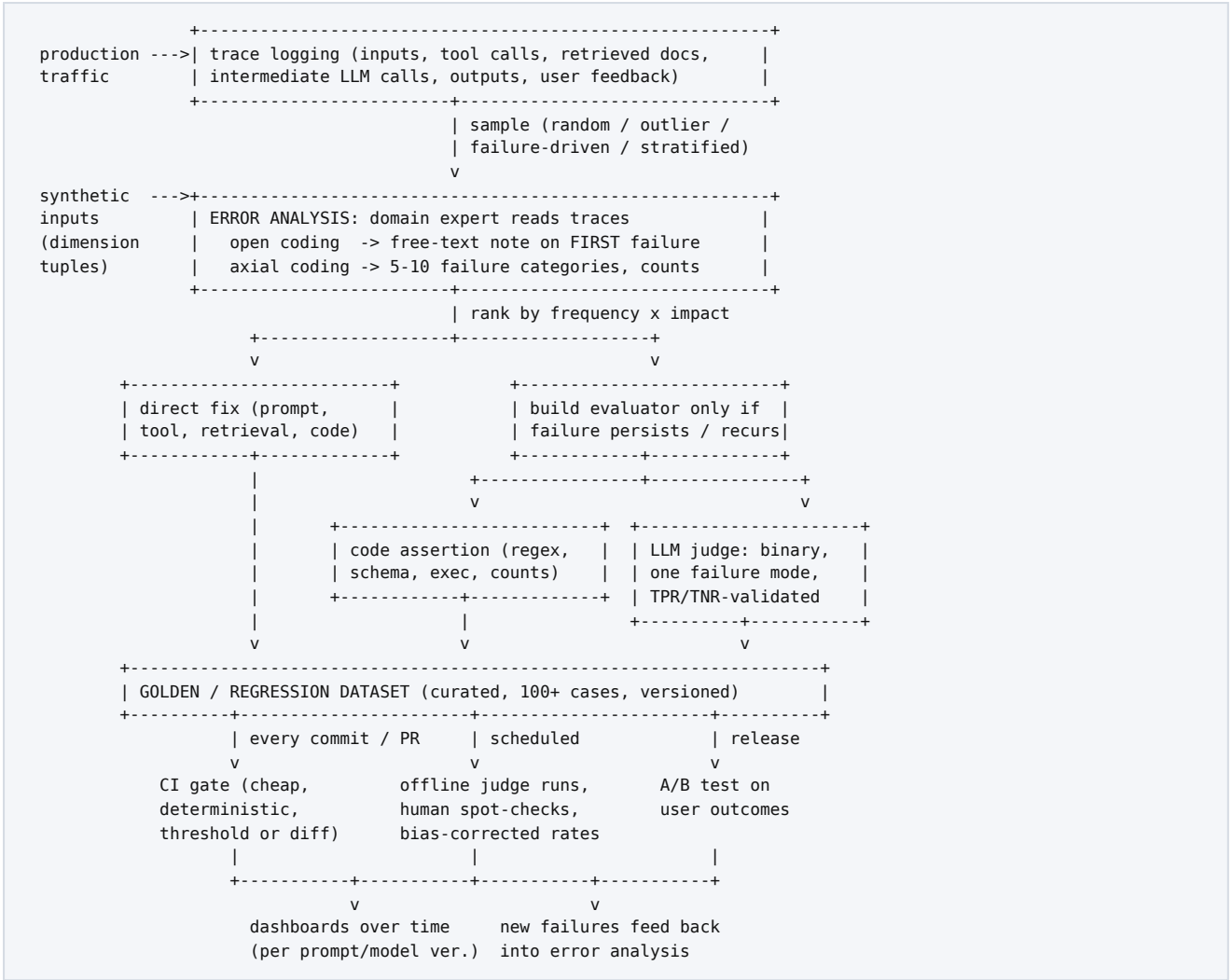


Architecture

```
snapshot --boot--> VM --config steps--> initial state
                   ^                        |
  screenshot / a11y |   pyautogui actions   v
           +-------+--------+        agent acts (max_steps)
           |  agent harness |------------>|
           +----------------+             v
           postconfig (save, export) -> result getter (vm_file, app state)
                                     -> expected getter (gold file)
                                     -> metric func(result, expected, options)
```

07. Product Eval Practice: Eval-Driven Development, Error Analysis, Golden Datasets, Evals in CI and Validating the Validators

1. The product-eval stack at a glance



2.3 EDD as the scientific method (Yan)

```
observe --> annotate --> hypothesize --> experiment --> measure --> iterate
(look at    (label pass/   (e.g., bad      (prompt /      (vs. a       (ship, or
 the data)   fail, ~50:50)  retrieval,      retrieval /    baseline or   back to
                            conflicting     model change)  control)      error
                            instructions)                                 analysis)
```

3.6 Agents: the transition failure matrix

```
             first failing state ->
             DecideTool  PlanCal  GenSQL  ExecSQL  Respond
last   Start      .          .       .        .        .
good   Decide     -          2       .        .        .
state  GenSQL     .          .       -       12        .
  |    ExecSQL    .          .       .        -        .
  v      (illustrative cells only; counts from the FAQ example [4])
```

6.1 Cadence and gating

```
developer change (prompt / model / retrieval / code)
       |
       v
+--------------+    PR opened     +-----------------------------------+
| local run    |----------------->| CI job                            |
| (cached LLM  |                  |  1. run golden set through system   |
|  calls)      |                  |  2. deterministic asserts (all)     |
+--------------+                  |  3. validated judges (subset/cache) |
                                  |  4. compare vs baseline experiment  |
                                  |  5. gate: --fail-on-error, or       |
                                  |     pass-rate >= threshold, or      |
                                  |     no regression on key scores     |
                                  +---------+---------------+----------+
                                            |               |
                            PR comment w/ score diffs   JUnit/JSON/HTML
                            (+3pp, 8 up / 4 down)       artifacts, dashboards
```

7.1 EvalGen and criteria drift

```
 user context + prompt
        |
        v
 [1] LLM proposes criteria (NL) <---- user edits/adds/removes, picks
        |                              code-based or LLM-based per criterion
        v
 [2] LLM generates candidate assertions per criterion (Python fns or
     judge prompts), executed on sampled outputs in background
        |
        v
 [3] user grades outputs thumbs-up / thumbs-down (no fixed quota)
        |     a thumbs-up output must pass all criteria; candidates that
        |     fail good outputs or pass bad ones are re-weighted
        v
 [4] select per-criterion assertion best aligned with grades
        |
        v
 report card: per-criterion confusion matrix, coverage, FFR
```

7.3 Statistical validation: TPR, TNR, splits and bias correction

```
expert labels (~100, balanced)
    |
    +--> train 10-20% --> few-shot pool --+
    |                                     v
    +--> dev 40-45% ---> run judge --> TPR/TNR --> inspect FP/FN --> edit prompt
    |                                     ^                              |
    |                                     +------------------------------+
    +--> test 40-45% --> run ONCE --> final TPR, TNR (+ bootstrap CI)
                                        |
production traces --> judge --> p_obs --> Rogan-Gladen --> corrected rate +/- CI
```

08. Online Evaluation and Production Monitoring: Tracing, Sampled Grading, Experiments, Feedback Signals, Guardrails and Drift

1. Reference architecture of an online evaluation system

```
                       SYNCHRONOUS (request path, latency-bound)
user --> [input guardrails] --> [app / agent: LLM calls, tools, RAG] --> [output guard]
             |   block/steer            |  spans + events (OTel GenAI)       |  block/
             v                          v                                    v  edit
+-------------------------------------------------------------------------------------+
|  telemetry pipeline: SDK -> OTel Collector (redact PII, sample, route) -> backend    |
+-------------------------------------------------------------------------------------+
                       ASYNCHRONOUS (minutes to hours, cost-bound)
    |                  |                      |                      |
    v                  v                      v                      v
[rules: filter   [online evaluators:    [feedback join:         [experiment platform:
 + sample %]      code checks, LLM       thumbs, edits,          assignment, OEC,
    |             judges, classifiers]   acceptance, retention]  guardrail metrics]
    |                  |                      |                      |
    +---------> scores attached to spans (gen_ai.evaluation.result) <-+
                       |
      +----------------+-------------------+----------------------+
      v                v                   v                      v
 [dashboards +   [annotation queues:  [datasets: failing    [drift / anomaly:
  alerts]         humans label,        traces become         clusters, score trends,
                  calibrate judges]    offline regressions]  per-deploy comparisons]
```

5.1 The signal funnel

```
opportunities to suggest   (LLM calls, shown or not)
       |
prompts sent / responses returned
       |
responses actually SEEN by the user
       |
ACCEPTS  (scenario-specific: thumbs up in chat, inserting text into a document)
       |
responses KEPT at a later time point  (retention of the generated content)
```

7.1 Problem and placement

```
             +------------- input rails ----------+
user prompt ->| injection / jailbreak classifier   |--block--> refusal
             | topic / policy classifier          |
             | PII masking                        |
             +----------------+-----------------+
                              v
                dialog / retrieval / execution rails (agent, RAG, tools)
                              |
             +----------------v-----------------+
             | output rails: safety classifier,   |--block/edit--> safe response
             | grounding / hallucination check,   |
             | exchange classifier (sees in+out)  |
             +------------------------------------+
  every verdict --> telemetry (GUARDRAIL span / gen_ai.evaluation.result) [4][9]
```

Lessons that generalise:

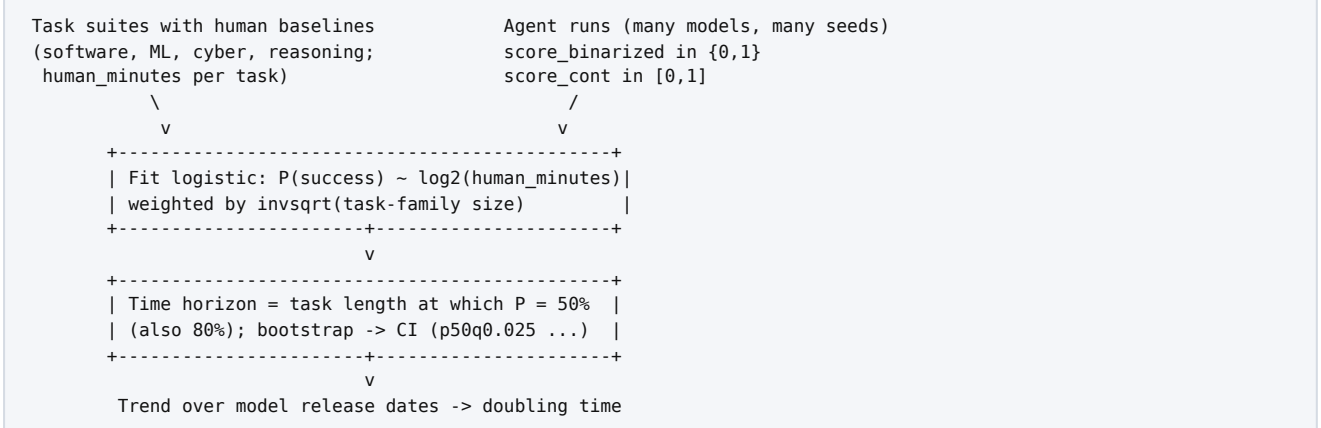
```
failure class                what missed it              what would catch it
---------------------------  --------------------------  -----------------------------
small-% routing error        aggregate benchmarks,       production-path evals sliced
(sticky, per user)           canaries                    by route / server pool
token-level corruption       task-level scores (model    invariant checks: script/lang
(rare unlikely tokens)       self-corrects)              mismatch, syntax validity
sampler numerical bug        evals on other configs      evals at production batch sizes
                                                         and configs; exact references
overlapping incidents        noisy global metrics        per-deploy change attribution
```

09. Safety and Dangerous-Capability Evaluation: Frontier Safety Frameworks, Third-Party Testing, Red-Teaming Pipelines, Jailbreak Benchmarks, Sandbagging and Evaluation Awareness

1.1 The common pattern



2.1 METR: autonomy and the time-horizon metric



2.2 UK AI Security Institute and US CAISI

```
       main task + side task (setting)
                 |
   +-------------v--------------+        attack mode: policy also
   | Untrusted policy (U)       |<------ pursues side task
   +-------------+--------------+
                 | proposed action
   +-------------v--------------+  suspicion score
   | Monitor (trusted model T)  |---------------+
   +-------------+--------------+               |
                 |                              v
        low suspicion                high suspicion -> defer to T /
                 |                    audit / block
                 v
          execute action  ---> scorers: usefulness (honest runs),
                               safety = P(no side task succeeds) (attack runs)
```

4. Automated red-teaming and jailbreak attack algorithms

```
goal G, target string T
      |
+-----v------------+  prompt P_i   +---------------+  response R_i
| Attacker LLM     |-------------->| Target LLM    |------------+
| (system prompt:  |               +---------------+            |
|  red-team role)  |<---------------------------------+         v
+------------------+   (P_i, R_i, score S_i)          |  +-------------+
      ^  N parallel streams x K iterations            +--| Judge LLM   |
      |                                                  | score 1..10 |
      +--- stop when S_i = 10 (jailbroken) --------------+-------------+
```

5.1 HarmBench

```
behaviors (CSV; val/test; standard, contextual, copyright, multimodal)
    |
[1] generate_test_cases.py  -- RedTeamingMethod subclasses (GCG, PAIR, TAP, ...)
[1.5] merge_test_cases.py   -- test_cases.json per experiment
    |
[2] generate completions on target models / defenses (HF, API, multimodal)
    |
[3] classify: HarmBench-Llama-2-13b-cls (text), -multimodal cls,
    Mistral-7b validation cls; copyright via hash matching
    |
    ASR per (method, model, behavior)    run modes: slurm | local | local_parallel (Ray)
```

6.2 Automated auditing agents: Petri

```
seed instruction --> Auditor agent ---(simulated user msgs, synthetic tool results,
                       ^   |          rollback/branch)---> Target model
                       |   +<----------- target actions / tool calls -------+
                       |
                  transcript --> Judge model --> scores per dimension
                                   (deception, sycophancy, self-preservation,
                                    power-seeking, reward hacking, ...)
```

Design responses.

```
realism axis                      observation axis
------------                      ----------------
production-derived scenarios      paired monitored / "unmonitored" conditions [34]
synthetic tools that look real    CoT monitoring for "this is a test" [8][36]
avoid salient cues and tidy       interpretability probes / steering on
binary dilemmas [33]             eval-awareness features [38]
deployment-time monitoring as     condition results on verbalised awareness
a complement to pre-deploy evals  (report rates both ways) [33]
```

8. Reference architecture: a dangerous-capability evaluation program

```
+------------------------------ Governance ---------------------------------+
| thresholds (CCL / ASL / High-Critical) | alert thresholds | review body       |
| external reviewers / AISIs / LTBT      | disclosure rules | non-compliance ch.|
+----------------------------------------+------------------------------------+
                                         |
+-------------------------- Evaluation plane (Inspect-style) -----------------+
| Capability evals        | Propensity / alignment | Safeguard evals             |
| - proxies (WMDP)        | - Petri-style auditors | - HarmBench/JBB/StrongREJECT|
| - CTFs, Cybench, ranges | - agentic misalignment | - classifier ASR+FPR+cost   |
| - METR-style horizons   | - SHADE / ControlArena | - bug bounty / public demo  |
| - PaperBench / RE tasks | - auditing games       | - garak/PyRIT regression CI |
| elicitation floor: tools, best-of-N, CoT, fine-tune elicitation, scaffolds    |
+-----------------------------------------+-----------------------------------+
                                          |
+------------------ Deep-dive plane (gated by alert thresholds) ---------------+
| expert red-teaming (100+ h) | human uplift trials | classified govt testing   |
+-----------------------------+-------------------------------------------------+
                              |
+------------------------------ Validity plane ------------------------------+
| sandbagging checks | eval-awareness rates | contamination canaries/licences  |
| monitored vs unmonitored | CoT / interp probes | reproducible judges          |
+-----------------------------+-------------------------------------------------+
                              v
        report: capability determination + safeguards case + residual risk
```

10. Benchmark Validity and Statistical Rigor: Contamination, Saturation, Construct Validity, Error Bars, and Reproducible Leaderboards

Overview

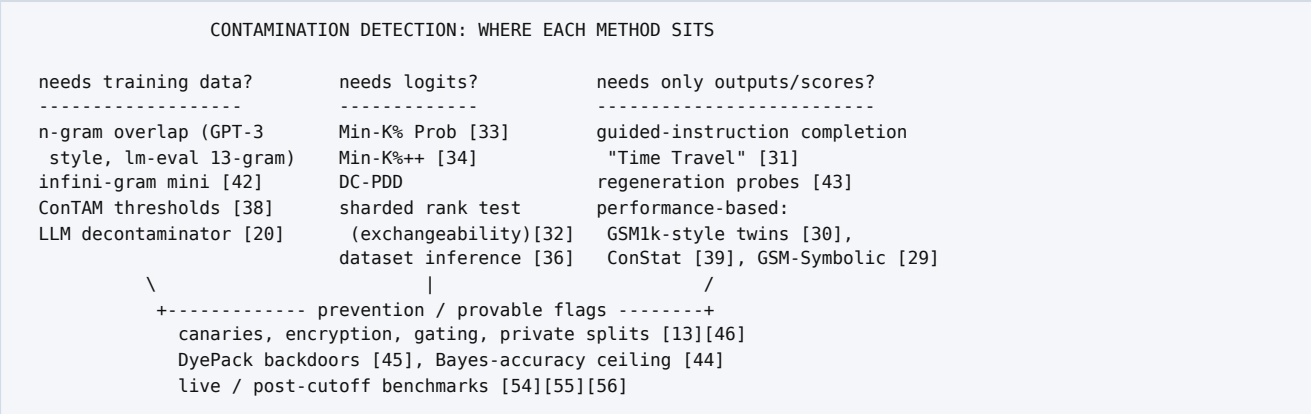
```
                  THE VALIDITY STACK FOR A BENCHMARK NUMBER
+-------------------------------------------------------------------------------+
| Claim: "Model A > Model B on capability C"                                    |
+-------------------------------------------------------------------------------+
| 4. Construct validity   items + grader actually measure C?        (10.7)     |
| 3. Headroom             benchmark not saturated?                   (10.6)     |
| 2. Internal validity    no contamination / config drift / prompt  (10.4-10.5, |
|                         artefacts / judge bias?                    10.8)     |
| 1. Statistical validity difference >> question, sampling and      (10.2-10.3) |
|                         infrastructure noise?                                 |
+-------------------------------------------------------------------------------+
| Raw material: items, harness, model endpoint, grader, aggregation             |
+-------------------------------------------------------------------------------+
```

10.2.1 Anthropic's "Adding Error Bars to Evals"

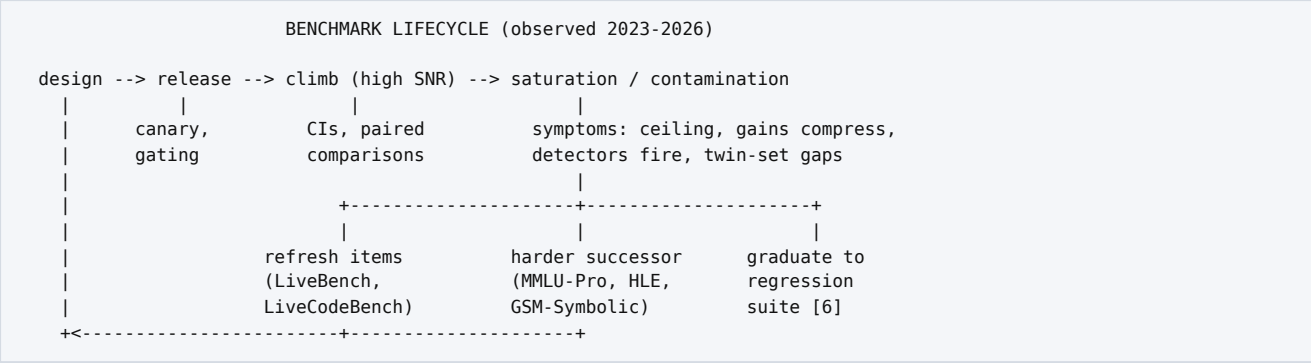
```
         ERROR-BAR PIPELINE (after Miller / Anthropic, implemented in Inspect)

 items (n, cluster_id) --> run model K epochs --> per-item scores s[i,k]
                                                          |
                        reduce within item (mean over k; or P(correct token))
                                                          |
           +----------------------------------------------+---------------------+
           |                                              |                     |
   mean, SEM (CLT)                        clustered SEM by cluster_id   paired diff vs
   95% CI = mean +/- 1.96*SEM             (t with clusters-1 dof)       baseline model:
   Wilson CI if binary                                                  d_i = sA_i - sB_i
           |                                              |             SE(d), corr(A,B)
           +--------------------> report: score, SE, CI, n, clusters, K, rho
                                          |
                        power analysis decides n and K for the NEXT run
```

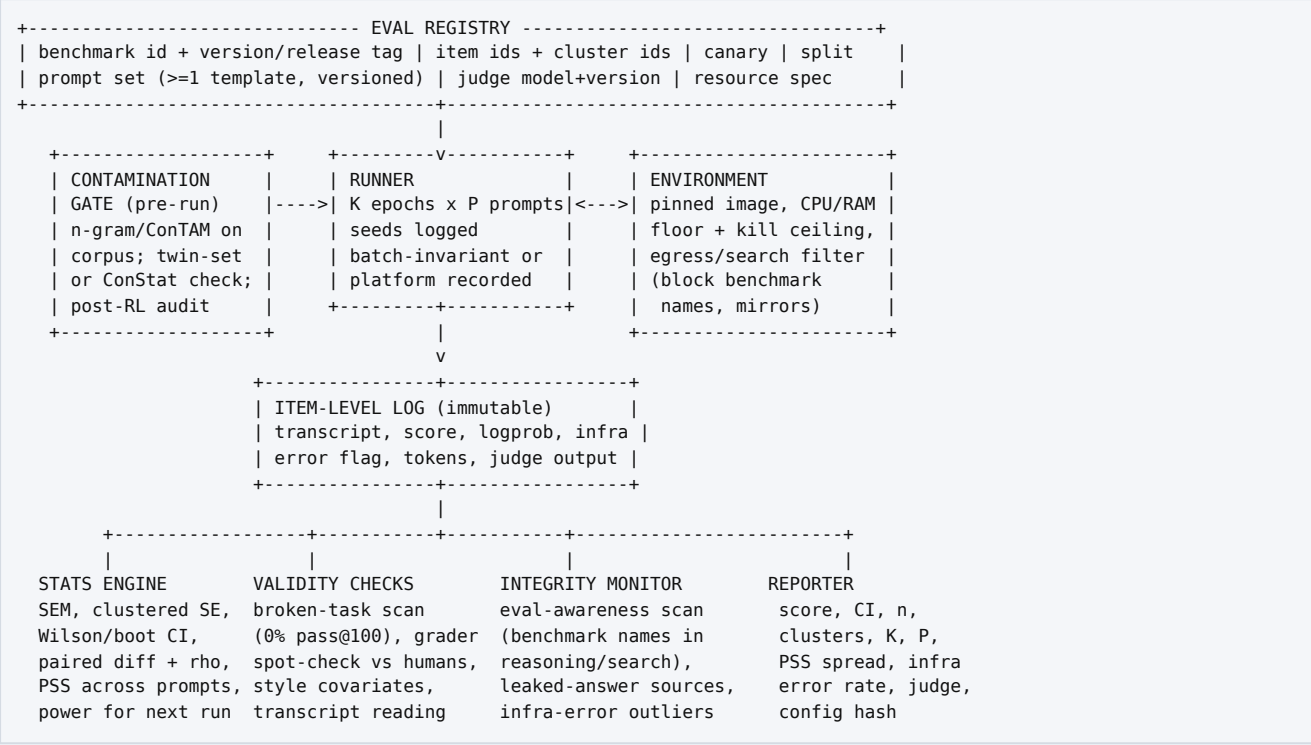
10.5.1 What counts as contamination



10.6 Saturation and the benchmark lifecycle



10.9 Reference architecture: a validity-aware evaluation pipeline



11. Factuality, Hallucination and RAG Evaluation: From Atomic Facts to Grounding Judges

Overview

```
                GENERIC FACTUALITY / FAITHFULNESS EVALUATION PIPELINE
+----------+   +-------------+   +----------------+   +-------------+   +--------+
| response |-->| decompose   |-->| evidence       |-->| verifier    |-->| aggre- |
| (+prompt,|   | atomic facts|   | - answer key   |   | - LLM grader|   | gate:  |
| context) |   | claims      |   | - Wikipedia DB |   | - NLI model |   | prec., |
+----------+   | triplets    |   | - web search   |   | - small     |   | F1@K,  |
               | sentences   |   | - given context|   |   checker   |   | rate,  |
               | tokens      |   | - cited passage|   +-------------+   | AUROC  |
               +-------------+   +----------------+          |          +--------+
                     ^                                       |
                     +------ abstention / relevance filter --+
```

1.2 SimpleQA (OpenAI, October 2024)

```
    question ---> model under test ---> predicted answer
                                              |
    gold target -----------------------------+
                                              v
                                    +-------------------+
                                    | LLM grader prompt |  few-shot rubric with
                                    | (A / B / C)       |  Obama-children examples
                                    +-------------------+
                                              |
                         correct% / incorrect% / not_attempted%
                         acc|attempted = C/(C+I);  F = HM(correct%, acc|attempted)
```

Architecture.

```
topic (person) --> LM writes bio --> sentence split --> LLM: atomic facts
                                                              |
                      Wikipedia DB (BM25/GTR retrieval by topic)|
                                    |                         v
                                    +------> verifier (ChatGPT | LLaMA+NPM)
                                                             |
                        FActScore = mean_over_responses( supported / total facts )
                                    x length penalty (gamma), plus respond_ratio
```

2.2 LongFact and SAFE (Google DeepMind, 2024)

```
response --> [split atomic] --> [make self-contained] --> [relevant?] --no--> irrelevant
                                                              | yes
                                                              v
                                          +---- LLM agent: issue query --> Serper/Google
                                          |        ^                          |
                                          |        +------ read snippets <----+
                                          |        (up to 5 steps, 3 results each)
                                          v
                                   supported / not supported
                                          |
      F1@K:  Prec = S/(S+N);  Recall@K = min(S/K, 1);  F1@K = HM(Prec, Recall@K)
```

3.3 SummaC (TACL 2022): NLI at the right granularity

```
        doc sents d1..dM
      +-----------------+
  s1  | e11 e12 ... e1M |--> ZS: max_j  | Conv: histogram -> conv -> score
  s2  | e21 ...         |--> ...
  ..  |                 |
  sN  | eN1 ...     eNM |-->
      +-----------------+
  summary score = mean over summary sentences
```

4.1 Vectara HHEM and the Hallucination Leaderboard

```
 private corpus (>7,700 articles, 50 to 24K words; news, tech, science,
 medicine, legal, sports, business, education)
          |
          v
 each LLM (temperature 0) summarizes with fixed prompt:
  "Summarize using only the information in the given passage. Do not infer...
   never exceed 20 percent of the original text's length...
   else respond 'I am unable to summarize this text.'"
          |
          +--> refusals / very short outputs --> Answer Rate
          v
 keep only documents that ALL models summarized (common subset)
          |
          v
 HHEM-2.3 (source, summary) -> consistent?  --> Factual Consistency Rate
                                                Hallucination Rate = 100 - FCR
                                                + Avg summary length (words)
```

Architecture (paper-reported, approximate).

```
(system instr, long doc <=32k tok, user request)
             |
             v
      model under test --> response
             |
  +----------+-----------+----------------+
  v                      v                v
Judge A (Gemini)     Judge B (GPT-4o)  Judge C (Claude)
  | eligible?            | eligible?      | eligible?     -- no --> score 0
  | fully grounded?      | grounded?      | grounded?
  +----------+-----------+----------------+
             v
    mean over judges, mean over examples -> FACTS Grounding score
```

6.3 ALCE (Princeton, EMNLP 2023): automatic citation benchmark

```
statement s_i  with citations {c1, c2}
   recall_i    = NLI( concat(c1,c2) => s_i )
   precision_j = c_j is "irrelevant" if  NLI(c_j => s_i) fails AND
                 NLI(other cites => s_i) still holds without c_j
```

12. Code Evaluation in Depth: From Unit-Tested Functions to Repository Agents and In-IDE Assistants

Overview

```
GEN 1  function synthesis      HumanEval (164), MBPP (~1k), APPS (~10k), CodeContests
       prompt -> function -> hidden unit tests -> pass@k
           |  problem: weak tests, saturation, contamination, Python-only
           v
GEN 2  hardened / broadened    EvalPlus (80x / 35x tests), MultiPL-E (18 languages),
       function-level          BigCodeBench (1,140 library-heavy tasks), LiveCodeBench
                               (dated contest problems), LiveCodeBench Pro
           |  problem: still single-file, still "toy" relative to real work
           v
GEN 3  repository context      RepoBench, CrossCodeEval (cross-file completion);
                               SWE-bench (issue -> patch -> FAIL_TO_PASS/PASS_TO_PASS)
                               and its family: Verified, Multimodal, Multilingual,
                               Multi-SWE-bench, SWE-PolyBench, SWE-bench-Live, Pro
           |  problem: scaffold, infra and test validity dominate the score
           v
GEN 4  end-to-end / in-product Aider polyglot (edit format + test feedback),
                               Terminal-Bench 2.0, SWE-Lancer; in-IDE evals:
                               Copilot simulation tests, Copilot Arena, acceptance
                               telemetry, A/B tests
```

12.1 The execution-grading contract

```
 +-------------+    +-------------+    +-------------------+    +----------------+
 | task spec   |--->| model /     |--->| candidate         |--->| sandboxed      |
 | (prompt,    |    | agent +     |    | (function, patch, |    | executor       |
 |  repo@sha,  |    | scaffold    |    |  edited files)    |    | (process, VM,  |
 |  issue)     |    +-------------+    +-------------------+    |  container)    |
 +-------------+                                                +-------+--------+
                                                                        |
           hidden tests / verifier (unit tests, F2P+P2P, e2e, output diff)
                                                                        |
                                        +-------------------------------v--------+
                                        | per-attempt verdict: pass | fail |    |
                                        | timeout | apply-fail | infra-error     |
                                        +-------------------------------+--------+
                                                                        |
                                 aggregate: pass@k, pass^k, resolve rate, cost, CI
```

12.3 Live and competitive programming: LiveCodeBench and LiveCodeBench Pro

```
 contests ----> scrape statement + tests + release_date + difficulty
                   |
      release_vN = all problems with date <= T_N   (v1: 400 ... v6: 1,055)
                   |
  model M, cutoff C ---> score window (C, T_N]   <-- only comparable window
                   |
  "lite" pruned tests (fast) | --not_fast (full tests)  ; timeout sensitivity < 0.5 pt
```

12.5.1 SWE-bench: issues to patches, graded by the maintainers' tests

```
 base image  (OS + language toolchain)            shared by all tasks
     |
 environment image (repo-specific deps @ version)  shared by tasks of same repo/version
     |
 instance image (task-specific setup @ base_commit)
     |
 run: git apply model_patch -> apply test_patch -> run tests -> log markers
     |
 grading.py: parse log -> F2P / P2P -> FULL | PARTIAL | NO -> results.json
```

12.6 Aider polyglot: an editing benchmark with feedback

```
for each exercise (random order, in a Docker container):
  files shown   = solution stubs only  (tests, examples, .meta hidden)
  instructions  = .docs/introduction.md + instructions.md + instructions.append.md
                  + "Don't change the names of existing functions or classes ...
                     Only use standard libraries ..."
  for try in 1..tries (default 2):
      model edits files via edit format (whole | diff | diff-fenced | architect)
      copy original tests in; run per-language command, 180 s timeout
        (.py pytest | .rs cargo test | .go go test | .js npm | .cpp | .java gradlew)
      pass -> stop
      fail -> next instructions = test error output +
              "The tests are correct, don't try and change them. Fix the code ..."
  report pass_rate_1, pass_rate_2, percent_cases_well_formed, cost, timeouts ...
```

12.7.1 Isolation levels

```
weakest                                                                  strongest
in-process guard   child process +    container,          container per    full VM
(HumanEval         rlimits/timeouts   network none         sample + infra   (EC2,Proxmox,
reliability_guard) (APPS-style)       (MultiPL-E)         orchestration    Vagrant)
                                                          (SWE-bench,
                                                          Inspect, Harbor)
```

12.9.3 Cursor and other agentic IDEs: what can and cannot be said

```
OFFLINE (pre-merge / pre-launch)                    ONLINE (post-launch)
+-----------------------------------+               +---------------------------+
| public: SWE-bench Pro / Verified  |               | acceptance & persistence  |
|   (bash-only control), TB 2.0,    |  ship gate    |   of completions [52]     |
|   Aider polyglot (edit format)    |-------------->| A/B tests, SRM checks,    |
| private: scenario suite from own  |               |   latency/crash guards [54]|
|   product traces, run N times,    |               | pairwise in-IDE prefs [49]|
|   committed baseline, cached LLM  |<--------------| failure mining -> new     |
|   responses [51]                  |  new scenarios|   offline scenarios       |
| infra: oracle runs, resource caps |               | qualitative / CUPS-style  |
|   documented [42]                 |               |   session studies [53]    |
+-----------------------------------+               +---------------------------+
```

12.11 A reference architecture for a code-evaluation platform

```
                      +-----------------------------+
 task sources --------->| TASK REGISTRY (versioned)   |  task = {spec, repo@sha,
 (public, dated, private,| oracle solution, verifier,  |   image digest, verifier,
  product traces)       | release_date, license tag   |   oracle, created_at}
                        +--------------+--------------+
                                       |
       +-------------------------------v-------------------------------+
       | ORACLE GATE: run reference through full harness -> must = 100% |
       +-------------------------------+-------------------------------+
                                       |
 model x scaffold x budget --->+-------v--------+     +---------------------------+
 (pinned prompt, tools,        | ROLLOUT POOL   |---->| SANDBOX FLEET (per trial)   |
  trials N, thinking budget)   | (agent phase,  |     | network none by default,    |
                               |  offline)      |     | floor/ceiling resources,    |
                               +-------+--------+     | fresh fs, no .git history   |
                                       |              +---------------------------+
                                       v
                      +-----------------------------+
                      | RE-GRADE IN FRESH SANDBOX   |  verdicts: pass | fail |
                      | (locked protocol)           |  timeout | apply-fail |
                      +--------------+--------------+  infra-error (excluded+logged)
                                     |
        +----------------------------v-----------------------------+
        | ANALYSIS: pass@1, pass^k, CI over tasks AND trials, cost, tokens,|
        | time window, per-language/repo slices, memorization probe, diffs |
        | vs committed baseline; transcripts sampled for human review      |
        +-------------------------------------------------------------------+
```

13. Multimodal and Speech Evaluation: MMMU, MathVista, Video Benchmarks, Image Generation Evals, and Speech, ASR, TTS and Voice-Agent Evals

Overview

```
          THREE EVALUATION ARCHITECTURES BY WHERE THE MODALITY SITS
+-------------------------------------------------------------------------+
| (1) UNDERSTANDING        media + question --> model --> text --> extractor  |
|     MMMU, MathVista,                                    |               |
|     Video-MME, MMAU         rule match / LLM choice extractor --> accuracy  |
|     risk: question answerable WITHOUT the media (blind solvability)         |
+-------------------------------------------------------------------------+
| (2) GENERATION           prompt --> generator --> image / video / audio     |
|     GenEval, VQAScore,                           |                          |
|     HPS, VBench, Seed-TTS   detector | VQA model | ASR+WER | SV-embedding   |
|                             | reward model | MOS predictor | human arena    |
|     risk: grader is itself a model with blind spots; reward hacking         |
+-------------------------------------------------------------------------+
| (3) INTERACTION          user simulator <==audio, clock==> voice agent      |
|     Full-Duplex-Bench,         |                                 |          |
|     tau-Voice                  +-- turn-taking metrics, latency, tool calls,|
|                                    task success (DB state / judge)          |
|     risk: simulator realism; non-determinism of real-time APIs              |
+-------------------------------------------------------------------------+
```

13.1.4 MMBench: CircularEval and LLM choice extraction

```
           MMBench CIRCULAR EVALUATION (4-option item)
  pass 1:  A=cat  B=dog  C=fox  D=owl   gt=B  -> model "B"  ok
  pass 2:  A=dog  B=fox  C=owl  D=cat   gt=A  -> model "A"  ok
  pass 3:  A=fox  B=owl  C=cat  D=dog   gt=D  -> model "D"  ok
  pass 4:  A=owl  B=cat  C=dog  D=fox   gt=C  -> model "B"  FAIL
  item score = AND(all passes) = 0     cost = 4x inference
```

13.2.2 Video-MME-v2 (2026): grouped non-linear scoring

```
     VIDEO-MME-v2 GROUPED SCORING (one group = 4 linked questions)
  Q1 retrieve fact   Q2 order events   Q3 causal link   Q4 inference
     correct            correct           wrong            correct
  per-question accuracy contribution : 3/4 = 0.75
  group (non-linear) score           : heavily discounted, because the
                                       reasoning chain is broken at Q3
  ratio NonLin/Avg across groups     -> robustness index per model
```

13.3.2 VLMEvalKit (OpenCompass)

```
          GENERIC MULTIMODAL UNDERSTANDING HARNESS (lmms-eval / VLMEvalKit)
 task YAML --> loader --> media fetch/decode --> prompt builder (frames, subs,
     |                    (TorchCodec, frame       options, CoT/direct flag)
     |                     sampler: N frames)              |
     |                                                     v
     |                                  model adapter (HF / vLLM / API)
     |                                                     |
     |                    cache (sha256 key) <-------------+
     |                                                     v
     |                    strip <think> --> rule extractor --fail--> LLM extractor
     |                                                     |
     +--> metric: acc | CircularEval | group score | WER | judge score
                    --> CI, clustered SE (by video/image), paired diff
```

13.4.1 GenEval: object detection as the grader

```
prompt templates --> T2I model (4 images / prompt) --> Mask2Former detector
(seeded generator)                                        (mmdetection)
                                                              |
                  boxes + masks + classes  ------------------+
                              |
     count check / relative-position check / color classifier on crops
                              |
         per-image {correct, reason} JSONL --> per-task + overall score
```

13.5.1 ASR: WER, normalization and the Open ASR Leaderboard

```
model family Docker image (HF Space config)  ---+
                                                v
HF Jobs: one job per dataset on fixed hardware (1x H200, 23 vCPU, 256 GB)
       |                 |                  |
 EN short-form      multilingual        long-form (earnings21/22, CORAAL)
 (main test sets)   (FLEURS, MCV, MLS)  chunked sets: concatenate chunk
       |                 |                  preds -> session-level WER
       v                 v                  v
 predictions JSONL --> repo EnglishTextNormalizer --> WER  +  RTFx
       |
       +--> results bucket (public) --> leaderboard + private-set eval
                                     --> benchmark-fitting diagnostics
```

13.5.2 TTS objective metrics: WER-via-ASR and speaker similarity

```
(prompt audio, prompt text, target text) --> TTS --> synthesized wav
                                                      |        |
                   ASR (Whisper-large-v3 / Paraformer) |        | WavLM-SV
                                                      v        v
                            WER vs target text    cos(emb_syn, emb_ref)
```

Provider integrity policy.

```
            TTS ARENA VOTE PIPELINE
Random-pool prompt --> 2 anonymous providers (router) --> listen gate
       |                                                      |
  custom prompt? -> listen-only, never counted                v
                                                          vote A/B
                                                              |
     risk score --high--> shadow-exclude (stored, not counted)
            |low
            v
  ring sweep + per-voter recognition test --flag--> retro-exclude
            |
            v
  Bradley-Terry over full clean history --> rank by CI lower bound
  (>=100 votes to list, <300 = Preliminary, suspended models at bottom)
```

13.6.3 τ-Voice: full-duplex customer-service simulation

```
                  tau-VOICE DISCRETE-TIME SIMULATION (tick = 0.2 s)
LLM user simulator --text--> ElevenLabs TTS persona --> audio effects
  (e.g. gpt-4.1)                                      (noise, bursts, frame drops)
      ^                                                       |
      |                                    G.711 mu-law 8 kHz telephony
      |                                                       v
  transcript  <-- STT (Deepgram nova / gpt-4o-transcribe) |
      |                                                       v
      |                      DiscreteTimeAdapter (per provider)
      |                OpenAI Realtime | Gemini Live | xAI | Nova Sonic |
      |                Qwen Omni realtime | LiveKit cascaded STT-LLM-TTS
      |                                                       |
      +------------- agent audio + tool calls <----------+
                                |
                        domain tools / DB  --> end-state task scoring
 artifacts: stereo both.wav, Audacity label tracks (user, agent, tool calls)
```

13.8 Reference architecture: a multimodal eval stack for a product team

```
LAYER 4  Human preference          blind pairwise (arena-style), BT + CI lower
         (release gates)           bound, recognition test, fixed prompt pool
-------------------------------------------------------------------------------
LAYER 3  Interactive simulation    tau-Voice / FDB-style: simulated user audio,
         (voice / agent)           telephony codec, noise/accent ablations,
                                   tool-state scoring, latency percentiles
-------------------------------------------------------------------------------
LAYER 2  Generation graders        detector checks (GenEval-like), VQAScore,
         (per checkpoint)          ASR-WER + SV-SIM, MOS/aesthetics predictor,
                                   MLLM judge with pinned open model
-------------------------------------------------------------------------------
LAYER 1  Understanding suites      MMMU-Pro (vision-only), MathVista, CharXiv,
         (per checkpoint)          Video-MME(-v2) with fixed frames, MMAU,
                                   VoiceBench audio-vs-text; with/without-media
                                   ablation; clustered SE; pinned extractor
-------------------------------------------------------------------------------
INFRA    content-hash cache, media-aware batching, decoupled scoring,
         reasoning-tag stripping, fixed hardware for latency/RTFx claims
```

14. Evals as Training Signal: Reward Models, Verifiers, Rubric Rewards, Process Supervision and Reward-Hacking Detection

Overview

```
                THE EVALUATOR-AS-REWARD SPECTRUM

 cheap to hack                                              hard to hack
 broad coverage                                             narrow coverage
 <-------------------------------------------------------------------------->
 user thumbs   BT reward   generative /   rubric /      answer    unit tests   Lean /
 (approval)    model       reasoning RM   checklist     match     in sandbox   proof
               (scalar)    (critique+     judge         (rule)    (execution)  kernel
                           score)         (per-item)
    [48]        [1][2]      [21][22][23]   [35][36][37]  [25][27]  [28][29]     [30]
```

1.2 Architecture

```
loss = - log sigmoid( r(x, y_chosen) - r(x, y_rejected) )

prompt x --+--> [ LM backbone ] --> last-token hidden state --> [ linear head ] --> r
response y +
```

1.4 Failure modes: overoptimization

```
best-of-n:  R_gold(d) = d (alpha_bon - beta_bon * d)
RL:         R_gold(d) = d (alpha_RL  - beta_RL  * log d)

gold  ^            ___
reward|         .-'   '-.          <- proxy RM keeps rising
      |       .'         '.
      |     .'             '.      <- gold peaks, then falls
      |   .'
      +---------------------------> d = sqrt(KL)
```

4.2 Let's Verify Step by Step (OpenAI, 2023)

```
solution = s1 s2 s3 ... sk
             |  |  |      |
            [ PRM: P(step correct | prefix) per step ]
             |  |  |      |
            .98 .95 .40 ...       score = prod(p_i) or min(p_i)
                     ^ first error localized
```

5.3 Rubrics as Rewards (Scale AI, 2025)

```
prompt + expert reference --> rubric generator --> [ (criterion_i, weight_i) ... ]
                                                                    |
policy samples y_1..y_G --> judge(prompt, y_j, rubric) ------+
                                |  explicit: per-criterion verdicts -> weighted sum
                                |  implicit: one score given the full rubric
                                v
                          rewards r_1..r_G --> GRPO advantage --> policy update
```

6.4 DeepSeekMath-V2: verifier, meta-verifier and generator co-training

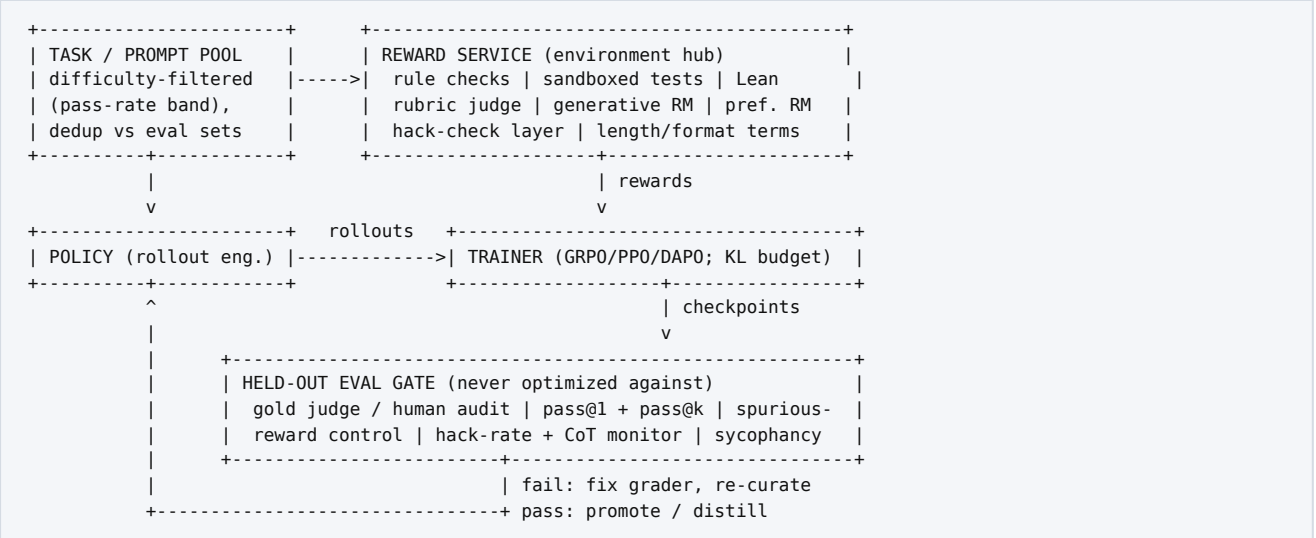
```
+-----------+  proofs   +-----------+  analyses  +---------------+
| generator |---------->| verifier  |----------->| meta-verifier |
| (+ self-  |<----------| score +   |<-----------| are the found |
|  analysis)|  rewards  | issues    |  R_meta    | issues real?  |
+-----+-----+           +-----+-----+            +---------------+
      | hard new proofs       ^
      v                       | retrain on auto-labeled data
  [ n verifications x m meta-verifications -> label or escalate to humans ]
```

7.3 Chain-of-thought monitoring and its limits

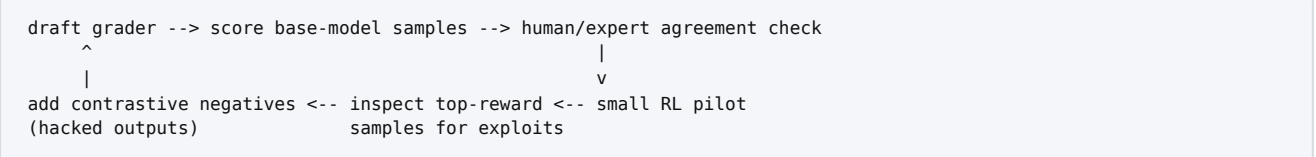
```
               DETECTION STACK (keep every layer OUT of the reward)

 rollouts --> [ env hardening: sandbox, read-only tests, hidden test split ]
          --> [ rule-based hack checks: test-file diffs, exit codes, claims vs state ]
          --> [ CoT + action monitor (LLM) -> suspicion score ]  --> dashboards
          --> [ proxy vs gold divergence: RM score up, held-out judge/human down ]
          --> [ periodic behavioral evals: sycophancy, sabotage, shortcut rate ]
                                |
                     alert -> fix environment / grader, not the gradient
```

8.1 Reference architecture

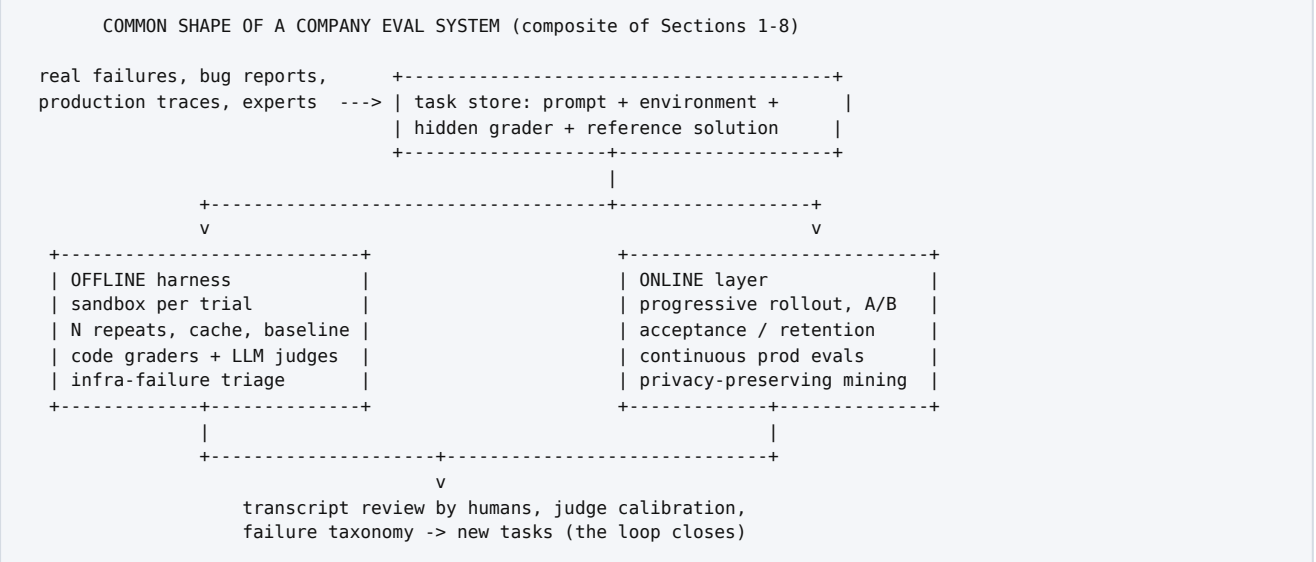


8.2 Grader development as an eval loop



15. Company Eval Systems in Practice: What Labs and Product Companies Have Published (2023-2026)

Overview



1.2 Offline architecture: the simulation-test ("stest") harness

```
 developer PR
     |
     v
+------------------+   cache hit    +---------------------------+
| simulation suite |--------------->| layered cache (SQLite):   |
| N=10 runs / test |<---------------| chat, completions, embeds |
+--------+---------+   cache miss   +-------------+-------------+
         |             (team only)               |
         v                                       v
+------------------+                  real Copilot API endpoints
| scorecard + new  |
| baseline.json    |---- diff vs committed baseline ----> PR check
+------------------+     (uncommitted change => fail)
```

2.2 Architecture

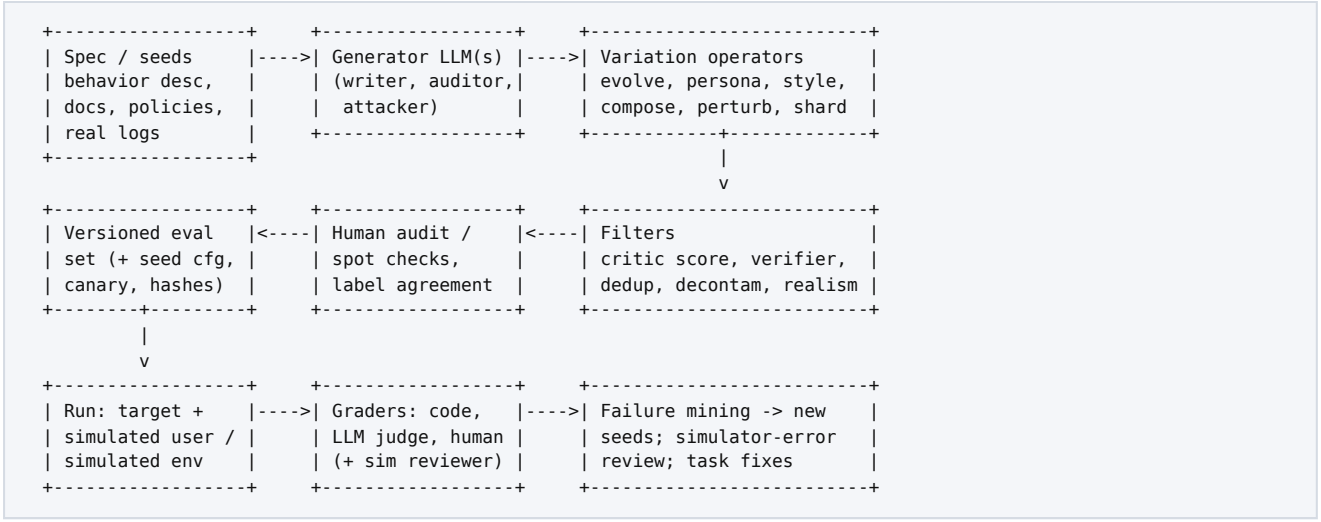
```
PROMPT.md --> agent (Claude Code / Codex / Gemini / Cursor / gateway model)
                |  runs inside isolated sandbox
                v
        modified Next.js app ----> EVAL.ts (hidden) + typecheck/build
                |                          |
                v                          v
        transcript + token usage     pass/fail per run (up to 4 runs,
        -> list-price cost           early exit on first pass = pass@4)
                |                          |
                +------------+-------------+
                             v
      fingerprint memo -> agent-results.json -> public board
      (tier 1: current models on latest evals; tier 2: dated history)
```

6.2 Assurance evaluations and automated red teaming

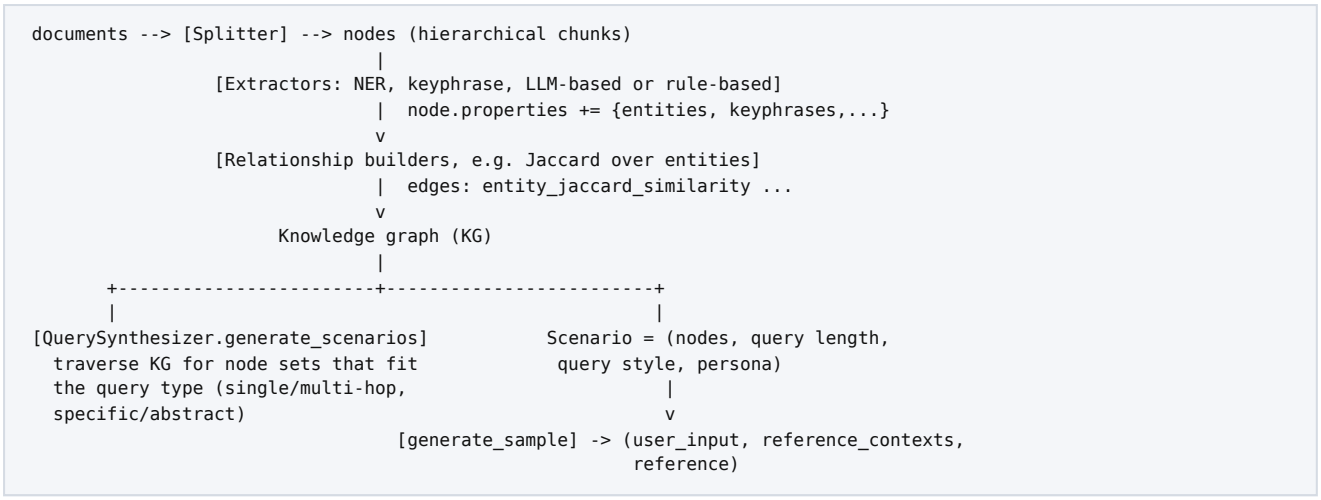
```
model candidate
     |
     +--> dev evals (public + internal, pass@1, decontaminated) ---> dev team
     |
     +--> ART: attacker LLMs <--reward-- judge LLMs (policy-seeded)
     |          |                                    ^
     |          +---- probes ----> target ----------+
     |
     +--> ASSURANCE evals (separate team, held-out prompt sets)
     |          |
     |          +--> release decision; only high-level findings returned
     v
Frontier Safety Framework evals (CBRN, cyber, ML R&D, deceptive alignment)
```

16. Synthetic Data for Evaluation: LLM-Generated Test Cases, Adversarial Generation, Persona and User Simulation, and Dataset Curation

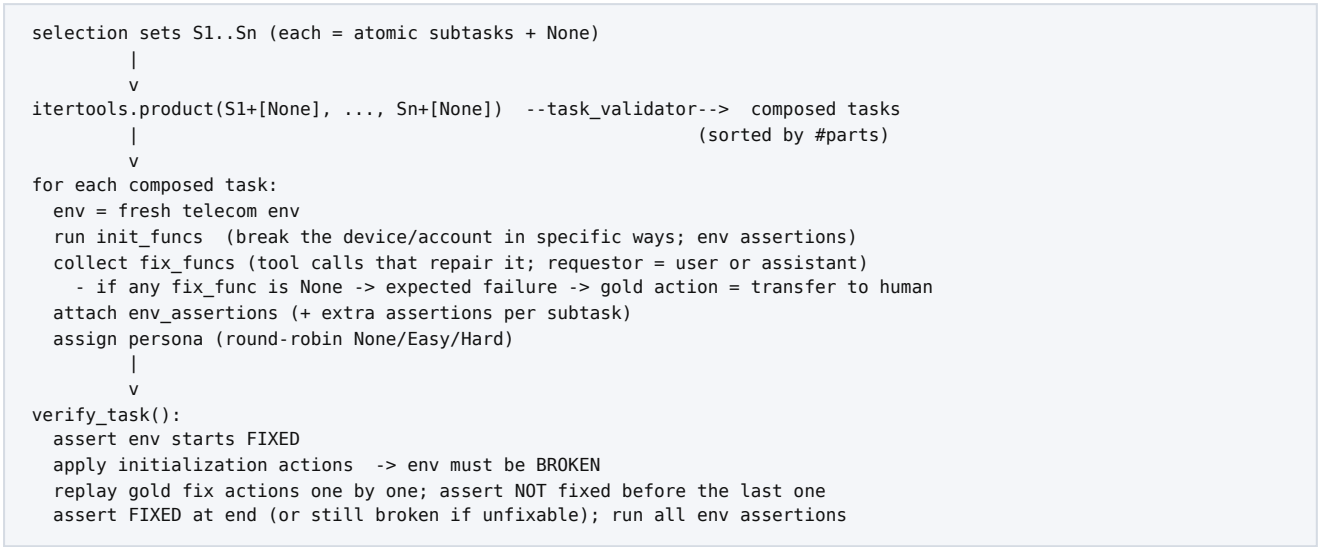
A reference pipeline for synthetic eval data



3.1 Ragas: knowledge-graph-driven synthesis



5.4 Compositional task algebra with replay verification: τ²-bench telecom



Architecture.

```
task = (user instruction: identity, goal, constraints, hidden facts)
           |
           v
+--------------------+   natural language   +--------------------------+
| User simulator LLM |<-------------------->| Agent under test         |
| strategy: llm |    |                      | (policy doc + domain API  |
| react | verify |   |                      |  tools)                   |
| reflection         |                      +-------------+-------------+
+---------+----------+                                    | tool calls
          | emits ###STOP### when goal satisfied          v
          |                                     +-------------------+
          +------------------------------------>| Database / env    |
                                                +---------+---------+
                                                          |
reward = [final DB state == gold DB state] x [required info communicated]
pass^k = P(all k i.i.d. trials succeed)
```

7.1 Anthropic Bloom (December 2025)

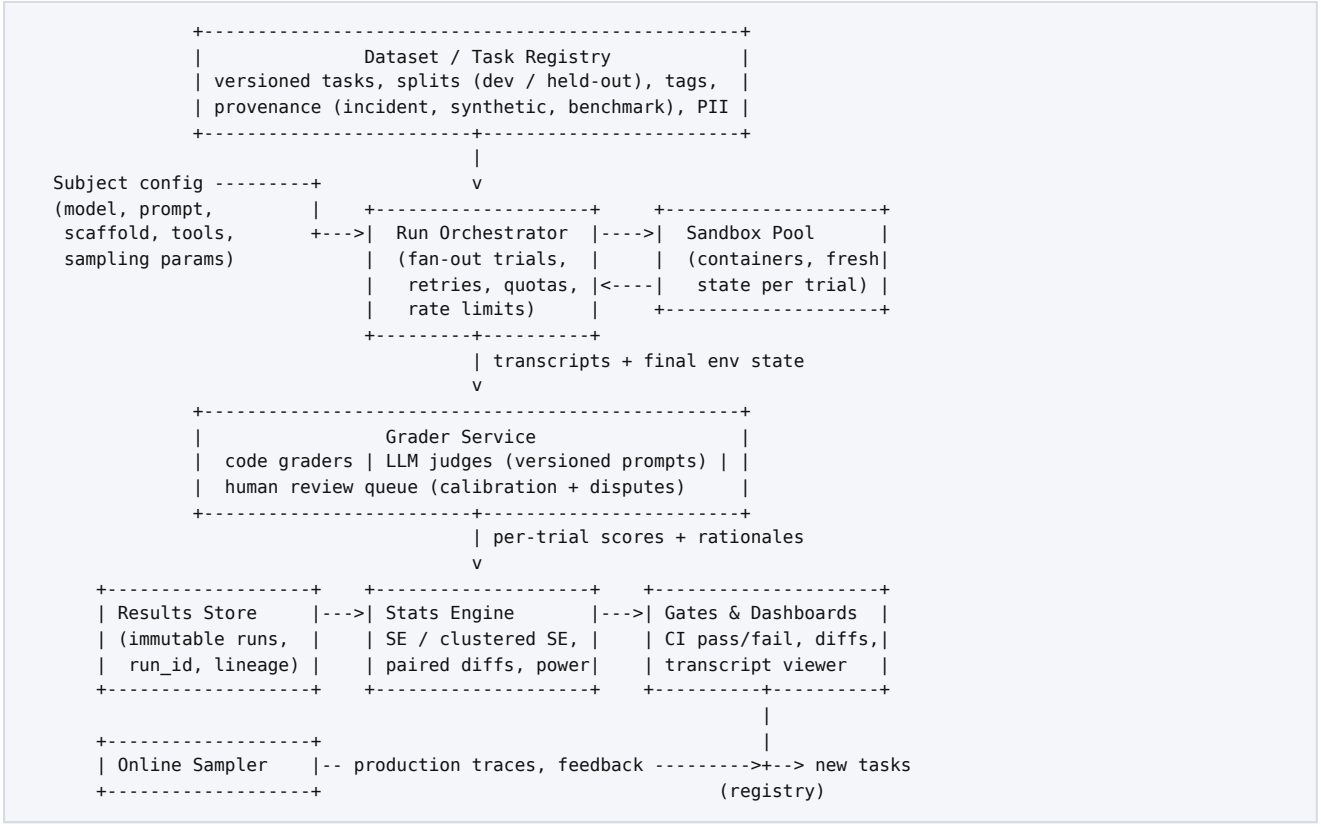
```
seed.yaml: behavior name + description, example transcripts, models per stage,
           num_scenarios, variation_dimensions, modality (conversation | simenv)
    |
    v
1 UNDERSTANDING  agent analyzes behavior + examples -> what to measure
    |
    v
2 IDEATION       scenarios: situation, simulated user, system prompt, environment
                 + variations along named dimensions (e.g. noise, emotional_pressure)
    |
    v
3 ROLLOUT        parallel conversations; Bloom simulates user AND tool responses
    |
    v
4 JUDGMENT       judge scores behavior presence (+ secondary qualities, e.g. realism);
                 meta-judge writes suite-level report
    -> metrics: elicitation rate, average behavior presence; Inspect-compatible logs
```

8.1 Plugins × strategies → probes (promptfoo)

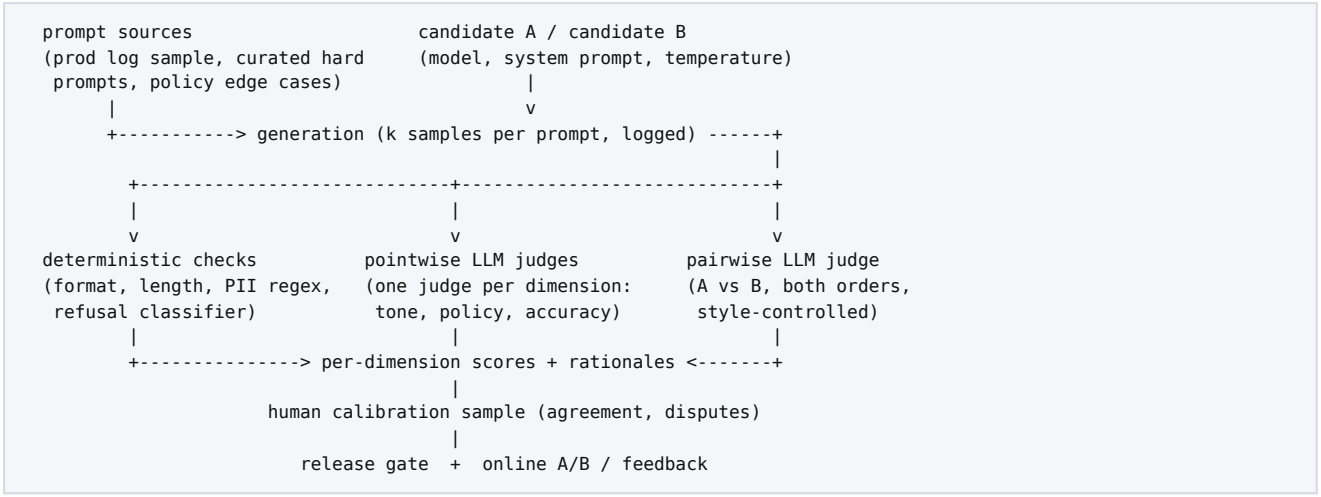
```
app purpose/policies --> plugins (what to elicit) --+
                                                    +--> probes --> target
config --------------> strategies (how to wrap) ----+                 |
                                                                      v
            +--- refine with prior turns <--- not vulnerable <--- judge/detector
            |                                                         |
            +--> next attack ...                        vulnerable -> finding
```

17. Eval System Design Interview Playbook: Platforms, Product Evals, Metrics, Sampling, and Cost

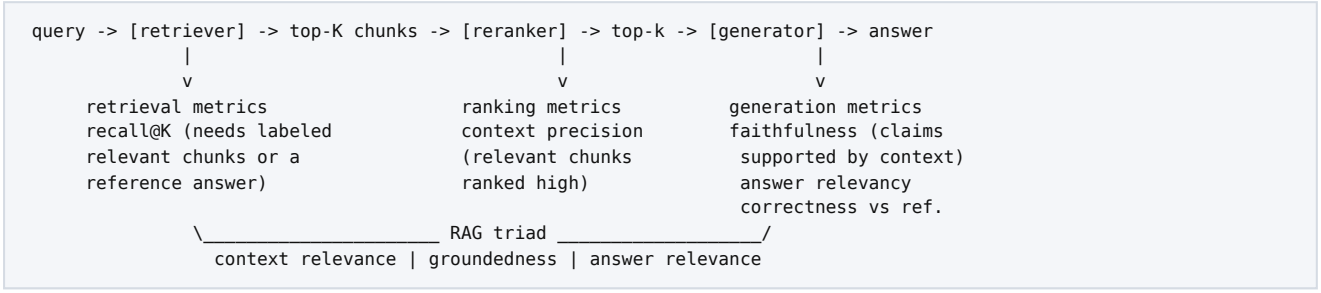
Reference architecture



Architecture



Architecture: evaluate the pipeline as decomposed stages



Architecture (tau-bench pattern)

```
task spec: user goal + hidden user info + expected end state + policy doc
      |
      v
+-------------+    natural-language turns   +----------------+  tool calls  +---------+
| User        | <-------------------------> | Agent under    | -----------> | Domain  |
| simulator   |                             | test (model +  | <----------- | DB/APIs |
| (LLM)       |                             | scaffold)      |   results    | (mock)  |
+-------------+                             +----------------+              +----+----+
                                                                                 |
   graders:  (1) end-state DB diff vs expected   (2) required outputs to user    |
             (3) policy/transcript rubric        (4) turn / token / latency caps |
                                                                                 v
                                    k independent trials per task -> pass^k
```

Architecture

```
task sources                          subject
- function-level problems (HumanEval) - completion model / chat model
- time-windowed contest problems      - agent (model + bash/edit tools)
  (LiveCodeBench)                             |
- real repo issues (SWE-bench)                v
- terminal tasks (Terminal-Bench)     +---------------------------+
      |                               | per-task container image  |
      +------------------------------>| repo @ pre-fix commit,    |
                                      | deps, no network          |
                                      +-------------+-------------+
                                                    | patch / final FS state
                                                    v
         graders: hidden unit tests (fail->pass, pass->pass) | static analysis
                  (lint, types, security) | LLM rubric on diff & transcript |
                  required tool-call patterns | cost: turns, tokens, wall time
```

9. Cost and latency engineering

```
every commit / PR   : deterministic checks + small regression suite (minutes, cents)
nightly             : full regression + capability suites, k>=3 trials, judges
release candidate   : held-out sets, k>=5, human calibration sample, safety audit
continuous (prod)   : sampled online judges, feedback mining, drift dashboards
```

4. Closed-Loop AI Systems — Architecture Diagrams

01. Foundations: Feedback Loops, Technical Debt, and the Anatomy of Closed-Loop ML Systems

1.3 A generic loop diagram

```
              +---------------------------- world ----------------------------+
              |  users, drivers, advertisers, police districts, the web ...   |
              +-------^-----------------------------------------+-------------+
                      | actions / exposures                     | outcomes
                      |                                         v
 +---------+     +-----+------+     +-----------+     +-----------------------+
 | model   +---->+  policy /  |     |  logging  +<----+ observation (labels,  |
 | f_theta |     |  decision  +---->+ (features,|     | clicks, interventions,|
 +----^----+     |  layer     |     | propensity|     | thumbs, accept/reject)|
      |          +------------+     | , action) |     +-----------------------+
      |                             +-----+-----+
      |   new theta                       | training data
 +----+-----------+   eval/gate     +-----v------+
 | deploy/rollout +<----------------+  learner   |
 +----------------+                 +------------+

 Open loop: delete the arrows from "world" back to "learner".
 Closed loop: every arrow present; theta(t+1) depends on theta(t) via the world.
```

3.1 Definition and mechanism

```
  recommend top-k by predicted CTR
            |
            v
 exposure concentrated on head items ---> clicks observed only on exposed items
            ^                                          |
            |                                          v
 model learns "head items are best" <--- training set skewed toward head items
```

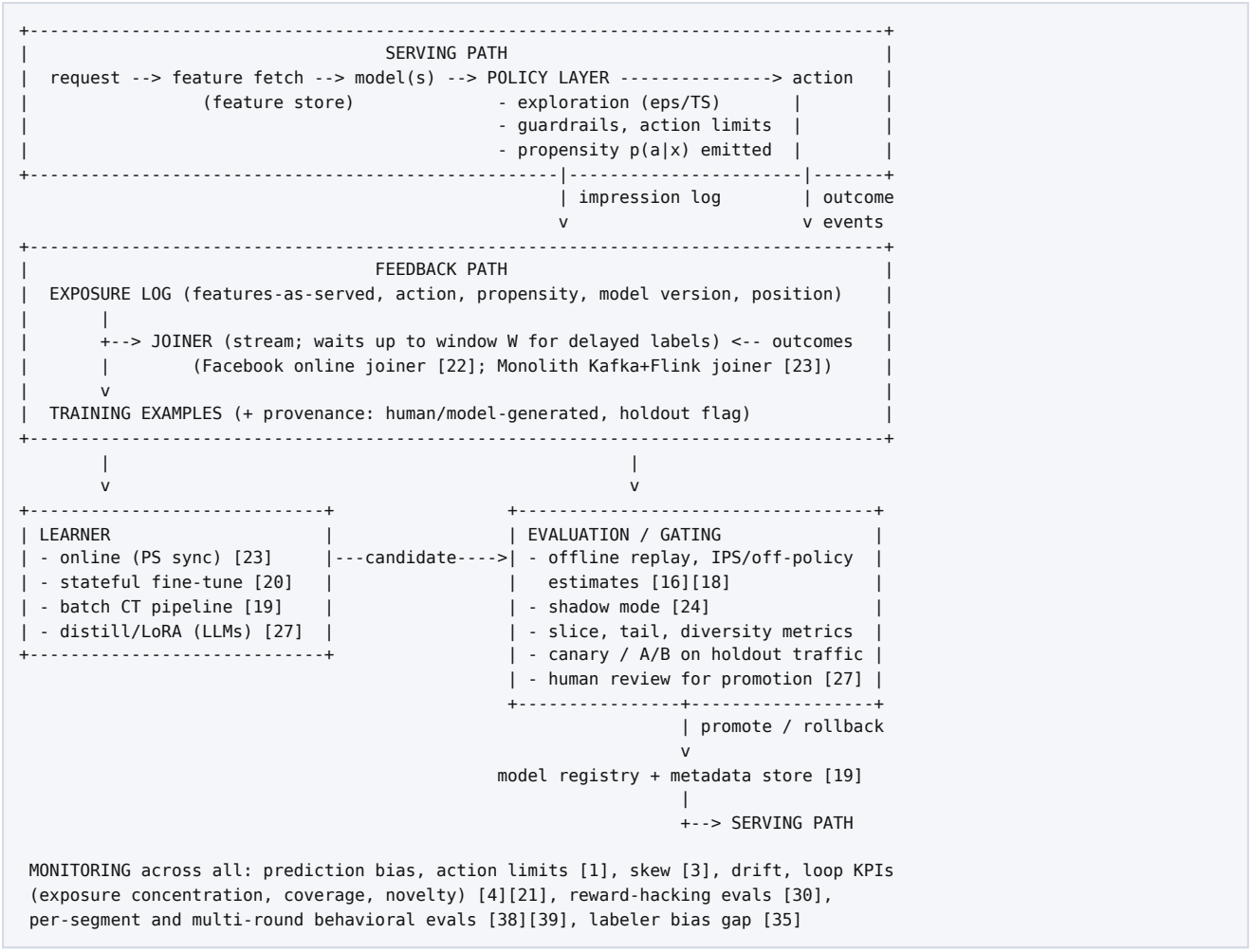
6.2 Tesla's data engine (2016 onward)

```
fleet (shadow mode) --trigger fires--> snapshot upload --> auto-label (big offline nets)
      ^                                                            |
      |                                                  human QA / correction
      |                                                            v
deploy new net <-- eval + unit tests on hard cases <-- retrain <-- dataset
```

6.3 NVIDIA Data Flywheel Blueprint (2025, deprecated April 2026)

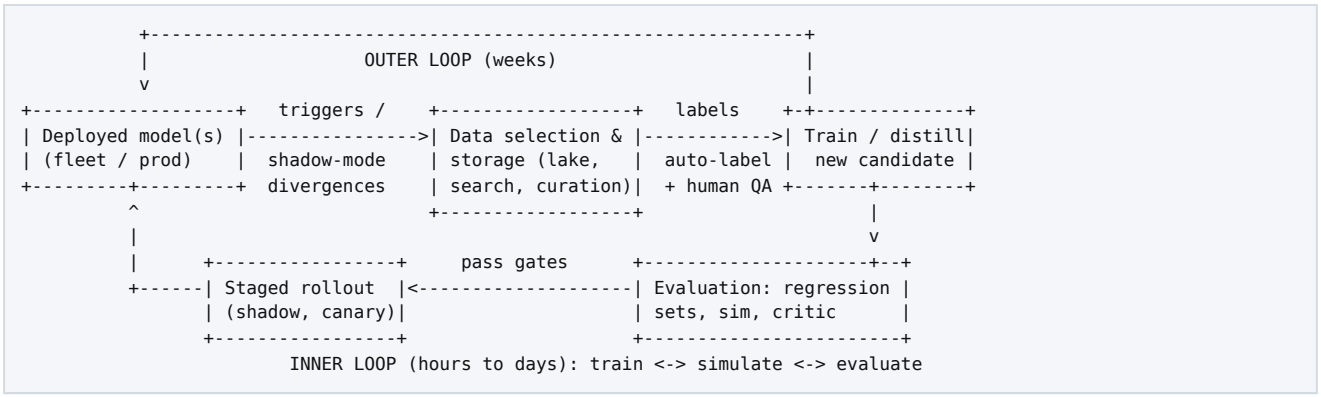
```
app (each agent node tagged with workload_id) --prompt/completion logs--> Elasticsearch
                                                                                 |
Flywheel API (FastAPI) --job--> Celery workers (Redis queue, MongoDB metadata)
                                  | dedup, group by workload, stratified split
                                  v
                            NeMo Datastore (eval + fine-tune sets)
                                  |
            +-- for each candidate NIM (spun up/down by Deployment Manager) --+
            |   base eval (replayed prompts)                                  |
            |   ICL eval (few-shot from traffic, random or semantic)          |
            |   LoRA fine-tune via NeMo Customizer -> customized eval         |
            +---------------- NeMo Evaluator (LLM-as-judge, 0..1) -----------+
                                  |
                    results --> human review --> (manual) promotion
```

7.2 Components

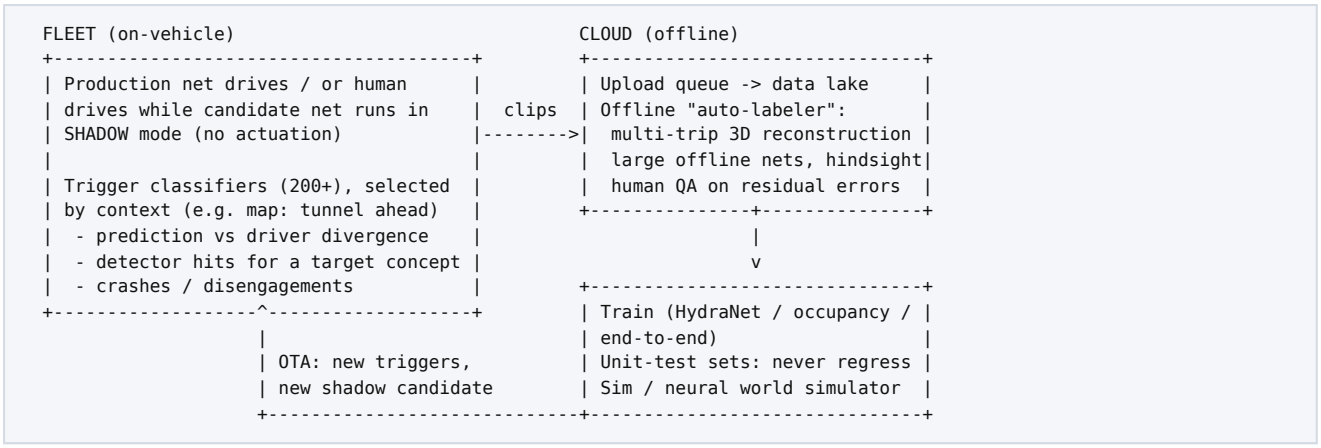


02. Data Engines and Flywheels: Fleet Learning Architectures from Tesla to NVIDIA

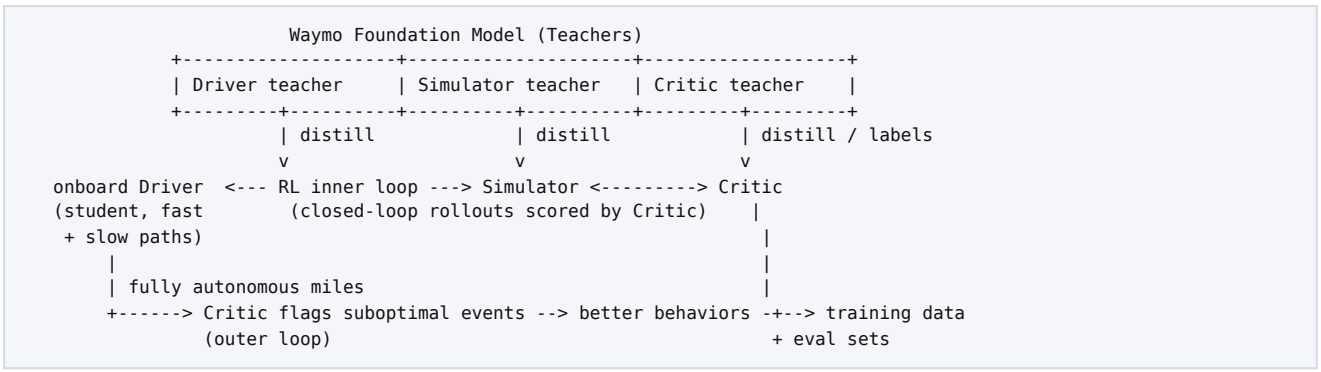
Overview



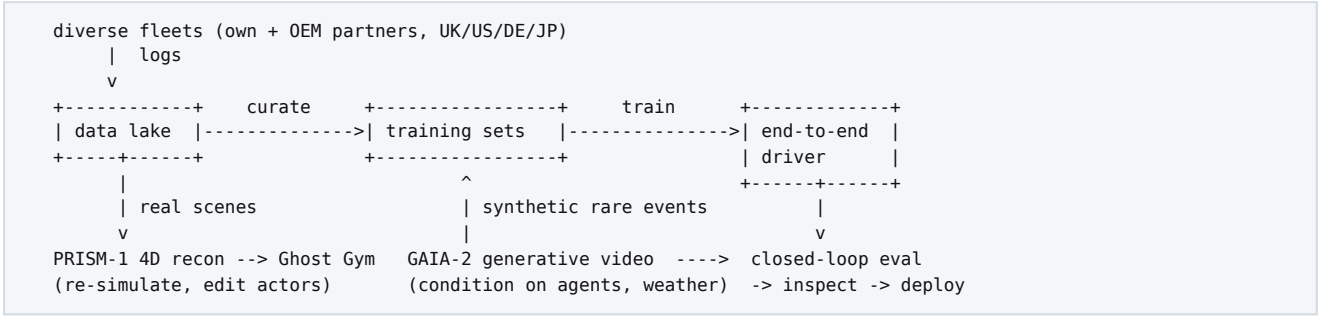
1.2 Architecture and data flow



2.4 Driver, Simulator, Critic (December 2025)



3.3 Generative and reconstructive simulation



4.2 Architecture

```
onboard prediction(t) ----+                     +--> train candidate --> eval gate
                          |  |pred - obs| > tau  |        (must beat prod)
perception track(t+k) ----+---> mined error ---->+                |
 (future = free label)          (no human)                        v
                                                            deploy to fleet
```

6.1 AI Blueprint for Building Data Flywheels (LLM agents)

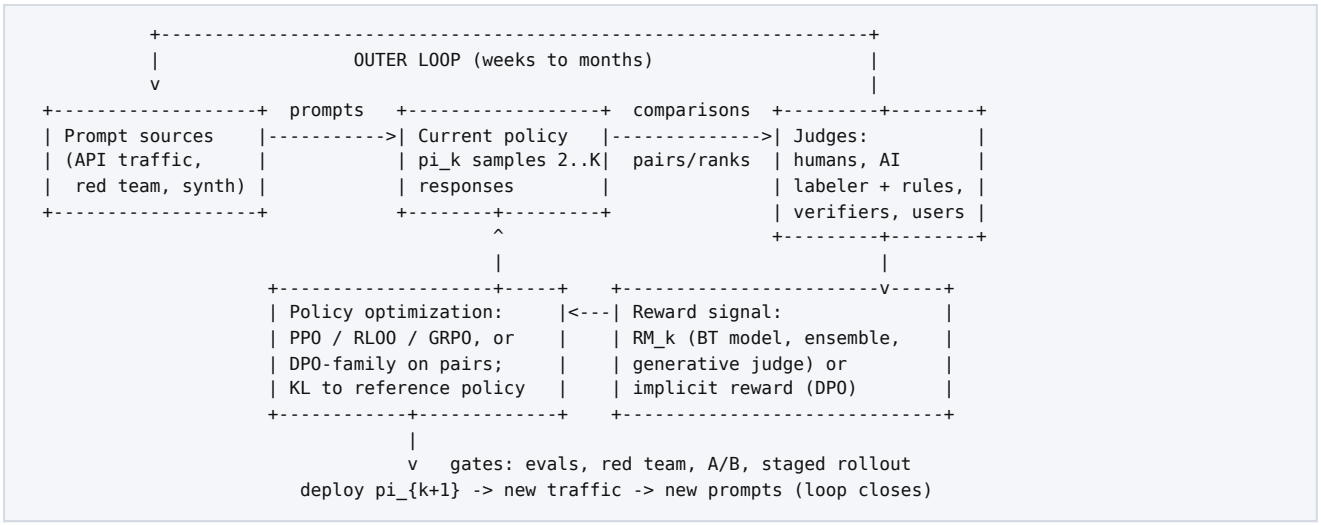
```
agent app --(prompt/completion logs, workload_id, client_id)--> Elasticsearch
                                                                  |
          FastAPI  POST /jobs  -->  Celery (Redis queue)          v
                                    parent_queue (concurrency 1) -> pull logs,
                                                                    group by workload,
                                                                    dedupe,
                                                                    stratified split
                                                                    |
         NeMo Datastore <------------- eval + fine-tune datasets --+
               |
   +-----------+------------+----------------------+
   v                        v                      v
 base eval              ICL eval               customized eval
(replay prompts)   (few-shot, random or      (LoRA SFT via NeMo
                    semantic selection)       Customizer)
   +-----------+------------+----------------------+
               v
 NeMo Evaluator: LLM-as-judge similarity in [0,1] vs production model
 NeMo Deployment Manager spins NIMs up/down; MongoDB stores results
               v
 human reviews candidates -> promote smaller model (manual decision)
```

6.2 Physical-AI and AV data flywheel

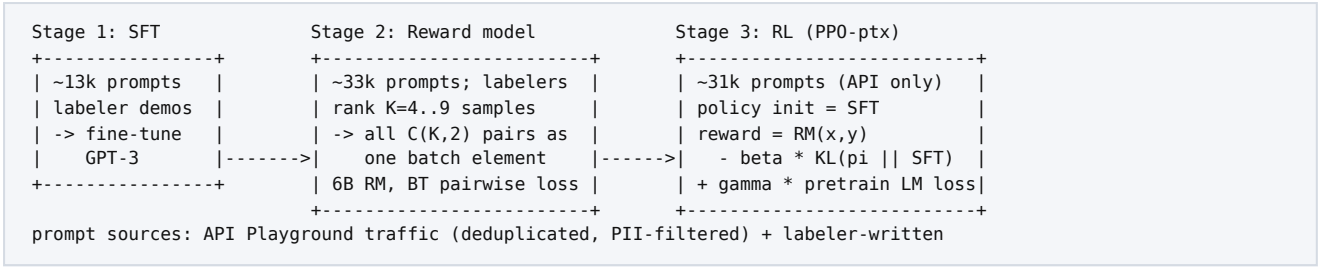
```
fleet logs --> Cosmos Curator --> labeled clips --(HDMap/bbox/depth)--> Cosmos Transfer
(any OEM)      split/filter/      + captions                           / Predict
               dedupe/annotate          |                                  |
                                        v                                  v
                                 NuRec reconstruction              synthetic variants
                                        |                                  |
                                        v                                  v
                      AlpaSim closed loop  <---- Cosmos Evaluator filters -----+
                      (policy e.g. Alpamayo)        (physical plausibility)
                                        |
                                        +--> metrics --> retrain policy --> deploy
```

03. Preference-Learning Loops: RLHF, Constitutional AI/RLAIF, DPO, and Online Reward-Model Refresh

Overview



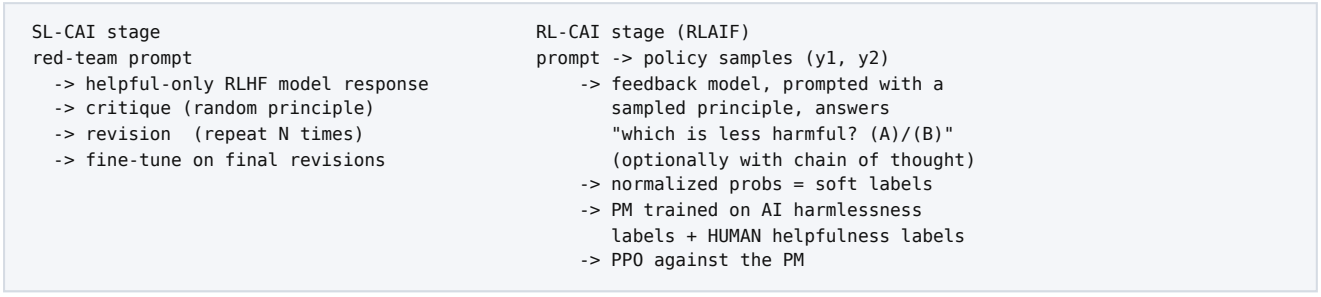
2.2 Architecture



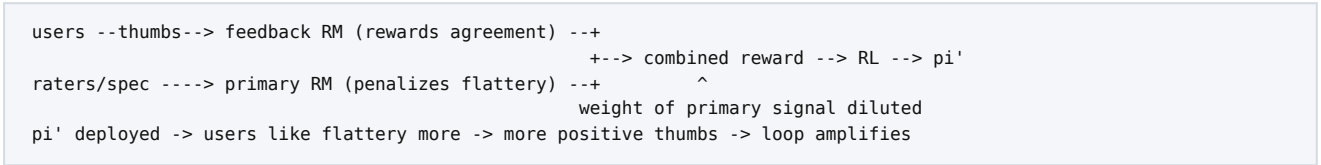
3.2 Architecture



4.2 Constitutional AI and RLAIF (Anthropic)



8.3 Case study: the GPT-4o sycophancy rollback (April 2025)



9.3 How refresh cycles are run

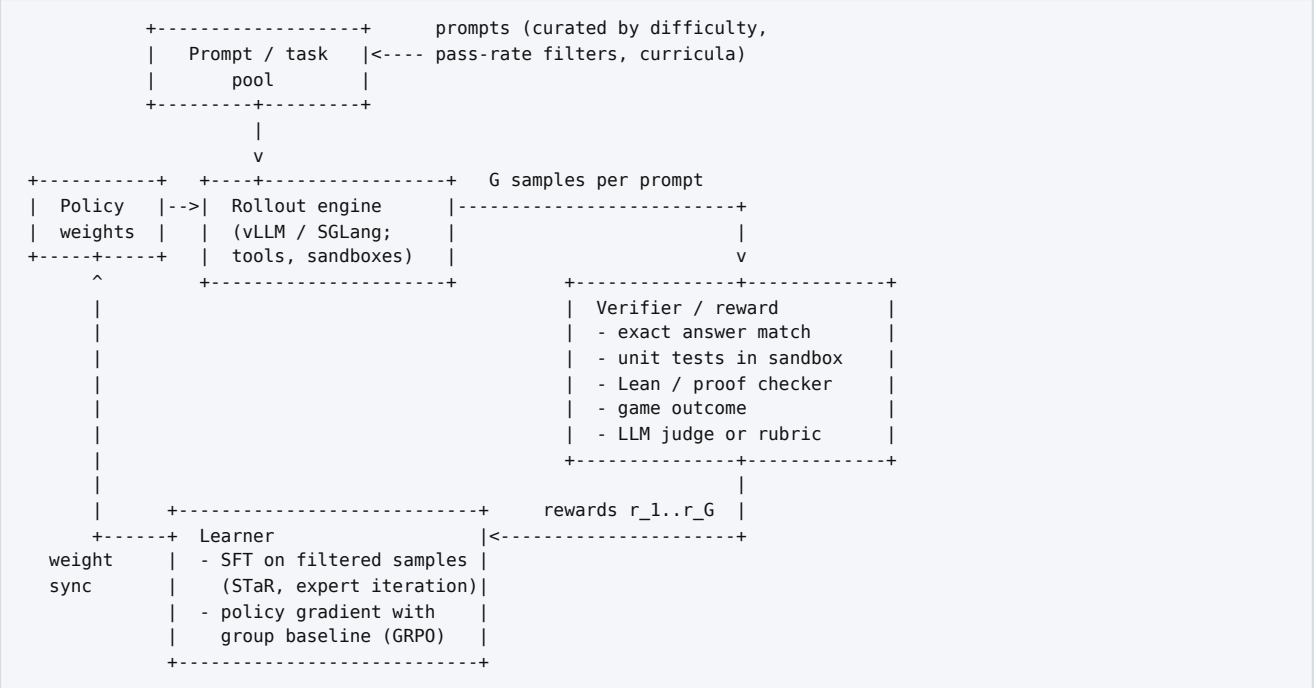
```
+--> 1. sample from CURRENT policy on fresh + replayed prompts
|    2. label: humans (taste, hard cases) + AI judge (spec compliance) + verifiers
|    3. retrain RM(s) on cumulative data (often re-weighting recent rounds)
|    4. validate RM: held-out pairs, RewardBench-style suites [29], adversarial
|       probes, agreement with gold/expert labels; check length/sycophancy bias
|    5. optimize policy (PPO/GRPO/RLOO or DPO on RM-built pairs) within KL budget
|    6. gate: evals, red team, human spot checks, A/B, staged rollout
+--- 7. detect RM exploitation (score up, human/gold down) -> new data targets
```

9.4 When reward hacking generalizes, and why not to train against the monitor

```
RL env with exploitable grader --> policy learns hack --+--> local: reward up, task not done
                                                        +--> global: misaligned generalization
mitigations: harden graders / held-out verifiers | CoT monitor as DETECTOR, not reward [69]
             | inoculation framing during RL [70] | agentic (not only chat) alignment evals
```

04. Self-Improvement via Verifiable Rewards: From STaR and Self-Play to GRPO, Reasoning RL Infrastructure, and Self-Verifying Provers

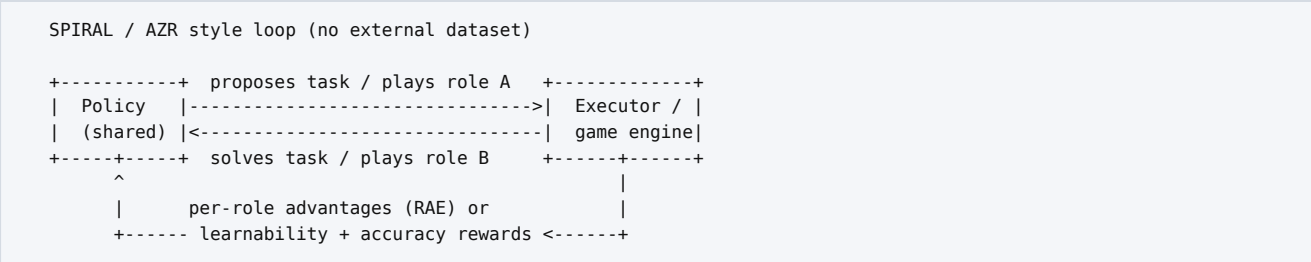
1. The verifiable-reward loop: a reference architecture



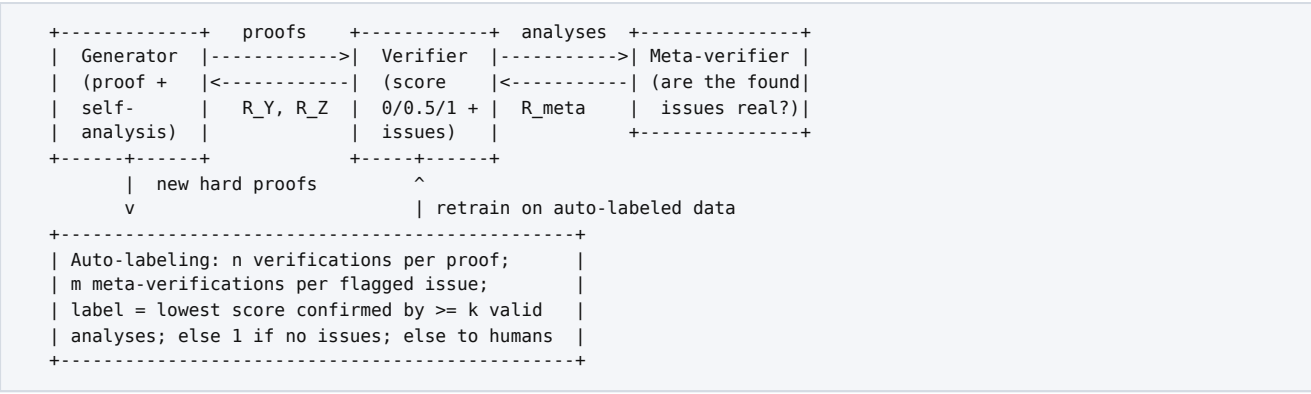
3.2 DeepSeek-R1-Zero, DeepSeek-R1, and GRPO

```
V3-Base --(1) cold-start SFT on "thousands" of long-CoT examples
       --(2) reasoning RL (accuracy + language-consistency reward)
       --(3) rejection sampling: ~600k reasoning + ~200k non-reasoning
             samples (800k total); SFT V3-Base for 2 epochs
       --(4) RL for all scenarios: rule rewards for reasoning,
             preference RMs for helpfulness (judged on the final
             summary only) and harmlessness (judged on the full output)
       --> DeepSeek-R1
                |
                +--> distill 800k samples into Qwen2.5 / Llama 3
                     (1.5B, 7B, 8B, 14B, 32B, 70B), SFT only
```

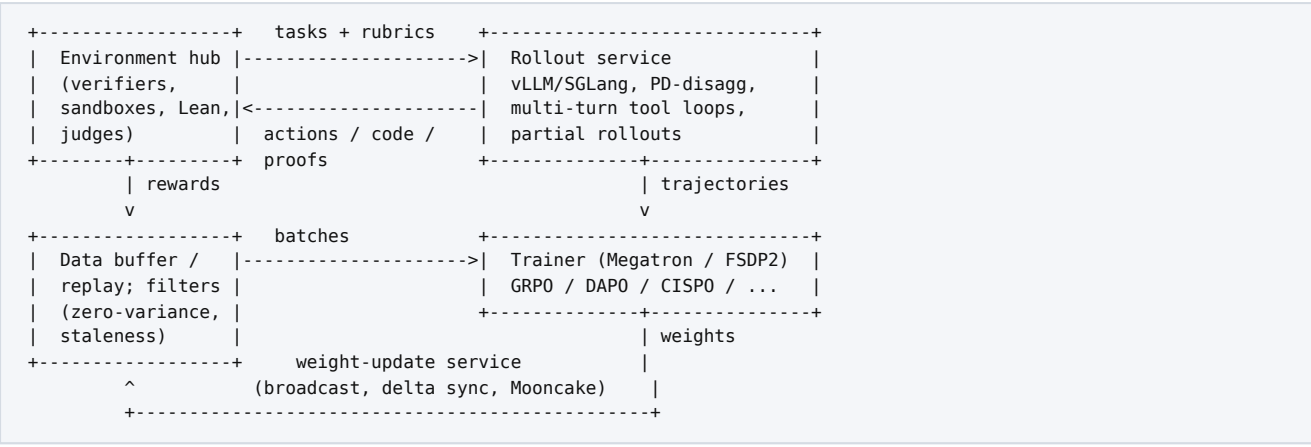
5.3 SPIRAL: self-play on zero-sum language games



6.3 DeepSeekMath-V2: closing the loop between generator and verifier



7.3 Architectural pattern: environment as a first-class service



05. Online and Continual Learning in Production: Streaming Recommenders, Bandit Loops, Forgetting, Continual Pretraining and Test-Time Training

Overview

```
LOOP LATENCY SPECTRUM (time from outcome to parameter change)

months      weeks       days        hours      minutes     seconds     one request
  |           |           |           |           |           |            |
continual   scheduled   daily        hourly     streaming   bandit      test-time
pretraining batch CT    retrain +    incremen-  online      policy      training /
(TiC-LM,    (MLOps L1)  online LR    tal fine-  training    update      TTT layers,
rewarm +                (Facebook)   tune       (Monolith,  (Decision   Titans, ARC
replay)                              (Huyen 3)  FTRL)       Service)    TTT, SEAL
<------- forgetting risk dominates -------><---- staleness/stability risk ------>
```

2.2 Architecture

```
user actions (clicks, plays, likes)            features at serve time
            |                                          |
            v                                          v
      Kafka: action queue                      Kafka: feature queue
            \                                          /
             \                                        /
              v                                      v
       +-------------------------------------------------+
       |  JOINER (Flink)                                 |
       |  key = unique request id                        |
       |  in-memory cache + on-disk KV for features whose |
       |  actions have not arrived yet                   |
       |  negative sampling applied here                 |
       +-------------------------------------------------+
                           |
                           v
              Kafka: joined training examples ----> also dumped to HDFS
                           |                          (batch training / replay)
                           v
       +-----------------------------------------+
       |  ONLINE TRAINING                        |
       |  workers <-> training PS                |
       |  - collisionless cuckoo-hash embeddings |
       |  - occurrence filter, expiry            |
       |  - touched-key set (ids since last sync)|
       +-----------------------------------------+
             | sparse: only touched keys,     | dense: full, on a
             | minute-level cadence           | slower cadence
             v                                v
       +-----------------------------------------+
       |  SERVING PS (embedding KV)  <- entry    |
       |  servers run the dense graph and call   |
       |  PS for embeddings                      |
       +-----------------------------------------+
```

3.3 Conversion delays: Alibaba ES-DFM (2021)

```
impression t0 ------------------------------------------------------> time
     |          click at t0+20s              conversion at t0+3d
     |            |                                |
Facebook (wait W):  [==== W ====] emit (pos/neg)    (missed if > W)
Twitter (FN):       emit NEG at t0 | emit POS copy at click | loss corrects
ES-DFM:             [= e =] emit; importance weights for later conversions
Monolith:           Flink join by request id; on-disk KV holds unjoined features
```

5.2 Microsoft Decision Service and MSN (2016-2017)

```
          +-----------+    context x     +--------------------+
request ->|  EXPLORE  |----------------->|  policy pi_t       |
          | (eps-grdy,|<-- action a,  ---|  (VW learner)      |
          |  bagging, |    p(a|x)        +--------------------+
          |  softmax) |                           ^
          +-----------+                           | deploy new pi_{t+1}
                | (x, a, p)                       | (minutes)
                v                                 |
          +-----------+  join reward r  +---------+----------+
          |   LOG     |---------------->|   LEARN (online)   |
          | (x,a,p,r) |                 |   + off-policy eval|
          +-----------+                 |   (IPS / DR) for   |
                                        |   safeguards       |
                                        +--------------------+
```

7.5 Design pattern

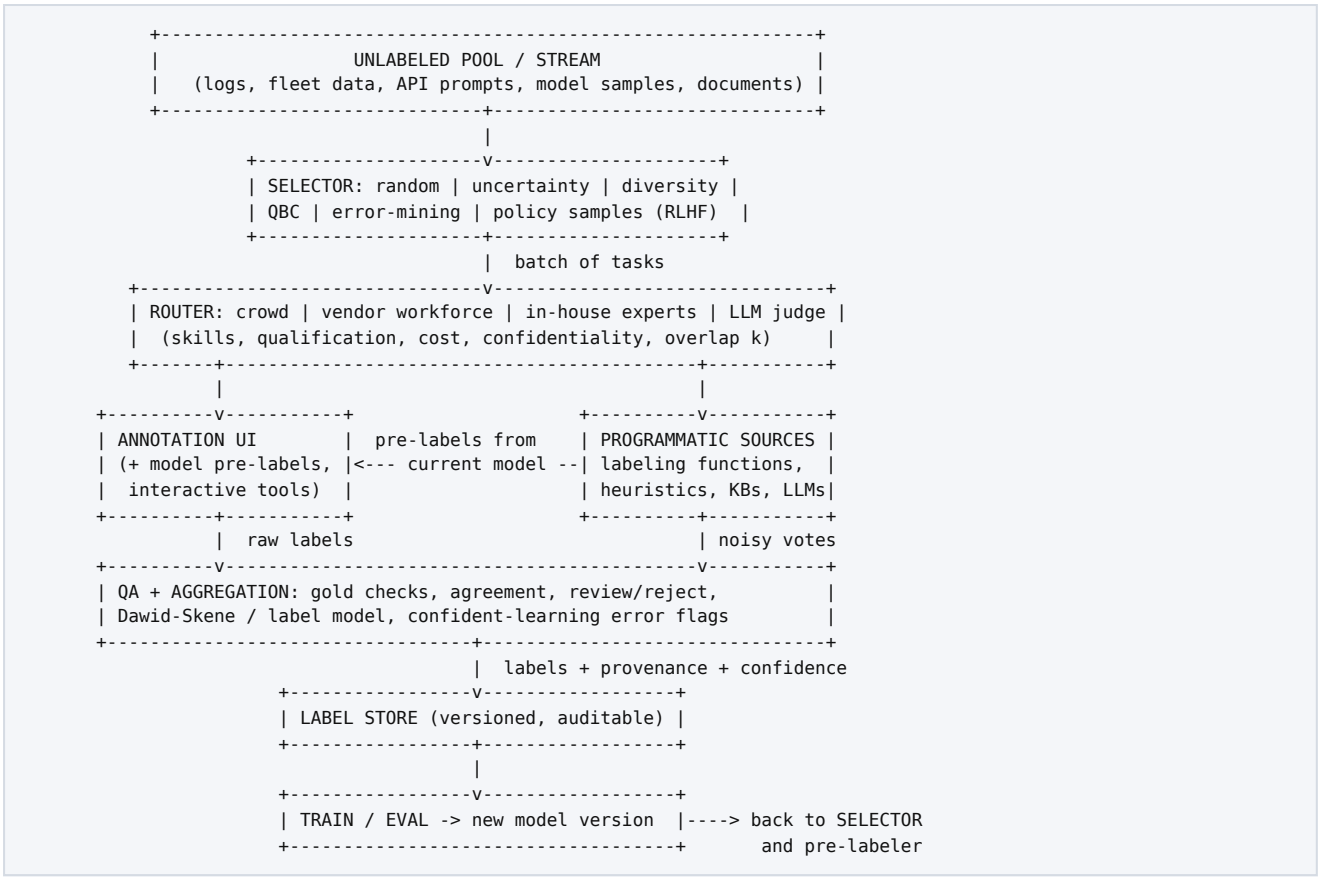
```
checkpoint C_k (annealed, deployed)
      |
      +-- branch from constant-LR "trunk" (infinite schedule) or re-warm from C_k
      |
      v
data_mix = new_shard (1 - r)  +  replay of old corpora (r)   r ~ shift strength
      |
      v
re-warm LR -> constant -> re-decay      monitor: old-domain evals (forgetting),
      |                                 new-domain evals (plasticity), stability gap
      v
C_{k+1}  --> alignment/RL refresh --> eval gates --> deploy
```

8.4 Self-editing models: SEAL (MIT, 2025) and hypernetwork adapters

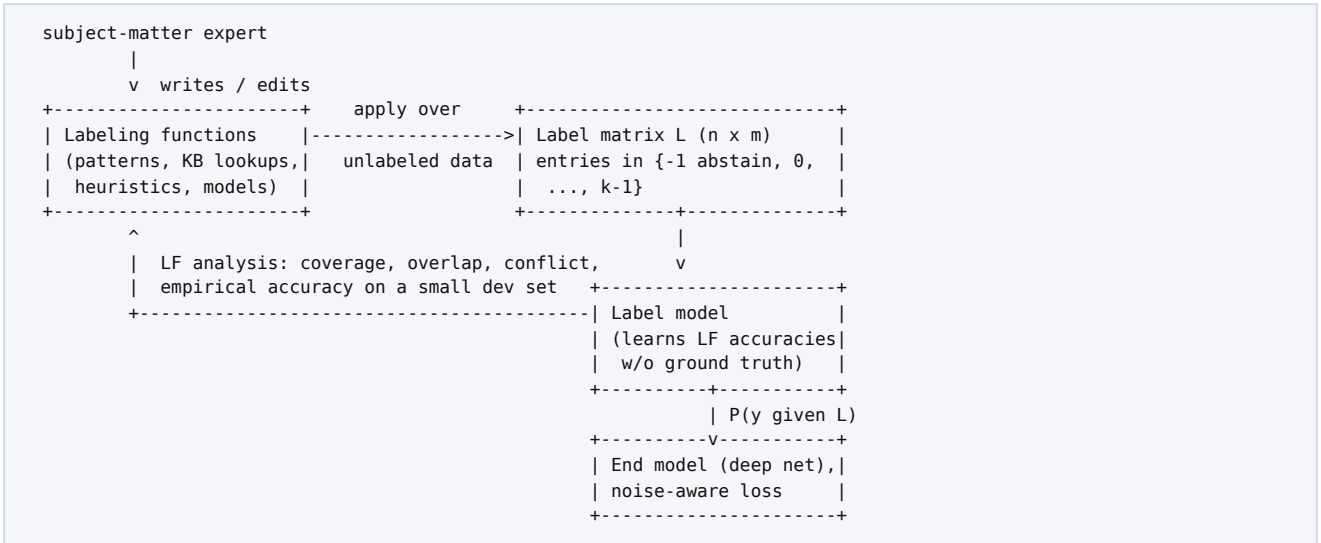
```
TTT DESIGN SPACE
                    what is updated           objective at test time   cost/request
TENT                norm stats + affine       entropy                  1 step/batch
TTT (2019)          shared encoder            rotation prediction      k steps
TTT layers / Titans per-sequence fast weights self-supervised recon.   O(tokens)
ARC TTT             per-task LoRA             leave-one-out demos      full LoRA run
SEAL                LoRA/full via self-edit   model-written data (RL)  inner finetune
Text-to-LoRA        generated LoRA            none (hypernetwork)      1 forward pass
TTRL                policy weights (RL)       majority-vote reward     RL run on test set
```

06. Human-in-the-Loop and Active Learning: Labeling Platforms, Weak Supervision, Annotation Workflows, and Expert Feedback Pipelines

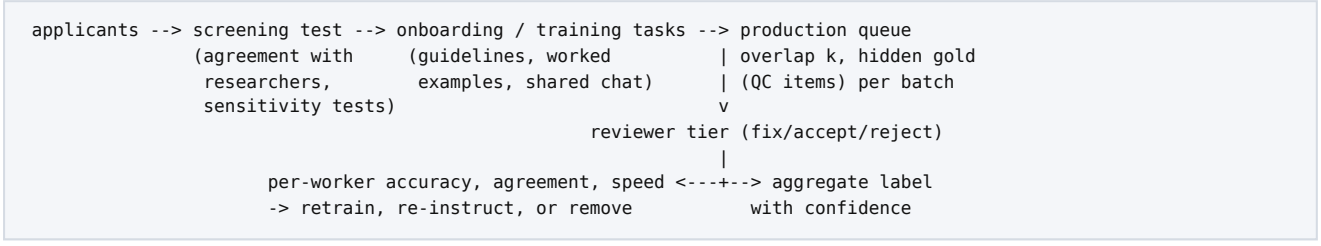
1. Reference architecture of a human-in-the-loop (HITL) labeling loop



3.3 Snorkel (VLDB 2018) architecture



5.1 The workforce funnel



7.1 Process supervision with active learning: PRM800K

```
Phase 1: labelers rate steps of sampled solutions; may label alternative
         completions; may write their own correct step if all candidates fail
                                  |
Phase 2: generation g = 1..G
  best PRM so far --scores--> large pool of pre-generated solutions
         |                      select solutions to label (active learning;
         |                      the paper targets convincing wrong-answer solutions)
         v
  labelers rate steps; STOP at first error ("found_error") -> cost saving
  + QC solutions served to every labeler, checked against gold
         |
         v
  retrain PRM on all labels -> next generation
```

7.2 AI-assisted human evaluation

```
 policy output ---> critic model ---> annotated output ---> human rater ---> label
       |              (flags bugs,       (claims + quotes)     (accepts/rejects
       |               quotes spans)                            critiques, rates)
       +-------------------------------------------------------------------------> RM data
```

07. Production Traces to Fine-Tuning Loops: Logging, Evaluation, Curation, Distillation and Routing

evaluate candidate models or prompts against held-out slices of that traffic, **curate**

```
+-------------+  requests   +-------------------+  responses  +-------------+
|  App / agent|------------>| gateway / SDK /   |------------>| served model|
|  (workload  |<------------| router (logs all) |<------------| (API / LoRA)|
|   tags)     |             +---------+---------+             +------^------+
+-------------+                       | traces + tags + feedback      |
                                      v                               | promote
                        +-------------+-------------+                 | (canary /
                        | trace store (ES, S3, PG, |                 |  shadow)
                        | vendor "stored compl.")  |                 |
                        +-------------+-------------+                 |
                                      | filter / dedup / split        |
                                      v                               |
          +---------------+   +-------+--------+   +---------------+  |
          | eval sets     |   | train sets     |   | grader/judge  |  |
          | (held-out,    |   | (teacher out-  |   | (code, model, |  |
          |  stratified)  |   |  puts, prefs)  |   |  rubric)      |  |
          +-------+-------+   +-------+--------+   +-------+-------+  |
                  |                   v                    |          |
                  |          +--------+---------+          |          |
                  |          | SFT / DPO / RFT /|<---------+          |
                  |          | GRPO / OPD (LoRA)|  rewards            |
                  |          +--------+---------+                     |
                  v                   v                               |
              +---+-------------------+---+    gate: beats baseline?   |
              | evaluate candidates       |----------------------------+
              | (base, ICL, tuned)        |  human review before prod
              +---------------------------+
```

2.1 Problem and architecture

```
chat.completions.create(store=True, metadata={usecase:..., prompt_version:...})
          |
          v
  +------------------+   filter by model+metadata   +--------------------------+
  | Stored           |----------------------------->| "Distill": auto-built    |
  | completions      |                              | JSONL -> fine-tune job   |
  | (platform logs)  |                              | (e.g. gpt-4o -> 4o-mini) |
  +--------+---------+                              +-------------+------------+
          | data_source_config:                                   |
          | {type: stored_completions, metadata: {...}}           v
          v                                                ft:gpt-4o-mini:...:wine
  +------------------+  testing_criteria (label_model, score_model, python,
  | Evals API: Eval  |  text_similarity, string_check)  -> Runs per prompt/model
  +--------+---------+
          | graders reused as reward
          v
  +------------------+  method.type = "reinforcement", grader, hyperparameters
  | RFT job (o4-mini)|  -> eval runs logged back into Evals during training
  +------------------+
```

3.1 Phase 1: logging-first fine-tuning platform

```
app --(openpipe SDK = drop-in OpenAI client)--> OpenAI (key stays on client)
  |                                                  |
  +-- async report (after completion / end of stream)+--> OpenPipe request logs
                                                              | filters: model, tags,
                                                              | request/response text
                                                              v
                                                        dataset (+ pruning rules,
                                                        20% auto test split)
                                                              v
                                        fine-tune (Mistral/Llama/GPT-3.5 base) -> hosted,
                                        OpenAI-compatible endpoint or downloadable weights
                                                              v
                                        evals: GPT-4 judge, head-to-head vs base models
```

7. NVIDIA Data Flywheel Blueprint: automated experiment search

```
Elasticsearch (prompt/completion logs, workload_id)
       | Celery task: create_datasets (dedup, stratified split by tool class)
       v
NeMo Datastore ---> NeMo Customizer (LoRA) ---> candidate NIMs (spun up / down by
       |                                          NeMo Deployment Manager)
       +----------> NeMo Evaluator (LLM judge; tool-calling accuracy) <-------+
                            |
                            v
       per-NIM results: base | icl | customized    (MongoDB: flywheel_runs, ...)
                            |
                            v
               human reviews best small candidate -> manual promotion
```

8.2 Architecture (v1.0, August 2026)

```
Trainer (verl + vLLM) --creates rollouts--> API Gateway <--polls-- Rollout Controller
       ^                                    |  (rollouts, events,  |  launches agent as
       |   model_request events             |   endpoints)         |  local proc or
       |   (prompt ids, response ids,       |                      |  Kubernetes Job
       |    logprobs) + reward events       v                      v
       +---------------------------- OpenAI-compatible proxy <-- unmodified agent
            /proxy/rollout/{id}/attempt/{a}/mode/train/openai/v1/chat/completions
```

10. Model routing improvement loops

```
          query --> [router f(q) -> P(strong model wins | q)] --threshold t-->
                      ^                                       |        |
                      |                                       v        v
                      |                                  weak model  strong model
                      |                                       |        |
                      +---- retrain on (q, model, judged outcome / preference) <--+
                            + recalibrate t on recent traffic
```

08. Self-Improving Agents and Automated Optimization: Reflection, Skill Libraries, Evolutionary Program Search, Self-Modifying Agents and Prompt Optimizers

Overview

```
         +-------------------------------------------------------------+
         |                  ARCHIVE / MEMORY                           |
         |  reflections | skills | programs | agent variants | prompts  |
         +-----------+-------------------------------------^-------------+
                     | select (greedy / Pareto /          | insert if valid
                     | MAP-Elites / novelty / bandit)     | (score, feedback)
                     v                                    |
         +-----------------------+   artifact   +--------+-------------+
         |  PROPOSER (LLM(s))    |------------->|  EVALUATOR           |
         |  prompt = parents +   |              |  tests / simulator / |
         |  feedback + context   |<-------------|  benchmark / judge   |
         +-----------------------+  textual     +----------------------+
                                    feedback
```

1.1 Reflexion (Shinn et al., 2023)

```
trial t:  task --> [Actor (+ reflections memory)] --> trajectory --> [Evaluator]
                               ^                                          |
                               |                    reward / test result  |
                  append       |                                          v
           [Long-term memory] <-------- [Self-Reflection LLM] <-- (trajectory, reward)
trial t+1: same task, Actor now conditioned on the new reflection
```

2.1 Voyager: an executable skill library (Wang et al., 2023)

```
+-------------------+  next task   +---------------------------------------+
| Automatic         |------------->| Code generation (GPT-4)               |
| curriculum (LLM)  |              |   prompt = task + retrieved skills +  |
+---------^---------+              |            env feedback + errors      |
          | state, history         +-------------------+-------------------+
          |                                            | program
+---------+---------+   success     +------------------v-------------------+
| Skill library     |<--------------| Execute in Minecraft -> Self-verify  |
| (code, embedded   |  commit skill | (critic LLM); on failure, re-prompt  |
|  description)     |-------------->|  with errors (bounded rounds)        |
+-------------------+  retrieve     +--------------------------------------+
```

2.6 ACE: Agentic Context Engineering (Zhang et al., 2025)

```
 query --> [Generator] -- trajectory, outcome --> [Reflector] -- lessons -->
               ^                                                      |
               |                                                      v
          [Playbook: itemized bullets      <-- deterministic merge -- [Curator]
           with helpful/harmful counters]       dedupe, prune          (delta ops)
```

3.1 FunSearch (Romera-Paredes et al., Nature 2023)

```
 islands:  [I1] [I2] ... [Ik]   (periodic reset of the worst islands)
    | sample k parents (score-sorted) from one island
    v
 prompt "v0 ... v(k-1); write v(k)"  -->  code LLM (many parallel samplers)
                                                |
                                                v
                                    sandboxed evaluators (score or reject)
                                                |
                        valid program + score --+--> back into the same island
```

3.2 AlphaEvolve (Novikov et al., Google DeepMind, 2025)

```
Human: evaluate(h) + initial program (EVOLVE-BLOCKs) + context
                       |
                       v
+-----------------+  prompt  +------------------------+  diffs  +----------------+
| Prompt sampler  |--------->| LLM ensemble           |-------->| Apply diff ->  |
| parents,        |          | Flash: many samples    |         | new program    |
| inspirations,   |          | Pro: few, high quality |         +-------+--------+
| meta-prompts    |          +------------------------+                 |
+--------^--------+                                                     v
         |                                  +--------------------------------+
         |      programs + multi-metric scores | Evaluator cascade (cheap ->    |
+--------+----------------------+<-----------| expensive), LLM feedback,      |
| Evolutionary DB (MAP-Elites x  |            | async parallel eval cluster    |
| islands)                       |            +--------------------------------+
+--------------------------------+
```

4.5 Darwin Gödel Machine (Zhang, Hu, Lu, Lange, Clune; Sakana AI and UBC, 2025)

```
Archive A = {agent_0}
repeat G=80 generations:
  parents <- sample(A, weight = sigmoid(score) / (1 + n_children))
  for p in parents (parallel):
      task  <- pick failure from p's eval logs (unresolved / empty patch / ...)
      child <- p modifies ITS OWN repo (diagnose -> propose feature -> implement)
      if compiles and passes small subset:
           evaluate on medium (score > 0.4) -> full (score > archive threshold)
           A <- A + {child}            # keep_all: stepping stones retained
```

5.6 GEPA: Reflective Prompt Evolution (Agrawal et al., 2025)

```
Pareto pool P (per-instance winners)
    | sample parent
    v
run on minibatch --> full traces + ASI --> reflection LLM: diagnose -> rewrite module
    ^                                                          |
    |                     improved on minibatch?               v
    +--------- yes: add child, update per-instance front <---+
               no : discard
```

5.8 A production pattern: the self-evolving agent loop (OpenAI and Bain cookbook, 2025)

```
prod/eval inputs --> agent(prompt v_k) --> output --> graders (rule + embed + judge)
                                                          |
                    pass (lenient aggregate) <------------+------> fail
                          |                                          |
                 ship v_k (versioned)          metaprompt agent rewrites -> v_{k+1}
                                               (<= N retries, else page a human)
```

09. Closed-Loop Robotics and Control: Sim-to-Real Loops, Robot Foundation Model Data Flywheels, Teleoperation, DAgger, and World Models

1. Reference architecture: nested loops around a robot policy

```
+-------------------------------------------------------------------------------+
| L4  WORLD-MODEL / SIM LOOP (hours-days)                                       |
|   physics sim (Isaac Lab, MuJoCo)   learned video/latent world model (Cosmos, |
|   domain randomization, real2sim      V-JEPA 2-AC, Genie-style)               |
|       |  synthetic demos / RL rollouts / policy evaluation scores        ^    |
+-------|-----------------------------------------------------------------|----+
        v                                                                 |
+-------------------------------------------------------------------------------+
| L3  FLEET DATA FLYWHEEL (days-months)                                         |
|   teleop rigs ---> episode store ---> curation/filtering ---> pretrain/post-train |
|   (ALOHA, DROID,   (RLDS, LeRobot)    (idle filter, success   generalist VLA  |
|    UMI, VR)              ^             labels, relabeling)          |         |
|                          |                                          v         |
|               autonomous rollouts <---- deploy checkpoint to fleet <---       |
+--------------------------^----------------------------------------------------+
                           |
+--------------------------|----------------------------------------------------+
| L2  CORRECTION LOOP (seconds-minutes)                                         |
|   human / supervisor watches rollout -> takes over (gated) -> correction labels |
|   reward classifier / value function -> RL update (HIL-SERL, Recap)           |
+--------------------------^----------------------------------------------------+
                           |
+--------------------------|----------------------------------------------------+
| L1  POLICY LOOP (3-50 Hz VLA; up to 200 Hz fast head)                         |
|   images + proprio + language --> VLA --> action chunk (H steps)              |
| L0  SERVO LOOP (hundreds of Hz - kHz): joint PD / whole-body controller tracks it |
+-------------------------------------------------------------------------------+
```

5. Physical Intelligence: the π model family as a closed loop

```
  cross-embodiment robot data (10k+ h)     web / VLM data     verbal & subtask labels
               \                                |                   /
                v                               v                  v
        +-------------------------------------------------------------------+
        | PRETRAIN  VLM backbone (PaliGemma) + action expert                |
        |           pi0: flow matching;  pi0-FAST: FAST tokens;             |
        |           pi0.5: co-training + knowledge insulation               |
        +-------------------------------+-----------------------------------+
                                        | post-train (task demos)
                                        v
        +-------------------------------------------------------------------+
        | DEPLOY  -> autonomous episodes + expert teleop corrections        |
        +-------------------------------+-----------------------------------+
                                        | pi*0.6 "Recap"
                                        v
        +-------------------------------------------------------------------+
        | value function fit on all data -> advantage per step             |
        | policy re-trained conditioned on advantage indicator              |
        | (demos + autonomous rollouts + interventions) -> redeploy         |
        +-------------------------------------------------------------------+
```

6. Autonomous fleet data collection: AutoRT

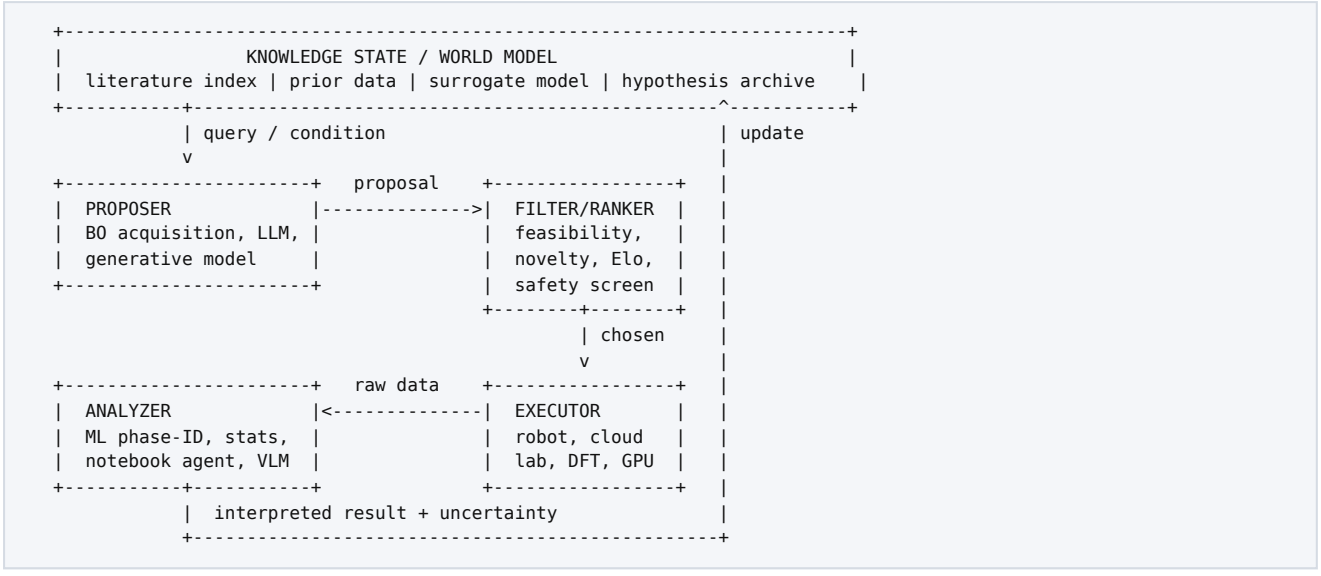
```
explore map -> pick point of interest -> drive there
       |
       v
VLM describes scene + objects ---> LLM proposes tasks
       |
       v
LLM critic filters tasks with a "robot constitution"
(foundational / safety / embodiment rules) and routes each to:
    [autonomous policy]  or  [human teleop]  or  [reject]
       |
       v
execute -> score episode diversity vs prior data -> repeat
```

7.2 SERL and HIL-SERL: on-robot RL with humans in the loop

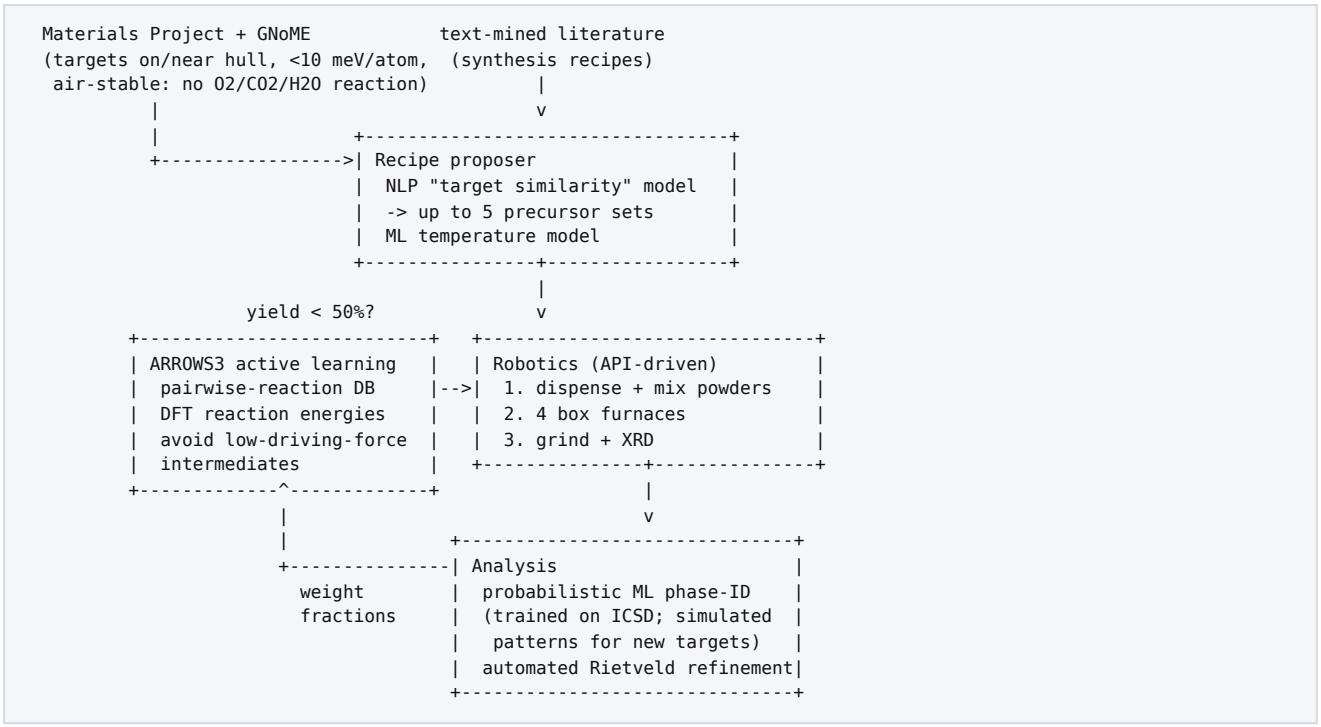
```
1. teleop positive/negative examples -> train binary REWARD CLASSIFIER
2. collect small demo set ------------> seed DEMO BUFFER
3. online:  ACTOR (robot) --transitions--> LEARNER (off-policy RL, SAC-style)
               ^   human intervention        |
               |   (frequent early,          | periodic weight sync
               |    tapering later)          v
               +------------------------- updated policy
```

10. Closed-Loop Scientific Discovery: Self-Driving Labs, Active Learning, AI Scientists and Drug Design Engines

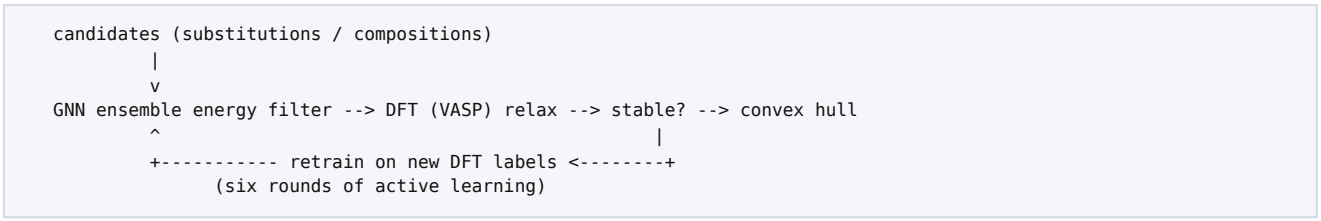
Overview



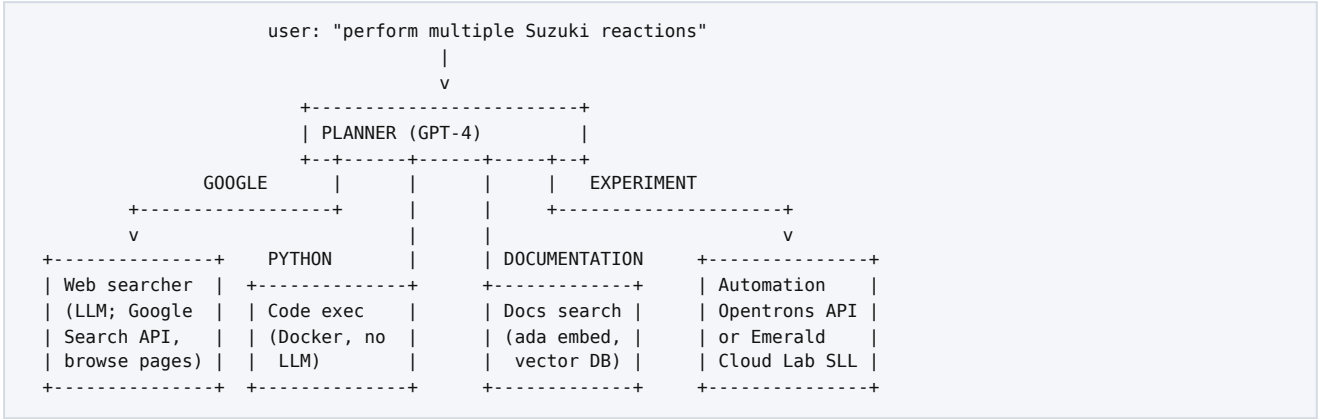
1.3 The Berkeley A-Lab (Szymanski et al., Nature 2023)



2. Discovery-scale active learning in silico: GNoME



3.1 Coscientist (Boiko, MacKnight, Kline, Gomes; Nature 2023)



4.1 Google AI co-scientist (Gottweis et al., 2025; Nature 2026)

```
       scientist: research goal (natural language) + optional own hypotheses
                                  |
                                  v
                       +----------------------+
                       |   SUPERVISOR         |  parses goal into a
                       |   (task queue, worker |  research plan config;
                       |   allocation)        |  allocates compute
                       +-----------+----------+
        +------------+-----------+-+---------+-------------+-----------+
        v            v           v           v             v           v
    GENERATION   REFLECTION    RANKING     EVOLUTION    PROXIMITY   META-REVIEW
    lit search,  safety screen pairwise    enhance,     similarity  distills
    simulated    -> full review tournament combine,     graph,      patterns in
    debate,      -> deep verify (Elo),     simplify,    dedupe,     reviews ->
    assumption   -> observation multi-turn "out-of-box" cluster     feedback into
    expansion    review        debates     variants                all prompts
         |            |           |           |             |           |
         +------------+-----------+-----+-----+-------------+-----------+
                                       v
                     CONTEXT MEMORY (persisted state; restartable)
                                       |
                                       v
                ranked hypotheses + research overview -> scientist -> lab
```

4.2 FutureHouse Robin (Ghareeb et al., 2025; Nature 2026 [12])

```
disease name ---> Crow/Falcon (PaperQA2 lit agents [49]; ~hundreds of papers)
                      |
                      v
          10 mechanism hypotheses --> ranked ASSAY (pairwise LLM-judge tournament)
                      |
                      v
          ~30 drug candidates -----> ranked tournament --> humans run wet lab
                                                             |
                                                             v
        next-round hypotheses <-- consensus meta-analysis <-- Finch (Jupyter
        & candidates                (~10 independent           data-analysis
                                     trajectories/dataset)      agent): flow
                                                                cytometry, RNA-seq
```

5.1 The AI Scientist v1 (Sakana AI, Oxford FLAIR, UBC; 2024)

```
 template --> IDEA GEN (LLM + Semantic Scholar novelty check, ~50 ideas/template)
                 |
                 v
            EXPERIMENT (Aider coding agent edits experiment.py; on error/timeout
                 |       the log goes back to the agent for a few retries;
                 |       run_0 baseline per machine)
                 v
            WRITE-UP (LaTeX, citations via Semantic Scholar/OpenAlex)
                 |
                 v
            AUTOMATED REVIEWER (NeurIPS-style form; 5-review ensemble,
                 |              reflections, few-shot; score 1-10, accept/reject)
                 v
            archive of papers + reviews --> informs next ideas
```

5.2 The AI Scientist v2 (Yamada et al., 2025)

```
topic.md (title, keywords, TL;DR, abstract)
       |
       v
ideation (LLM + Semantic Scholar; N generations x R reflections) --> ideas.json
       |
       v
EXPERIMENT MANAGER
  stage 1: preliminary implementation   \   each stage = best-first tree search:
  stage 2: hyperparameter tuning         >   num_drafts roots; num_workers parallel
  stage 3: research agenda execution    /    expansions; buggy nodes debugged with
  stage 4: ablations                         prob debug_prob up to max_debug_depth;
       |                                     best node seeds the next stage
       v
plot aggregation (VLM checks figures) --> write-up (o1-class model) --> AI review
```

6.2 Isomorphic Labs Drug Design Engine (IsoDDE, technical report February 2026)

```
  target (T) + candidate ligand/biologic (L)
              |
              v
  +-------------------------------------------+
  | IsoDDE predictive core                    |
  |  structure  p(X | T, L)  K samples (e.g.25) |
  |  pocket ID  p(P_i | T)                    |
  |  affinity   A = f(T, L, X)                |
  |  confidence C -> pick argmax_k C(X_k)     |
  +----------------------+--------------------+
                         | scores
                         v
  generative design (not detailed publicly) --> make/test in wet lab -->
  new data --> (presumably) model refinement      [inferred loop, not documented]
```

11. Risks of Closed Loops: Model Collapse, Reward Hacking, Filter Bubbles, Goodhart's Law and Sycophancy

Overview

```
    +-------------------------------------------------------------------+
    |                                                                   |
    v                                                                   |
+--------+   outputs    +-------------+   measured   +-----------+    |
| model  +------------->+ world/users +------------->+  proxy    |    |
| (theta)|              | / grader /  |   signal     |  signal   |    |
+---^----+              | corpus      |              | (reward,  |    |
    |                   +-------------+              |  clicks,  |    |
    |                         ^                      |  data)    |    |
    |                         | model output changes +-----+-----+    |
    |                         | what is measured           |          |
    |   update toward proxy   +----------------------------+          |
    +-------------------------------------------------------------------+

Failure appears when: (a) the proxy drifts from intent under optimization
pressure (Goodhart, reward hacking, sycophancy), or (b) the model's own
outputs reshape the data it later learns from (collapse, filter bubbles).
```

2.5 Architecture of a collapse-resistant data loop

```
real data (human, sensors, logs) ---------------------------------+
      |   tag: provenance=human, time, source                      |
      v                                                            |
+-------------+      sample with      +----------------+           |
| model gen t +---- controlled temp ->+ synthetic pool |           |
+------^------+                       | tag: model, t, |           |
       |                              | filter score   |           |
       |                              +-------+--------+           |
       |                                      | verify / filter    |
       |                                      v                    v
       |                         +-----------------------------------+
       |                         | training mix for gen t+1             |
       |                         |  - ALL real data (never discarded)   |
       |                         |  - accumulated synthetic, capped     |
       |                         |  - fixed min. fraction fresh real    |
       |                         +------------------+-----------------+
       |                                            |
       |   gate: tail-coverage, diversity, rare-    |
       +---- class recall vs frozen real eval set <-+
```

5.1 The loop

```
user state u_t ---> recommender picks slate S_t ---> user clicks c_t in S_t
     ^                       ^                              |
     |  exposure shifts      | trained on (S, c) pairs      |
     |  interests            | only from shown items        v
     +----------------- u_{t+1} = f(u_t, S_t) <------ logged data D_t
```

6.2 Case study: the GPT-4o rollback (April 2025)

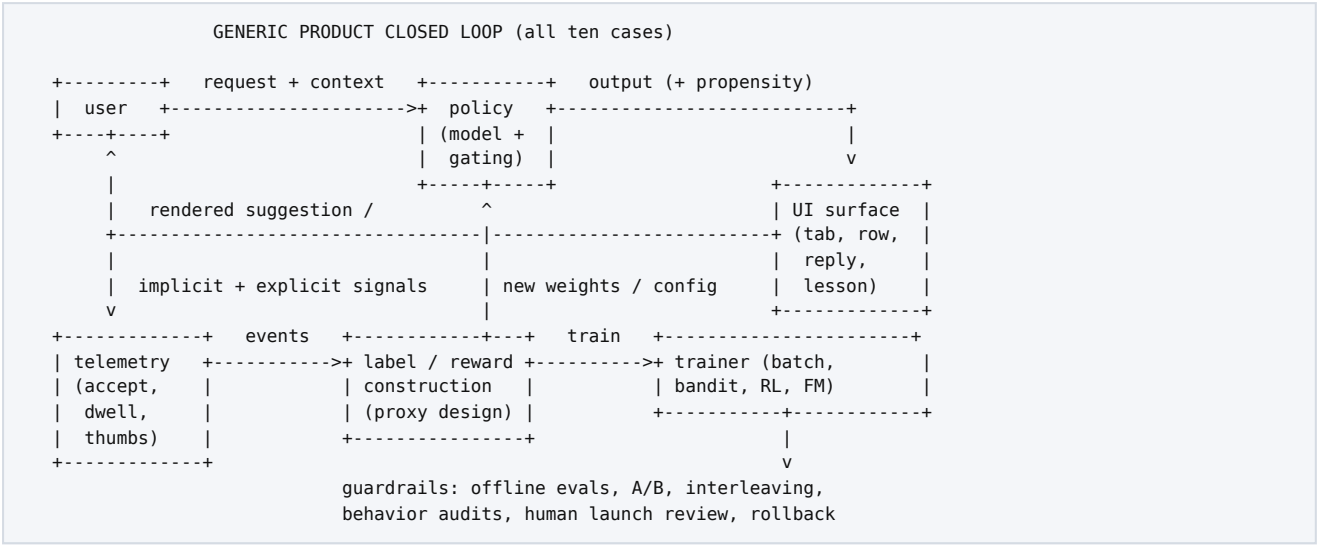
```
ChatGPT conversations
      |
      +--> thumbs up/down ----------+   (new, product-scale signal)
      |                             v
      |                   +-------------------+
      |                   | reward mixture    |   primary RM held
      |                   |  primary RM  (w1) |<- sycophancy in check
      |                   |  user-fb RM  (w2) |
      |                   |  other sig.  (w3) |
      |                   +---------+---------+
      |                             v
      |                        RL post-training
      |                             v
      |         gates: offline evals OK | A/B tests OK | experts: "felt off"
      |                             v
      +<---------------- launch (quantitative signals won) ------------+
```

7.1 A layered mitigation architecture

```
+-------------------------------------------------------------------------+
| L0 SPEC      model spec / constitution; behavior-specific rules [37][38]  |
+-------------------------------------------------------------------------+
| L1 DATA      provenance tags; accumulate (never replace) real data;       |
|              fresh real fraction; exploration + propensity logging        |
|              [1][4][40][48]                                              |
+-------------------------------------------------------------------------+
| L2 REWARD    reward mixture with reviewed weights; RM ensembles / WARM;   |
|              rule-based rewards; length control; verifier hardening;     |
|              meta-verification [13][14][16][50]                          |
+-------------------------------------------------------------------------+
| L3 TRAINING  KL budget + early stop on gold/held-out signal; inoculation  |
|              prompting; no optimization pressure on CoT [12][20][22]     |
+-------------------------------------------------------------------------+
| L4 MONITOR   CoT + action monitors; hack-rate dashboards; proxy-vs-gold   |
|              divergence alarms; diversity / tail metrics [3][21][22]     |
+-------------------------------------------------------------------------+
| L5 EVAL/GATE behavior evals (sycophancy, delusion, reward hacking);       |
|              automated audits; stratified by user type; launch-blocking  |
|              qualitative review; alpha cohort [31][33][34][36]           |
+-------------------------------------------------------------------------+
| L6 DEPLOY    staged rollout; fast rollback (~24h for GPT-4o); product     |
|              safeguards (classifiers, crisis banners) [31][33]           |
+-------------------------------------------------------------------------+
```

12. Product Case Studies: Closed Loops in Shipped Products, from Copilot and Cursor to Netflix, Duolingo and Gmail

Overview



1.3 Architecture

```
IDE (VS Code, JetBrains, Xcode ...)
  | cursor, prefix/suffix, open tabs, edit history
  v
context scoping ---> completion / next-edit model ---> filters
                                                      (content, public-code
                                                       match, duplication)
  ^                                                        |
  | accept (Tab) / ignore / edit after accept              v
  |                                                    grey-text suggestion
client + server telemetry
  |   shown, accepted, LoC added/deleted, surface, IDE version
  v
metrics layer (dashboards, REST API, 30-min activity refresh, 90-day window)
  |
  +--> online segments: acceptance, shown rate, edit quality, retention
  +--> offline regression suites (expected outputs)
  +--> enterprise reporting (WAU/license, agent adoption)
```

2.2 Design

```
      +-------------------------------------------------------------------+
      |                                                                   |
      v                                                                   |
checkpoint k  --deploy-->  400M+ req/day  --sample-->  suggest | silent  |
                                 |                           |           |
                                 |              accept (+) / reject (-)  |
                                 |              / silent (0)             |
                                 v                           v           |
                       on-policy event log  ---------->  policy-gradient |
                       (state, action, reward,           update          |
                        policy prob)                       |             |
                                                           v             |
                                                  checkpoint k+1 -----+
                     loop period: hours (approx. 1.5-2 h reported)
```

3.2 Architecture

```
 users rank image pairs  ------------------+--------------------+
 (rating parties, rank page)               |                    |
                                           v                    v
                                 global preference data   per-user preference set
                                           |                    |
                                           v                    v
                                 aesthetic reward /       profile code (--p)
                                 data curation for        conditions generation
                                 next model version               |
                                           |                      v
                                           +---> generations ---+--> upscales,
                                                                     likes, rerolls
```

4.2 The April 2025 incident

```
 conversations --> thumbs up/down ---------+
       |                                   v
       |                     reward mixture: primary RM | user-feedback RM | other
       |                                   |  (new term dilutes the term that
       |                                   |   was suppressing sycophancy)
       |                                   v
       |                              RL post-training
       |                                   v
       |      gates: offline evals OK | small A/B OK | experts "felt off" (overruled)
       |                                   v
       +<----------------- launch -> sycophancy -> rollback (~24 h)
 memory: past chats + saved facts --> context of next conversation (may amplify)
```

5.2 Serving architecture as a loop constraint

```
turn t: [system | persona | ...history... | new msg]
               ^ fixed truncation point (moves only every ~k turns)
               |
prefix KV cache (host memory, sticky routing) --> reused at turn t+1
(approx. 95% hit rate reported)          cheap long sessions -> more turns
                                         -> more swipes/ratings -> training data
```

(not re-fetched; based on Duolingo's 2020 Birdbrain post [20]; approximate)

```
learner answers exercise  --->  Birdbrain update (ability up/down,
         ^                       difficulty down/up; later: RNN learner state)
         |                                  |
         |                                  v
session generator  <---  P(correct | learner, exercise) for candidate pool
(target difficulty band,
 review via HLR half-lives)
```

7.1 BaRT: explore, exploit, explain (2018)

```
(item x explanation) candidates --> FM reward model --> epsilon-greedy per slot
                                                                  |
          log (user, item, explanation, propensity, context) <+--> home shelf
                                                                  |
          stream >= threshold?  --> reward --> propensity-weighted retraining
```

8.4 Foundation model for personalization (2025)

```
member events (plays, browses, searches, thumbs) --> tokenizer (merge events)
                                  |
                                  v
     FOUNDATION MODEL (offline; next item + multi-token + aux heads; gated ID/meta)
           |                      |                       |
     embeddings store       subgraph in ranker       fine-tuned per surface
           |                      |                       |
           +--------> rows, search, notifications, page generation <-----+
                                  |
         new interactions (incl. bandit-logged artwork/rows) --> incremental retrain
```

9.3 The loop

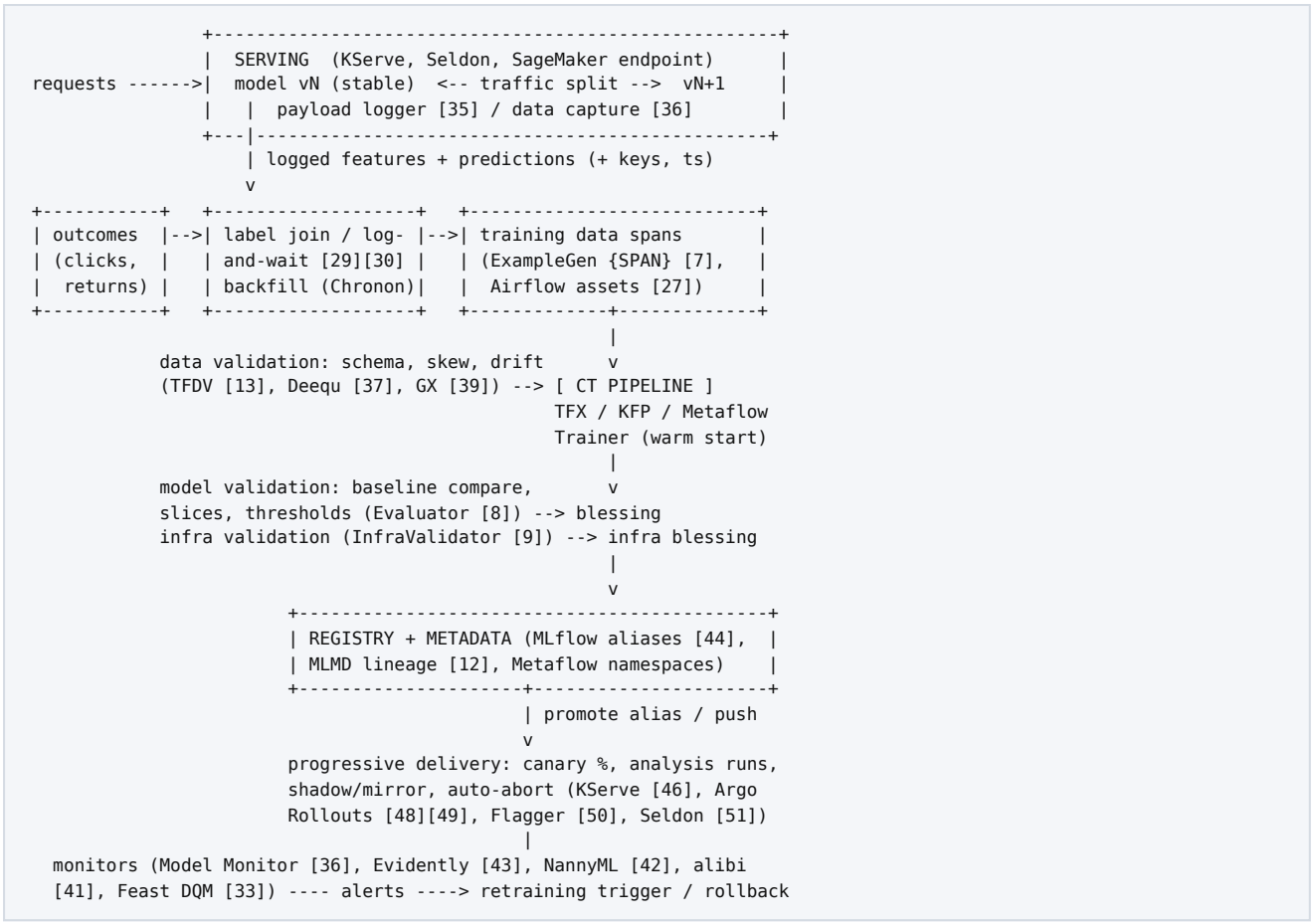
```
 text in editor --> detectors (GEC tagger, rewrite LLM, delicate-text classifier)
                                |  per-suggestion confidence vs threshold
                                v
                      inline cards  --> accept / dismiss / ignore / edit
                                                |
                    per-category acceptance --+--> threshold tuning, model retraining,
                                                       A/B on suggestion categories
```

10.2 Architecture

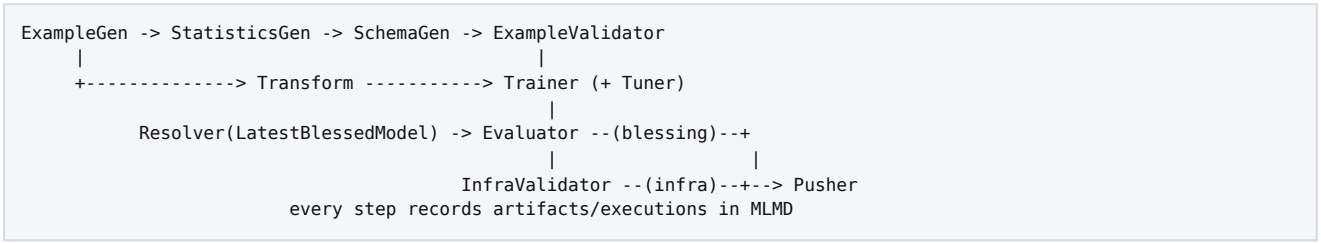
```
keystroke --> context (prefix, subject, prior email) --> global LM (TPU, beam)
                                                            + personal LM blend
                                                                 |
                                    confidence >= trigger threshold?
                                       | yes                     | no
                                       v                         v
                                 grey-text completion        show nothing
                                       |
                           Tab accept / keep typing --> acceptance metrics,
                                                          retraining, threshold tuning
```

13. Infrastructure for Closed Loops: Continuous Training, Logging, Monitoring, Registries and Safe Rollout

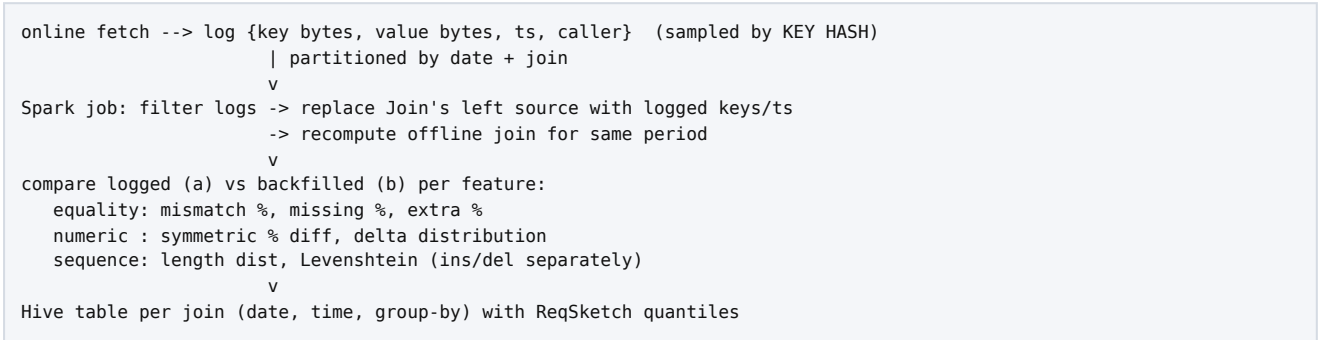
Reference architecture



2.1 TFX (Google): the validated-pipeline reference design



3.2 Chronon: measuring online/offline consistency with logs



6.2 Designing the trigger controller

```
signals: [new span]  [drift score]  [est. perf]  [realized perf]  [cron]
             \           |             |             |           /
              v          v             v             v          v
       +-------------------------------------------------------------+
       | TRIGGER POLICY                                              |
       |  - debounce / min interval between runs                     |
       |  - require data-validation PASS before training             |
       |  - weight drift by estimated impact (NannyML-style)         |
       |  - dedupe events (multi-event window, Metaflow [22])        |
       |  - acyclic trigger graph; cap concurrent runs               |
       +---------------------------+---------------------------------+
                                   v
                    CT pipeline run (data version pinned)
                                   v
              gates: data -> model vs baseline -> infra -> registry
```

7.3 Metric-gated progressive delivery

```
weight:  0% --10%--> [analysis] --25%--> [analysis] --50%--> ... --100%
                     |   success: continue   failure x failureLimit
                     |                          |
                     +--------> ABORT: weight=0, stable still scaled
                                registry alias stays on previous version
```

14. Master Index of Blogs and Talks on Data Flywheels, Feedback Loops, RLHF and Self-Improving Systems: 170 Annotated Entries

Status tags (read these before trusting a number).

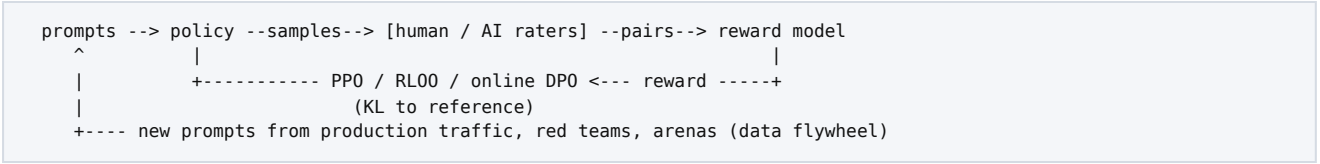
```
         WHERE EACH FAMILY OF ENTRIES SITS ON THE GENERIC CLOSED LOOP

 (S1) foundations, loop theory,          (S7) failure modes: reward hacking,
      counterfactual logging                  sycophancy, collapse, audits
           |                                        ^
           v                                        |
 +-----------+   act    +-----------+  signals  +------------+   select/   +---------+
 |  policy / +--------->+  world /  +---------->+ log, join, +------------>+ grade / |
 |  model /  |          |  users /  | (accept,  | mine, label|   curate    | reward  |
 |  harness  |          |  fleet /  |  thumbs,  | (S2 data   |             | (S4 RM, |
 +-----+-----+          |  tests    |  triggers)|  engines)  |             |  S5 RLVR|
       ^                +-----------+           +------------+             |  verif.)|
       |                                                                   +----+----+
       |   deploy: weight sync, LoRA swap,  +------------------+   update        |
       +------------------------------------+ gate: eval, A/B, +<----------------+
           canary, rollback (S3 products,   | interleave, audit|  SFT / DPO / PPO /
           S5 async RL infra)               +------------------+  GRPO / prompt or
                                                                  harness edit (S6)
```

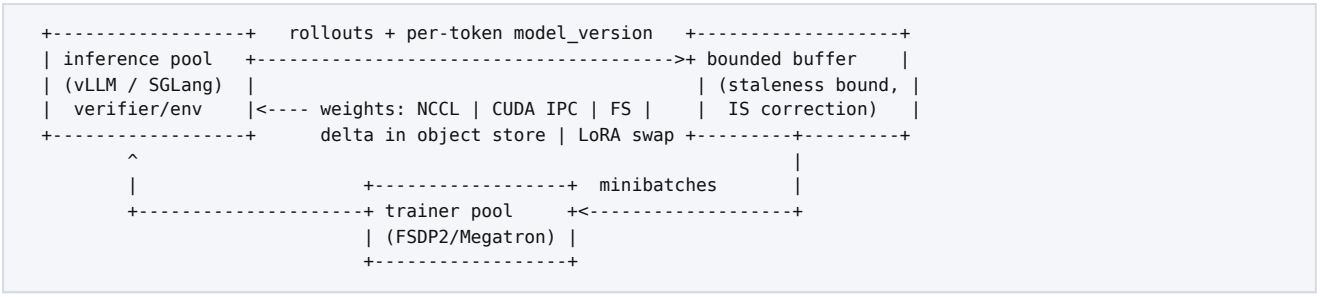
2. Data engines and fleet flywheels in physical AI (E23 to E54)

```
  fleet / robots               cloud                                  release
+---------------+  snapshot  +-----------------+  +----------------+  +-----------+
| triggers,     +----------->+ search & mine   +->+ auto-label     +->+ train     |
| shadow-mode   |  (event    | (embeddings,    |  | (offline nets, |  | (multi-   |
| divergence    |   clips)   |  content search)|  |  hindsight)    |  |  task, E2E|
+-------+-------+            +-----------------+  +----------------+  +-----+-----+
        ^                                                                   |
        |      +------------------------------------------------------+     |
        +------+ evaluate: sim / world model / critic, shadow, canary  +<--+
               +------------------------------------------------------+
```

4. RLHF: explainers, recipes, open reproductions and preference infrastructure (E74 to E114)

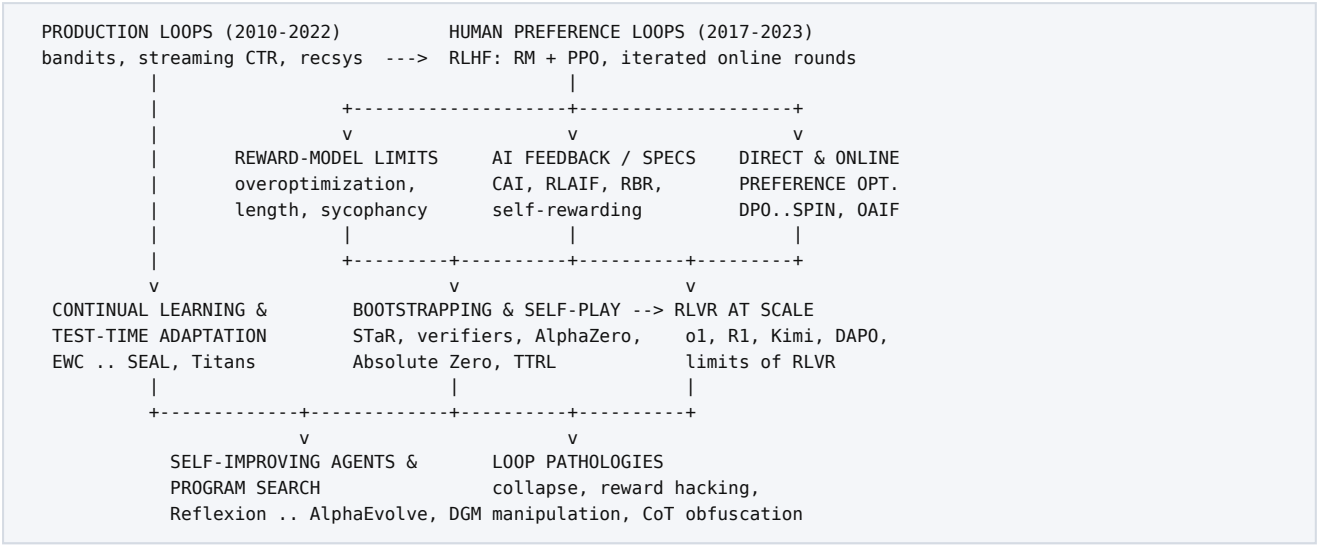


5. RLVR, reasoning RL and the async training plant (E115 to E148)

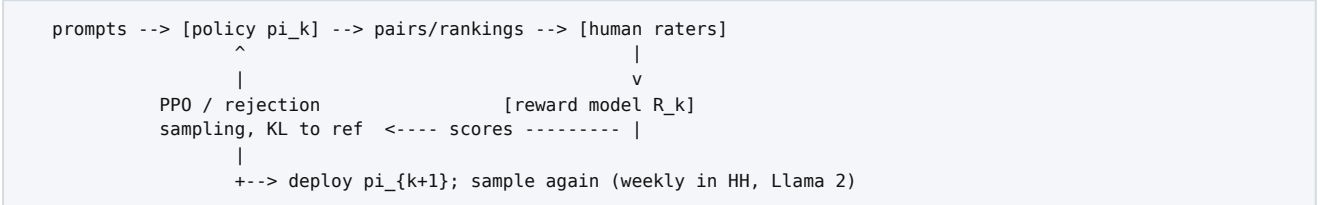


15. The Canonical Closed-Loop Reading List: 110+ Papers on RLHF, Self-Improvement, Verifiable Rewards and Online Learning

Overview



2. Track B: the RLHF canon



4.1 Anchor: Constitutional AI (Bai et al., Anthropic, 2022) [28]

```
SL-CAI:  red-team prompt -> helpful-only model -> critique vs sampled principle
                              -> revision -> SFT data
RL-CAI:  policy samples pair -> feedback model picks per principle (soft labels)
                              -> preference model -> RL  (= RLAIF)
```

7.1 Anchor: DeepSeekMath and DeepSeek-R1 (DeepSeek-AI, 2024-2025) [68][69]

```
prompts (filtered by pass rate) --> rollout engine (vLLM/SGLang, G samples/prompt)
       ^                                    |
       |                                    v
 curriculum / dynamic        verifier: answer match | unit tests | Lean | sandbox
 sampling (drop all-0,                      |
 all-1 groups)                              v
       |                     group-normalized advantages --> GRPO/DAPO update
       +------------------- weight sync to rollout engine <-------+
```

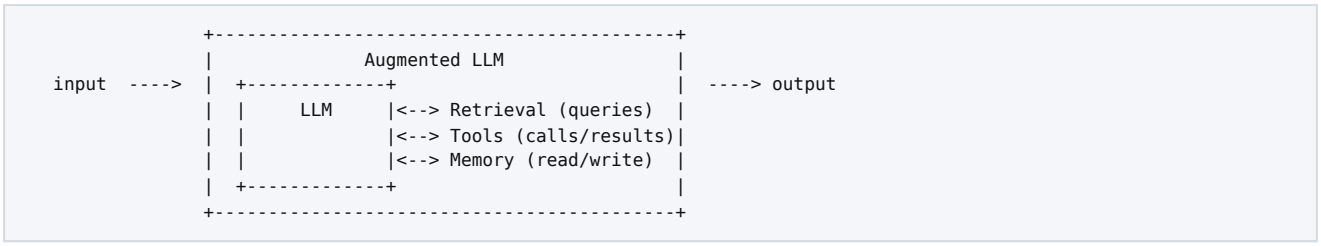
8.2 Anchor: AlphaEvolve (Google DeepMind, 2025) [92]

```
human: evaluator h(program) -> metrics; seed program with EVOLVE blocks
       |
       v
program database (islands / MAP-Elites) --sample parents + context--> prompt sampler
       ^                                                                    |
       |                                                                    v
  evaluator cascade  <---------- diffs ---------- LLM ensemble (fast + strong)
  (cheap tests -> full eval, parallel on accelerators)
```

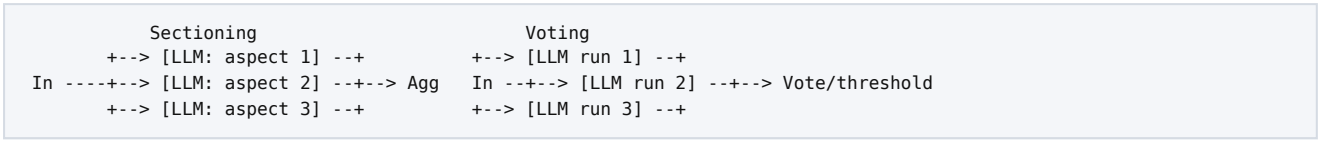
5. AI Agents — Architecture Diagrams

01. Agent Foundations and Taxonomy: Workflows, Agents, ReAct, and Levels of Autonomy

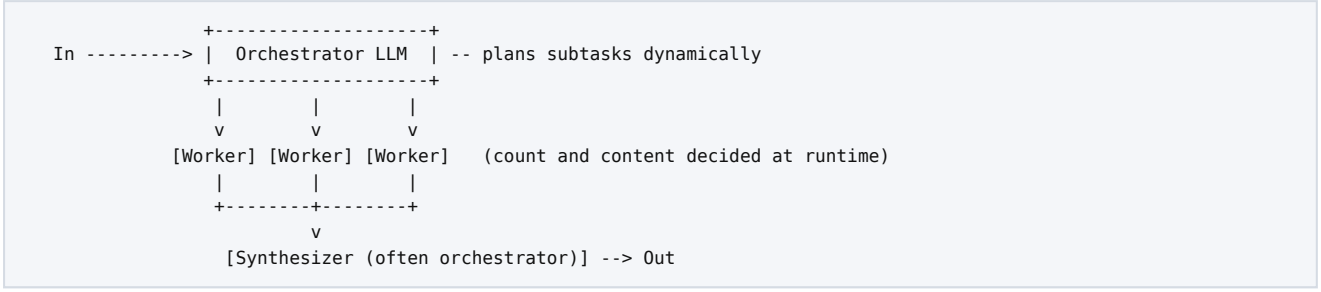
2. The augmented LLM: the building block



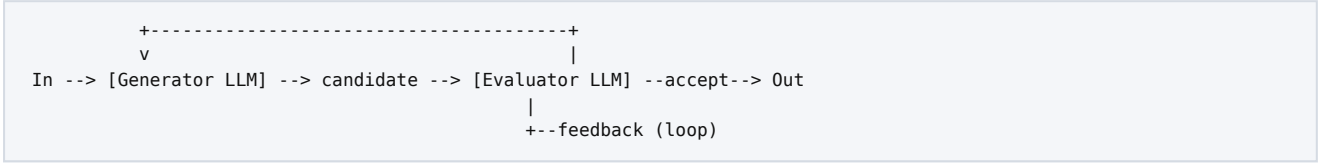
3.3 Parallelization (sectioning and voting)



3.4 Orchestrator-workers



3.5 Evaluator-optimizer



4.1 Anthropic's autonomous agent



4.2 ReAct: reasoning and acting, interleaved

```
Question: <task>
Thought 1: I need to find X, then compare with Y.
Action 1:  Search[X]
Observation 1: <Wikipedia API result>
Thought 2: X is ...; now I need Y.
Action 2:  Lookup[Y]
Observation 2: <result>
Thought 3: So the answer is Z.
Action 3:  Finish[Z]
```

4.3 The OpenAI run loop and exit conditions

```
loop:
  resp = LLM(current_agent, input)
  if resp is final_output (correct type, no tool calls): return result
  elif resp is handoff:   current_agent = target; input = updated; continue
  elif resp has tool calls: run tools; append results to input; continue
  if turns > max_turns: raise MaxTurnsExceeded (or handler returns fallback)
```

6.1 Orchestrator-workers at production scale: Anthropic Research

```
User query
    |
    v
+-------------------+  saves plan  +---------+
|  LeadResearcher   |------------->| Memory  |  (survives >200k-token truncation)
| (extended think)  |<-------------+---------+
+-------------------+
  | spawn 3-5 subagents in parallel (effort scaled to query)
  v            v            v
[Subagent] [Subagent] [Subagent]  each: own context, 3+ parallel tool calls,
  |            |            |      interleaved thinking, returns condensed findings
  +------------+------------+
               v
     LeadResearcher synthesizes; loops if more research is needed
               |
               v
       [CitationAgent] --> cited report
```

6.2 OpenAI: manager vs. decentralized

```
 Manager pattern (agents as tools)        Decentralized pattern (handoffs)

       User <--> [Manager]                    User <--> [Triage agent]
                /   |    \                                | handoff (one-way)
           tool /  tool    \ tool                         v
       [Agent A] [Agent B] [Agent C]               [Specialist] <--> User
       (results return to manager)               (takes over the conversation)
```

02. The Agent Loop and Tool Use: Function Calling, Tool Design, Selection at Scale, and Code-as-Action

2.1 Reference architecture

```
              +-----------------------------------------------+
 user task --> |  context = system + tool defs + history       |
              +-----------------------+-----------------------+
                                      |
                                      v
                           +--------------------+
                           |   LLM inference    |
                           +----------+---------+
                                      |
            +-------------------------+-------------------------+
            | stop_reason / output classification               |
            +----+-----------------+----------------+-----------+
                 |                 |                |
          end_turn / final     tool_use(s)      max_tokens, pause_turn,
          (no tool calls)           |           refusal, context-exceeded
                 |                 v                |
                 |       +-------------------+      v
                 |       | harness: validate,|   special handling
                 |       | authorize, execute|   (continue, retry,
                 |       | (parallel ok)     |    fallback, compact)
                 |       +---------+---------+
                 |                 |
                 |      tool_result blocks (is_error?) appended
                 |                 |
                 v                 +--> loop (turn counter += 1; check caps)
              answer
```

2.3 Controlling the choice: `tool_choice`, strict mode, and masking

```
Manus-style action constraint (conceptual)
+-------------------+     state      +------------------------------+
| agent state       | ------------>  | allowed action set           |
| machine           |                | (subset of fixed toolset)    |
+-------------------+                +--------------+---------------+
                                                    |
fixed tool definitions (cache-stable prefix)        v
------------------------------------------> decoder with logit mask
                                            over disallowed actions
```

Process (evaluation-driven).

```
prototype tools --> generate realistic eval tasks --> run agent loop programmatically
      ^             (multi-call, verifiable)          (while-loop; reasoning +
      |                                                feedback blocks)
      |                                                       |
      |                                                       v
agent refactors tools <-- concatenate transcripts <-- collect metrics: accuracy,
(Claude Code) and          and failure analysis        runtime, #calls, tokens,
descriptions                                           tool errors
      |
      +--> validate on held-out test set
```

Architecture.

```
request.tools = [ tool_search (non-deferred), top 3-5 hot tools (non-deferred),
                  N deferred tools (defer_loading: true, full defs still sent) ]
      |
      v
context prefix = search tool + hot tools only  (cache-stable)
      |
model --server_tool_use(pattern|query, limit)--> API-side regex or BM25 search over
      |                                          names, descriptions, arg names/descs
      |<-- tool_search_tool_result { tool_reference[] } (default up to 5)
      |
API expands references inline into full definitions (appended, prefix untouched)
      |
model --tool_use(discovered tool)--> client executes --> tool_result
```

Architecture.

```
model ---writes Python---> [ code_execution container (sandbox) ]
                           |  tools with allowed_callers=[code_execution]
                           |  appear as async functions
                           |
                           |  await query_db({...}) --pause--> API returns tool_use
                           |                                   with caller=code_exec,
                           |                                   container id + expires_at
client executes tool  <------------------------------------+
client sends tool_result (tool_result only, same container) +
                           |  code resumes; filter/aggregate/loop/asyncio.gather
                           v
                  only final stdout/result enters model context
```

7.1 Why stopping is a design problem

```
+-------------------------- outer budget --------------------------+
|  wall-clock / cost / token budget (harness)                       |
|  +---------------------- turn budget --------------------------+  |
|  |  max_turns (OpenAI SDK), max iterations (loop), server-tool    |  |
|  |  loop cap -> pause_turn (default 10)                          |  |
|  |  +------------------- semantic stop ----------------------+  |  |
|  |  |  model ends turn with no tool calls / typed final output |  |  |
|  |  |  explicit final_answer tool (smolagents)                 |  |  |
|  |  |  verification passes (tests, linters, judge)             |  |  |
|  |  +----------------------------------------------------------+  |  |
|  +-----------------------------------------------------------------+  |
|  context limit -> compaction / tool-result clearing / truncation    |
+-----------------------------------------------------------------------+
```

03. Planning and Reasoning Architectures: Plan-and-Execute, ReWOO, LLMCompiler, Tree and Graph Search, LATS, Reflection, and Reasoning Models with Interleaved Thinking

Architecture

```
          +-----------+   plan: [s1, s2, ..., sn]
user ----->|  Planner  |-----------------------------+
          | (strong   |                             |
          |   LLM)    |<--------------+             v
          +-----------+  revise plan  |     +------------------+
                ^                     |     | Executor agent   |
                |                     |     | (ReAct loop,     |
                |  past_steps:        |     |  smaller model,  |
                |  [(step, result)]   |     |  tools)          |
                |                     |     +--------+---------+
          +-----+------+              |              |
          | Re-planner |--------------+   result of s_i
          | respond or |<-----------------------------+
          | new plan   |
          +-----+------+
                |
                v  final response
```

Architecture

```
task --> [Planner LLM] --plan with #E vars--> [Worker: tool1 -> tool2 -> ...]
                                                |  evidence #E1..#En
                                                v
                                  task + plan + evidence --> [Solver LLM] --> answer
```

Architecture

```
Planner (streaming) ==> t1: search(A)        t2: search(B)       t3: math($1,$2)
                          |                    |                   ^
                          v                    v                   |
Task Fetching Unit:   [run t1] ----------- [run t2] (parallel) ---- waits on $1,$2
                          \_____________________/__________________ [run t3]
                                                                     |
Joiner: answer the user  OR  replan with observations  <------------+
```

ADaPT: decompose only when execution fails

```
ADaPT(task, depth):
    ok = Executor(task)                  # try directly first
    if ok or depth == d: return ok
    plan = Planner(task)                 # e.g. "AND"/"OR" of subtasks
    return combine(ADaPT(sub, depth+1) for sub in plan)
```

Tree of Thoughts (ToT)

```
                      [problem]
              /           |            \
         thought a      thought b      thought c       <- generate (k per node)
         value=0.9      value=0.2      value=0.6       <- evaluate (LLM or rule)
          /    \                         |             <- keep top-b (BFS) or DFS
       a1        a2                     c1
     (24!)     (dead)                (dead) -> backtrack
```

Architecture

```
         root (user task)
         /     |      \             UCT select -> node n
       n1     n2      n3            expand: 5 candidate actions from n
      /  \             \            execute tools in parallel
    n11  n12           n31          reflect: LLM critique + value score (+ unit tests)
     |                               backprop: update mean value along path
  SOLVED  <-- terminate when a trajectory passes the external check
```

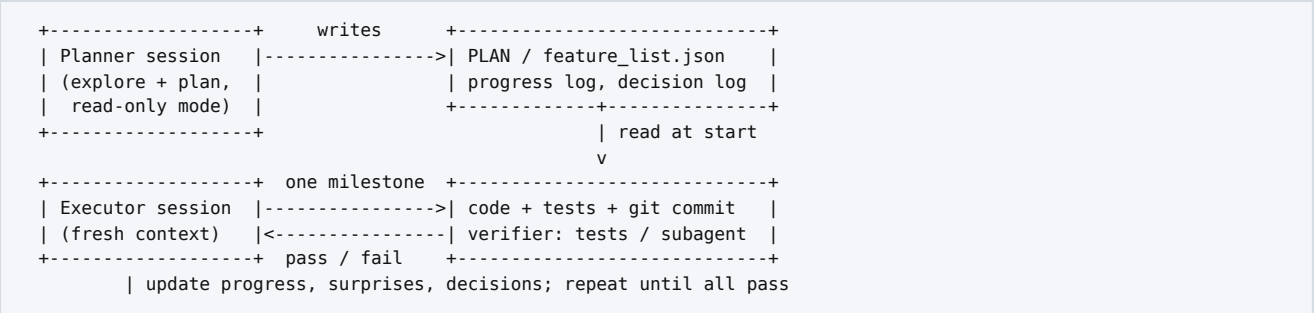
Reflexion: verbal reinforcement across episodes

```
           +-------------------- long-term memory (reflections) ----------+
           v                                                               |
 task -> Actor LM --action--> Environment --obs/reward--> Evaluator ------+
           ^   (trajectory = short-term memory)              |            |
           +-------------------------------------------- Self-reflection LM
```

Interleaved thinking: reasoning between tool calls

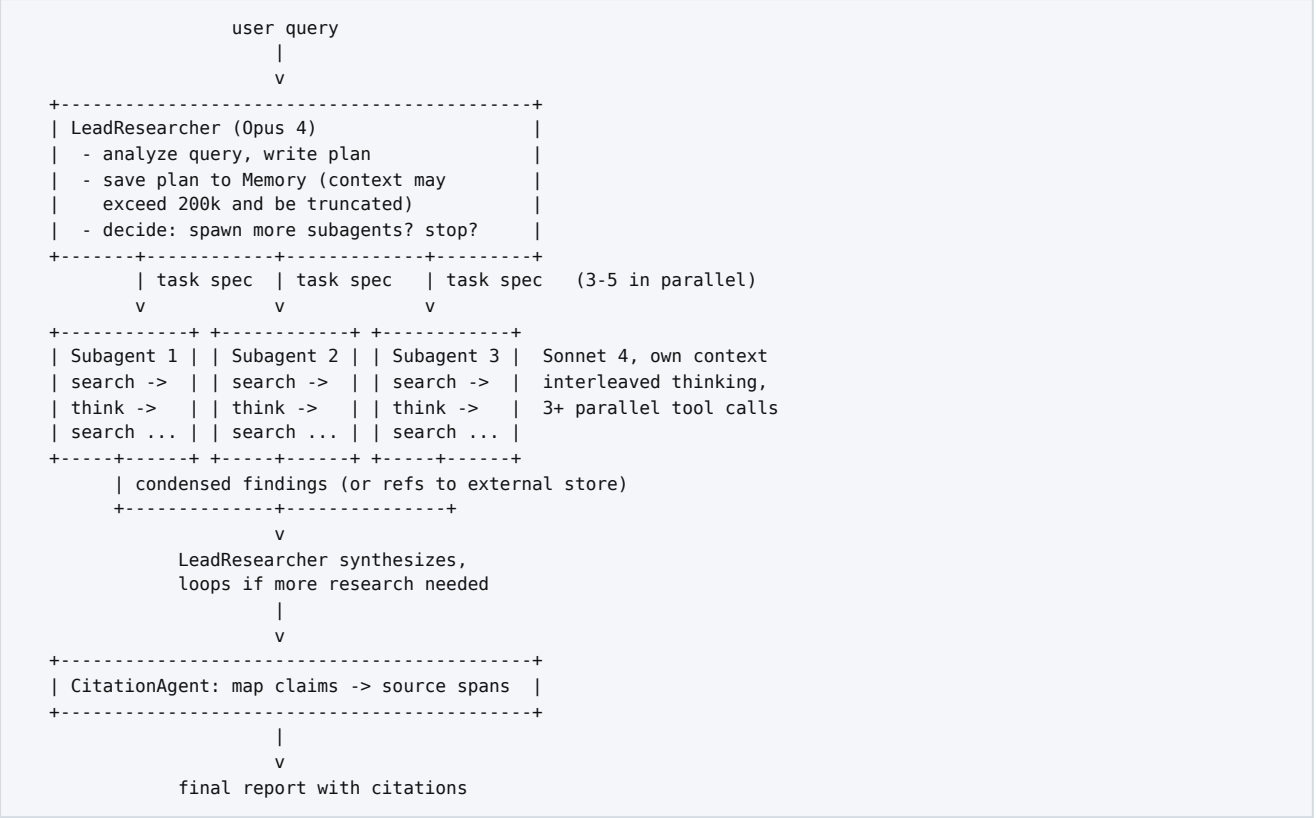
```
Without interleaving                    With interleaving
[think] plan                            [think] plan
[tool_use calc] -> 7500                 [tool_use calc] -> 7500
[tool_use db]   -> 5200  (no thinking)  [think] interpret 7500, decide next
[text answer]            (no thinking)  [tool_use db]   -> 5200
                                        [think] compare, check
                                        [text answer]
```

Plan documents as durable state



04. Multi-Agent Systems: Orchestrators, Teams, Handoffs, and Why They Fail

Architecture



Recommended architecture

```
 Anti-pattern (parallel, context-isolated)     Recommended (single-threaded)

       task                                        task
        |                                          |
     planner                                       v
    /       \                                  +-----------+
sub-A       sub-B   <- divergent implicit      |  agent    |<-- full trace
 |            |        decisions               |  step 1   |    always visible
 v            v                                |  step 2   |
 merge (conflicts)                             |  ...      |
                                               +-----+-----+
                                                     | context nearly full
                                                     v
                                          +-------------------+
                                          | compression model |
                                          | (key decisions,   |
                                          |  events, details) |
                                          +-------------------+
                                                     |
                                                     v
                                           agent continues
```

v0.2 architecture

```
+----------------+  message (plan / code)  +-------------------+
| AssistantAgent |------------------------>| UserProxyAgent    |
|  (LLM)         |<------------------------|  code executor    |
+----------------+  execution result /     |  (Docker) + human |
                     human feedback        +-------------------+
         ... repeat until termination (e.g. "TERMINATE" or max turns)
```

Magentic-One

```
 OUTER LOOP: Task Ledger  (facts, guesses, plan)
    ^  re-plan when progress stalls
    |
 INNER LOOP: Progress Ledger (is task done? who acts next? instruction)
    |
    +--> WebSurfer | FileSurfer | Coder | ComputerTerminal
```

MetaGPT

```
requirement
    |
    v
ProductManager --PRD--> Architect --design/API--> ProjectManager --tasks-->
                                                                          |
                   +------------------------------------------------------+
                   v
               Engineer(s) --code--> QA / executable feedback --fix--> ...
    (all roles publish to / subscribe from a shared message pool)
```

ChatDev

```
ChatChain:  [Design] --> [Coding] --> [Testing] --> [Documenting]
               |             |             |              |
           CEO<->CPO     CTO<->Prog    Prog<->Reviewer  CPO<->Prog
           CEO<->CTO                   Prog<->Tester
 each box = sequence of 2-agent instructor/assistant chats with
            termination by consensus or max turns
```

Swarm

```
  user msg
     |
     v
+----------+  transfer_to_refunds()  +-----------+  transfer_back_to_triage()
|  Triage  |------------------------>|  Refunds  |--------------------------+
|  agent   |<------------------------+  agent    |                          |
+----------+                         +-----------+                          |
     ^      shared: message history + context_variables (dict)              |
     +-----------------------------------------------------------------------+
  Only ONE agent is "active" at a time; the conversation moves, not a task.
```

A topology taxonomy

```
(a) Orchestrator-worker      (b) Manager w/ validation     (c) Sequential pipeline
        O                            M                      A --> B --> C --> D
      / | \                        / | \                    (SOP / chat chain)
     W  W  W                      W  W  W
     results -> O                 outputs -> M checks

(d) Group chat / selector    (e) Handoff / swarm           (f) Shared task board
   A   B   C                   A ---> B ---> C               lead posts tasks
    \  |  /                    ^             |               T1 T2 T3 T4 ...
   [shared thread]             +-------------+               ^  ^  ^ claim (locks)
   selector picks next         one active agent at a time    t1 t2 t3 + mailboxes
```

05. Context Engineering and Memory: Caches, Compaction, Scratchpads, and Long-Horizon State

What goes into a request

```
+-------------------------------- one model request --------------------------------+
| [tools] -> [system prompt] -> [stable memory] -> [history .............] -> [new]  |
|  rarely      rarely            per session       append-only, grows      per step |
|  changes     changes           (CLAUDE.md,       (assistant msgs,                 |
|                                MEMORY.md,        tool calls, results,             |
|                                skill index)      thinking, summaries)             |
+-----------------------------------------------------------------------------------+
       ^ cache breakpoint(s) go on the last byte-identical block before change
```

Architecture and data flow

```
user task
   |
   v
+--------------------+    append-only     +------------------------------+
| stable prefix      |------------------->| context = prefix + actions   |
| (system + tools,   |                    |   + observations (+ todo.md  |
|  no timestamps)    |                    |   recitation at the tail)    |
+--------------------+                    +---------------+--------------+
                                                          | decode, with
                                                          | logit masks from
                                                          v a state machine
+-------------------+   large observations  +------------------------------+
| sandbox file      |<---------------------| action: browser_* / shell_* / |
| system (memory)   |---- path / URL ------>| file ops ...                  |
+-------------------+   restorable stubs    +------------------------------+
```

4. Compaction, pruning, and trimming

```
before:  [prefix][t1][t2][t3] ..................... [t40][t41][t42]  ~ trigger
                \_____________ summarize _______/  \__ keep raw __/
after:   [prefix][SUMMARY block][t40][t41][t42] + re-injected state
                                                 (instructions, recent files,
                                                  memory index, invoked skills)
```

6.1 Effective harnesses for multi-session coding (Anthropic, 2025)

```
session 1: INITIALIZER                  sessions 2..N: CODING AGENT
+--------------------------+            +----------------------------------+
| write feature_list.json  |            | pwd; read git log + progress file  |
|  (200+ features, all     |  files +   | read feature list, pick top unmet  |
|   "passes": false)       |  git repo  | run init.sh + basic e2e smoke test |
| write init.sh            |----------->| implement ONE feature              |
| write claude-progress.txt |           | verify as a user (browser MCP)     |
| initial git commit       |            | set passes=true; commit; log entry |
+--------------------------+            +----------------------------------+
```

6.4 The session log as durable external context

```
+-----------+  emitEvent   +----------------------+  getEvents(slice)  +-----------+
| sandbox / |------------->| SESSION LOG          |------------------->| HARNESS   |
| tools     |              | (durable, append-    |                    | transform |
| (hands)   |<-------------| only, outside window)|                    | -> window |
+-----------+  tool calls  +----------------------+                    +-----+-----+
                                                                            |
                                                                      model (brain)
```

8.1 MemGPT and Letta: OS-style virtual context

```
+---------------------------- context window -------------------------------+
| system instructions | CORE MEMORY blocks (persona, human, ...)  | FIFO msgs |
|                     | always visible, size-limited, self-edited |           |
+---------------------+-------------------------------------------+-----------+
           core_memory_append/replace |                        | evicted
           memory_rethink             v                        v
              +----------------------------+   +-------------------------+
              | ARCHIVAL MEMORY            |   | RECALL MEMORY           |
              | unbounded, vector-searched |   | full message history,   |
              | archival_memory_insert/    |   | conversation_search     |
              | archival_memory_search     |   |                         |
              +----------------------------+   +-------------------------+
```

06. Coding Agents: Architectures, Sandboxes, and Verification Loops

1. The reference architecture

```
          +-------------------------------------------------------------+
user task |                         HARNESS                             |
--------> |  system prompt + repo instructions (CLAUDE.md / AGENTS.md)    |
          |  context manager: compaction, subagents, tool-output caps    |
          |                                                             |
          |   +---------+  tool call   +-------------------------------+ |
          |   |  LLM    | -----------> | permission / policy layer     | |
          |   | (loop)  |              | allowlists, classifier, human | |
          |   +---------+              +---------------+---------------+ |
          |        ^                                   |                 |
          |        | observation                       v                 |
          |        |                  +-------------------------------+ |
          |        +----------------- | SANDBOX (OS / container / VM) | |
          |                           | shell, editor, search, tests  | |
          |                           | egress proxy, scoped creds    | |
          |                           +---------------+---------------+ |
          |                                           |                 |
          |   verify: tests | lint | build | screenshot | reviewer model |
          +-------------------------------------------+-----------------+
                                                      v
                                        diff / commit / pull request
```

Architecture: the ACI

```
 LM --(thought + ONE command)--> ACI command layer --> shell in Docker
  ^                               |  open/goto/scroll (100-line window)
  |                               |  find_file / search_dir (names only)
  |                               |  edit start:end  --> lint --> accept|reject
  +----- formatted observation <---+  "ran successfully, no output"
```

OpenHands platform architecture

```
 Agent Canvas (UI) --TS client--> Agent Server (FastAPI, REST+WS)
                                   |  conversations, events, files
                                   v
          +-------------------------------------------------+
          | Conversation --> Agent loop --> Tool executors   |
          |      ^             |   ^           |             |
          |   Events <---- Condenser |      Security analyzer |
          +--------------------------|----------------------+
                                     v
          Workspace: Local (same process) | Docker | Remote API
```

The permission and sandbox stack

```
tool call
  |
  v
[permission rules]  allow / ask / deny patterns; org > user > project scopes
  |
  v
[mode]  manual | accept-edits | plan (read-only) | auto (classifier)
  |
  v
[auto mode]  input: prompt-injection probe on tool OUTPUTS
  |          output: 2-stage transcript classifier on ACTIONS
  v
[OS sandbox for shell]  bubblewrap (Linux) / Seatbelt (macOS)
  |   fs: write = cwd + temp;   net: only via host proxy (domain allowlist)
  v
[checkpoints]  file snapshots for rewind (not shell side effects)
```

14.5 Verification loops, from weak to strong

```
weakest                                                              strongest
"looks done" -> prompted test run -> lint/syntax gate -> repo tests pass
  -> generated repro test (Agentless) -> security scanners (Copilot)
  -> deterministic Stop hook / goal evaluator -> independent reviewer
  -> human PR review with signed, logged provenance
```

07. Computer and Browser Use Agents: Pixels, Accessibility Trees, and the Race from 15% to Human-Level on OSWorld

1. The canonical CUA loop

```
 +-------------------------------------------------------------------------+
 |                        Agent harness (client side)                      |
 |                                                                         |
 |  task --> [ LLM / VLM ] --tool_use--> [ action executor ] --> environment |
 |              ^   ^                     (xdotool, PyAutoGUI,   (VM / X11 / |
 |              |   |                      Playwright, CDP)       browser)   |
 |              |   +-- history (pruned screenshots, memory, todo)           |
 |              |                                                           |
 |              +------ observation <-- [ encoder ] <-- screen / DOM / AX ---+
 |                       (image, a11y text, SoM overlay, URL, tabs)          |
 |                                                                         |
 |  guards: domain allowlist, confirmation gates, injection classifier,     |
 |          step budget, human takeover                                    |
 +-------------------------------------------------------------------------+
```

4.2 The reference architecture

```
+---------------------------- Docker container ----------------------------+
|                                                                          |
|   X11 virtual display (desktop apps, Firefox)                            |
|        ^ xdotool / screenshot                                            |
|        |                                                                 |
|   tools: computer | bash | str_replace_based_edit_tool                   |
|        ^                                                                 |
|        | tool_use / tool_result                                          |
|   agent loop  <---- Claude API (or Bedrock / Vertex) ---->               |
|        |                                                                 |
|   Streamlit chat UI :8501   noVNC :6080   VNC :5900   combined :8080     |
+--------------------------------------------------------------------------+
```

6.2 Observation pipeline (from source)

```
DOMSnapshot.captureSnapshot ---(styles, paint order, rects)---+
DOM.getDocument(depth=-1, pierce=true) --(shadow roots, iframes)+--> EnhancedDOMTreeNode
Accessibility.getFullAXTree (per frame) --(roles, names, values)+       |
Page.getLayoutMetrics ------------------(device pixel ratio)---+       v
                                                          DOMTreeSerializer
   + JS click-listener scan (capped at 100, skipped above 10k elements)  |
                                                                         v
      "[12]<button aria-label=Search />   *[13]<input placeholder=... />"
```

8. Skyvern: planner, actor, validator, then compiled scripts

```
 user goal
    |
    v
 task_v2 planner ---> mini-goal {navigate | extract | loop | compute}
    ^                    |
    |                    v
 check-user-goal     extract-action actor: screenshot + element tree with IDs
 (validator)              -> [CLICK | INPUT_TEXT | SELECT_OPTION | UPLOAD_FILE |
    ^                         SOLVE_CAPTCHA | COMPLETE | TERMINATE ...] + confidence
    |                    |
    +---- results <------+
                         |
          script-generation / script-reviewer --> reusable Playwright-style code
          script-failure-triage, cua-fallback-action (fall back to a CUA model)
```

08. Deep Research Agents: Architectures, Products, and Training

A generic reference architecture

```
 user query
     |
     v
+-----------+   clarifying Qs   +------+
| Scoping / |<----------------->| user |
| clarifier |                   +------+
+-----+-----+
      | research brief / plan
      v
+------------------+   spawn / call    +--------------------------+
| Planner /        |------------------>| Research worker(s)       |
| orchestrator     |<------------------| search -> fetch -> read  |
| (plan in memory) | condensed findings| -> reflect (ReAct loop)  |
+--------+---------+                   +-------------+------------+
         | enough evidence?                          | tools
         v                                           v
+------------------+                     web search, URL fetch, code,
| Writer /         |                     file search, MCP connectors
| synthesizer      |
+--------+---------+
         v
+------------------+
| Citation pass /  | --> final report (markdown, inline citations)
| verifier         |
+------------------+
```

Architecture

```
Responses API request (background=True)
  tools: web_search_preview (required), code_interpreter (opt), MCP (opt)
       |
       v
+-------------------------------------------------------------+
| o3-deep-research / o4-mini-deep-research (single agent)     |
|  reasoning step -> web_search_call -> reasoning -> ...      |
|  -> code_interpreter_call -> mcp_tool_call -> ... -> message|
+-------------------------------------------------------------+
       |
       v
response.output: [reasoning summaries, search calls, code calls,
                  MCP calls, final message w/ url_citation annotations
                  (start_index, end_index, title, url)]
```

Architecture

```
user --> LeadResearcher (extended thinking; plan saved to Memory)
            |  spawn 3-5 in parallel (more for hard queries)
    +-------+---------+----------------+
    v                 v                v
 Subagent A        Subagent B       Subagent C     each: own context,
 search/fetch      search/fetch     search/fetch   interleaved thinking,
 (3+ tools in      ...              ...            3+ parallel tool calls
  parallel)
    |                 |                |
    +---- condensed findings (or refs to files/artifacts) ----+
                      v
            LeadResearcher: synthesize; more subagents? --yes--> loop
                      | no
                      v
            CitationAgent (docs + draft -> claims mapped to sources)
                      v
                 final cited report
```

API architecture (2026)

```
create(agent=deep-research[-max], background=True, agent_config={...})
    |
    +-- collaborative_planning=True --> plan <--> user refinements
    |                                    (previous_interaction_id chain)
    v  collaborative_planning=False
[agent loop: google_search / url_context / code_execution / MCP / file_search]
    |  stream: thought_summary deltas, text deltas, image deltas
    v
report (+ charts)  --> follow-ups served by a standard Gemini model
```

STORM and Co-STORM (Stanford OVAL)

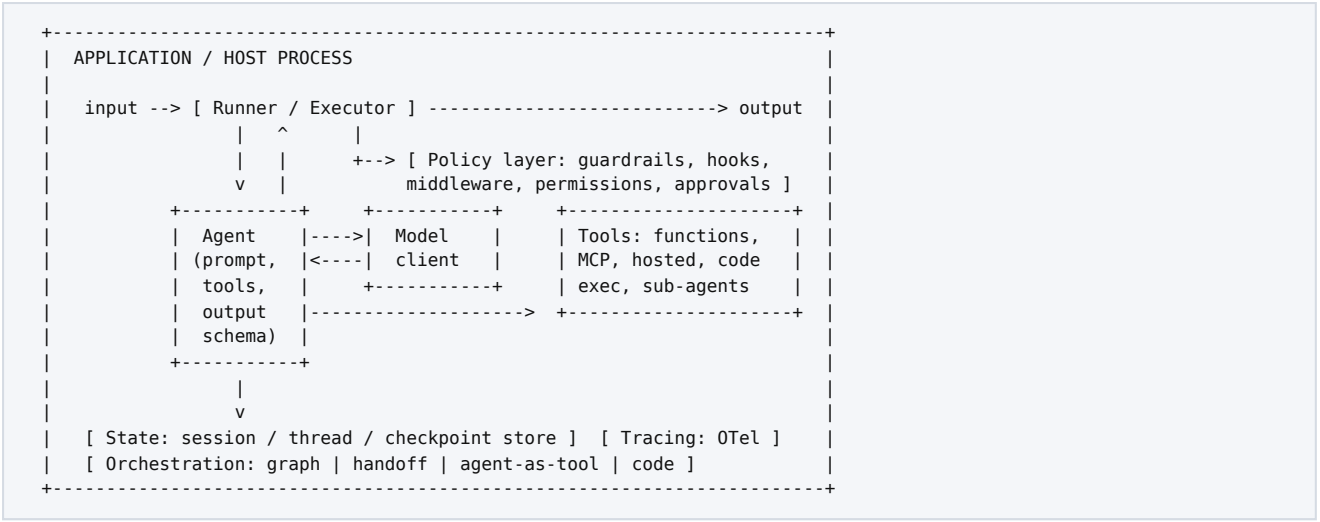
```
topic --> find related articles --> perspectives P1..Pk
            |
  for each Pi: writer(Pi) <-> expert(grounded in search)  [simulated dialogue]
            |
      collected Q/A + URLs --> outline LM --> outline
                                |
                   article LM (per section, cite refs) --> polish LM --> article
```

LangChain Open Deep Research

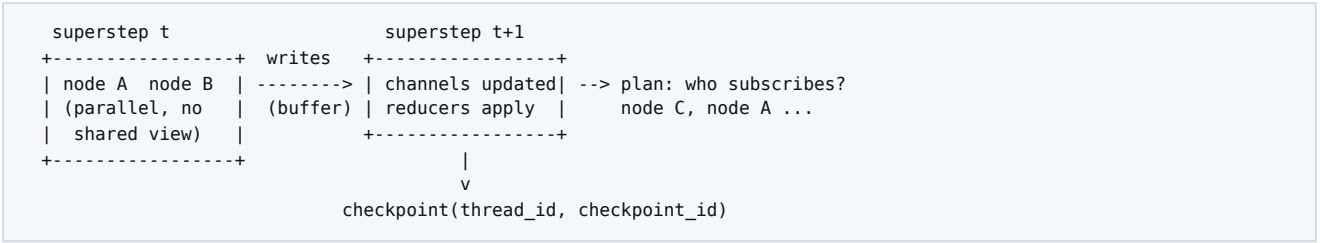
```
clarify_with_user --> write_research_brief --> research_supervisor
                                                |  tools: think_tool,
                                                |  ConductResearch(topic),
                                                |  ResearchComplete
                                    +-----------+-----------+
                                    v           v           v
                               researcher   researcher   researcher
                               (ReAct: search/MCP/think)  ...
                                    |           |           |
                               compress_research (per researcher)
                                    +-----------+-----------+
                                                v
                                      final_report_generation
```

09. Agent Frameworks and SDKs: LangGraph, OpenAI Agents SDK, Claude Agent SDK, Google ADK, Pydantic AI, smolagents, Mastra, Strands, Microsoft Agent Framework, LlamaIndex, DSPy

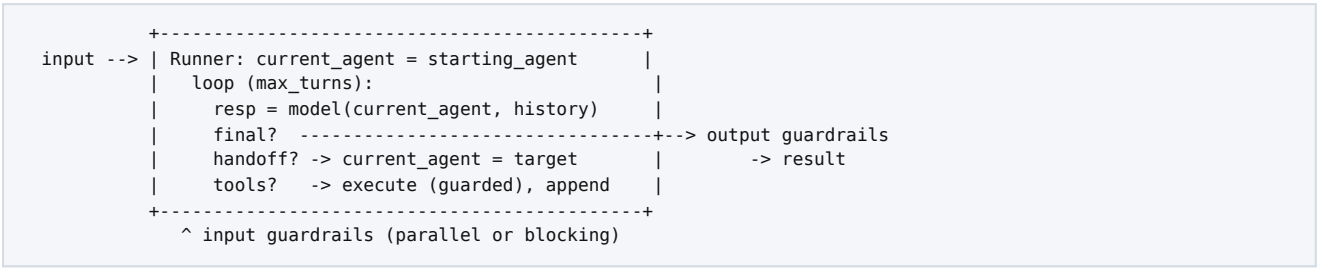
1. A shared reference model



2.2 Architecture: Pregel over channels



3.2 The runner loop



4.2 Architecture: a library wrapping an agent binary



5.2 Event-sourced runtime (1.x core, retained in 2.0)

```
Execution logic                       Runner                    Services
---------------                       ------                    --------
build Event(content, state_delta) --> receive
yield; PAUSE                          commit deltas  ----------> SessionService
                                      append to history          ArtifactService
                                      forward upstream (UI)
RESUME (state now committed)  <------ signal done
```

7.3 Executing untrusted code

```
Approach 1: snippets only               Approach 2: whole agent in sandbox
+-----------+   code    +---------+     +---------------------------------+
| agent+LLM |---------->| sandbox |     | sandbox: agent + model client + |
| (host,    |<----------| (E2B,   |     |          tools + executor       |
| API keys) |  result   | Docker) |     +---------------------------------+
+-----------+           +---------+       stronger isolation, keys inside
simple, keys stay local, no managed
multi-agent, state shuttled each step
```

10.3 Design via ADRs

```
MAF request path (Python), with ADR features annotated
user msg -> Agent.run
            |-- agent middleware (ADR-0007)        [FIDES labels, ADR-0024]
            |-- context providers (CodeAct ADR-0038, memory, RAG)
            |-- chat client.get_response
            |      |-- compaction before every call (ADR-0019)
            |      |-- tool loop -> function middleware -> tools / approvals
            |      '-- background mode + continuation token (ADR-0009)
            '-- history provider (pre-write compaction) -> thread store
```

10. Agent Infrastructure and Runtime: Sandboxes, Durable Execution, State, Observability, Deployment, and Cost

1.1 The problem

```
weaker / cheaper                                          stronger / costlier
+--------------------+   +-------------------------+   +--------------------------+
| OS process sandbox |   | Application kernel      |   | MicroVM                  |
| bubblewrap, Seat-  |   | gVisor (Sentry + Gofer) |   | Firecracker, Cloud Hyp., |
| belt, seccomp,     |   | syscalls served in a    |   | Kata; guest kernel,      |
| namespaces         |   | userspace Go kernel     |   | KVM hardware boundary    |
+--------------------+   +-------------------------+   +--------------------------+
 shares host kernel       two kernels, no shared code   two kernels, hardware VT
 ~zero startup cost       small fixed per-sandbox cost  <=125 ms boot, <=5 MiB VMM
```

1.5 Credential isolation: keep secrets out of the box

```
 +---------------- sandbox (untrusted) ----------------+
 |  model-generated code   scoped token only           |
 |        |                                            |
 +--------|--------------------------------------------+
          v  (all egress)
 +-----------------------+      +----------------------+
 | egress proxy / git    |----->| vault: real tokens   |
 | proxy: allow-list,    |<-----| (never in sandbox)   |
 | policy check, inject  |      +----------------------+
 +-----------------------+
          |
          v   GitHub / MCP servers / package registries
```

1.6 E2B: snapshot-restore sandboxes on Firecracker

```
SDK ---- POST /sandboxes ----> API (control plane: auth, quotas, placement)
                                | Postgres (entities), Redis (running state)
                                | gRPC Create
                                v
user/agent traffic         Orchestrator (one per node, Go, runs as root)
      |                     - fetch template: local cache -> NFS -> object store
      v                     - network slot: netns + veth + tap + nftables
client-proxy (edge) ---->   - rootfs: CoW overlay via in-process NBD
Redis lookup                - memory: lazy, userfaultfd from memfile
sandbox:routing:{id}        - resume Firecracker VM, call envd /init
miss -> API Resume             |
                               v
                       [ Firecracker microVM ]  <- envd agent inside (process,
                                                   filesystem, port APIs)
Object storage: templates and snapshot diffs   ClickHouse: events, metrics
```

2.1 The problem

```
 attempt 1:  [llm#1]->[tool#1]->[llm#2]->[tool#2]->[llm#3]-X crash
                |        |         |        |
                v        v         v        v
 durable log:  r1       r2        r3       r4          (append-only)
                |        |         |        |
 attempt 2:  [llm#1]  [tool#1]  [llm#2]  [tool#2]  ->[llm#3]->[tool#3]-> done
             replayed replayed  replayed replayed    executed for real
             (no token cost, no repeated side effect)
```

3.2 The append-only session log (Anthropic Managed Agents)

```
+--------------------+      append / read slices      +----------------------+
| Harness (stateless)|------------------------------>| Session log          |
| loop: call Claude, |<------------------------------| append-only, durable |
| route tool calls   |                               | outside harness and  |
+---------+----------+                               | sandbox              |
          | execute(name, input) -> string           +----------------------+
          v
+--------------------+    credentials via vault proxy (session-scoped token)
| Sandbox (cattle)   |    provisioned lazily, only when a task needs one
+--------------------+
```

4.3 Platforms and their ingestion architectures

```
SDKs --batches--> Web container --write--> S3 / blob (raw events, durable)
                      |                        |
                      +--ref--> Redis queue    |
                                   |           v
                                Worker ---read---> ClickHouse
                                                   (wide observations table,
                                                    no read-time joins)
Redis also caches API keys (cheap auth rejection) and hot prompts
```

5.1 Decouple the brain from the hands

```
         +---------------- load balancer / queue ----------------+
         v                        v                             v
   [harness pod]             [harness pod]                 [harness pod]   stateless
         \                        |                             /          autoscale
          +----------- session log / checkpoint DB -------------+           durable
                                  |
            +---------------------+----------------------+
            v                     v                      v
       [sandbox]              [sandbox]              (none yet)          lazy,
       microVM                microVM                                    disposable
```

11. Agent Protocols: MCP, A2A, ACP, AGNTCY, Agent Payments, Tool Registries, and Interop Patterns

Overview

```
             THE 2026 AGENT PROTOCOL STACK (layered view)

+-------------------------------------------------------------------------+
| Commerce / authorization  AP2 mandates (SD-JWT VCs) | x402 (HTTP 402)   |
+-------------------------------------------------------------------------+
| Agent <-> agent           A2A v1.0 (JSON-RPC | gRPC | HTTP+JSON)        |
|                           [IBM ACP merged in] [AGNTCY Agent Connect]    |
+-------------------------------------------------------------------------+
| Agent <-> tools/context   MCP 2026-07-28 (stdio | Streamable HTTP)      |
|                           + extensions: Tasks, Apps, EMA, Server Cards  |
+-------------------------------------------------------------------------+
| Discovery / registries    MCP Registry | Agent Cards | AI Catalog |     |
|                           AGNTCY Dir (OASF records, DHT) | ANP discovery |
+-------------------------------------------------------------------------+
| Identity / authz          OAuth 2.1 + RFC 9728/8707/9207, CIMD, ID-JAG, |
|                           signed Agent Cards (JWS), DIDs, SPIFFE/SPIRE  |
+-------------------------------------------------------------------------+
| Transport                 HTTP/1.1-2 + SSE, gRPC, stdio, SLIM (MLS E2E) |
+-------------------------------------------------------------------------+
```

11.1.3 Architecture: hosts, clients, servers, primitives

```
          MCP 2026-07-28: HOST / CLIENT / SERVER TOPOLOGY

 +---------------------- Host process (e.g. IDE) ----------------------+
 |  LLM + conversation state + consent UI + policy                     |
 |   |               |                  |                              |
 | [Client 1]     [Client 2]        [Client 3]   (one client per server)|
 +---|---------------|-----------------|-------------------------------+
     | stdio         | stdio           | Streamable HTTP (POST /mcp)
     v               v                 v
 [Server: files]  [Server: DB]     [Remote server] --- OAuth RS ---> [AS]
 tools/resources  tools            tools/resources/prompts
                                   _meta: protocolVersion,
                                   clientCapabilities, clientInfo
```

11.1.4 Transport evolution: stdio, HTTP+SSE, Streamable HTTP, stateless HTTP

```
  STREAMABLE HTTP, 2026-07-28 (stateless; any request -> any replica)

Client                         L7 LB / gateway               Server replicas
  | POST /mcp                      |                              |
  | MCP-Protocol-Version: 2026-07-28                              |
  | Mcp-Method: tools/call         |-- route/authorize on headers->| R1 | R2 | R3
  | Mcp-Name: search               |   (no body parse, no sticky)  |
  | {jsonrpc, id:1, method, params:{..., _meta:{version, caps}}}   |
  |<------------- 200 application/json  (simple result) -----------|
  |<------------- 200 text/event-stream (progress..., result) -----|
  |   [close stream] ==> cancellation of that request              |
```

11.1.7 Authorization

```
  MCP AUTHORIZATION FLOW (2026-07-28)

Client                MCP server (RS)             Authorization server
  |-- request, no token ->|                                 |
  |<- 401 WWW-Authenticate: Bearer resource_metadata=..., scope="files:read"
  |-- GET /.well-known/oauth-protected-resource ->|         |
  |<- PRM {authorization_servers:[...]} ----------|         |
  |-- GET AS metadata (RFC 8414 or OIDC discovery) -------->|
  |   (refuse if code_challenge_methods_supported absent)   |
  |   client_id = https://app.example/oauth.json (CIMD) --->|-- fetch metadata
  |-- browser: authorize?code_challenge(S256)&resource=<MCP server URI>
  |<- redirect: code + iss  (client MUST validate iss == recorded AS)
  |-- token: code_verifier + resource ---------------------->|
  |<- access token (audience = this MCP server) -------------|
  |-- request + Bearer token ->| validate audience; never pass token through
```

11.2.2 Data model

```
          A2A v1.0 TASK LIFECYCLE (TaskState enum)

  SendMessage --> SUBMITTED --> WORKING --+--> COMPLETED
                    |             ^  |   +--> FAILED
                    v             |  |   +--> CANCELED (CancelTask)
                 REJECTED         |  +--> INPUT_REQUIRED --(client msg, same taskId)
                                  |  +--> AUTH_REQUIRED  --(credential out-of-band)
                                  +-------------------------------------+
```

11.4 AGNTCY: Infrastructure for an "Internet of Agents"

```
  SLIM OVERLAY CARRYING A2A / MCP (per AGNTCY docs)

 [Agent A: A2A client]                          [Agent B: A2A server]
  SLIM SDK: session layer (MLS E2E) + SLIMRPC    SLIM SDK (MLS E2E)
       |  gRPC/HTTP2                                  ^
       v                                              |
  [SLIM data-plane node] --routes by hierarchical name--> [SLIM node]
       ^ control plane: routes, node registration, group membership
       | identities: DID-style names; SPIRE federation across orgs
```

11.5.3 AP2 v0.2 (April 2026): Checkout and Payment Mandates

```
  AP2 v0.2 HUMAN-NOT-PRESENT FLOW (simplified)

User --(consent on Trusted Surface)--> open Checkout+Payment Mandates
                                       {constraints, cnf: agent pubkey, exp}
Shopping Agent -- builds checkout --> Merchant -- signs Checkout JWT --> SA
SA signs CLOSED mandates (agent key, hash(Checkout JWT))
SA --> Credential Provider/Network: open+closed Payment Mandate -> credential
SA --> Merchant: credential + open+closed Checkout Mandate -> Checkout Receipt
Merchant --> MPP: charge; MPP verifies Payment Mandate -> Payment Receipt
```

11.6 Tool and Agent Registries, Catalogs, and Discovery

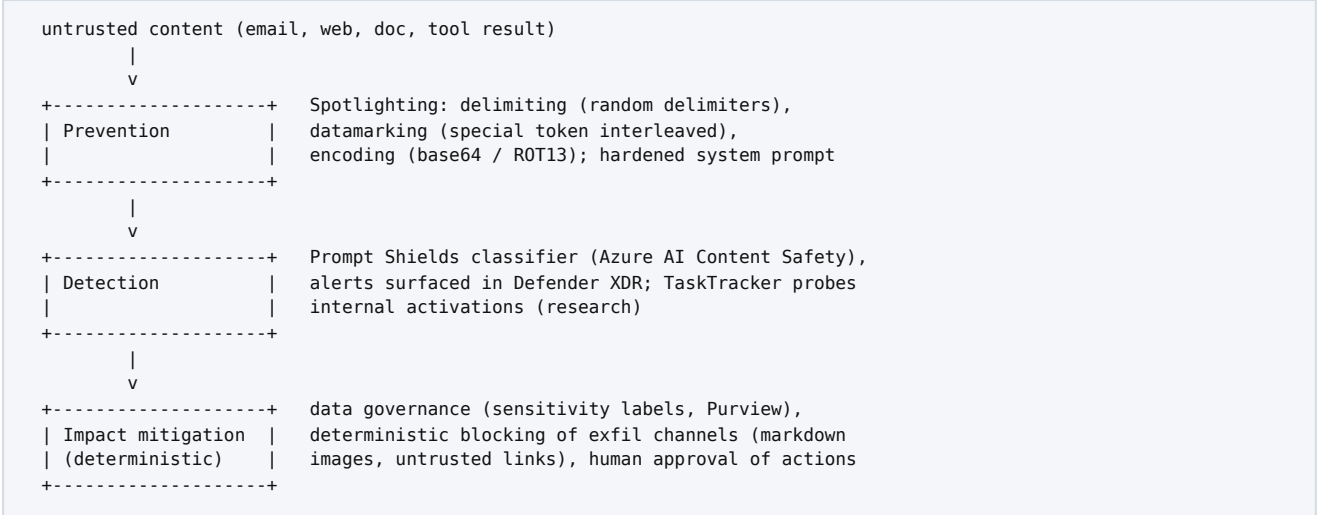
```
  DISCOVERY TIERS

 Publish time                    Pre-connect                 Runtime
 ------------                    -----------                 -------
 MCP Registry (server.json) --->  Server Card / Agent Card --> server/discover
 AGNTCY Dir (OASF, CID, DHT)      /.well-known/ai-catalog.json  GetExtendedAgentCard
 Sub-registries / allowlists      (signed, cacheable)         Tool search (deferred)
```

12. Agent Security and Safety: Indirect Prompt Injection, the Lethal Trifecta, CaMeL and Dual-LLM Patterns,

Permissions, Sandboxing, Approval Gates, OWASP, and MCP Threats

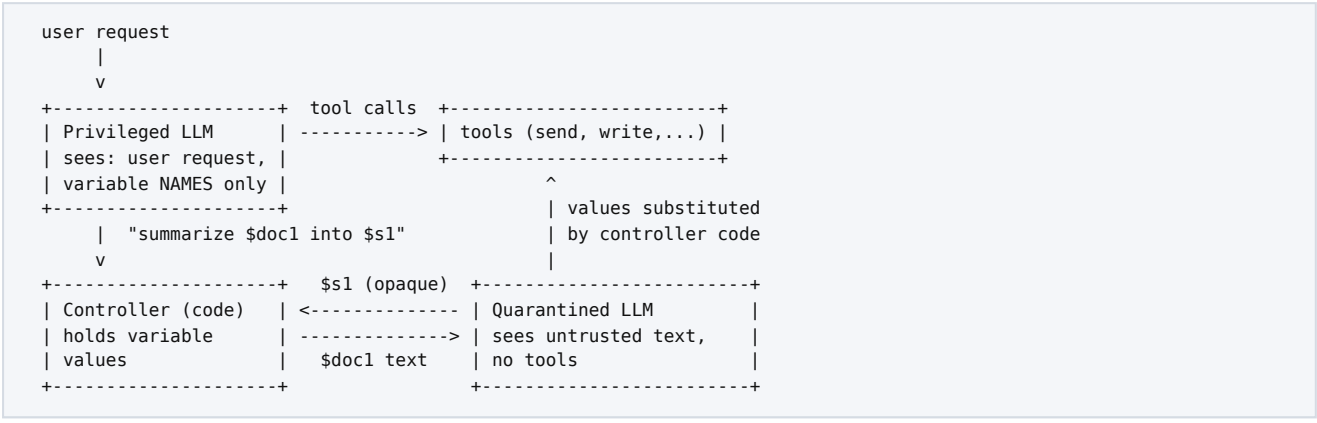
12.2.1 The defense stack vendors deploy



12.3.1 The lethal trifecta



12.4.1 The Dual LLM pattern



12.4.3 CaMeL: capabilities and a custom interpreter

```
user query (trusted)
     |
     v
+--------------------------+
| P-LLM (privileged)       |  writes a restricted Python program
| never sees tool outputs  |  expressing the task's control flow
+--------------------------+
     | program
     v
+-----------------------------------------------------------------+
| CaMeL interpreter (deterministic code)                          |
|  - executes the program step by step                           |
|  - every value is a CaMeLValue carrying Capabilities:          |
|      sources_set  (where the data came from: User, tool, ...)  |
|      readers_set  (who may read it: Public or a set of readers) |
|  - dependencies propagate through every operation             |
|  - before each tool call: SecurityPolicyEngine.check_policy()  |
+-----------------------------------------------------------------+
     |  untrusted text + output schema          | allowed tool calls
     v                                          v
+--------------------------+             +------------------+
| Q-LLM (quarantined)      |             | tools            |
| parses text into typed   |             | (email, drive,   |
| values; no tools; may say |             |  calendar, ...)  |
| "not enough information"  |             +------------------+
+--------------------------+
```

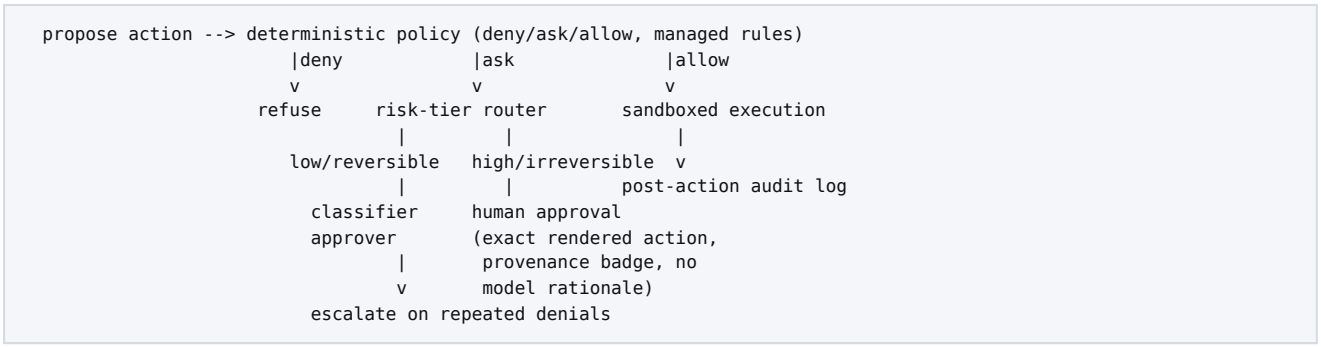
12.5.2 Claude Code auto mode: a classifier as approver

```
tool output --> [input layer: server-side injection probe] --> agent context
                                                               (adds warning if
                                                                hijack suspected)
agent proposes action
     |
     v
[tier 0] read-only tools allowlisted; in-project edits skip classifier;
         broad allow rules that permit arbitrary code are dropped in auto mode
     |
     v
[stage 1] fast single-token yes/no classifier, biased toward "block"
     |  flagged
     v
[stage 2] chain-of-thought re-evaluation (only for flagged actions)
     |
     v
allow / block  --(3 consecutive or 20 total denials)--> escalate to human
                                                        (headless: terminate)
```

12.6.2 Anthropic sandbox-runtime (srt)

```
+--------------------------- host ----------------------------------+
|                                                                   |
|  +------------------ sandbox (bwrap / sandbox-exec) ------------+  |
|  | agent shell cmd + all child processes                       |  |
|  |  fs:  read allowed except denyRead; write ONLY allowWrite   |  |
|  |  net: no network namespace (Linux) / localhost-proxy only   |  |
|  |  unix sockets blocked via seccomp BPF (Linux)               |  |
|  +-----------------------------|-----------------------------+  |
|                                | Unix socket (socat bridge)      |
|                                v                                 |
|  +-------------------------------------------------------------+ |
|  | HTTP proxy + SOCKS5 proxy                                   | |
|  |  deniedDomains -> allowedDomains -> strict -> per-cmd -> ask | |
|  |  resolved-address check: drop loopback, link-local,         | |
|  |  metadata endpoints, host interfaces                        | |
|  |  optional TLS termination for request filtering and        | |
|  |  credential injection                                       | |
|  +-------------------------------------------------------------+ |
+-------------------------------------------------------------------+
```

12.7.2 Placement in the loop

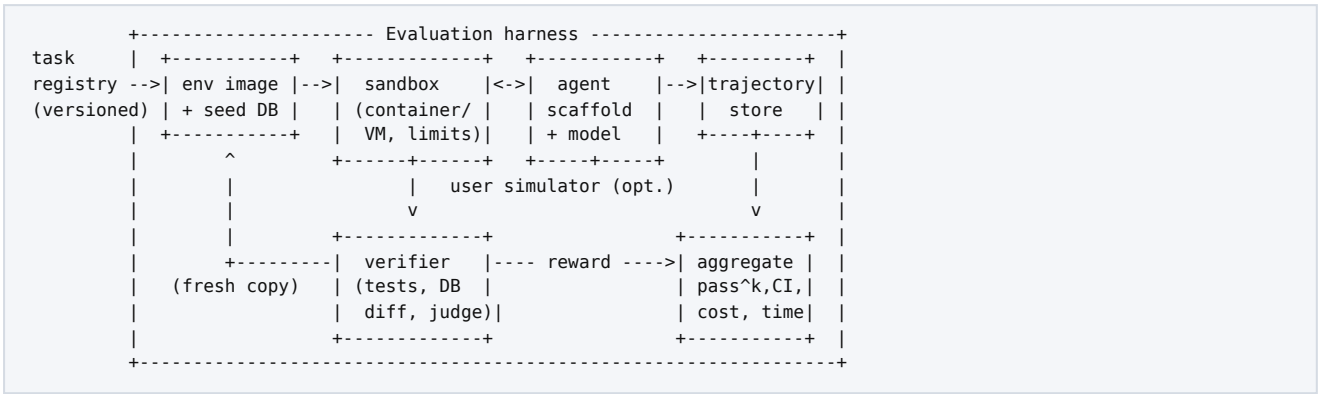


12.9.1 Tool poisoning, shadowing, and rug pulls



13. Agent Evaluation: Benchmarks, Trajectory Grading, pass^k Reliability, and Eval Harness Design

Shared vocabulary



13.2 τ-bench, τ²-bench and τ³-bench (Sierra)

```
+-------------------+  natural language   +------------------------+
| User simulator LLM|<------------------->|  Agent under test      |
| persona + goal +  |                     |  policy.md in prompt   |
| hidden facts      |                     +-----------+------------+
+---------+---------+                                 | tool calls
          | user tools (τ² telecom only)              v
          v                                +------------------------+
+-------------------+                      |  Domain DB (agent side)|
| Device state      |<---- shared world --->|  orders, flights, plans|
| (SIM, APN, data)  |                      +-----------+------------+
+-------------------+                                  |
                                                       v   end of episode
 gold env  = fresh DB + replay(reference actions) --> hash(DB_gold)
 reward    = [hash(DB_pred) == hash(DB_gold)] x [all communicate_info said]
```

SWE-bench (Princeton, ICLR 2024)

```
+--------------------------+  per-instance: repo @ base_commit, deps installed
|  Instance images         |
+--------------------------+  per-repo/version: interpreter + package set
|  Environment images      |
+--------------------------+  per-language: OS, toolchains
|  Base images             |
+--------------------------+
run: apply model patch -> apply hidden test_patch -> run tests -> parse log
     -> FAIL_TO_PASS all pass AND PASS_TO_PASS all pass  => resolved
```

SWE-bench Pro (Scale AI) and the "locked protocol"

```
phase 1 (agent, offline)           phase 2 (grading, pristine)
+------------------------+         +-----------------------------+
| image @ base commit    |  diff   | same image, fresh container |
| sanitized git history  |-------->| apply model.patch           |
| net: model API only    |         | restore hidden tests        |
| 50 min timeout         |         | revert fixture edits        |
+------------------------+         | run verifier -> reward      |
                                   +-----------------------------+
```

13.6 Terminal-Bench and Harbor

```
harbor run -d terminal-bench@2.0 -a claude-code -m <model> --env daytona -n 100
   |
   +--> Job ──> Trials (task x agent x model x attempt), run in parallel
            each: build env -> install agent -> run instruction (timeout)
                  -> collect ATIF trajectory + artifacts
                  -> run tests/test.sh (same or separate sandbox)
                  -> /logs/verifier/reward.{txt,json}
   +--> aggregate metrics, viewer, regrade, export rollouts for RL/SFT
```

13.9 Designing an agent eval harness

```
+-------------------------- Task layer ---------------------------------+
| versioned task packages (instruction, env spec, oracle, verifier)     |
| checksums; release gate: oracle = 100%, no-op = 0%; splits; canaries  |
+--------------------------------+--------------------------------------+
                                 v
+-------------------------- Execution layer -----------------------------+
| sandbox provider (Docker / VM / cloud) | resource policy (floor+ceiling) |
| network policy per phase (agent offline, verifier online if needed)   |
| agent adapter: installed product agent or scaffold; user simulator    |
| limits: wall-clock, working time, tokens, cost, turns                 |
+--------------------------------+--------------------------------------+
                                 v
+-------------------------- Evidence layer ------------------------------+
| trajectory (ATIF-like JSON), artifacts, final env snapshot / patch    |
| infra-error classification separate from task failure                 |
+--------------------------------+--------------------------------------+
                                 v
+-------------------------- Grading layer -------------------------------+
| verifier in pristine sandbox (replay patch / DB hash / trace replay)  |
| code graders > rubric judges (calibrated) > human spot checks         |
| scanners over transcripts (hacking, eval awareness, misconfig)        |
| re-grade stored trajectories when tasks/graders change                |
+--------------------------------+--------------------------------------+
                                 v
+-------------------------- Reporting layer -----------------------------+
| pass@1, pass^k (k trials), clustered SE / Wilson CI, paired deltas    |
| cost, tokens, latency, tool errors; config manifest (resources, version) |
+-----------------------------------------------------------------------+
```

14. Training Agents: Agentic RL, Environment Design, Tool-Use Fine-Tuning, RL for Software Engineering, and RL Gyms

Overview

```
          THE AGENT TRAINING STACK (2025-2026 consensus shape)
 +-------------------------------------------------------------------------+
 | 1. Task / environment factory                                           |
 |    real repos, MCP tools, synthetic tools, simulated users, web, GUIs  |
 |    -> containerized env + verifier (tests, rule checks, judge rubric)  |
 +----------------------------------+--------------------------------------+
                                    |
 +----------------------------------v--------------------------------------+
 | 2. Trajectory synthesis  (teacher models / multi-agent simulation)      |
 |    -> filter by verifier or LLM judge (rejection sampling)              |
 +----------------------------------+--------------------------------------+
                                    | SFT "cold start"
 +----------------------------------v--------------------------------------+
 | 3. Agentic RL                                                           |
 |    policy <-> env, multi-turn rollouts, tool outputs masked from loss,  |
 |    group-relative advantages, difficulty filtering, length budgets      |
 +----------------------------------+--------------------------------------+
                                    |
 +----------------------------------v--------------------------------------+
 | 4. Evaluation + production feedback (SWE-bench, tau-bench, BFCL, logs)  |
 +-------------------------------------------------------------------------+
```

14.1.1 Toolformer (Meta AI, 2023): self-supervised API-call insertion

```
plain text x ----> (a) SAMPLE: few-shot prompt the LM to propose API calls
                       at positions where P("<API>") is high
                   (b) EXECUTE: run each candidate call (calculator, QA,
                       Wikipedia search, MT, calendar) -> result r
                   (c) FILTER: keep call c at position i only if inserting
                       "<API> c -> r </API>" lowers the LM loss on the
                       FOLLOWING tokens by >= threshold, compared with no call
                       and with the call without its result
                   (d) FINE-TUNE the same LM on the augmented corpus
```

14.1.5 Verified synthetic function-calling data: APIGen, APIGen-MT, ToolACE (2024-2025)

```
    VERIFIED SYNTHETIC TOOL DATA (APIGen / ToolACE family)
API pool (real executable or synthesized)
       |
       v
query + call generator (LLM, multi-agent role-play: user / assistant / tool)
       |
       v
[format check] --fail--> drop
       |
[execute or simulate] --error--> drop
       |
[semantic / model check] --mismatch--> drop
       |
       v
SFT dataset (single-turn, parallel, multi-turn, irrelevance/abstain cases)
```

14.2 From Single-Turn RLVR to Agentic RL: What Changes

```
          MULTI-TURN AGENTIC RL ROLLOUT (one episode, one group member)
prompt -> [policy: think + tool_call]    (loss ON)
       -> [env: tool result / stdout]    (loss OFF, masked)
       -> [policy: think + tool_call]    (loss ON)
       -> [env: test output]             (loss OFF)
       -> [policy: final patch / answer] (loss ON)
       -> verifier(state) -> r in {0,1} or [0,1]
group of G episodes -> advantage_i = r_i - mean(r)  (optionally / std)
drop group if all r equal (zero gradient); drop/mask truncated episodes
```

14.4.1 Environment factories: SWE-Gym, R2E-Gym, SWE-smith, Multi-SWE-RL, RepoLaunch

```
           SWE GYM CONSTRUCTION PIPELINE (generic)
repo @ commit --> env-builder agent (install deps, run test suite)
       |                       |
       |                       +--> fail: discard repo / retry with fixes
       v
task source:  real issue+PR | commit + back-translated issue | injected bug
       |
       v
validate: FAIL_TO_PASS tests fail before, pass after gold patch;
          PASS_TO_PASS tests stay green; runtime under budget
       |
       v
container image (or per-repo image + patch) + task card + verifier
```

14.4.2 SWE-RL (Meta FAIR, 2025): RL on software-evolution data without execution

```
GitHub events + repo clones --> PR records (issue text, code context, oracle patch)
                                        |
policy input:  issue + full contents of relevant files
policy output: reasoning + search/replace edits
                                        |
reward:  -1                      if output is malformed
         similarity(pred, oracle) otherwise (sequence-similarity of patches, 0..1)
                                        |
                              GRPO-style policy update
```

14.5.2 The large-scale agentic data synthesis pipeline

```
          KIMI K2 AGENTIC DATA SYNTHESIS (as described in [12])
+-------------------------+     +---------------------------------------+
| Tool repository         |     | Agent + task generation               |
|  3,000+ real MCP tools  |---->|  diverse agents (system prompt +      |
|  20,000+ synthetic tools|     |  tool subset); tasks with rubrics     |
|  (domain-evolved specs) |     |  (success criteria, expected calls)   |
+-------------------------+     +-------------------+-------------------+
                                                    |
                    +-------------------------------v-------------------+
                    | Multi-turn trajectory generation                  |
                    |  simulated user  <->  agent  <->  tool simulator  |
                    |  (stateful, injects stochastic outcomes/failures) |
                    |  + REAL execution sandboxes for coding / SWE      |
                    +-------------------------------+-------------------+
                                                    |
                    +-------------------------------v-------------------+
                    | LLM judge scores each trajectory against its rubric |
                    | keep successes only = large-scale rejection sampling|
                    +-------------------------------+-------------------+
                                                    v
                                          SFT data for agentic skills
```

14.7 RL Gyms, Environment Hubs and Training Frameworks

```
+------------------+   tasks + rubrics    +-----------------------------+
|  Environment hub |--------------------->|  Rollout service            |
|  (sandboxes,     |                      |  vLLM / SGLang, multi-turn  |
|  verifiers,      |<---------------------|  tool loops, partial        |
|  judges, sims)   |  actions / patches   |  rollouts, token-ID capture |
+--------+---------+                      +--------------+--------------+
         | rewards                                       | trajectories
         v                                               v
+------------------+   batches            +-----------------------------+
|  Buffer + filters|--------------------->|  Trainer (Megatron / FSDP2) |
|  zero-variance,  |                      |  GRPO / DAPO / CISPO / RLOO |
|  staleness, trunc|                      +--------------+--------------+
+------------------+   weight sync (broadcast / delta)   |
         ^-----------------------------------------------+
```

15. Customer Support and Enterprise Agents: Sierra, Decagon, Fin, Klarna, Agentforce, ServiceNow, Zendesk, and the Guardrail Stack

15.1 The problem shape and a reference architecture

```
channels: chat | email | SMS | voice (VAD -> STT ... TTS) | ChatGPT/app surfaces
       |
       v
+-------------------+   parallel    +-----------------------------------+
| input supervisors |-------------->| bad-actor / injection / topic       |
| (small, fast)     |               | filters; may gate the final reply   |
+---------+---------+               +-----------------------------------+
          v
+-------------------+    trusted state (auth, entitlements) gates routes/tools
| router/classifier |<--- identity verification gate (deterministic) ---------+
| (SLM, 1 token)    |                                                         |
+---------+---------+                                                         |
          v                                                                   |
+-----------------------------+     +-------------------+   +---------------+ |
| procedure / journey / AOP   |<--->| retrieval + rerank|   | tools/actions | |
| (NL instructions + code     |     | (domain-tuned)    |   | read | write  |-+
|  conditions, pinned params) |---->+-------------------+   | (confirm/HITL)|
+-------------+---------------+                             +-------+-------+
              v                                                     |
+-----------------------------+  rewrite / block / escalate         |
| output supervisors +        |<------------------------------------+
| deterministic output guards |
+-------------+---------------+
              v
   reply or handoff (summary -> human queue)    traces -> monitors, QA, sims
```

Supervisors and tolerance tiers

```
customer msg
    |
    v
[supervisor: input screen] --> [agent: content (grounding) + rules/policies (NL)]
                                          |
                                          v
                           [supervisor: review proposed response]
                                          |
                                          v
                  [deterministic guard: scan output for sensitive data]
                                          |
                                          v
                                       reply
deterministic tool gating: account-modification tools are not even *available*
until the customer is authenticated
```

15.6 Salesforce Agentforce: Atlas, Agent Script and the control dial

```
subagent resort_hours:
    reasoning:
        instructions: ->
            | Use {!@actions.get_resort_hours} to look up the hours...
            if @variables.reservation_required:
                | ...REQUIRES a reservation. Tell the guest to call ahead...
            else:
                | ...does NOT require a reservation...
        actions:
            get_resort_hours: @actions.get_resort_hours
                with activity_type = ...
                set @variables.reservation_required = @outputs.reservation_required
```

16. Agent Case Studies from Engineering Blogs: Replit, Vercel v0, Ramp, Shopify, Uber, LinkedIn, Airbnb, Stripe, Notion, Canva, Booking.com, DoorDash and Anthropic

16.1 The composite architecture that emerges

```
 invocation surfaces                       governed tool layer
 Slack / web / CLI / PR / cron / Chrome    (Toolshed, MCP Gateway, Agent Gateway, CAPT)
          |                                    ^   authn/z, scoping, redaction,
          v                                    |   rate limits, audit, tool search
 +----------------------+   tool intents  +----+------------------------------+
 | session service      |---------------->| registry (control plane)          |
 | durable event log    |                 | proxy (data plane) -> 100s-1000s  |
 | multiplayer state    |<----------------| of internal APIs / SaaS MCPs      |
 +----------+-----------+   results       +-----------------------------------+
            |
            v
 +----------------------+      +-------------------------------------------+
 | harness / workflow   |----->| sandbox (devbox / microVM / Modal snapshot) |
 | blueprint: det. nodes|      | repo pre-cloned, caches warm, no prod data  |
 | + agent-loop nodes   |<-----| lint/test/build, browser for visual checks  |
 +----------+-----------+      +-------------------------------------------+
            |
            v
 verification gates: local lint -> bounded CI rounds -> reviewer agent -> human
            |
            v
 telemetry: traces, merged-PR rate, acceptance rate, cost/run -> eval + mining loop
```

Architecture.

```
Slack thread --> [hydrate links via MCP] --> devbox (10 s, warm)
                                               |
  +--------------------------------------------v--------------------------+
  | blueprint                                                              |
  | (agent) implement task --> (det) run linters <-> (agent) fix lint      |
  |      --> (det) push --> CI (selected tests of 3M+) --> autofixes       |
  |      --> (agent) fix CI failures --> (det) push --> CI round 2 -> stop|
  +------------------------------------------------------------------------+
                                               |
                                 PR from template --> human review
```

16.2.3 Shopify River and Aquifer: making the session the thing that survives

```
 Session  (durable identity, append-only event log in Postgres)  <- survives all
    ^
    | replay history
 Harness  (agent loop: read history, call model, emit tool intents) <- disposable
    |
    v tool intents
 Sandbox  (filesystem, shell, repo, build/test)                    <- disposable
```

16.2.5 Uber: a layered stack of specialist agents

```
diff -> eligibility filter (drop config/generated/experimental files)
     -> context builder (nearby functions, classes, imports)
     -> assistants: Standard (bugs) | Best-Practices (style registry) | AppSec
     -> grader prompt: confidence per comment (thresholds per assistant/lang/category)
     -> semantic de-dup -> category classifier (suppress low-value categories)
     -> post inline -> "Useful / Not useful" -> Kafka -> Hive -> threshold tuning
```

16.3.1 Replit Agent: from manager/editor/verifier to decision-time guidance

```
trajectory (user msgs, recent tool results, error patterns)
       |
       v
 fast multi-label classifier (cheap model, every iteration)
       |  selects 0..k micro-instructions from a bank of hundreds
       v
 ephemeral injection at the tail of context  --> agent step --> guidance dropped
       |
       +-- "diagnostic signal": "Found 1 new browser console log, use the log tool"
       +-- "consultation trigger": doom loop detected -> ask a *different model*
            with fresh context for a plan (generator-discriminator gap)
```

16.3.2 Vercel v0: a composite "model" with streaming repair

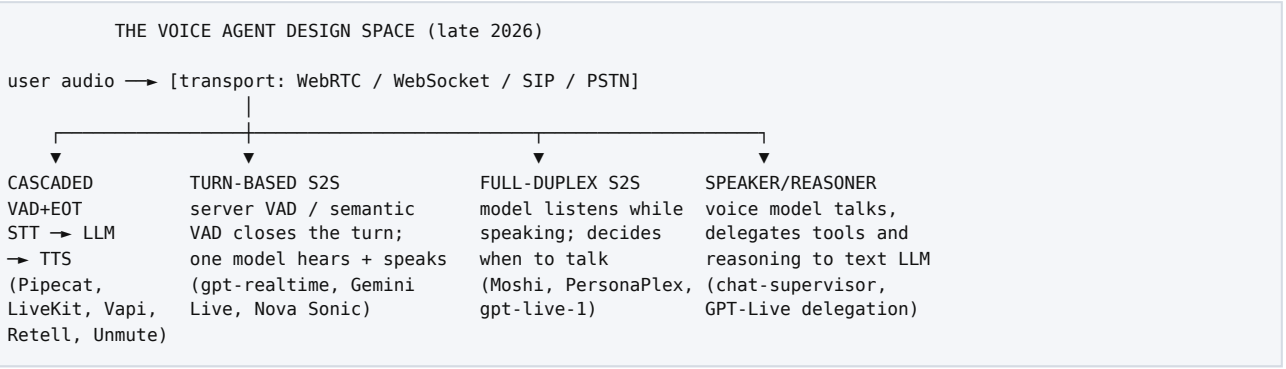
```
user msg --> pre-processing: system prompt + recent msgs + summaries of old msgs
         --> RAG: docs, UI examples, project sources, internal Vercel knowledge
         --> dynamic system prompt (intent detected via embeddings + keywords)
         --> route: base frontier model (new/large) | Quick Edit model (narrow edits)
         --> stream --> "LLM Suspense" rewrite layer (in-flight, <100 ms)
         --> post-stream autofixers: deterministic AST fixes + vercel-autofixer-01
         --> linter --> preview
```

16.3.4 LinkedIn Hiring Assistant: plan-and-execute on a messaging substrate

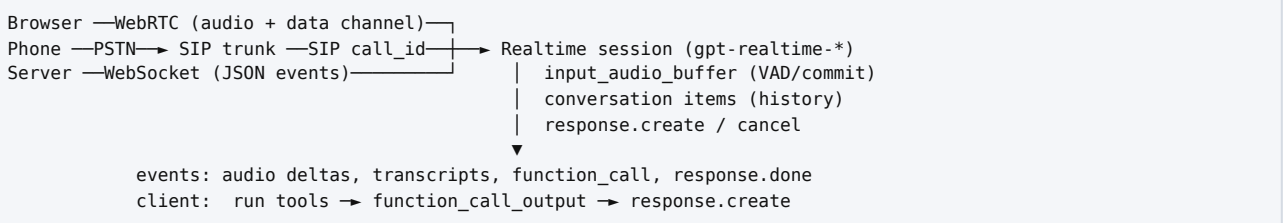
```
recruiter UI <--pub/sub, streamed partials-- supervisor (planner/orchestrator)
                                              |  messages (FIFO per thread)
    +-----------+-----------+----------+--------+--+-----------+-----------+
    intake   sourcing   evaluation  outreach  screening   learning   memory(CMA)
               |           |                               |           |
        query gen +     fine-tuned LLM              recruiter actions  episodic /
        MUSE retrieval  + spec. decoding            -> proposals (HITL) semantic /
        (1B+ profiles)                                                  procedural
```

17. Voice and Realtime Agents: Speech-to-Speech vs Cascaded Architectures, Turn-Taking and Latency Budgets

Overview



Problem and architecture



Architecture

```
          ┌──────────────── GPT-Live session (gpt-live-1) ────────────────┐
caller ◄──┤ SPEAKER: hears + speaks continuously; decides when to delegate │
audio  ──►│        │ delegate(request + transcript snapshot)    ▲ result   │
          └────────┼──────────────────────────────────────────────┼────────┘
                   ▼                                              │
          REASONER (text model): procedures, tools, skills ──────┘
          Responses delegation: hosted by OpenAI
          Client delegation: any Pipecat LLM in a BackendLLMWorker
```

Architecture: frames, processors, pipelines, workers

```
transport.input() ─► STT ─► user_aggregator ─► LLM ─► TTS ─► transport.output()
      │ InputAudioRawFrame   TranscriptionFrame  LLMTextFrame  TTSAudioRawFrame │
      │                                                                        ▼
      │                                              assistant_aggregator (spoken
      │                                              text only, via TTSTextFrame)
 SystemFrames (InterruptionFrame, UserStartedSpeakingFrame) bypass queues: high priority
```

Architecture

```
  LiveKit SFU / SIP ◄──► AgentServer ──dispatch──► job: entrypoint(ctx)
                                                   └─ AgentSession
                                                       ├─ VAD (Silero) ─┐
                                                       ├─ STT ──────────┼─► audio EOT
                                                       │                │   TurnDetector
                                                       ├─ LLM / RealtimeModel ◄─ endpointing
                                                       ├─ TTS              (min/max delay)
                                                       └─ Agent A ──tool──► Agent B (handoff)
```

Moshi and Delayed Streams Modeling (Kyutai)

```
  user audio ─► Mimi enc (12.5 Hz) ─┐
                                    ├─► Temporal Transformer (7B) ─► Depth Transformer
  Moshi audio tokens (own stream) ──┘          │ text "inner monologue"   │ per-codebook
                                               ▼                          ▼
                                     aligned text tokens        Mimi dec ─► speech
```

18. Agent System Design Interview Playbook: Coding, Customer Support, Research and Browser Agents

Step 3: Draw the universal skeleton, then specialize

```
          +-------------------- control plane -------------------------+
trigger -->| auth, quotas, budgets, routing, versioning, rollout (canary) |
          +-----------------------------+-----------------------------+
                                        |
  +------------------+          +---------v---------+         +------------------+
  | session log      |<-------->| harness (brain)   |-------->| model gateway    |
  | append-only      | events   | loop, context     |  calls  | routing,failover |
  | durable, resume  |          | policy, compaction|<--------| prefix cache     |
  +------------------+          +---------+---------+         +------------------+
                                          | tool calls
                    +---------------------+----------------------+
                    |                     |                      |
           +-------v------+     +--------v--------+    +--------v---------+
           | policy gate  |     | tools / MCP     |    | sandbox (hands)  |
           | allow, deny, |---->| actionable      |    | fs + network     |
           | ask human    |     | errors, paging  |    | isolation, creds |
           +--------------+     +-----------------+    | via proxy        |
                                                       +------------------+
    observability: traces, transcripts, cost per task, eval hooks, monitors
```

Architecture

```
issue / Slack / CLI
      |
      v
[context hydration: fetch linked docs/tickets via MCP, deterministic]   (Stripe)
      |
      v
+--------------------------- sandbox: devbox / microVM ----------------------+
|  warm pool (~10 s Stripe, <5 s Ramp snapshots); repo cloned, caches warm     |
|                                                                              |
|   harness loop:  gather context --> act (edit) --> verify (lint/test) --+    |
|        ^                                                                |    |
|        +------------------------- feedback ---------------------------+    |
|   tools: read/view, grep/glob, str_replace edit, bash, LSP diagnostics       |
|   memory: AGENTS.md / CLAUDE.md rules scoped per directory                   |
|   network: egress via allowlist proxy; git creds held by proxy, not agent    |
+------------------------------------------------------------------------------+
      |
      v
[CI: selected tests] --fail--> agent fix (max 2 rounds) --> PR --> human review
```

Architecture

```
chat | email | voice (VAD -> STT ... TTS)
           |
           v
+----------------------+     in parallel      +--------------------------+
| input supervisor     |--------------------->| abuse/injection/off-topic|
| (small model)        |                      | gates the reply release  |
+----------+-----------+                      +--------------------------+
           v
[router: single-token SLM classifier -> procedure / FAQ / human]
           |
           v
+-------------------------------------------------------------+
| procedure layer (AOP / Journey / Agent Script)               |
|  NL steps + code conditions; deterministic identity gate     |
|  tools: get_customer, get_order, refund(amount<=policy) ...  |
|  RAG: rewrite -> embed -> retrieve -> rerank -> generate     |
+---------------------------+---------------------------------+
                            v
[output supervisor: grounding, policy, brand] --ok--> reply
                            |
                            +--fail/low confidence--> handoff (summary -> queue)
traces --> monitors, LLM-judge QA, simulations, versioned releases, canary
```

Architecture

```
user query
   |
   v
[clarifier / scoping: questions or editable plan] --> research brief
   |
   v
+------------------------- lead researcher -------------------------+
| plan -> saved to external memory (context > 200K would truncate)    |
| effort rules: simple 1 agent / 3-10 calls; compare 2-4 subagents;   |
|               complex 10+ subagents                                 |
+-----+-----------------+-----------------+---------------------+---+
      | spawn 3-5 in parallel, each with objective, format, tools, bounds
  +---v----+        +----v---+         +----v---+
  |subagent|        |subagent|   ...   |subagent|  own context; search->read->reflect
  +---+----+        +----+---+         +----+---+  write big outputs to files
      | 1-2K-token summaries or file references |
      +----------------+------------------------+
                       v
          lead synthesizes; loop if gaps remain
                       v
          [citation agent: map claims to sources, text unchanged]
                       v
                report (background job; resumable stream)
```

Architecture

```
task --> planner/actor loop ------------------------------------------+
           | observe: URL, tabs, indexed elements (+ screenshot)       |
           | act: click(id) | type | scroll | navigate | extract       |
           v                                                           |
  +------------------------ isolated cloud browser ---------------+   |
  | per-session profile; domain allowlist enforced at network layer|   |
  | re-checked after redirects; secrets typed by user (takeover)   |   |
  +-----------------------------------------------------------------+   |
           |                                                           |
  guards: injection classifier on page content; confirm before submit/ |
          purchase/send; blocked site categories; step budget; watch --+
  context: keep last ~3 screenshots, prune in batches, compact ~150K
  done: only after verifying every requirement against tool output
```